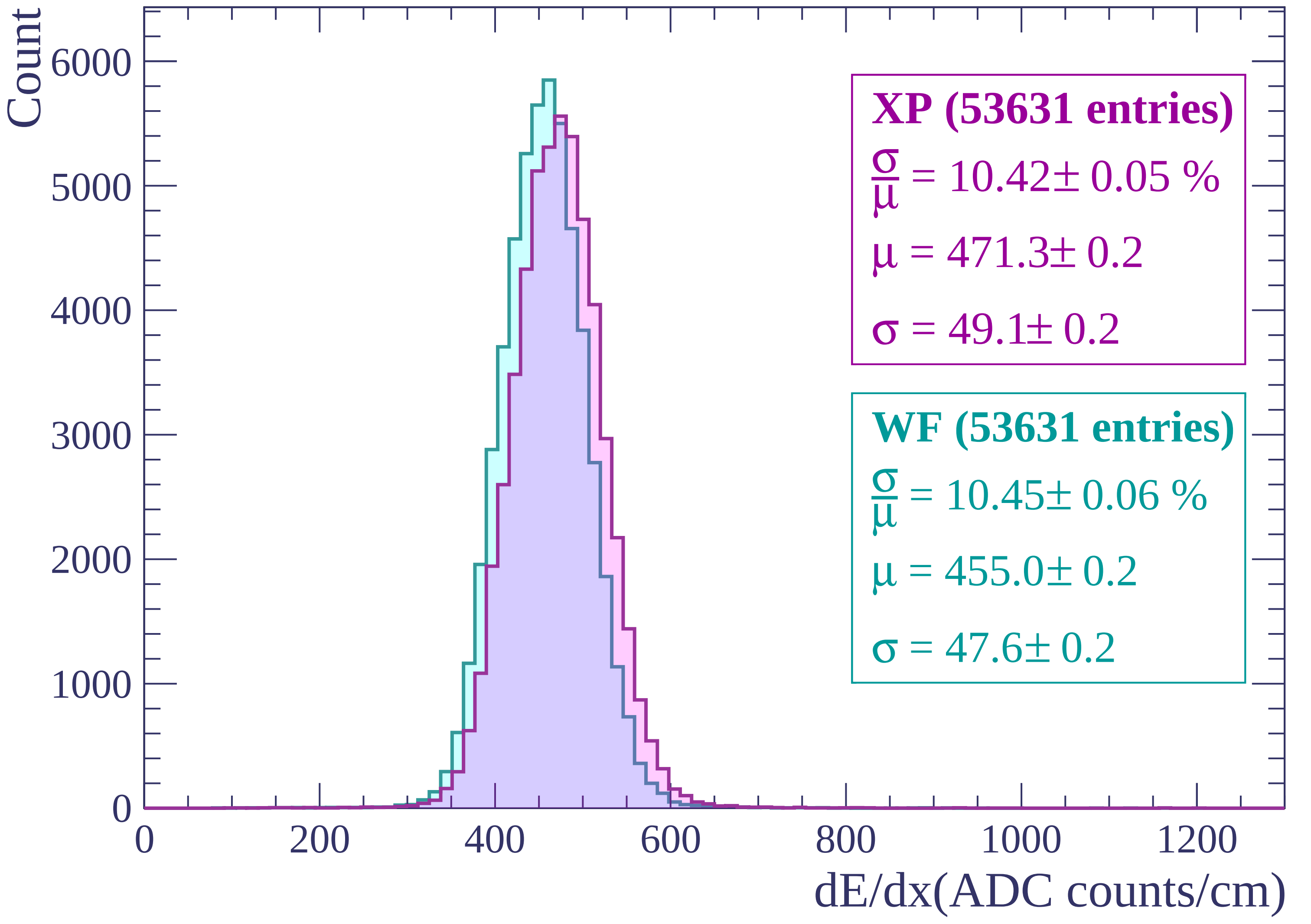
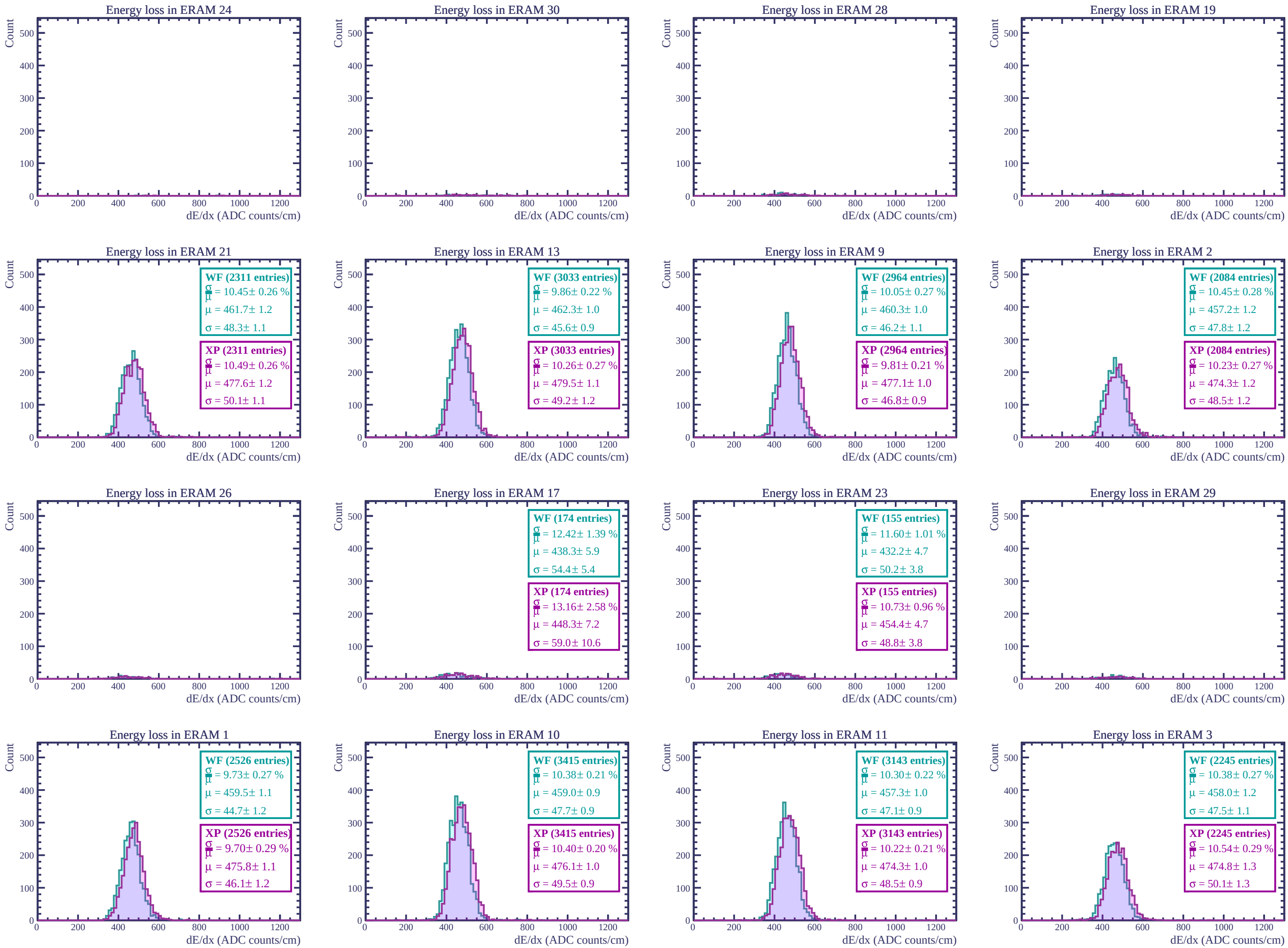
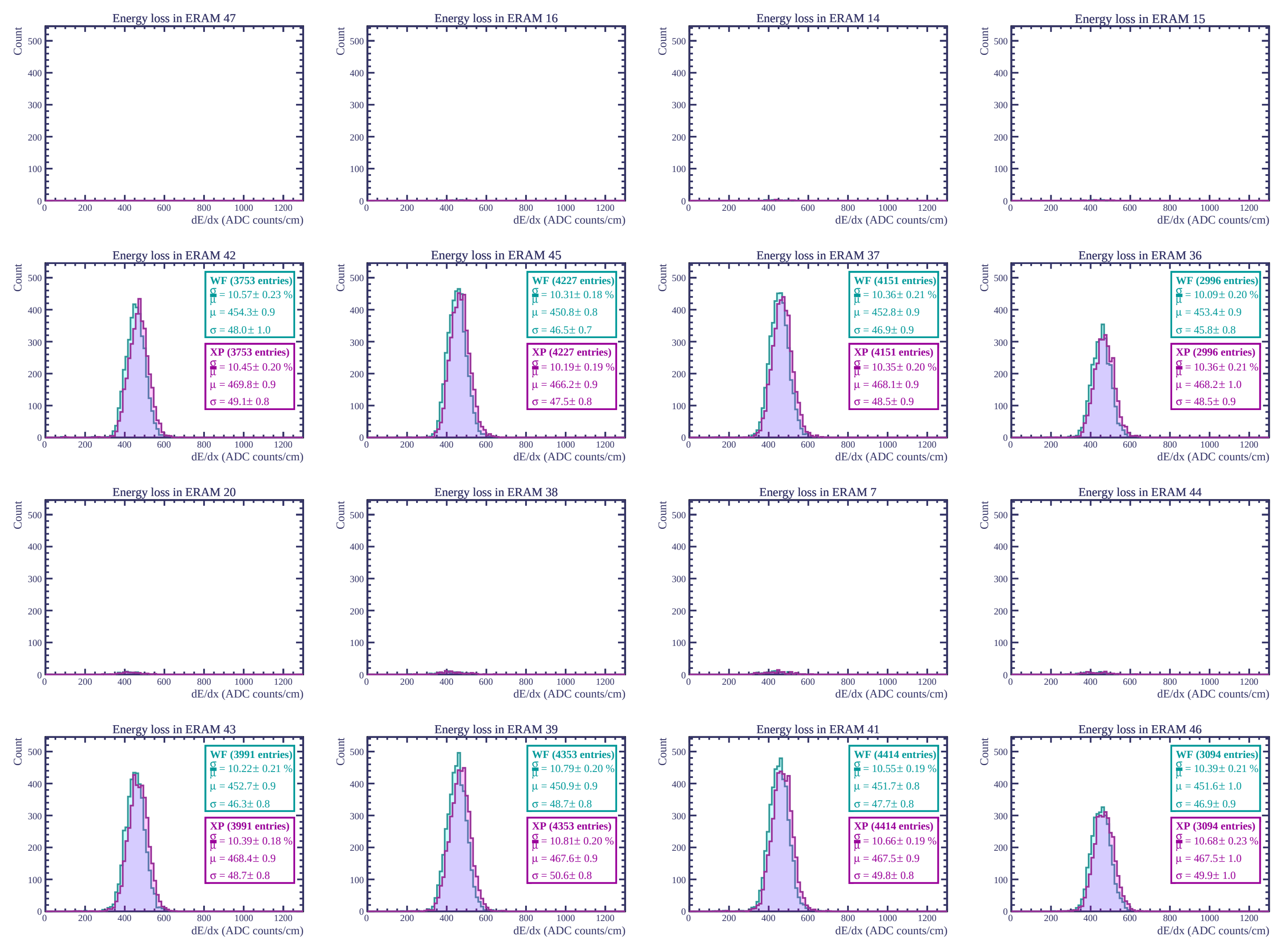
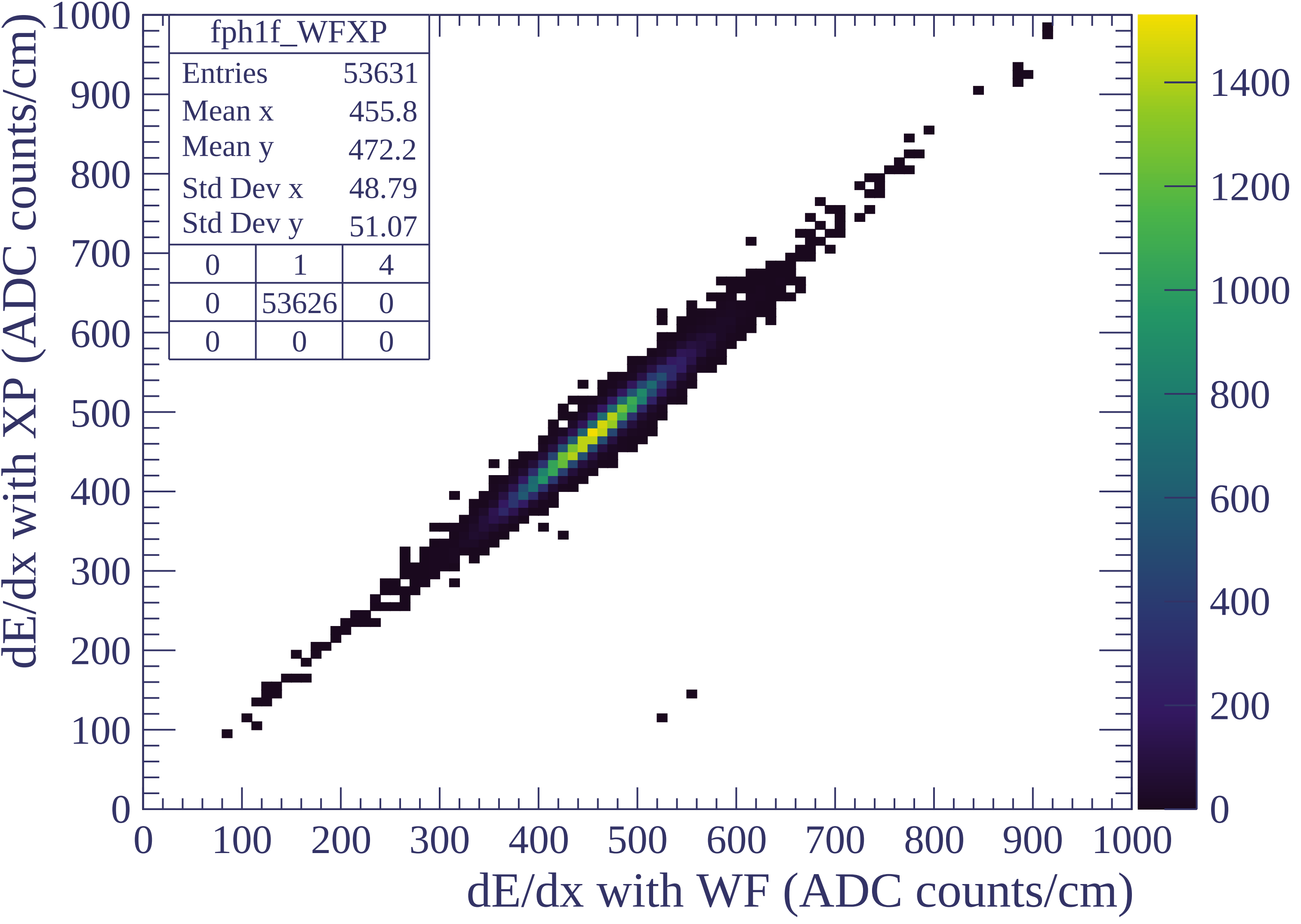
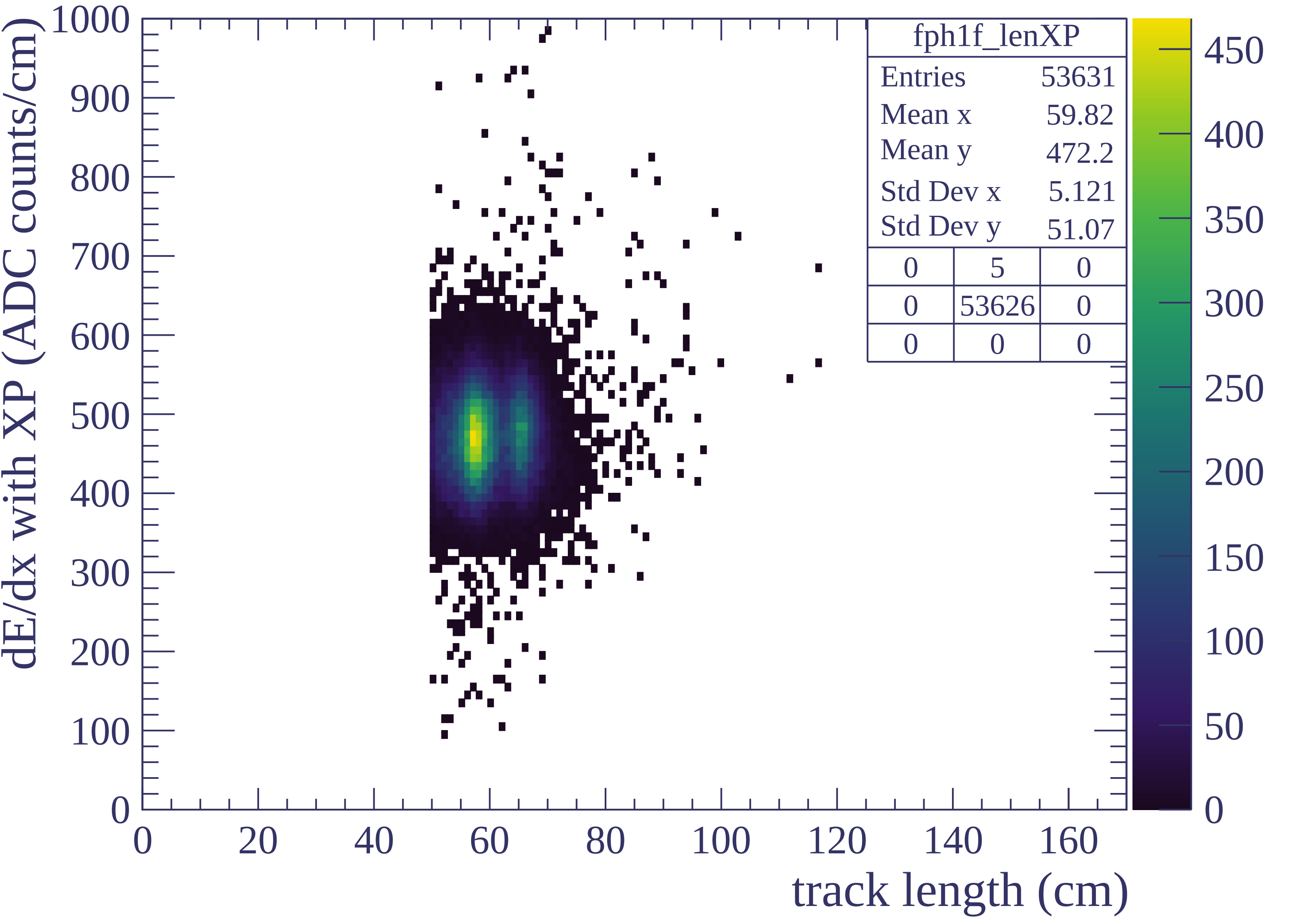


Count
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (53631 entries)
$\frac{\sigma}{\mu}$ = 10.42± 0.05 %
μ = 471.3± 0.2
σ = 49.1± 0.2
WF (53631 entries)
$\frac{\sigma}{\mu}$ = 10.45± 0.06 %
μ = 455.0± 0.2
σ = 47.6± 0.2

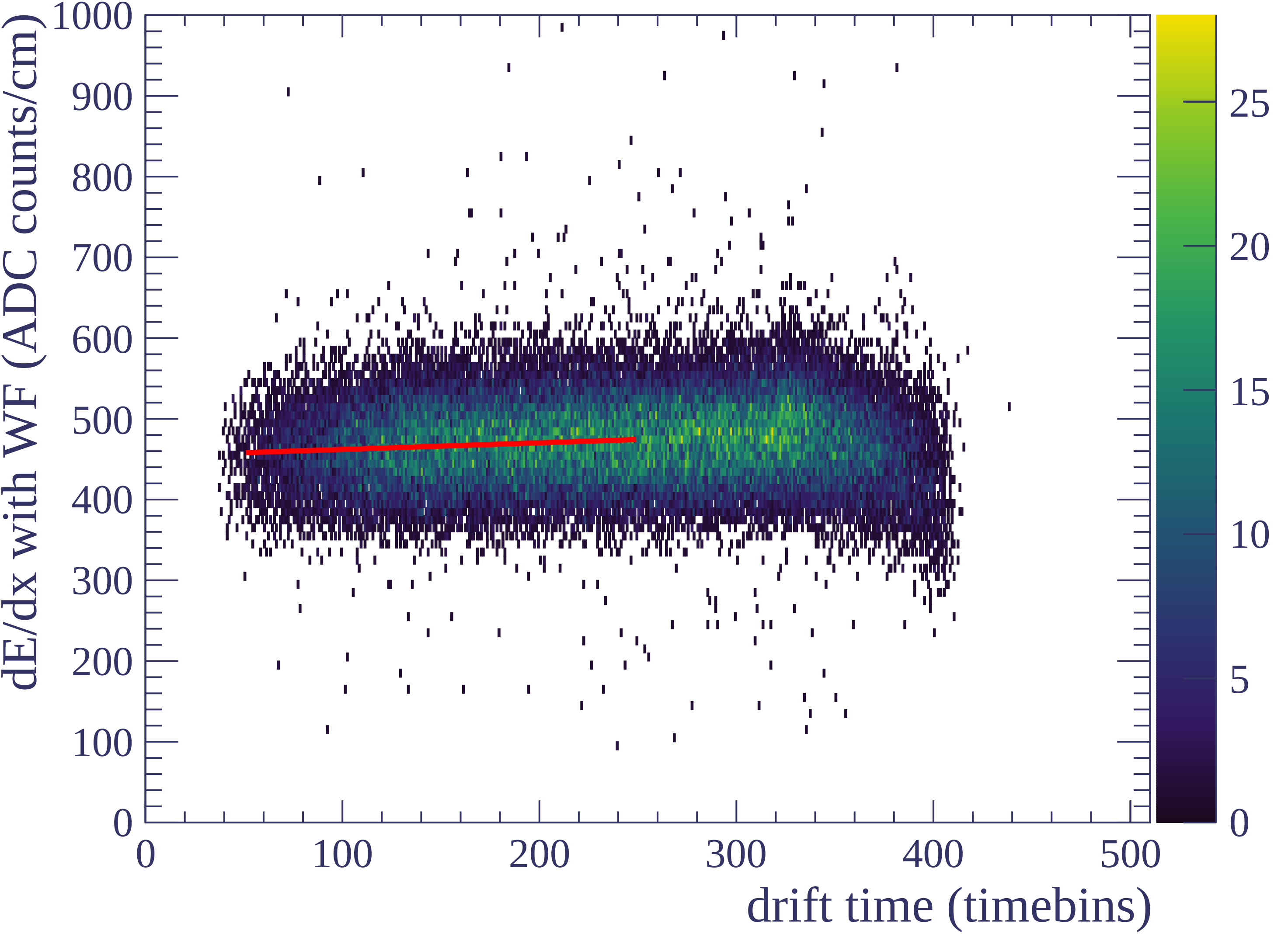

Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
WF (2311 entries)
σ/μ = 10.45± 0.26 %
μ = 461.7± 1.2
σ = 48.3± 1.1
XP (2311 entries)
σ/μ = 10.49± 0.26 %
μ = 477.6± 1.2
σ = 50.1± 1.1
Energy loss in ERAM 13
WF (3033 entries)
σ/μ = 9.86± 0.22 %
μ = 462.3± 1.0
σ = 45.6± 0.9
XP (3033 entries)
σ/μ = 10.26± 0.27 %
μ = 479.5± 1.1
σ = 49.2± 1.2
Energy loss in ERAM 9
WF (2964 entries)
σ/μ = 10.05± 0.27 %
μ = 460.3± 1.0
σ = 46.2± 1.1
XP (2964 entries)
σ/μ = 9.81± 0.21 %
μ = 477.1± 1.0
σ = 46.8± 0.9
Energy loss in ERAM 2
WF (2084 entries)
σ/μ = 10.45± 0.28 %
μ = 457.2± 1.2
σ = 47.8± 1.2
XP (2084 entries)
σ/μ = 10.23± 0.27 %
μ = 474.3± 1.2
σ = 48.5± 1.2
Energy loss in ERAM 26
Energy loss in ERAM 17
WF (174 entries)
σ/μ = 12.42± 1.39 %
μ = 438.3± 5.9
σ = 54.4± 5.4
XP (174 entries)
σ/μ = 13.16± 2.58 %
μ = 448.3± 7.2
σ = 59.0± 10.6
Energy loss in ERAM 23
WF (155 entries)
σ/μ = 11.60± 1.01 %
μ = 432.2± 4.7
σ = 50.2± 3.8
XP (155 entries)
σ/μ = 10.73± 0.96 %
μ = 454.4± 4.7
σ = 48.8± 3.8
Energy loss in ERAM 29
Energy loss in ERAM 1
WF (2526 entries)
σ/μ = 9.73± 0.27 %
μ = 459.5± 1.1
σ = 44.7± 1.2
XP (2526 entries)
σ/μ = 9.70± 0.29 %
μ = 475.8± 1.1
σ = 46.1± 1.2
Energy loss in ERAM 10
WF (3415 entries)
σ/μ = 10.38± 0.21 %
μ = 459.0± 0.9
σ = 47.7± 0.9
XP (3415 entries)
σ/μ = 10.40± 0.20 %
μ = 476.1± 1.0
σ = 49.5± 0.9
Energy loss in ERAM 11
WF (3143 entries)
σ/μ = 10.30± 0.22 %
μ = 457.3± 1.0
σ = 47.1± 0.9
XP (3143 entries)
σ/μ = 10.22± 0.21 %
μ = 474.3± 1.0
σ = 48.5± 0.9
Energy loss in ERAM 3
WF (2245 entries)
σ/μ = 10.38± 0.27 %
μ = 458.0± 1.2
σ = 47.5± 1.1
XP (2245 entries)
σ/μ = 10.54± 0.29 %
μ = 474.8± 1.3
σ = 50.1± 1.3
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
WF (3753 entries)
σ/μ = 10.57± 0.23 %
μ = 454.3± 0.9
σ = 48.0± 1.0
XP (3753 entries)
σ/μ = 10.45± 0.20 %
μ = 469.8± 0.9
σ = 49.1± 0.8
Energy loss in ERAM 45
WF (4227 entries)
σ/μ = 10.31± 0.18 %
μ = 450.8± 0.8
σ = 46.5± 0.7
XP (4227 entries)
σ/μ = 10.19± 0.19 %
μ = 466.2± 0.9
σ = 47.5± 0.8
Energy loss in ERAM 37
WF (4151 entries)
σ/μ = 10.36± 0.21 %
μ = 452.8± 0.9
σ = 46.9± 0.9
XP (4151 entries)
σ/μ = 10.35± 0.20 %
μ = 468.1± 0.9
σ = 48.5± 0.9
Energy loss in ERAM 36
WF (2996 entries)
σ/μ = 10.09± 0.20 %
μ = 453.4± 0.9
σ = 45.8± 0.8
XP (2996 entries)
σ/μ = 10.36± 0.21 %
μ = 468.2± 1.0
σ = 48.5± 0.9
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
Energy loss in ERAM 43
WF (3991 entries)
σ/μ = 10.22± 0.21 %
μ = 452.7± 0.9
σ = 46.3± 0.8
XP (3991 entries)
σ/μ = 10.39± 0.18 %
μ = 468.4± 0.9
σ = 48.7± 0.8
Energy loss in ERAM 39
WF (4353 entries)
σ/μ = 10.79± 0.20 %
μ = 450.9± 0.9
σ = 48.7± 0.8
XP (4353 entries)
σ/μ = 10.81± 0.20 %
μ = 467.6± 0.9
σ = 50.6± 0.8
Energy loss in ERAM 41
WF (4414 entries)
σ/μ = 10.55± 0.19 %
μ = 451.7± 0.8
σ = 47.7± 0.8
XP (4414 entries)
σ/μ = 10.66± 0.19 %
μ = 467.5± 0.9
σ = 49.8± 0.8
Energy loss in ERAM 46
WF (3094 entries)
σ/μ = 10.39± 0.21 %
μ = 451.6± 1.0
σ = 46.9± 0.9
XP (3094 entries)
σ/μ = 10.68± 0.23 %
μ = 467.5± 1.0
σ = 49.9± 1.0
Count
dE/dx (ADC counts/cm)

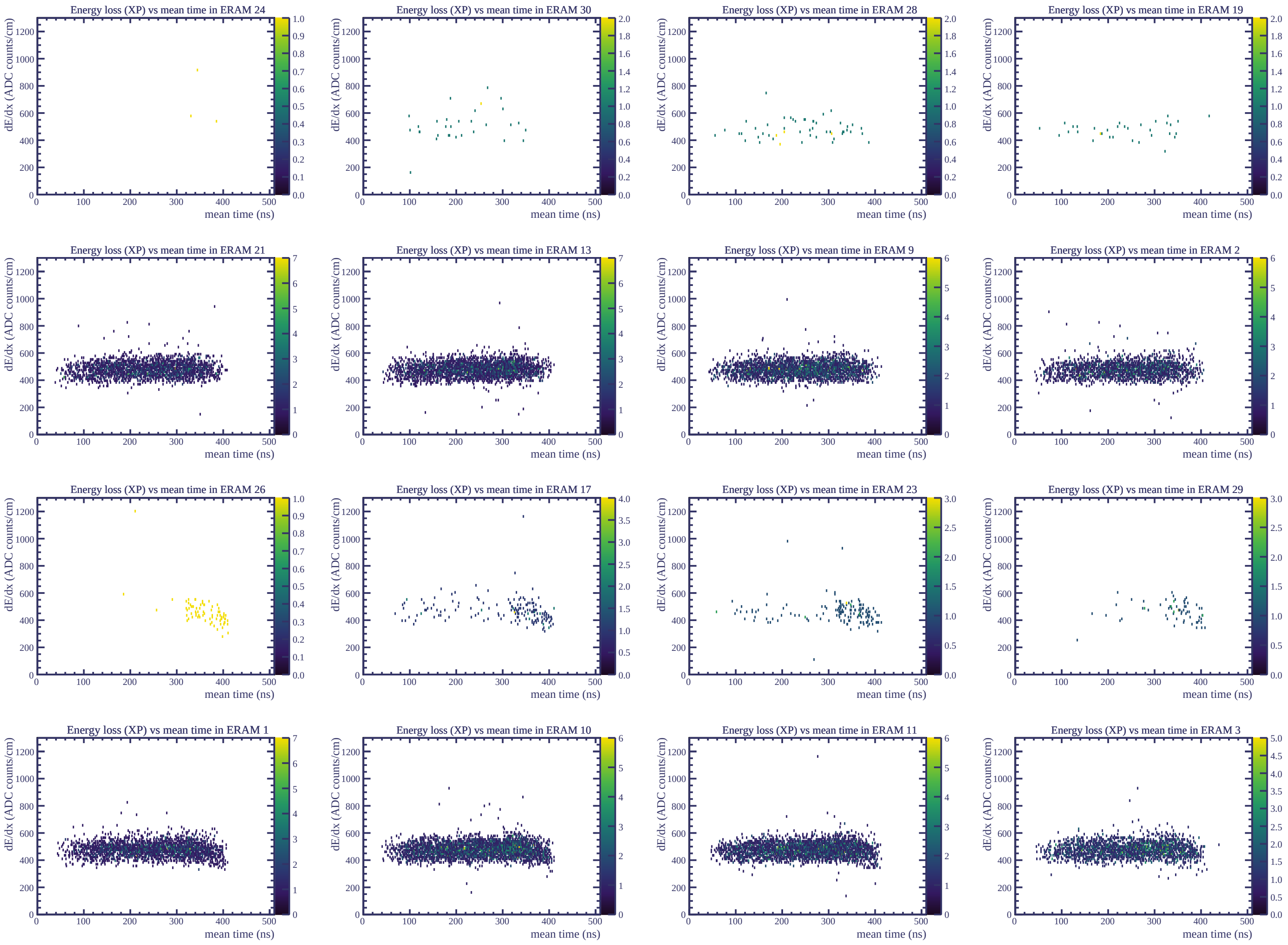

fph1f_WFXP
Entries 53631
Mean x 455.8
Mean y 472.2
Std Dev x 48.79
Std Dev y 51.07
0 1 4
0 53626 0
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400

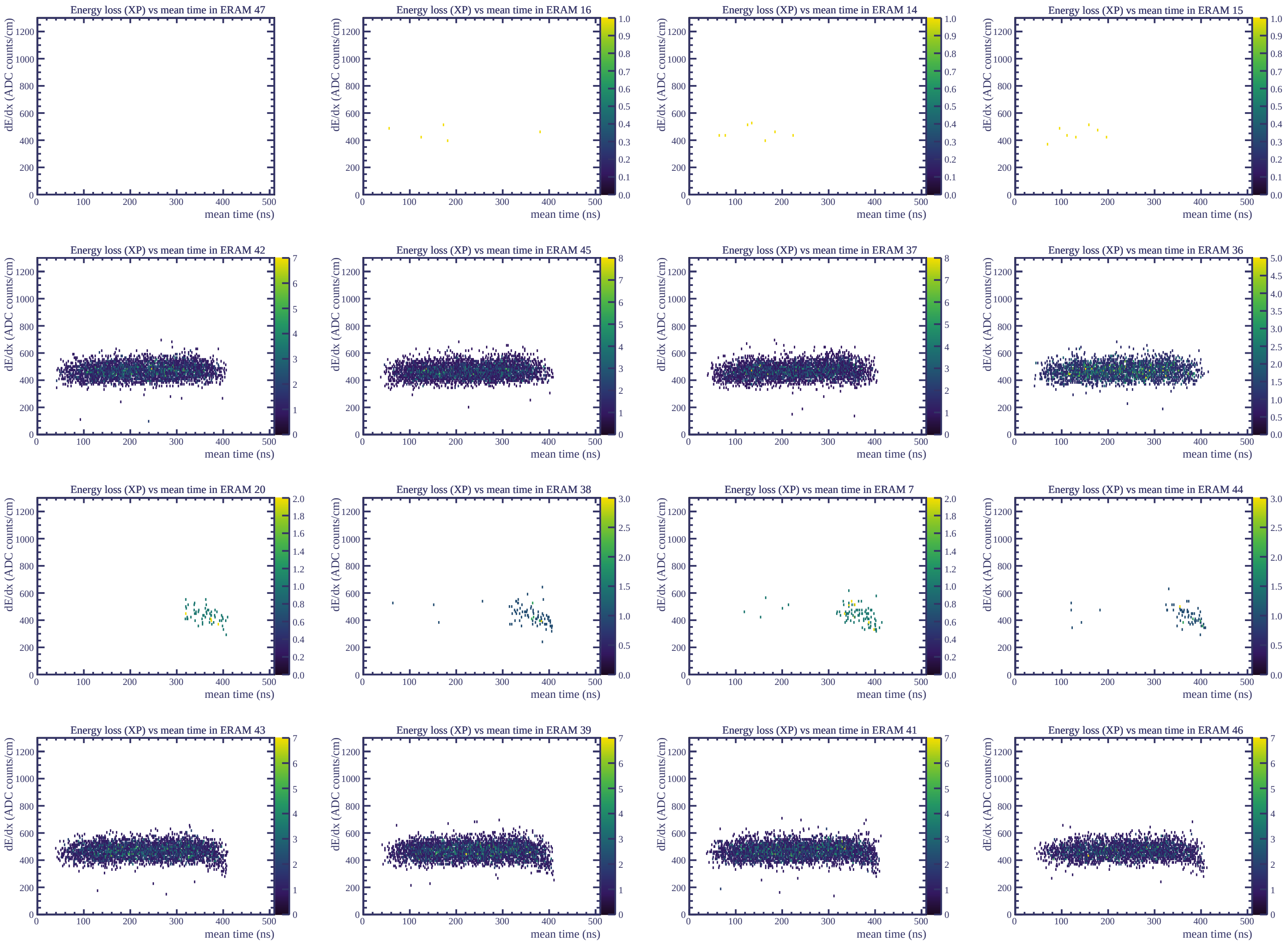


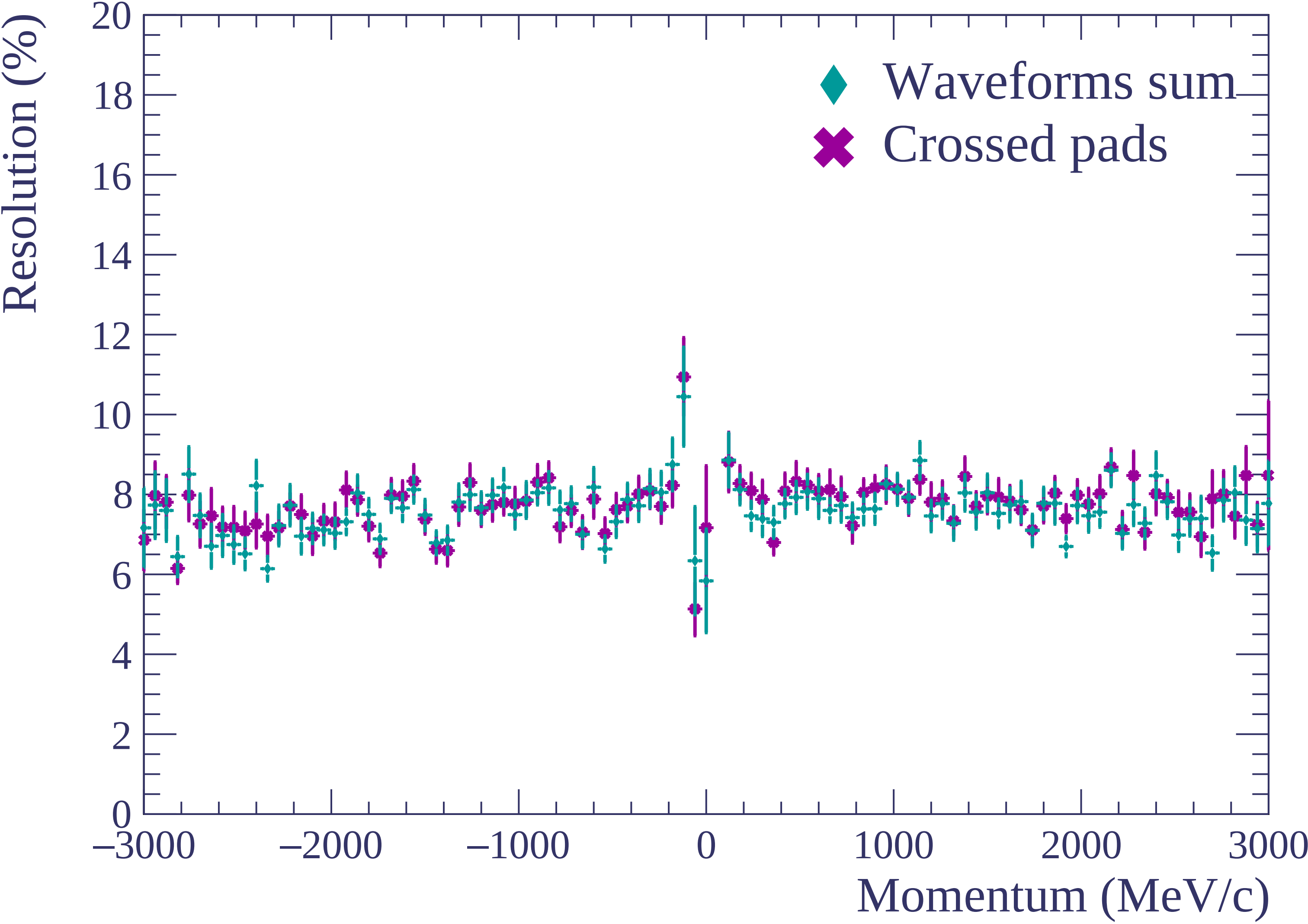
fph1f_lenXP
Entries 53631
Mean x 59.82
Mean y 472.2
Std Dev x 5.121
Std Dev y 51.07
0 5 0
0 53626 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
50
150
250
350
450

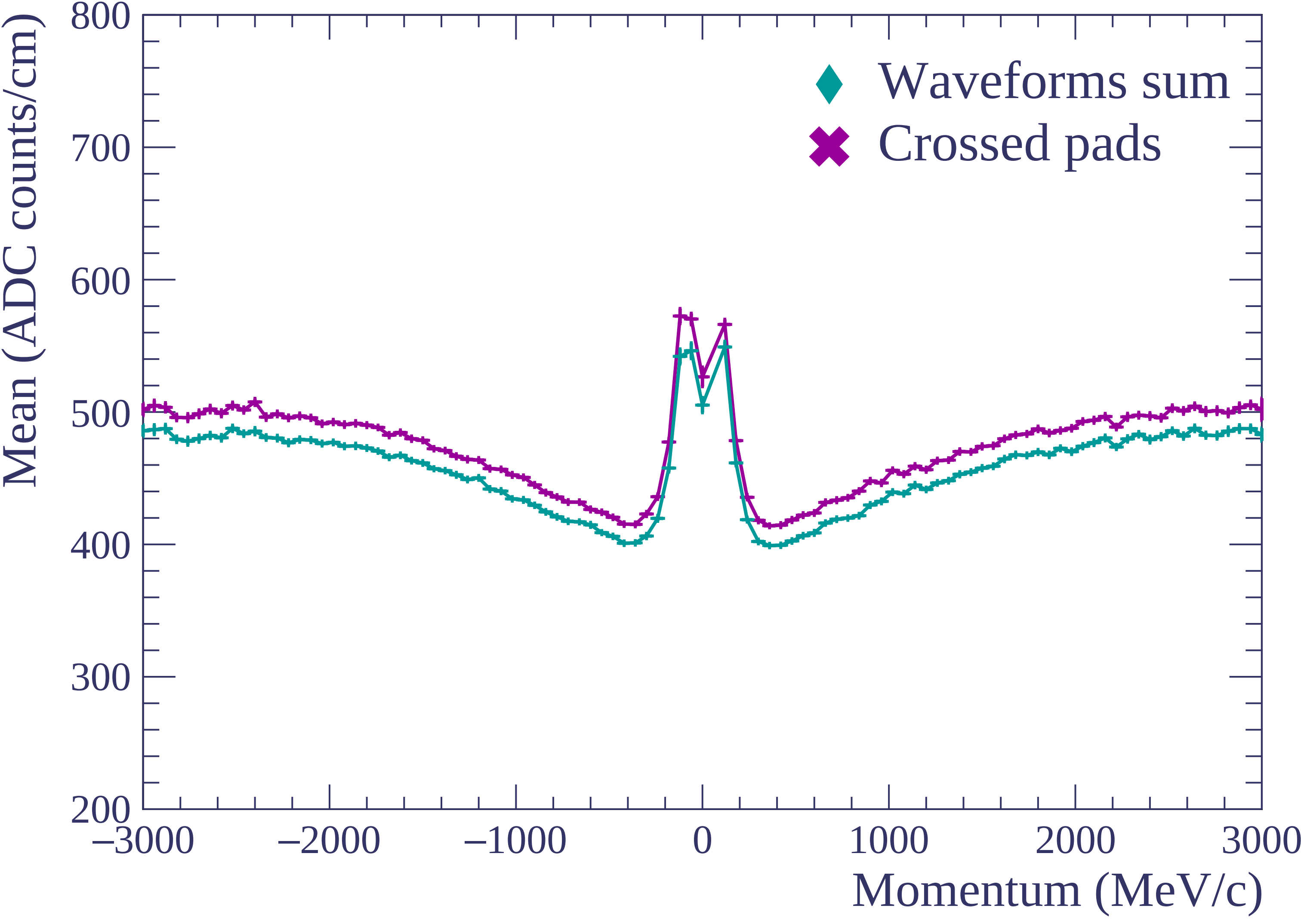


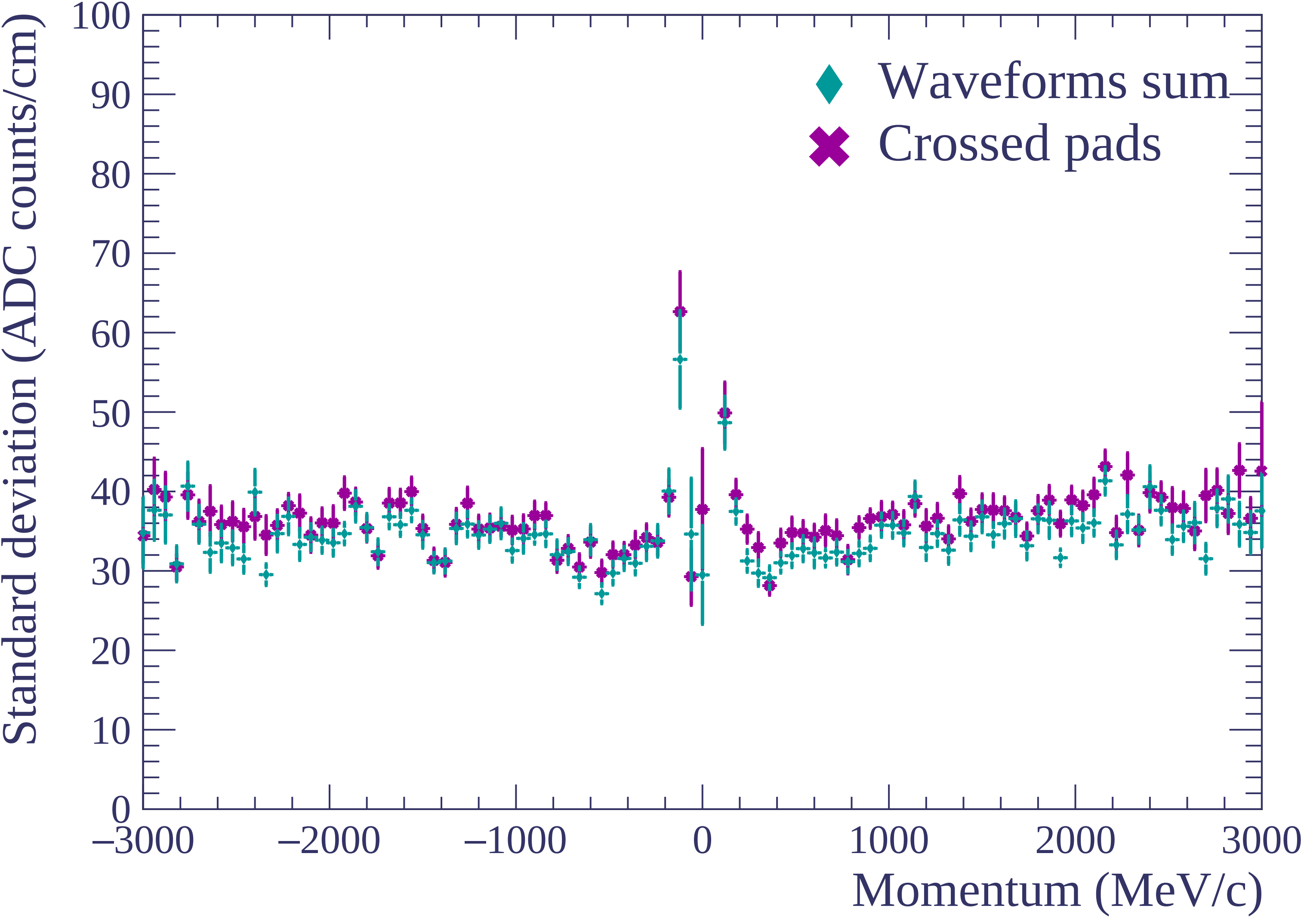
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
drift time (timebins)
25
20
15
10
5
0


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

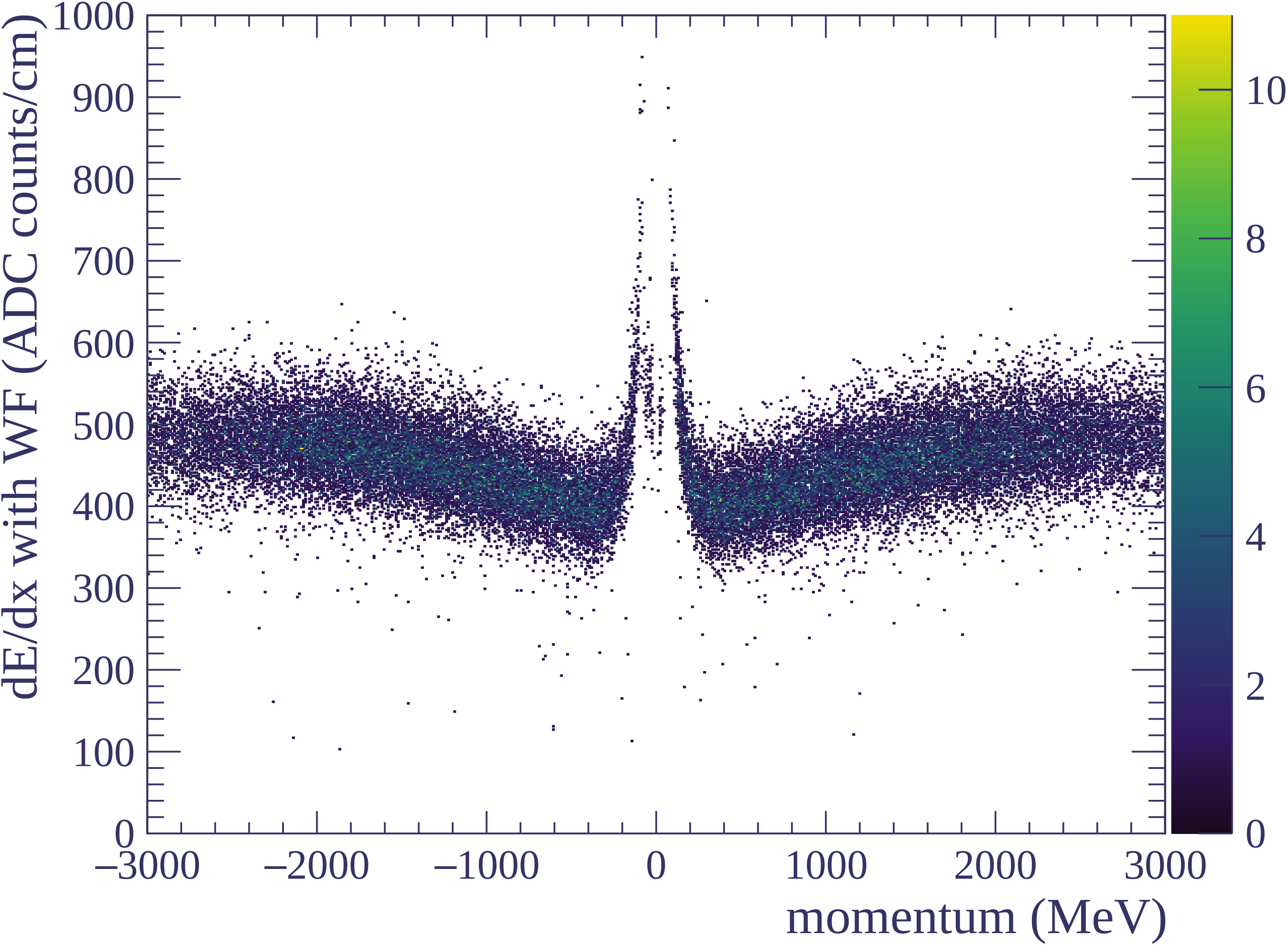

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

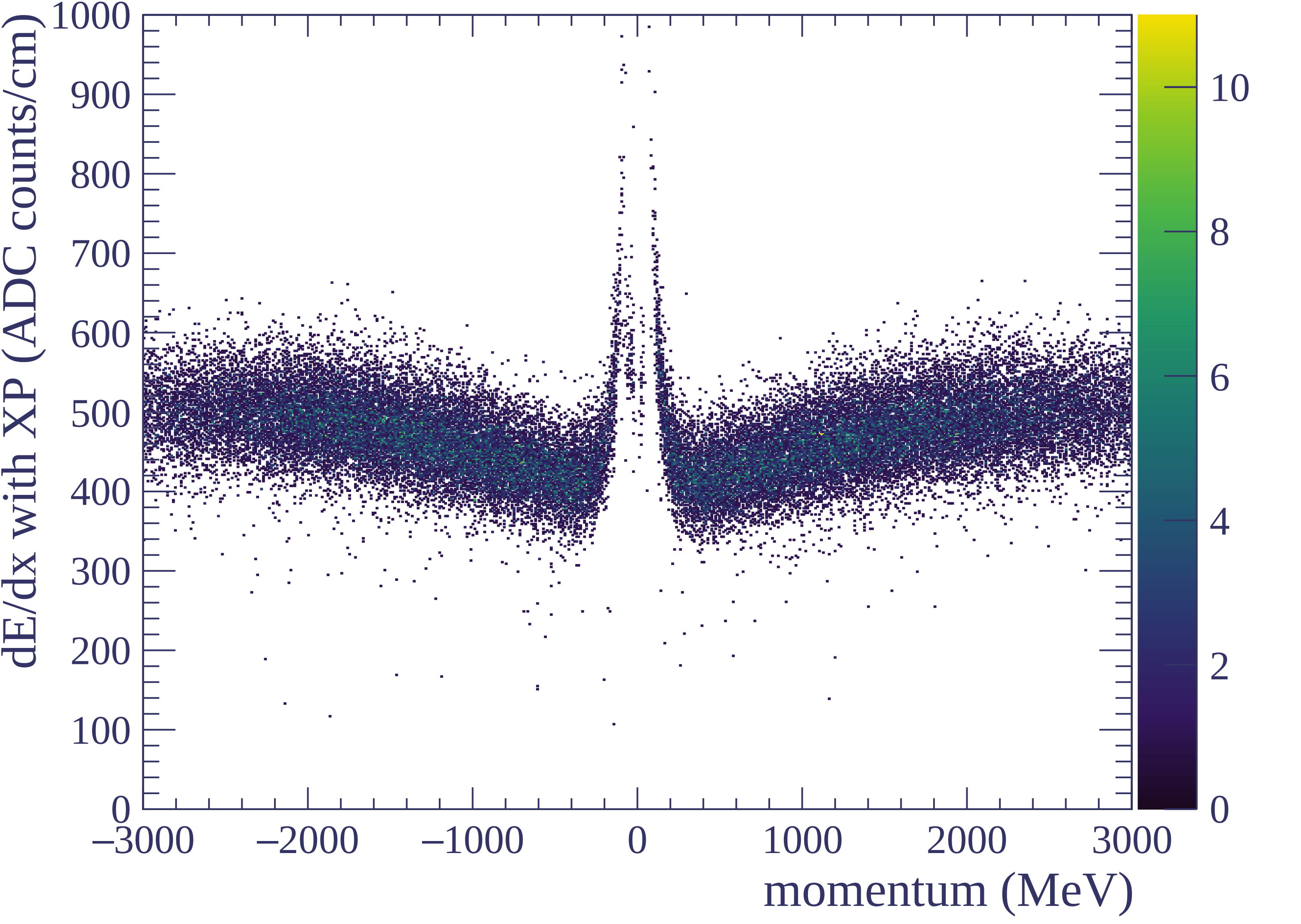

Resolution (%)
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

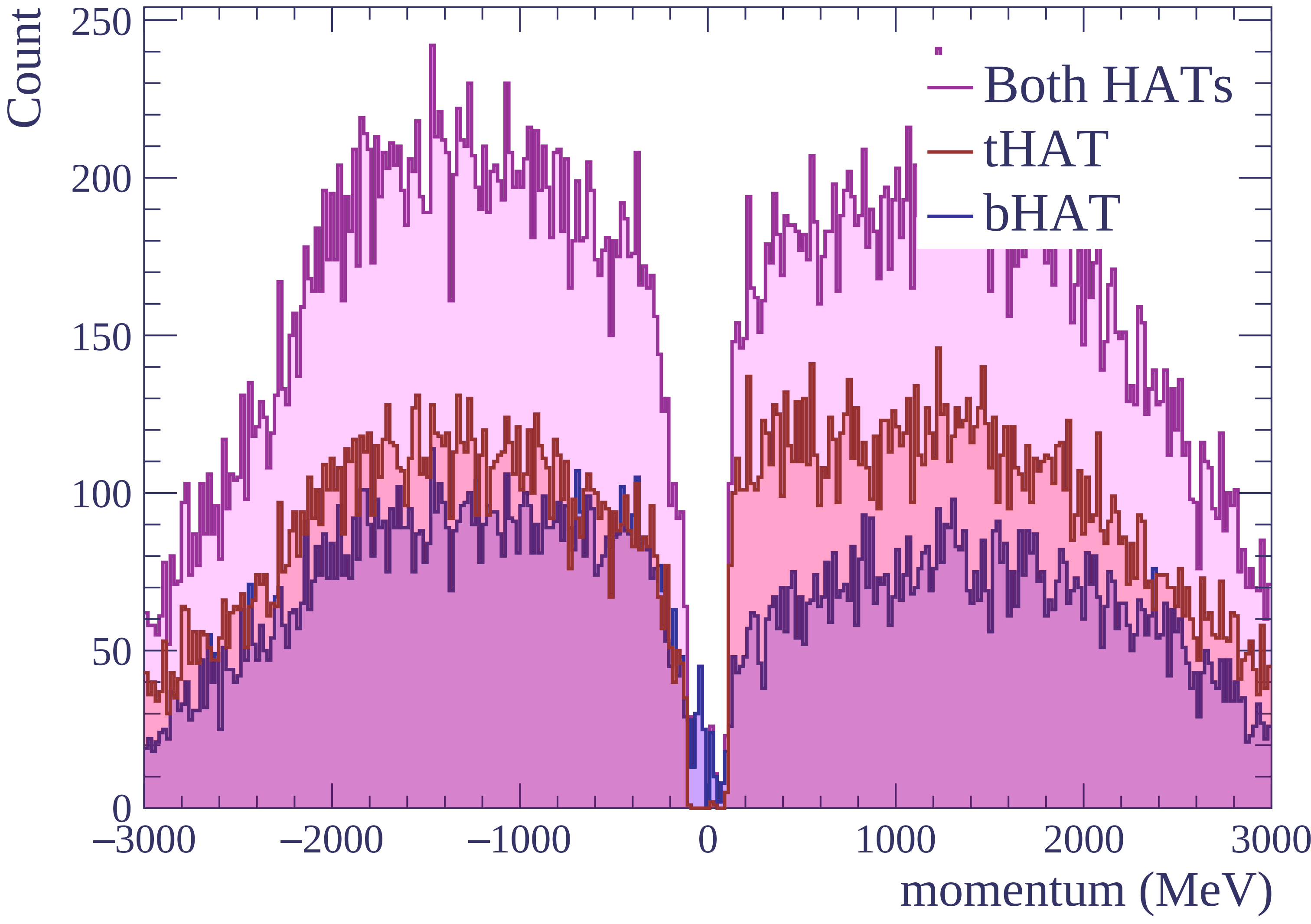


Standard deviation (ADC counts/cm)
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

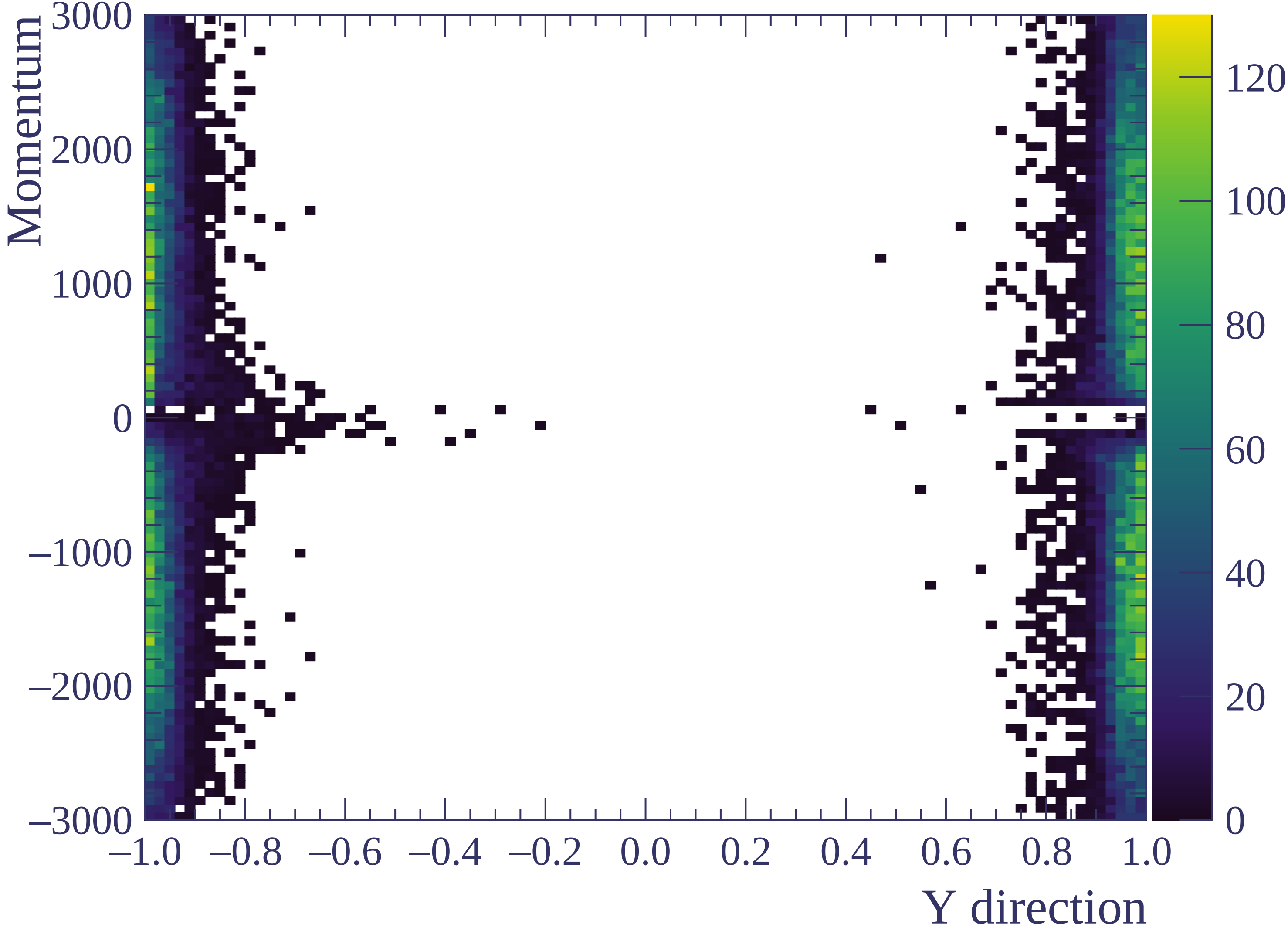

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
10
8
6
4
2
0

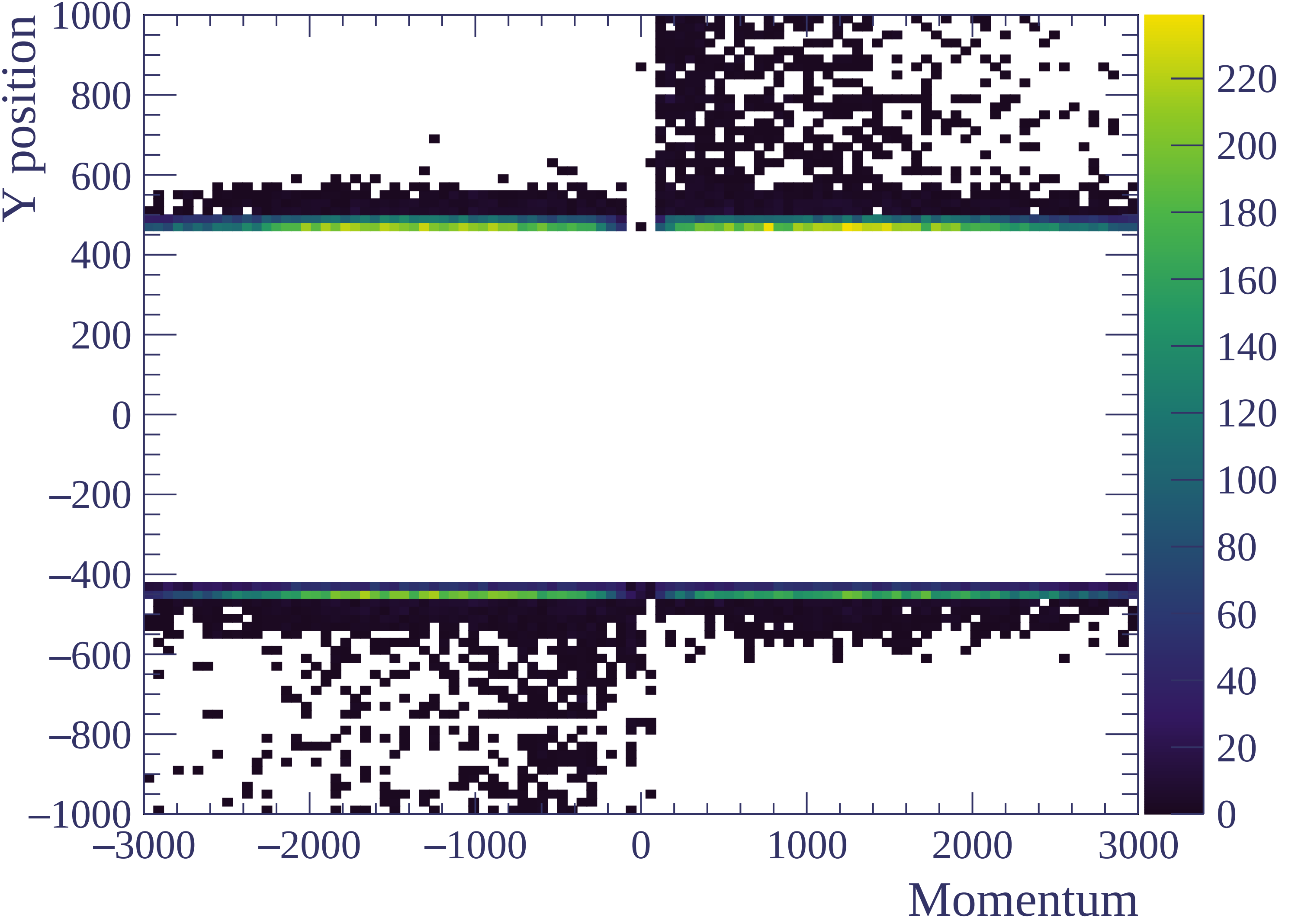


dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
2
4
6
8
10

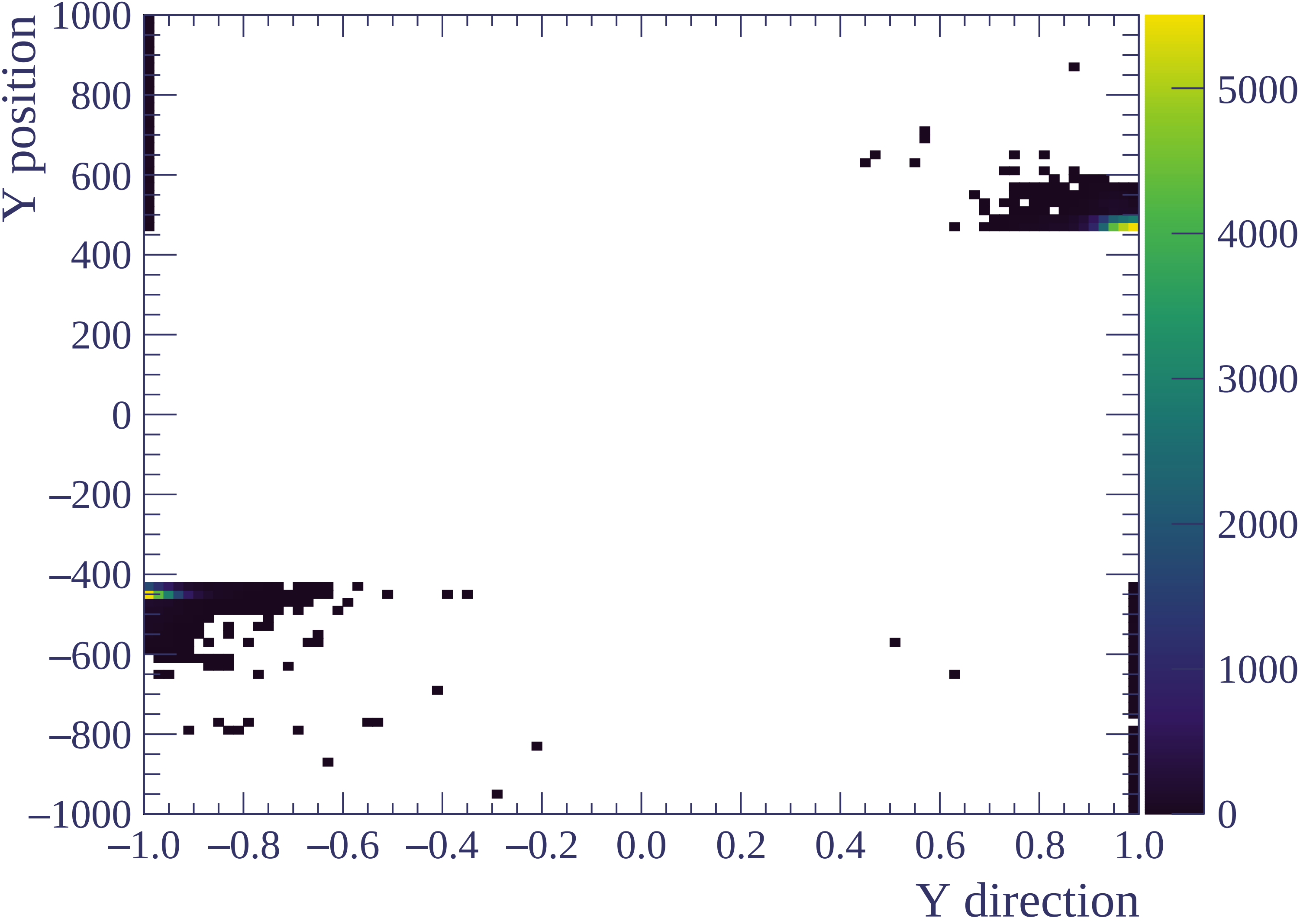


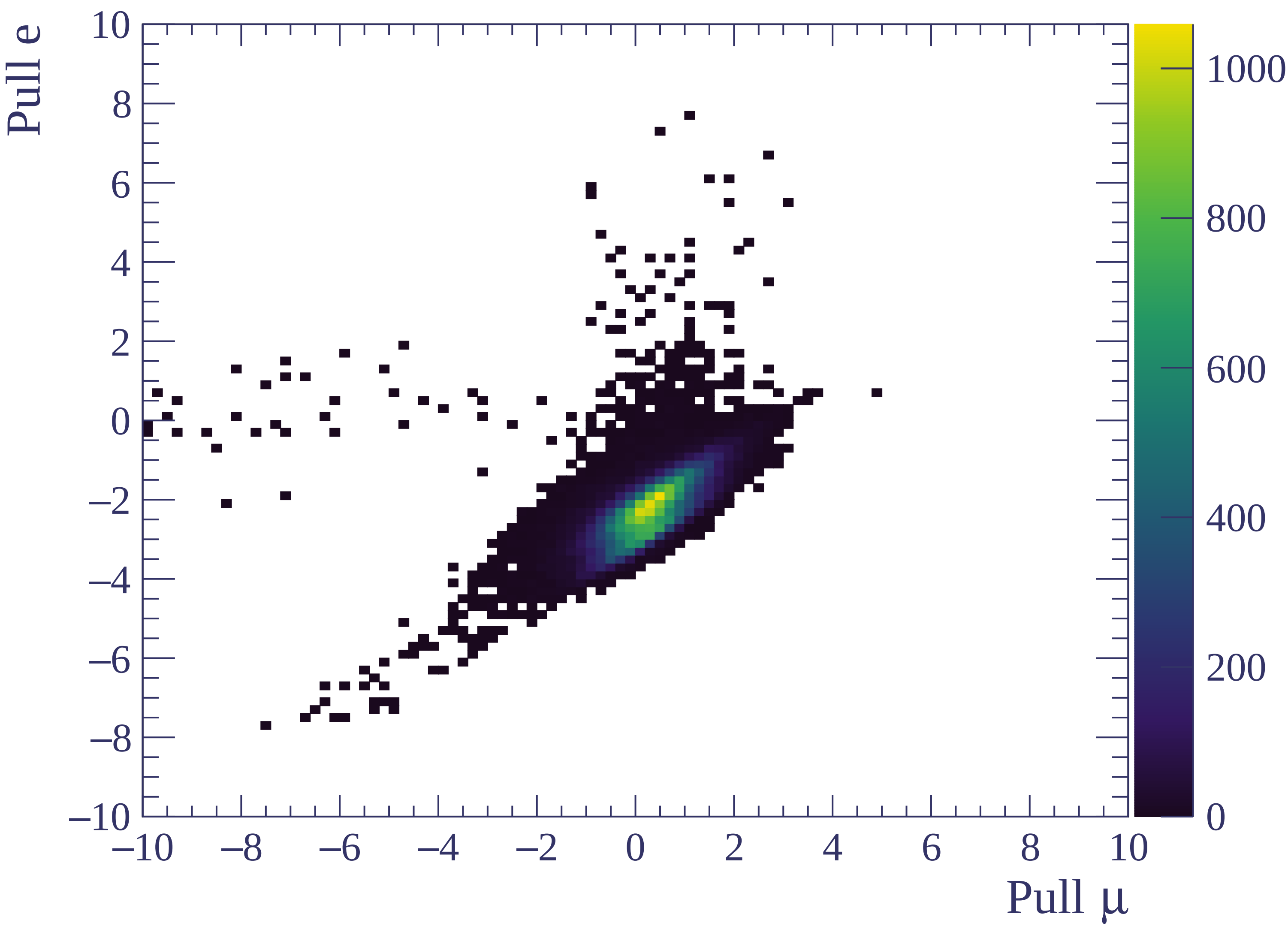
Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
50
100
150
200
250
−3000
−2000
−1000
0
1000
2000
3000

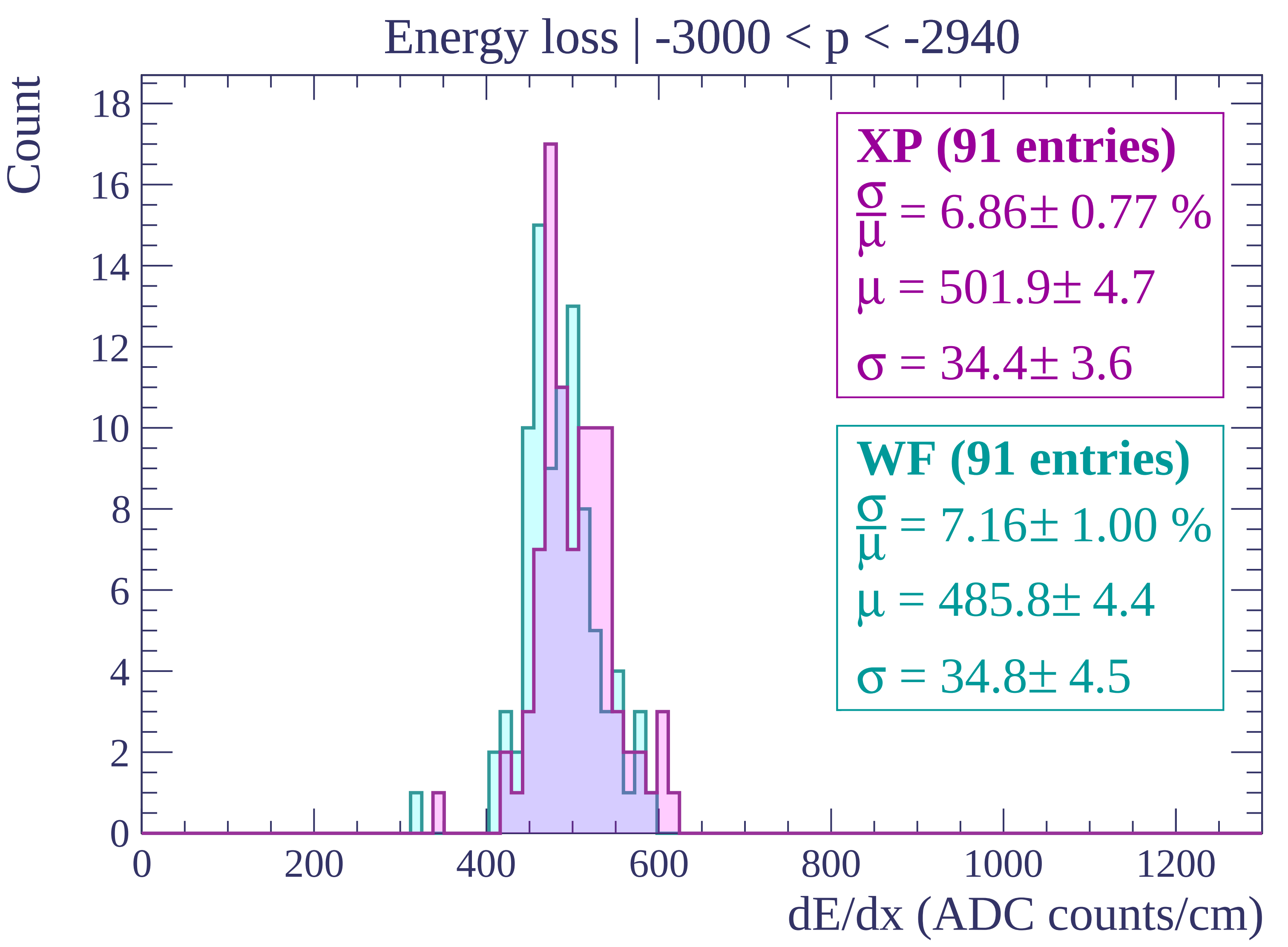

Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
120
100
80
60
40
20
0

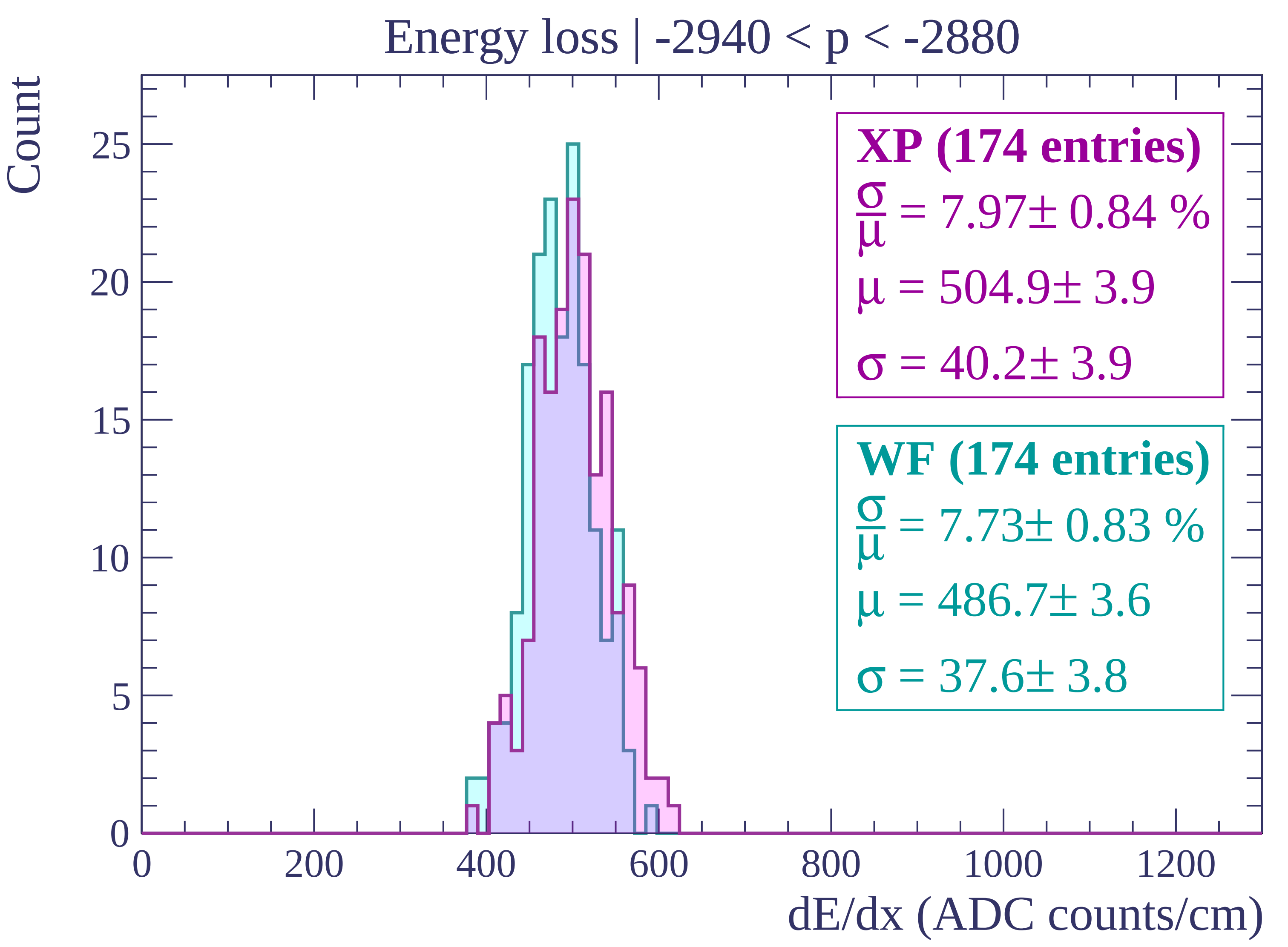


Y position
Momentum
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
220
200
180
160
140
120
100
80
60
40
20
0

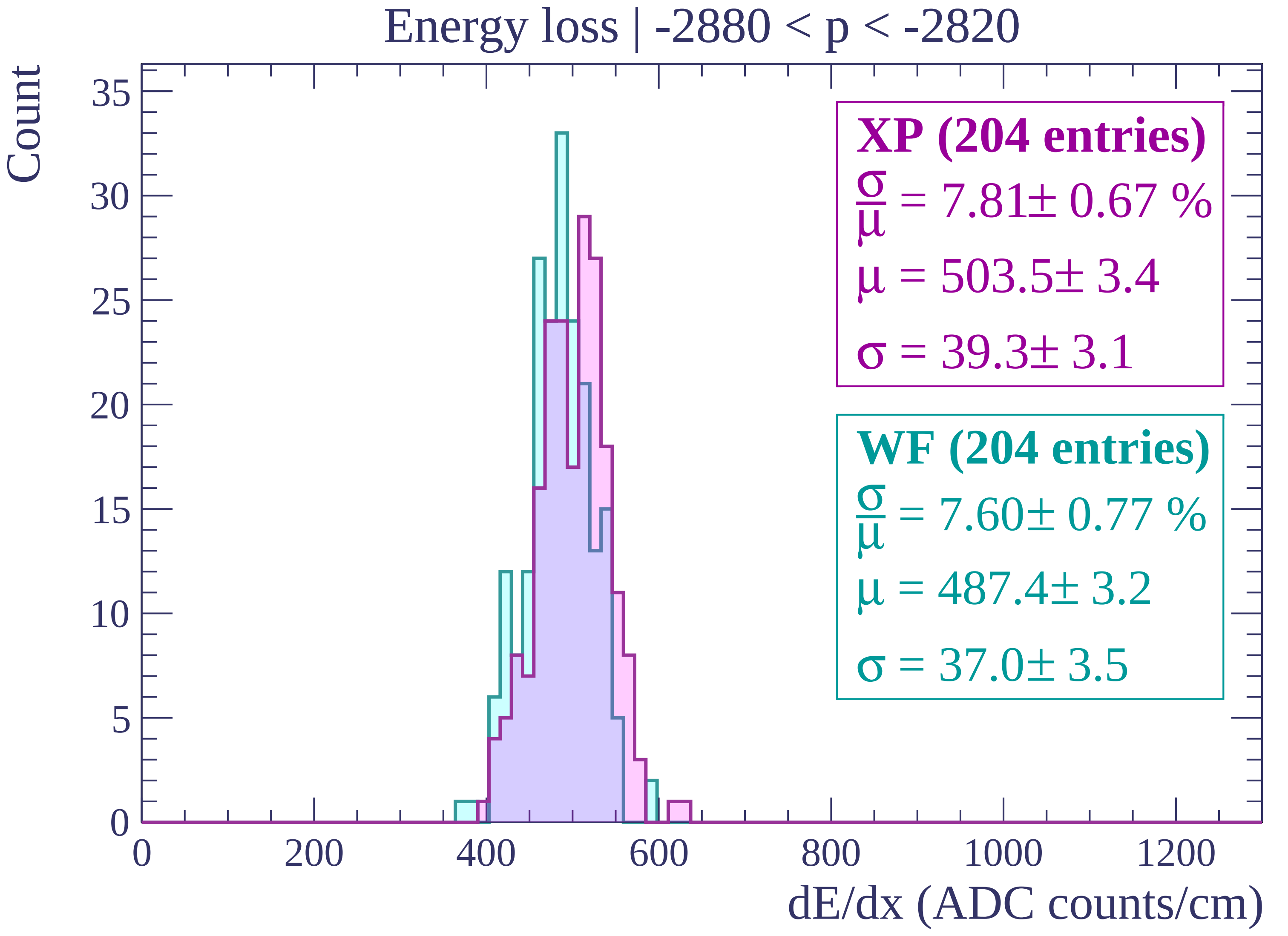

Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
5000
4000
3000
2000
1000
0


Pull e
10
8
6
4
2
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0
2
4
6
8
10
Pull μ
1000
800
600
400
200
0

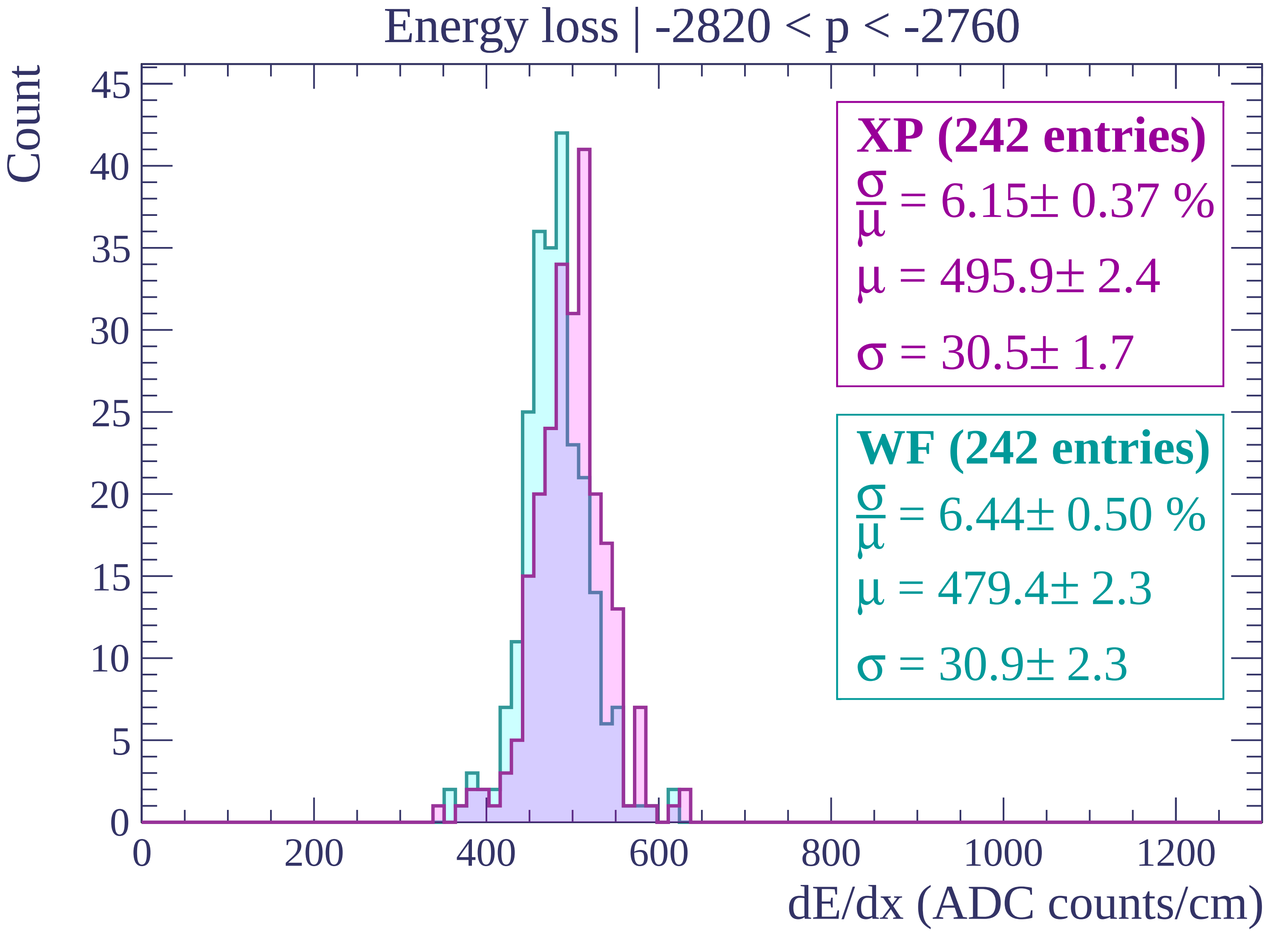

Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
XP (91 entries)
σ/μ = 6.86± 0.77 %
μ = 501.9± 4.7
σ = 34.4± 3.6
WF (91 entries)
σ/μ = 7.16± 1.00 %
μ = 485.8± 4.4
σ = 34.8± 4.5

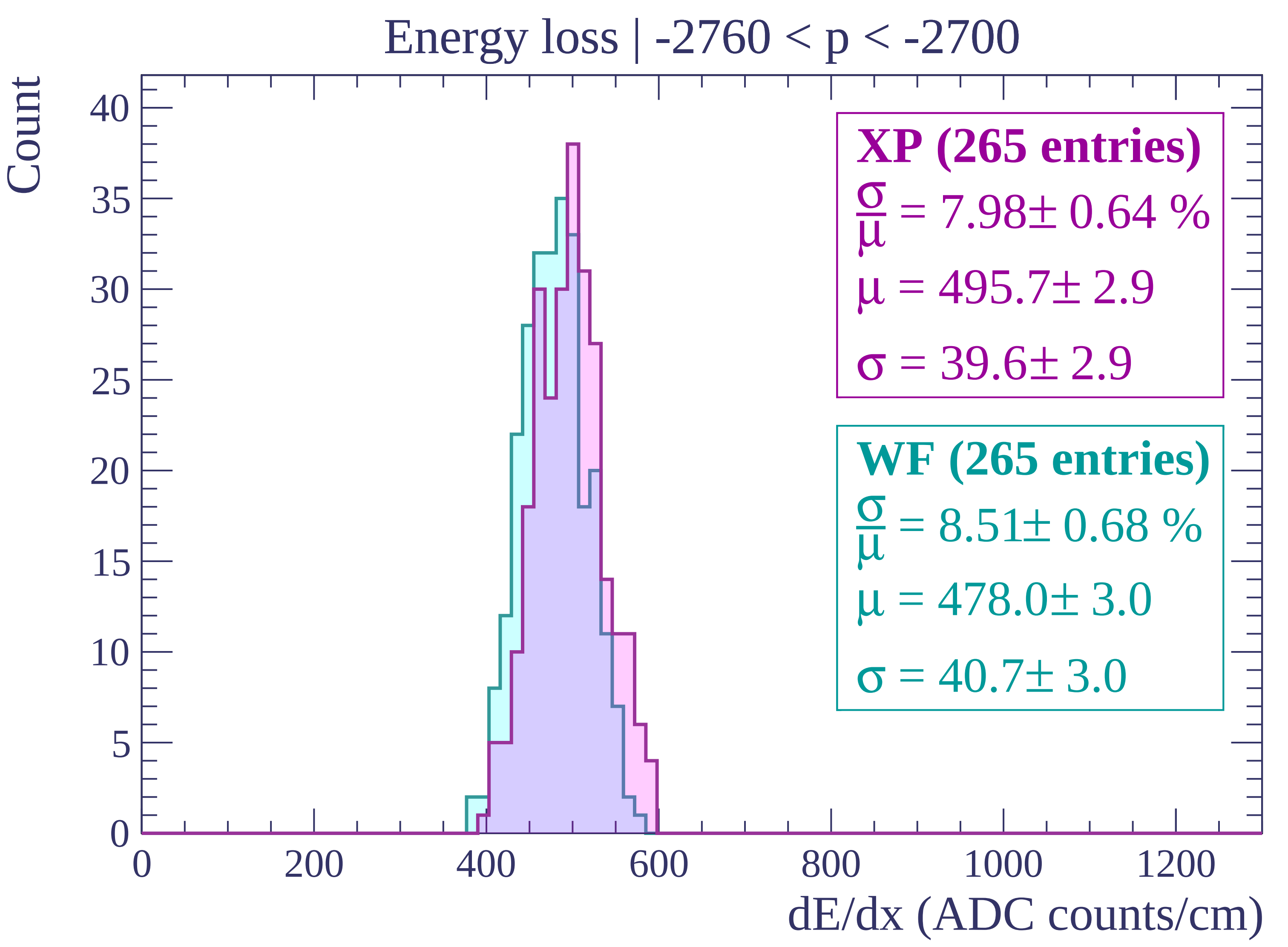


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP (174 entries)
σ/μ = 7.97± 0.84 %
μ = 504.9± 3.9
σ = 40.2± 3.9
WF (174 entries)
σ/μ = 7.73± 0.83 %
μ = 486.7± 3.6
σ = 37.6± 3.8

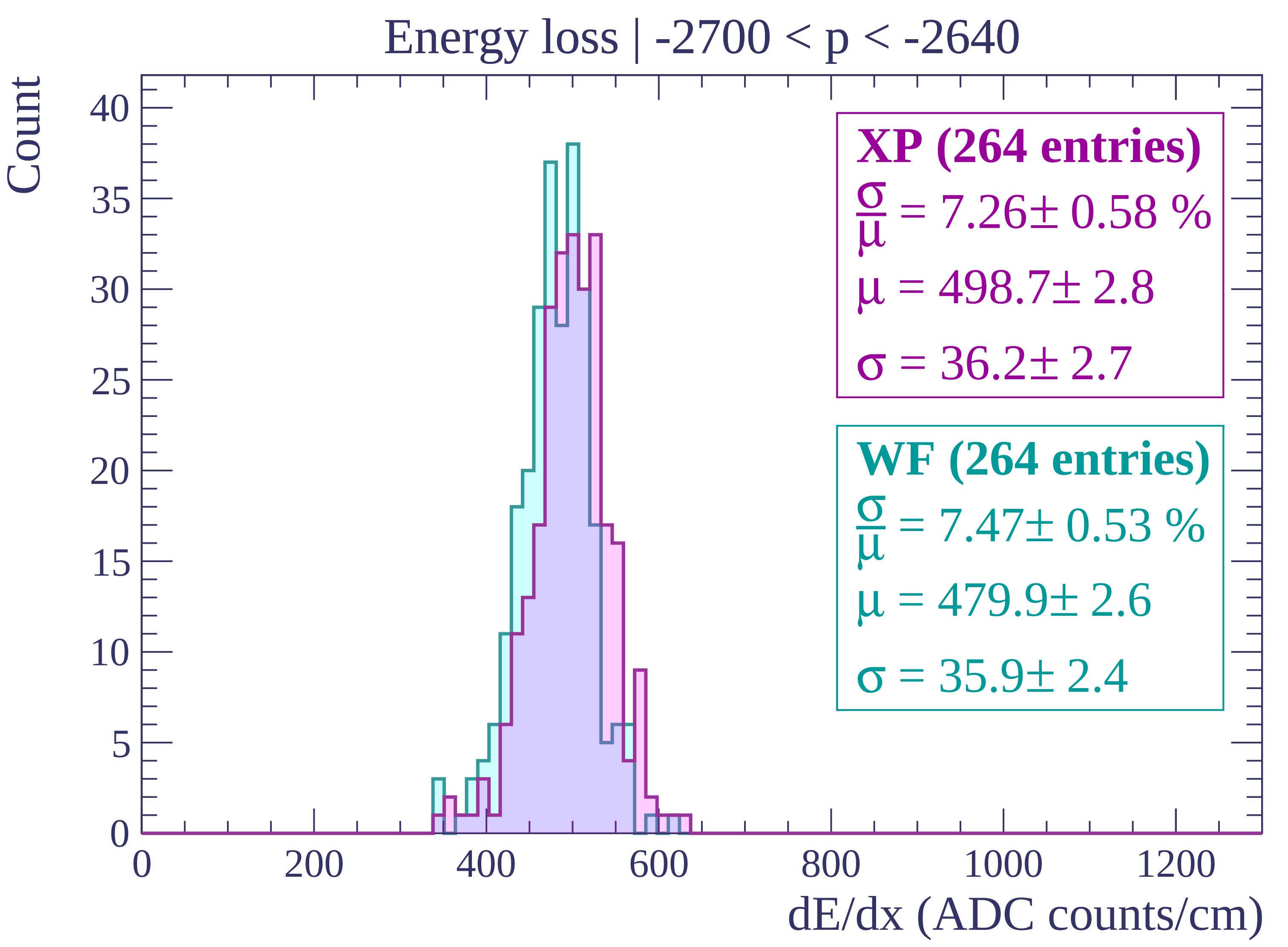


Energy loss | -2880 < p < -2820
Count
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (204 entries)
σ/μ = 7.81± 0.67 %
μ = 503.5± 3.4
σ = 39.3± 3.1
WF (204 entries)
σ/μ = 7.60± 0.77 %
μ = 487.4± 3.2
σ = 37.0± 3.5

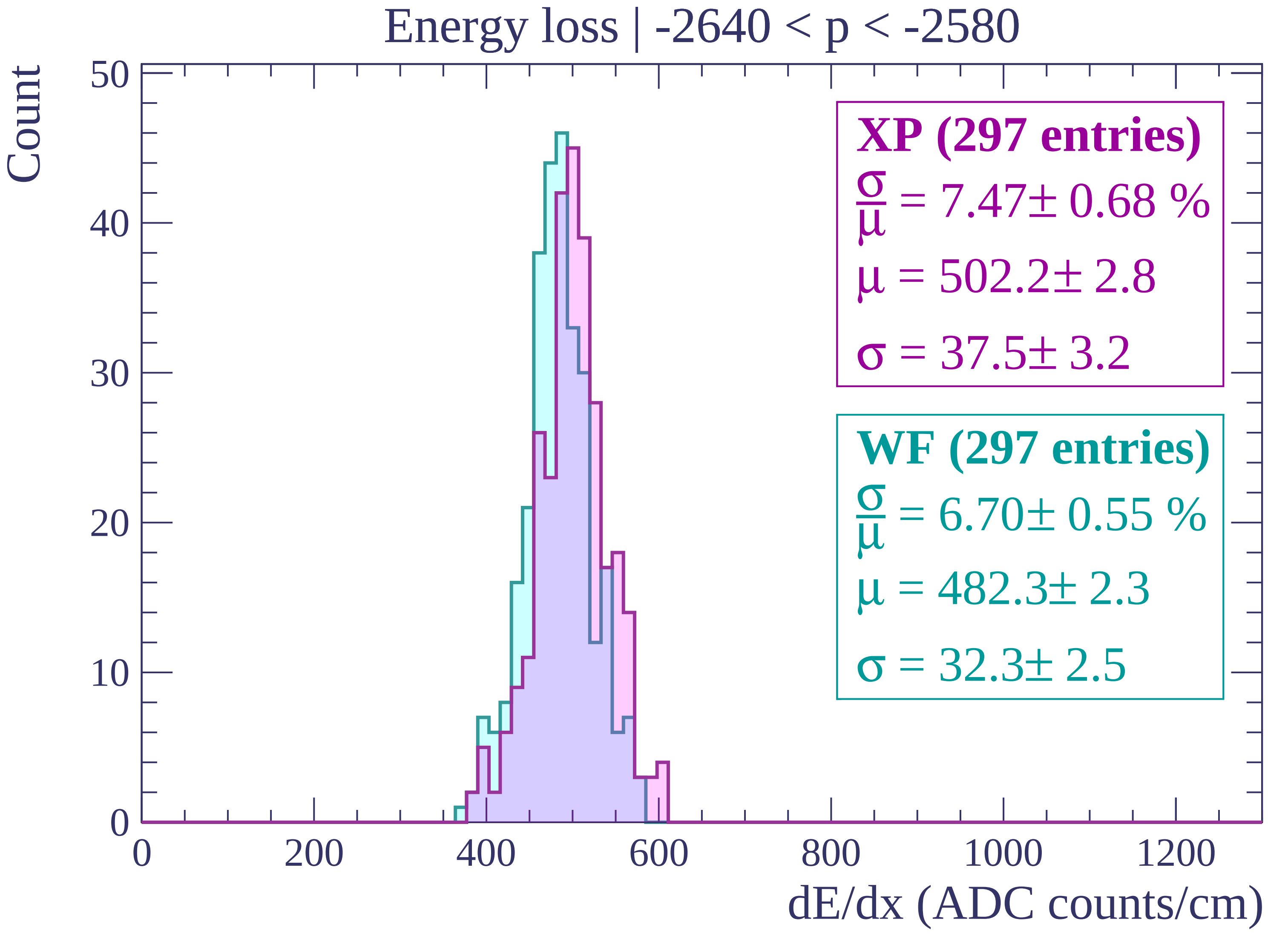

Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP (242 entries)
σ/μ = 6.15± 0.37 %
μ = 495.9± 2.4
σ = 30.5± 1.7
WF (242 entries)
σ/μ = 6.44± 0.50 %
μ = 479.4± 2.3
σ = 30.9± 2.3

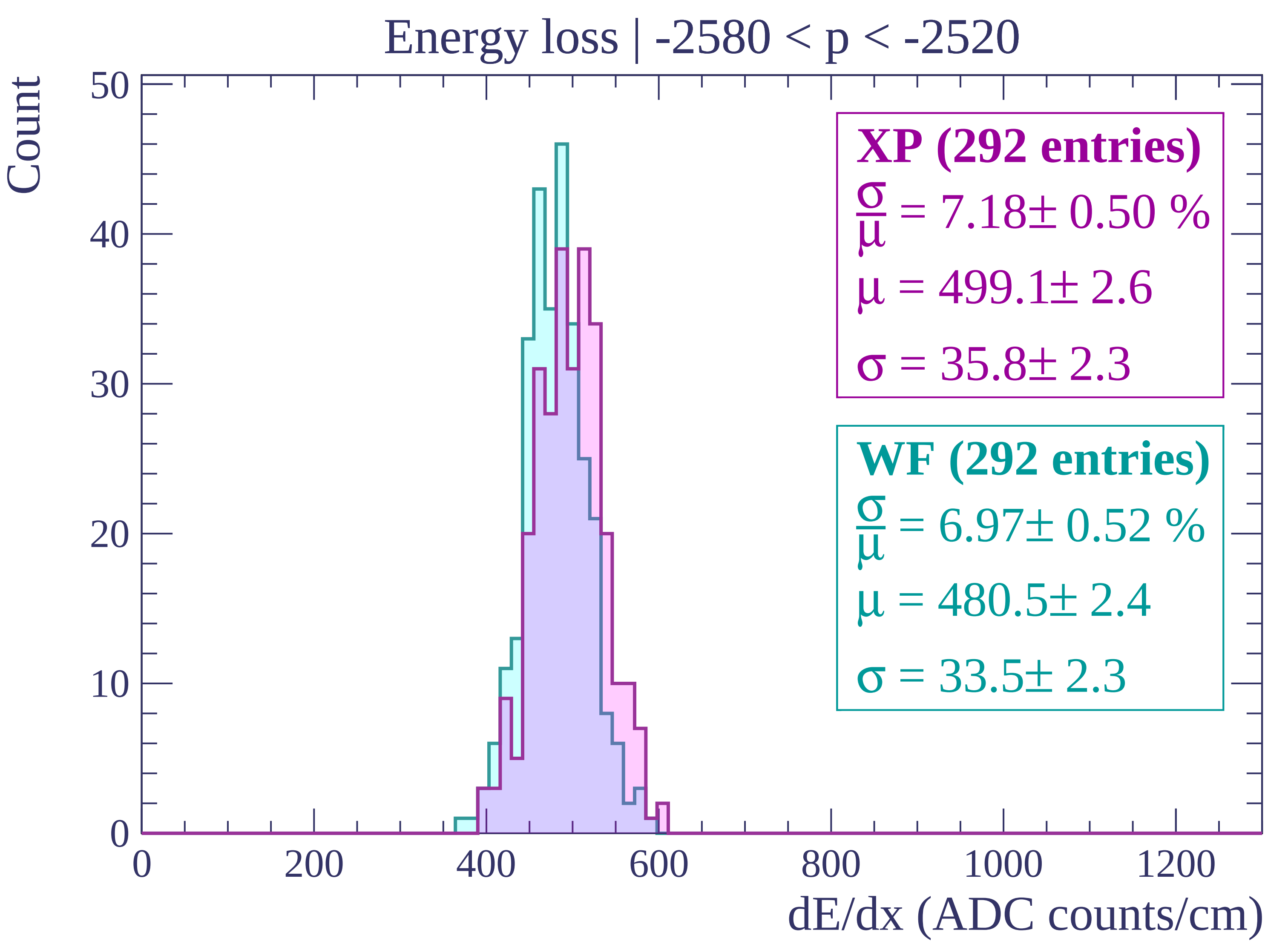


Energy loss | -2760 < p < -2700
Count
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (265 entries)
σ/μ = 7.98± 0.64 %
μ = 495.7± 2.9
σ = 39.6± 2.9
WF (265 entries)
σ/μ = 8.51± 0.68 %
μ = 478.0± 3.0
σ = 40.7± 3.0

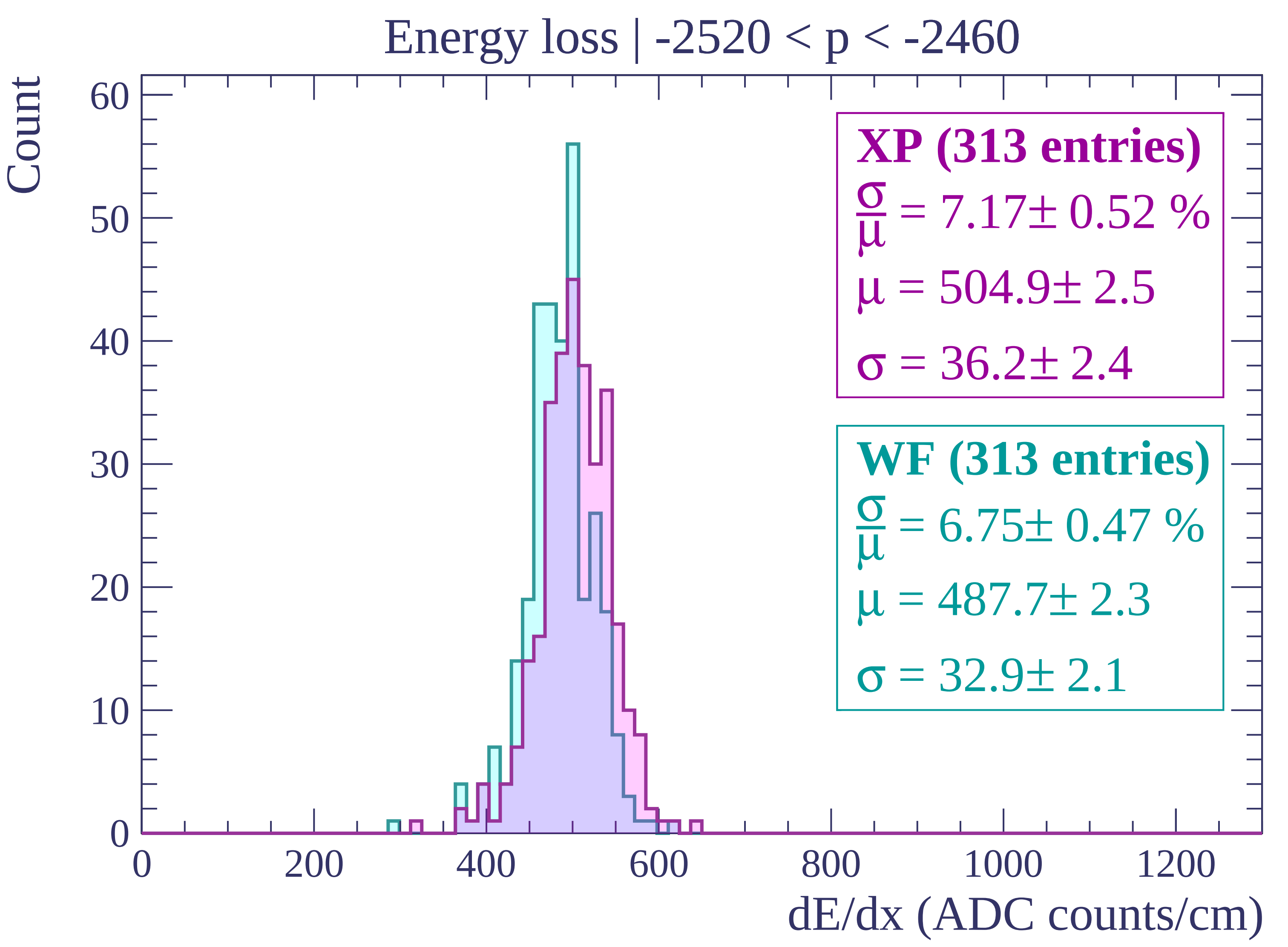


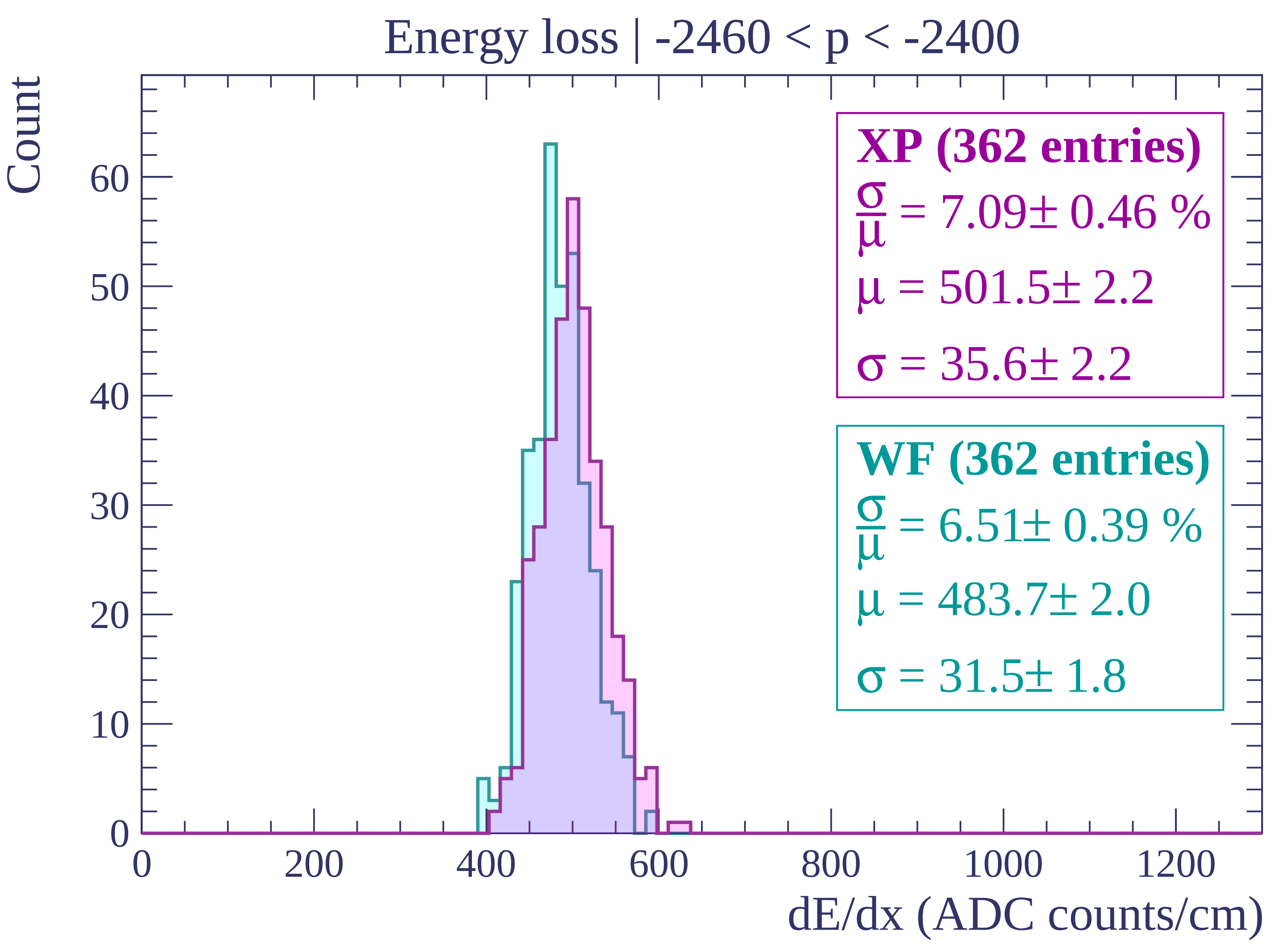
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
XP (264 entries)
$\frac{\sigma}{\mu}$ = 7.26± 0.58 %
μ = 498.7± 2.8
σ = 36.2± 2.7
WF (264 entries)
$\frac{\sigma}{\mu}$ = 7.47± 0.53 %
μ = 479.9± 2.6
σ = 35.9± 2.4

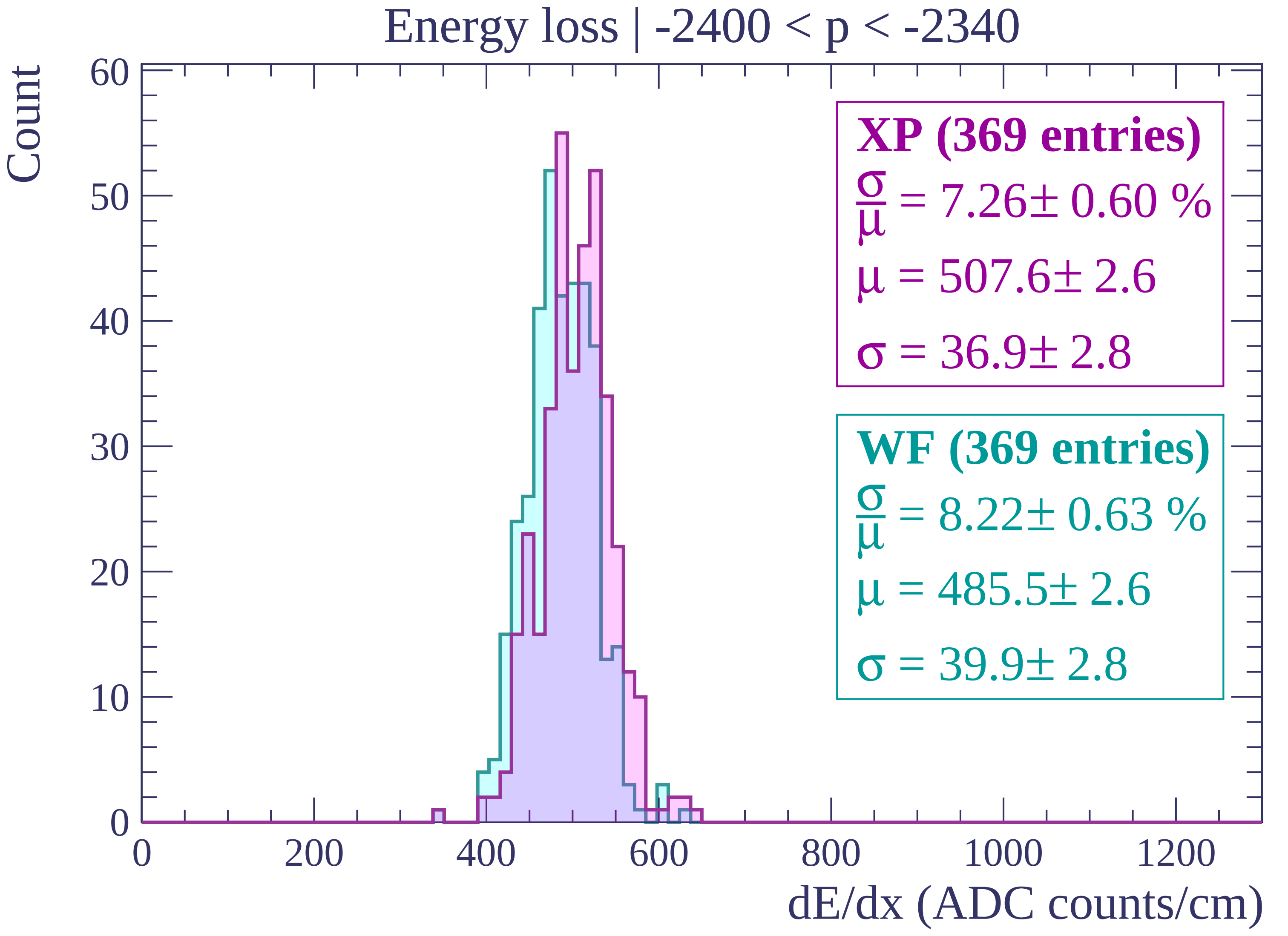

Energy loss | -2640 < p < -2580
Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (297 entries)
σ/μ = 7.47± 0.68 %
μ = 502.2± 2.8
σ = 37.5± 3.2
WF (297 entries)
σ/μ = 6.70± 0.55 %
μ = 482.3± 2.3
σ = 32.3± 2.5



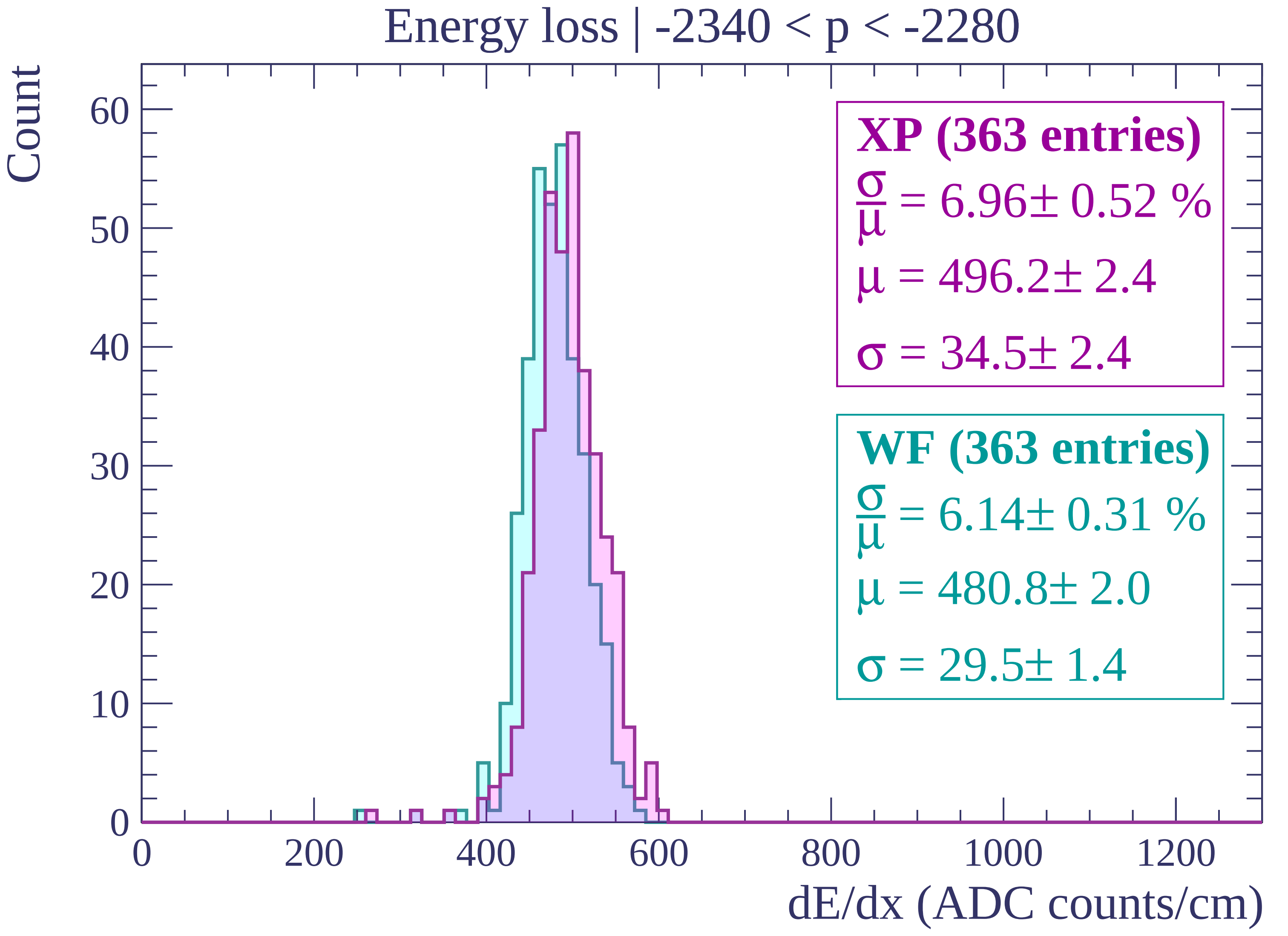
Energy loss | -2580 < p < -2520
Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (292 entries)
σ/μ = 7.18± 0.50 %
μ = 499.1± 2.6
σ = 35.8± 2.3
WF (292 entries)
σ/μ = 6.97± 0.52 %
μ = 480.5± 2.4
σ = 33.5± 2.3

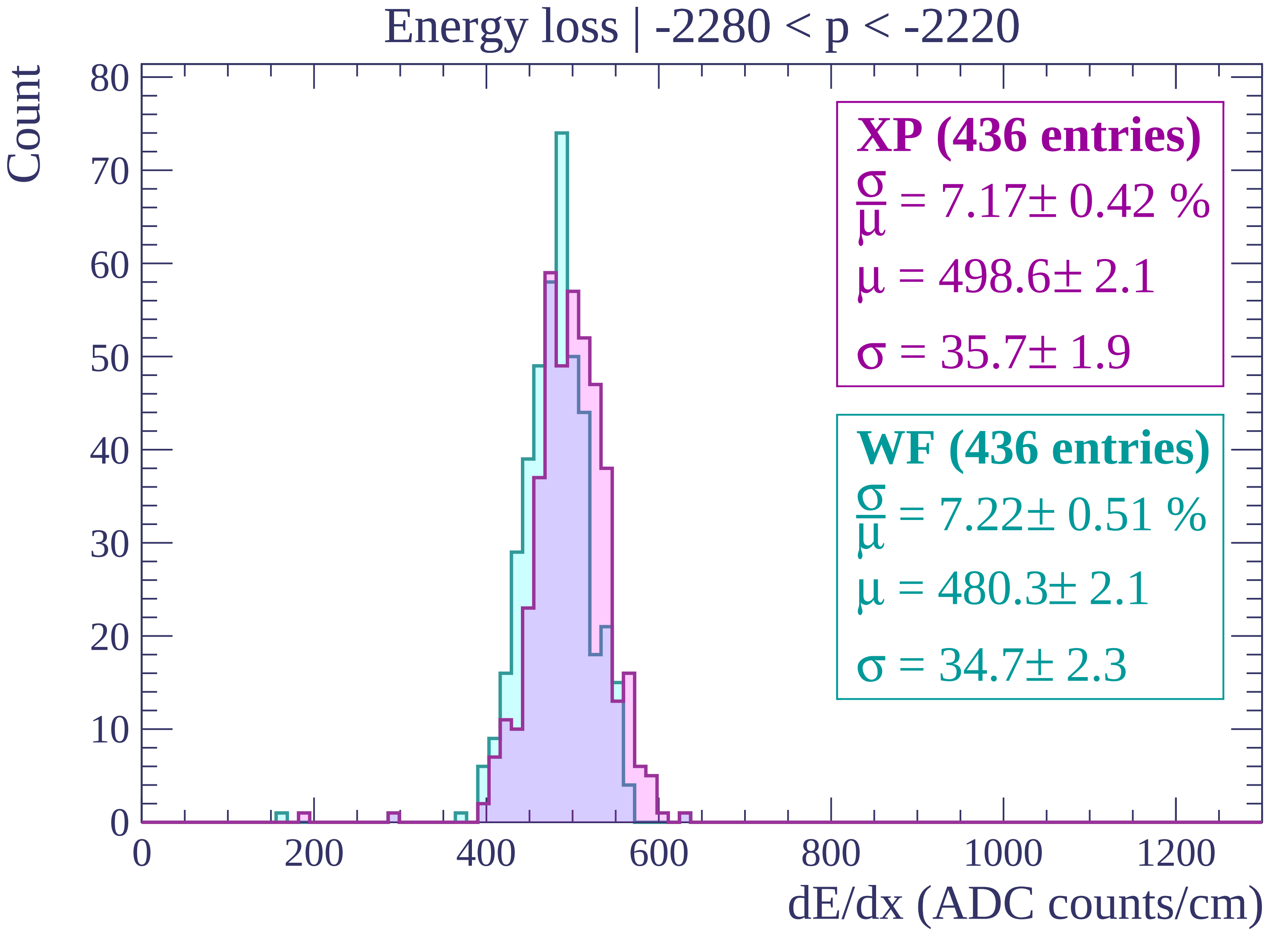

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (313 entries)
σ/μ = 7.17± 0.52 %
μ = 504.9± 2.5
σ = 36.2± 2.4
WF (313 entries)
σ/μ = 6.75± 0.47 %
μ = 487.7± 2.3
σ = 32.9± 2.1

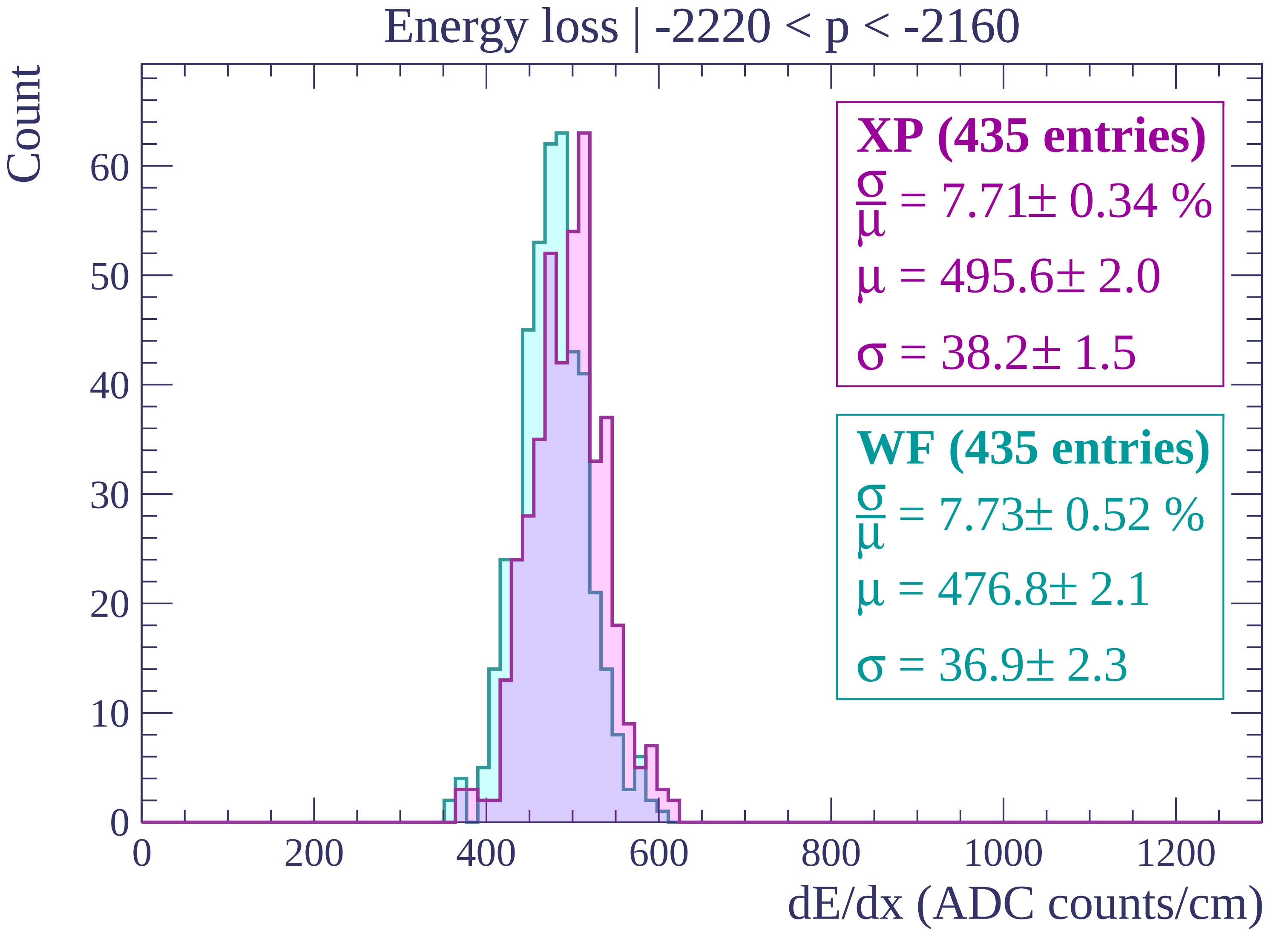

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (362 entries)
$\frac{\sigma}{\mu}$ = 7.09± 0.46 %
μ = 501.5± 2.2
σ = 35.6± 2.2
WF (362 entries)
$\frac{\sigma}{\mu}$ = 6.51± 0.39 %
μ = 483.7± 2.0
σ = 31.5± 1.8
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200



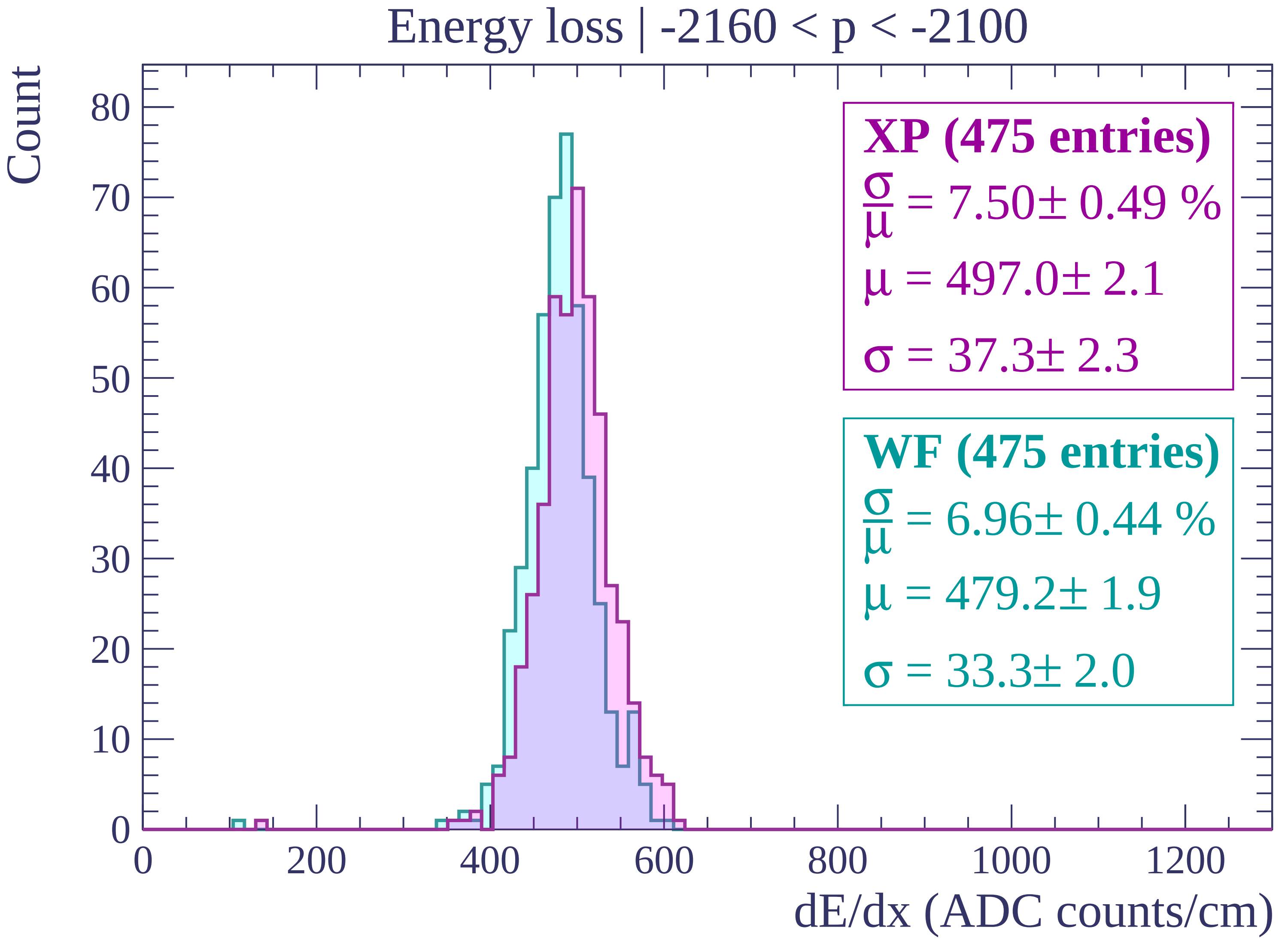
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (369 entries)
$\frac{\sigma}{\mu}$ = 7.26± 0.60 %
μ = 507.6± 2.6
σ = 36.9± 2.8
WF (369 entries)
$\frac{\sigma}{\mu}$ = 8.22± 0.63 %
μ = 485.5± 2.6
σ = 39.9± 2.8


Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (363 entries)
$\frac{\sigma}{\mu}$ = 6.96± 0.52 %
μ = 496.2± 2.4
σ = 34.5± 2.4
WF (363 entries)
$\frac{\sigma}{\mu}$ = 6.14± 0.31 %
μ = 480.8± 2.0
σ = 29.5± 1.4

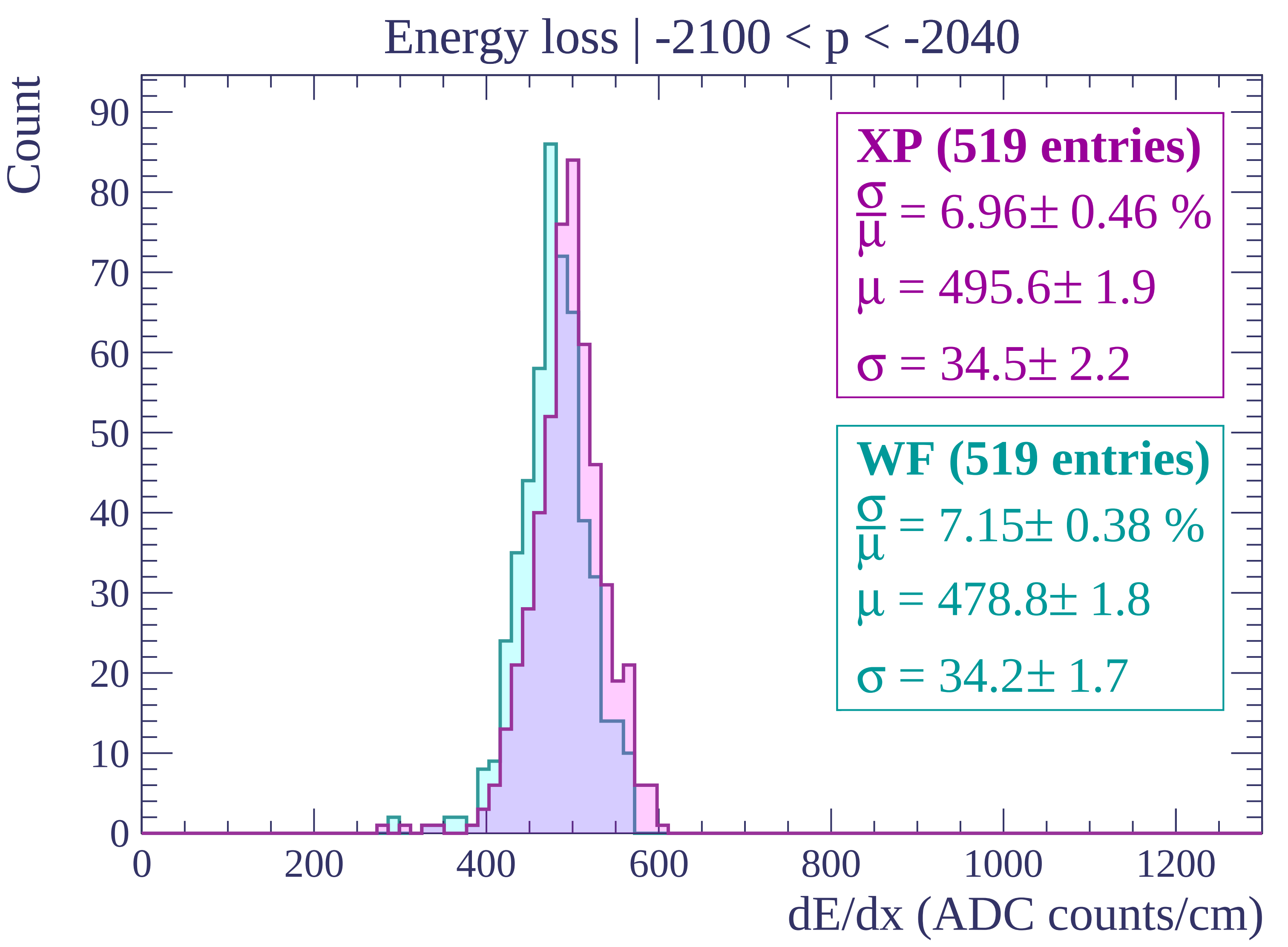

Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (436 entries)
σ/μ = 7.17± 0.42 %
μ = 498.6± 2.1
σ = 35.7± 1.9
WF (436 entries)
σ/μ = 7.22± 0.51 %
μ = 480.3± 2.1
σ = 34.7± 2.3



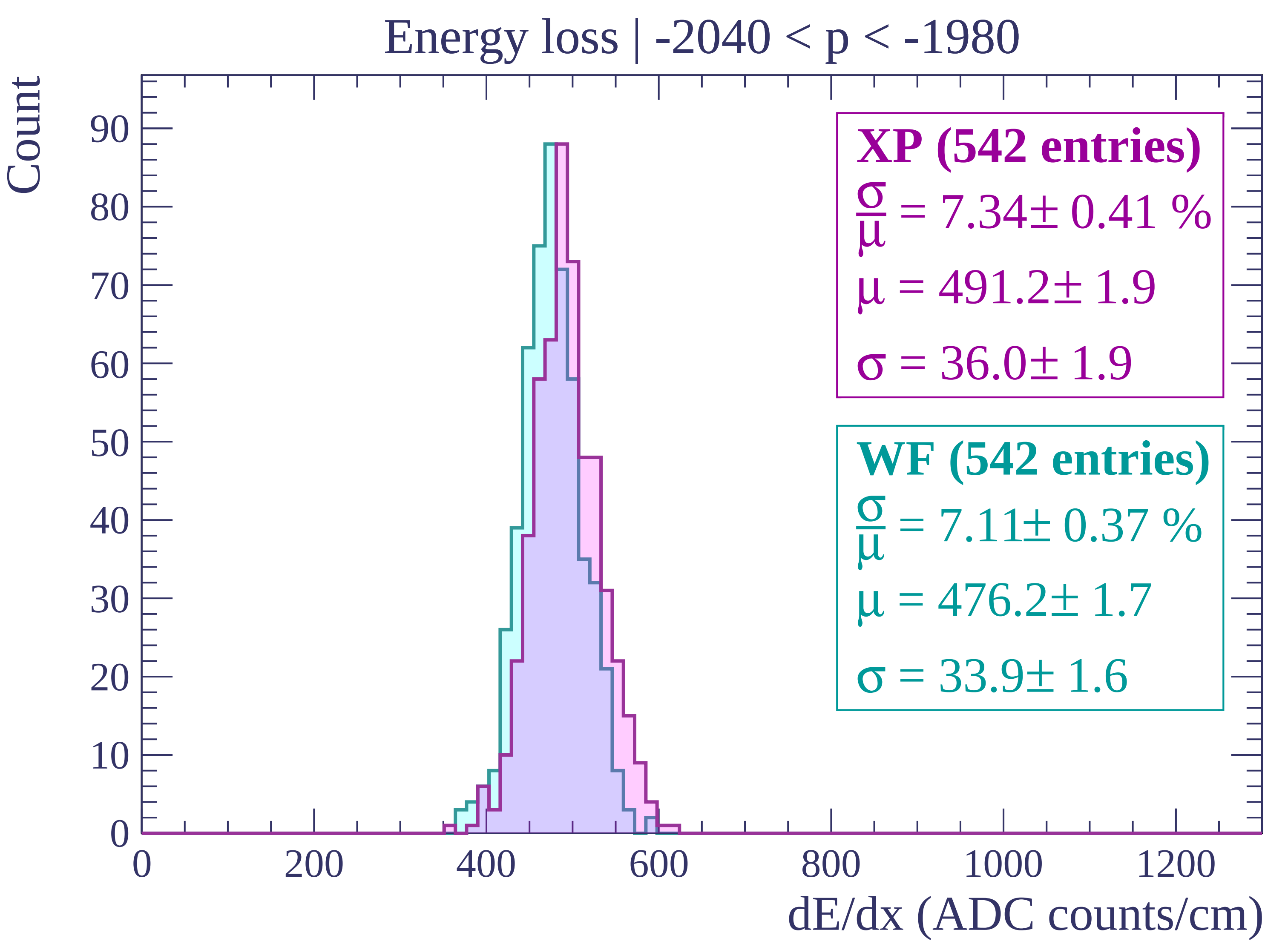
Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (435 entries)
σ/μ = 7.71± 0.34 %
μ = 495.6± 2.0
σ = 38.2± 1.5
WF (435 entries)
σ/μ = 7.73± 0.52 %
μ = 476.8± 2.1
σ = 36.9± 2.3

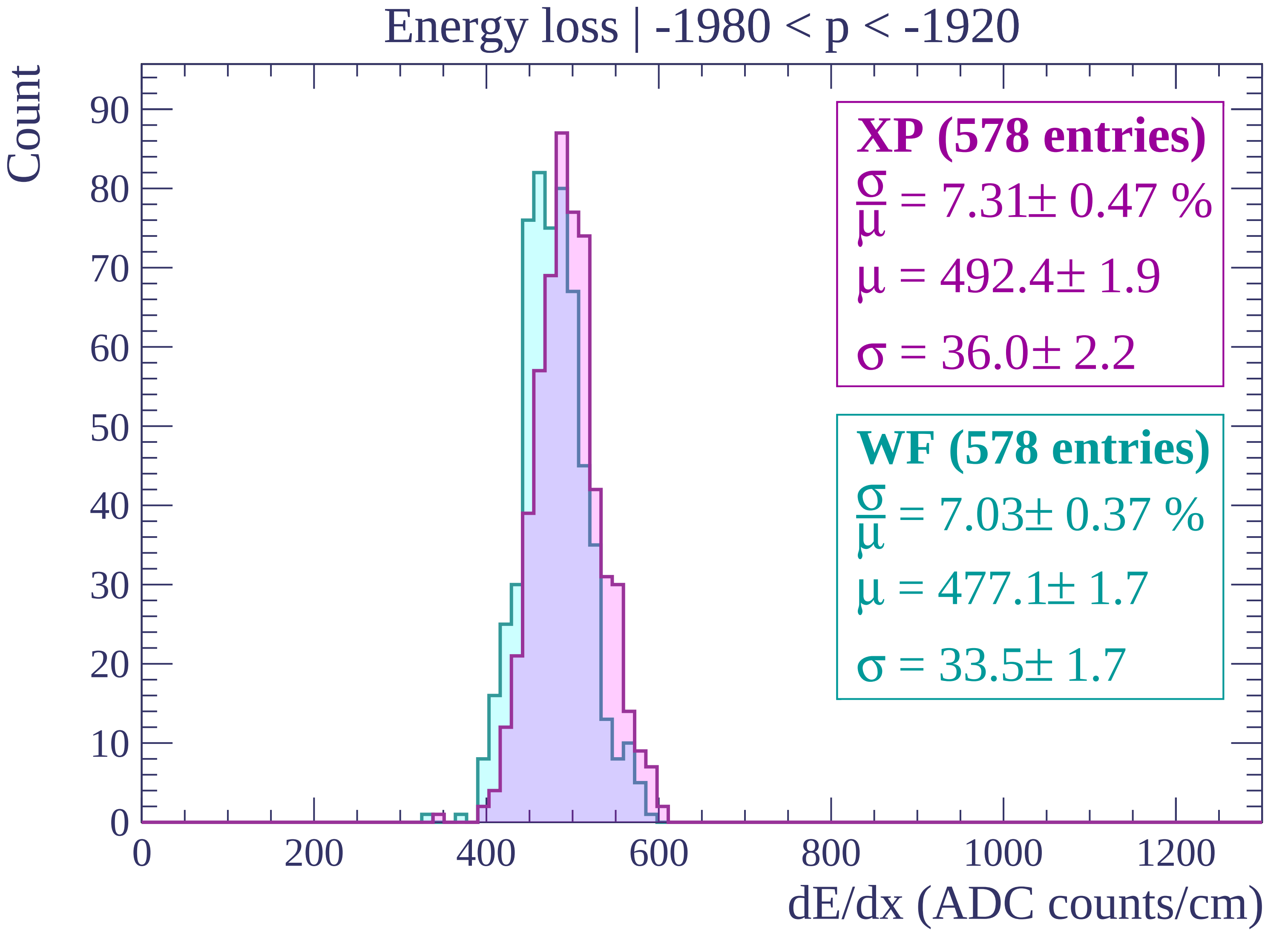


Energy loss | -2160 < p < -2100
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (475 entries)
$\frac{\sigma}{\mu}$ = 7.50± 0.49 %
μ = 497.0± 2.1
σ = 37.3± 2.3
WF (475 entries)
$\frac{\sigma}{\mu}$ = 6.96± 0.44 %
μ = 479.2± 1.9
σ = 33.3± 2.0

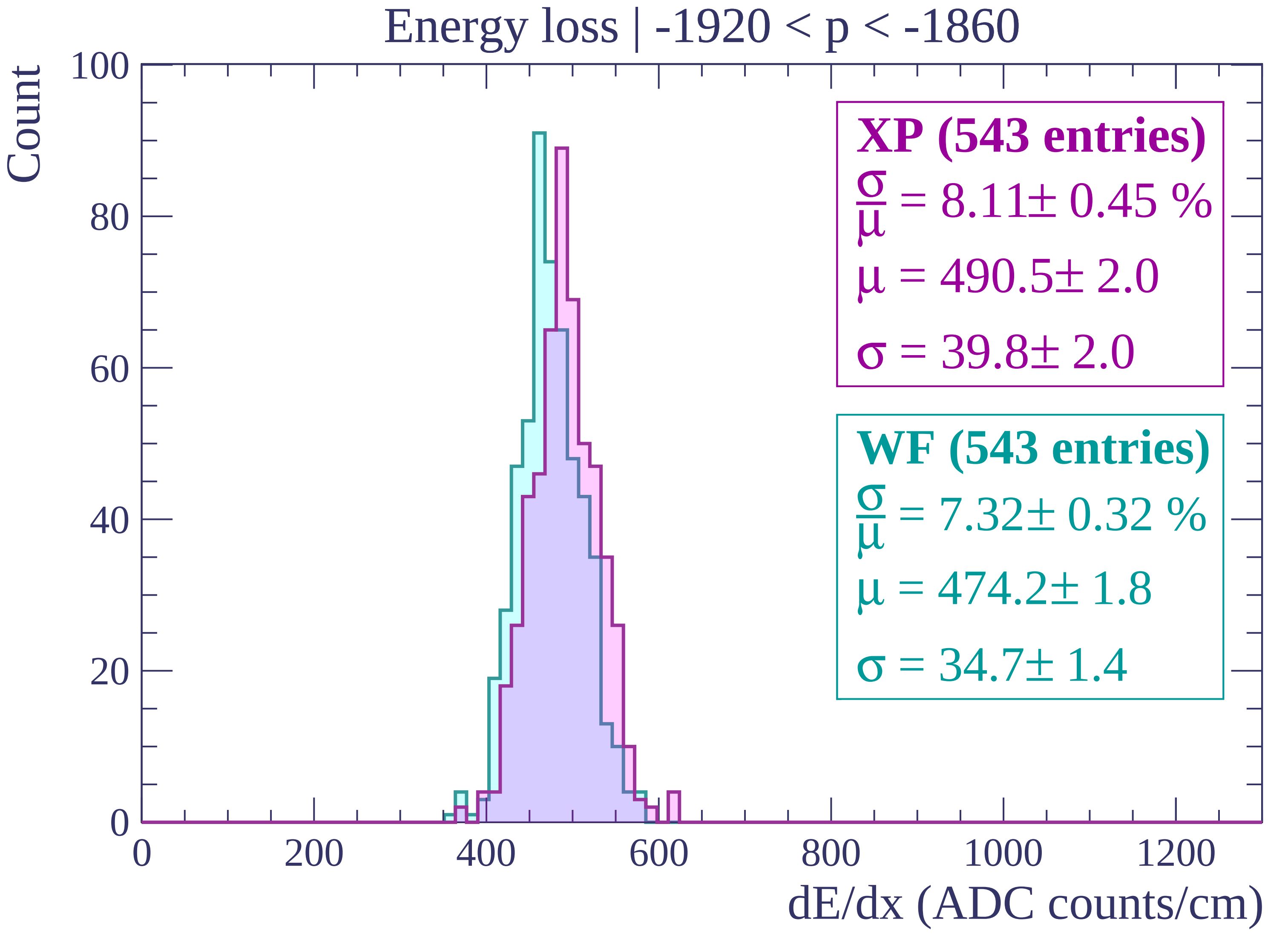


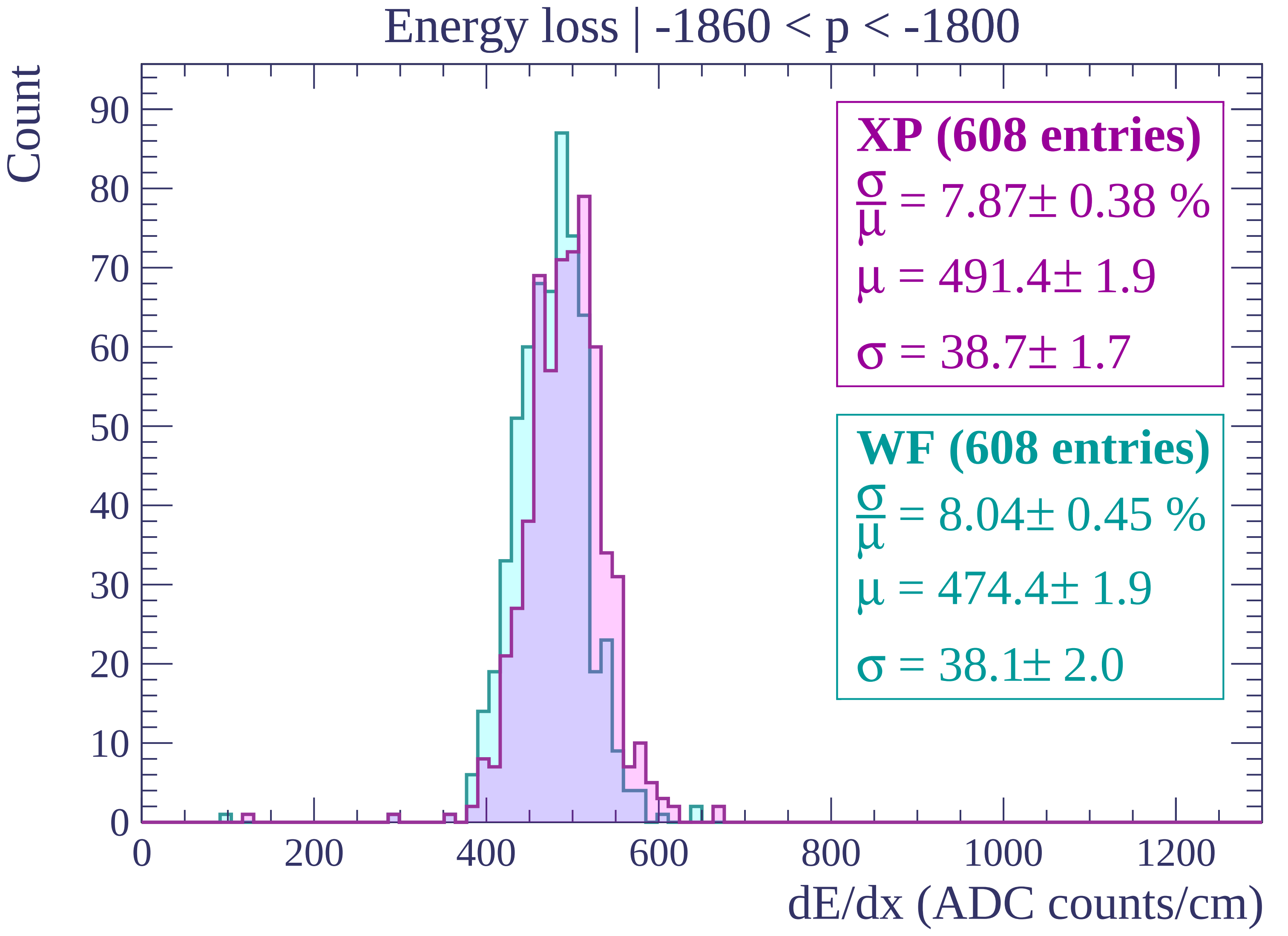
Energy loss | -2100 < p < -2040
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (519 entries)
σ/μ = 6.96± 0.46 %
μ = 495.6± 1.9
σ = 34.5± 2.2
WF (519 entries)
σ/μ = 7.15± 0.38 %
μ = 478.8± 1.8
σ = 34.2± 1.7


Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (542 entries)
$\frac{\sigma}{\mu}$ = 7.34± 0.41 %
μ = 491.2± 1.9
σ = 36.0± 1.9
WF (542 entries)
$\frac{\sigma}{\mu}$ = 7.11± 0.37 %
μ = 476.2± 1.7
σ = 33.9± 1.6

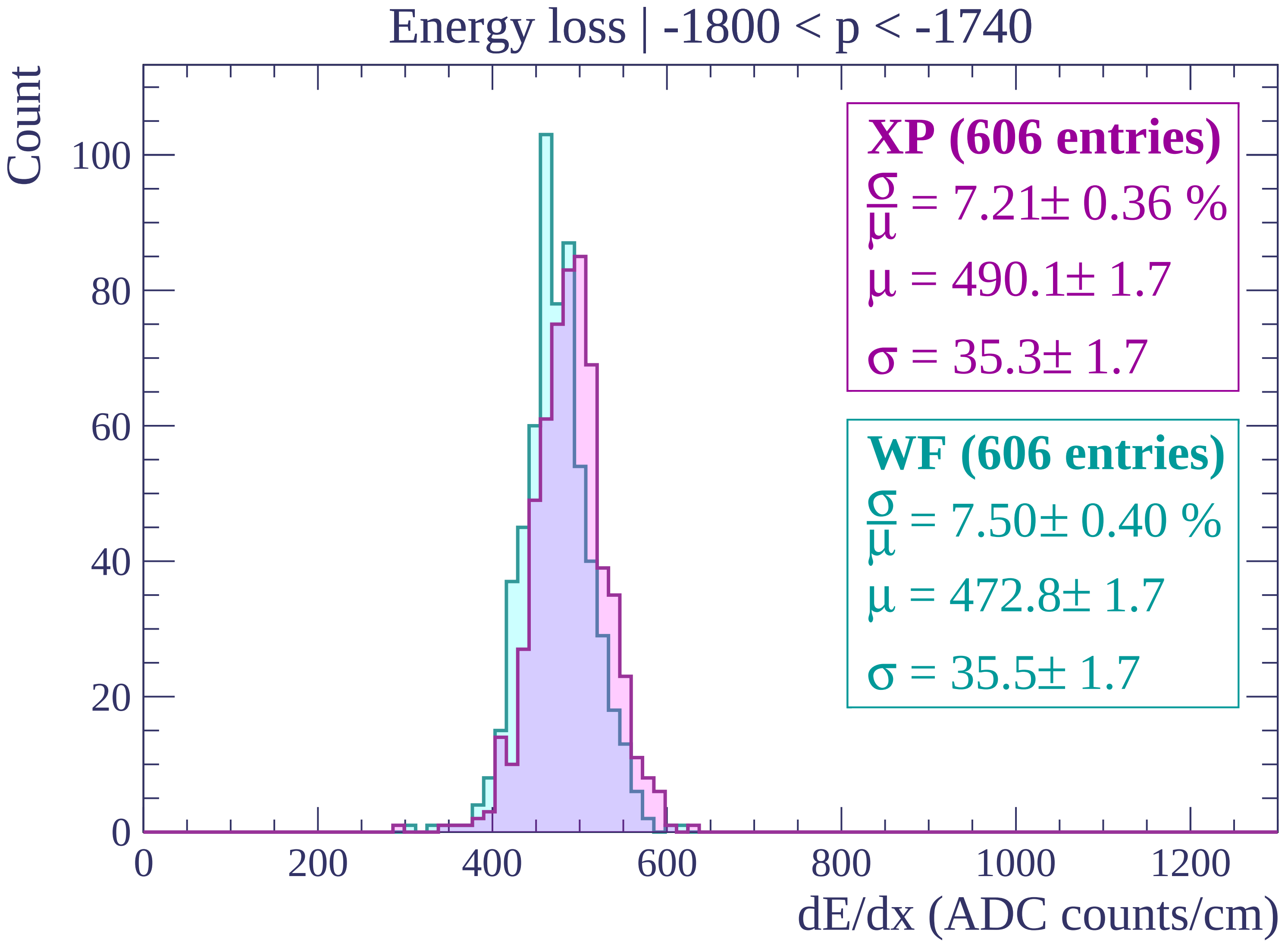


Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (578 entries)
σ/μ = 7.31± 0.47 %
μ = 492.4± 1.9
σ = 36.0± 2.2
WF (578 entries)
σ/μ = 7.03± 0.37 %
μ = 477.1± 1.7
σ = 33.5± 1.7

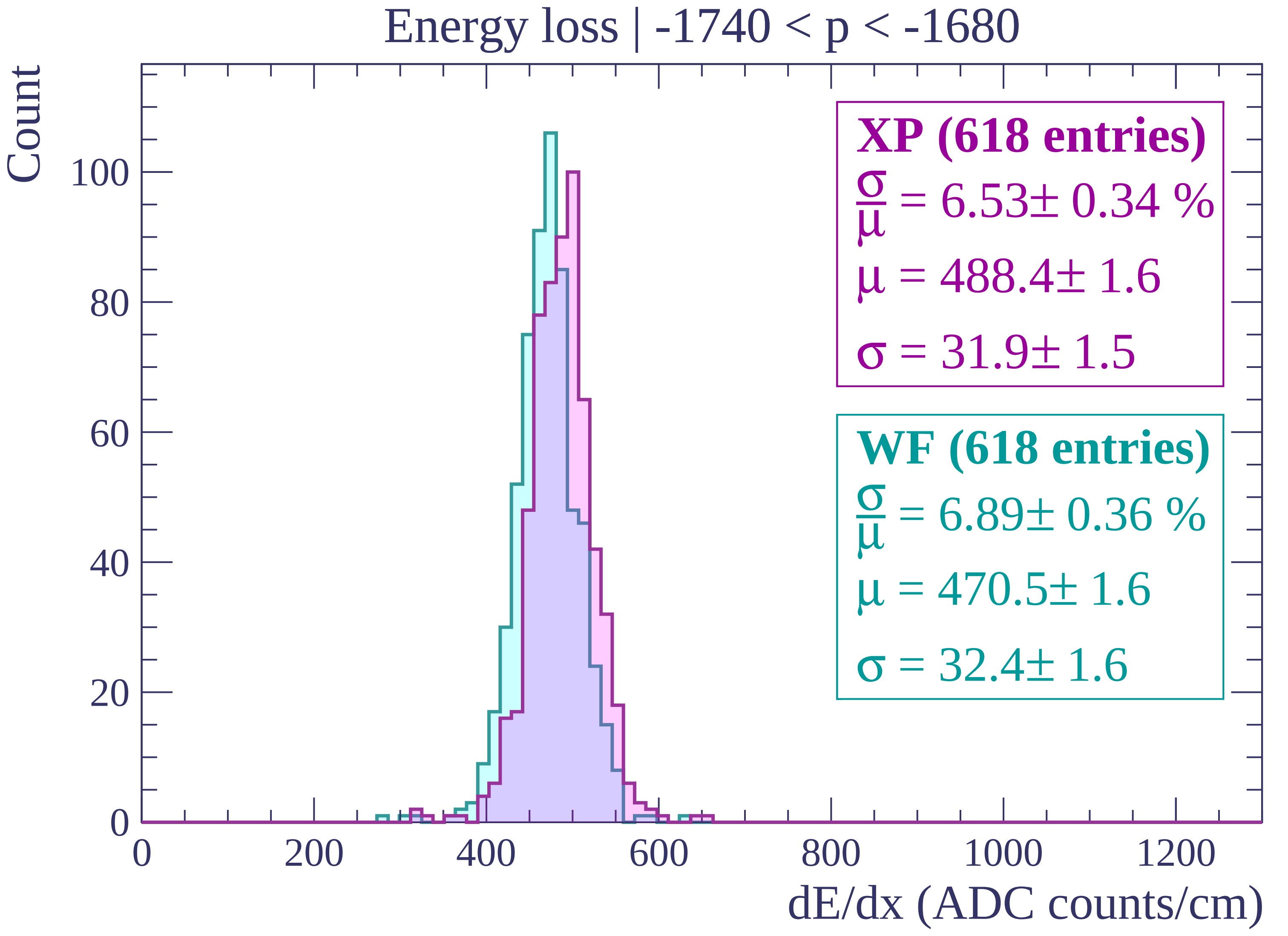

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (543 entries)
σ/μ = 8.11± 0.45 %
μ = 490.5± 2.0
σ = 39.8± 2.0
WF (543 entries)
σ/μ = 7.32± 0.32 %
μ = 474.2± 1.8
σ = 34.7± 1.4

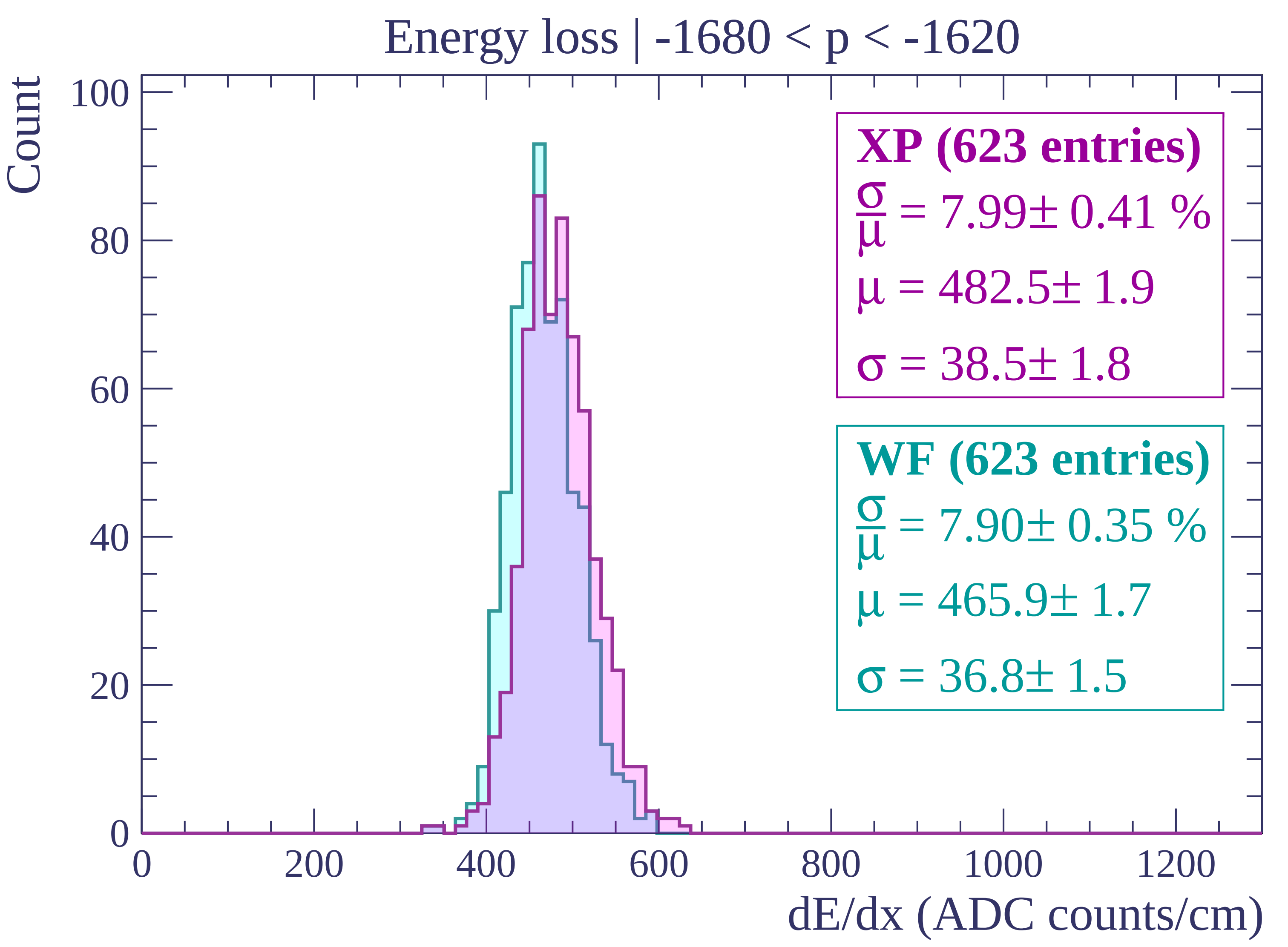


Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (608 entries)
$\frac{\sigma}{\mu}$ = 7.87± 0.38 %
μ = 491.4± 1.9
σ = 38.7± 1.7
WF (608 entries)
$\frac{\sigma}{\mu}$ = 8.04± 0.45 %
μ = 474.4± 1.9
σ = 38.1± 2.0

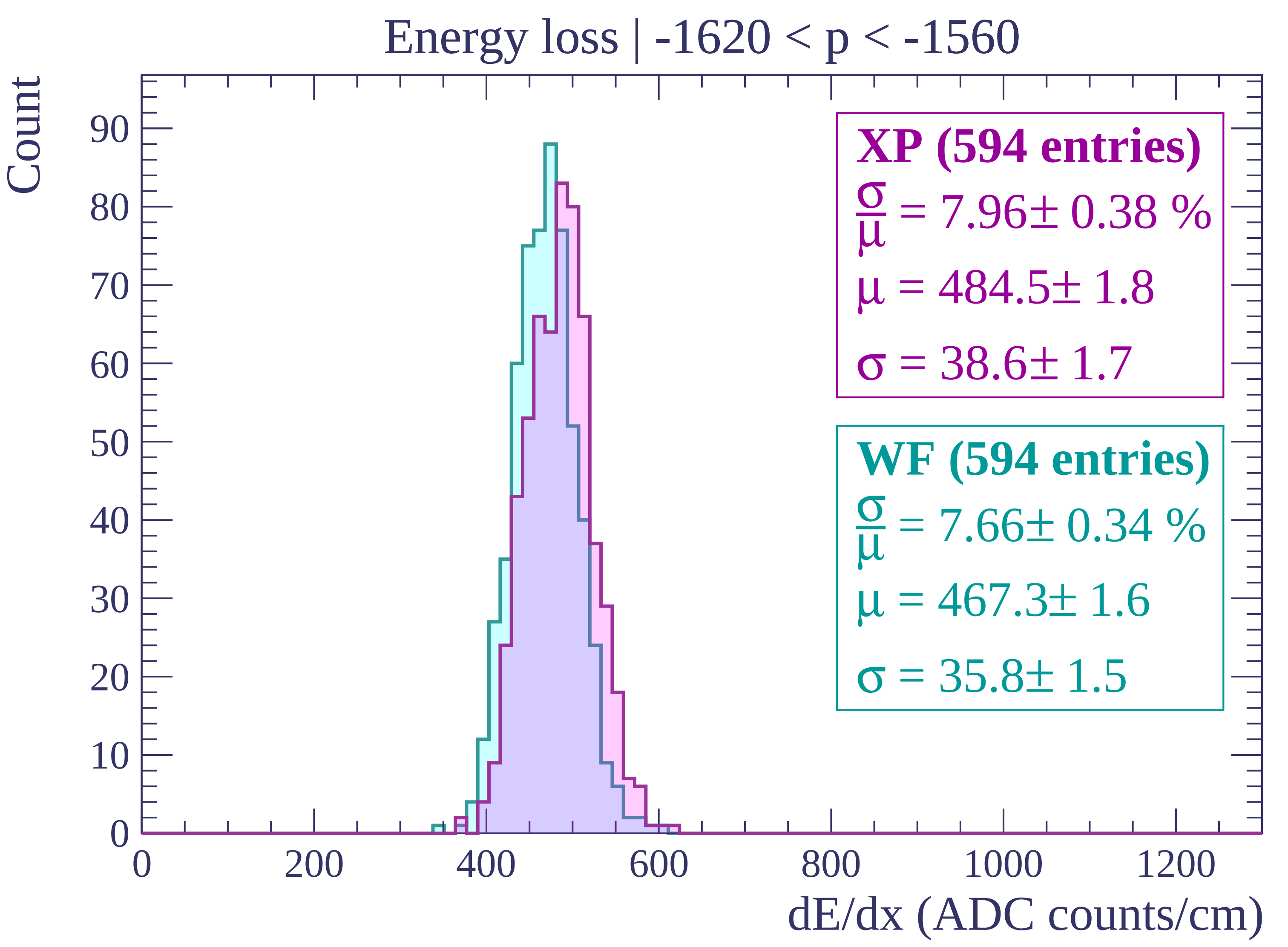


Energy loss | -1800 < p < -1740
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (606 entries)
σ/μ = 7.21± 0.36 %
μ = 490.1± 1.7
σ = 35.3± 1.7
WF (606 entries)
σ/μ = 7.50± 0.40 %
μ = 472.8± 1.7
σ = 35.5± 1.7

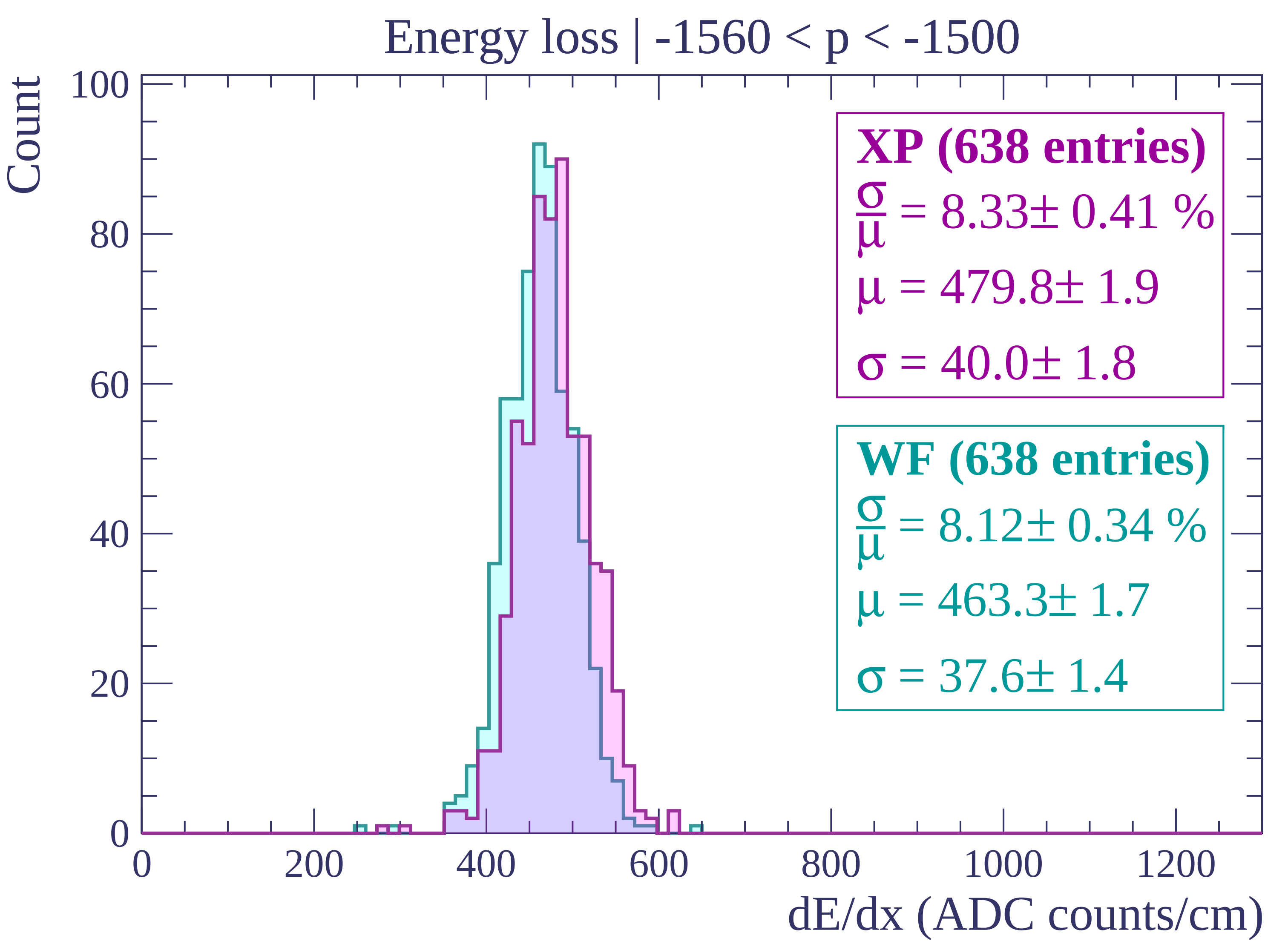

Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (618 entries)
σ/μ = 6.53± 0.34 %
μ = 488.4± 1.6
σ = 31.9± 1.5
WF (618 entries)
σ/μ = 6.89± 0.36 %
μ = 470.5± 1.6
σ = 32.4± 1.6



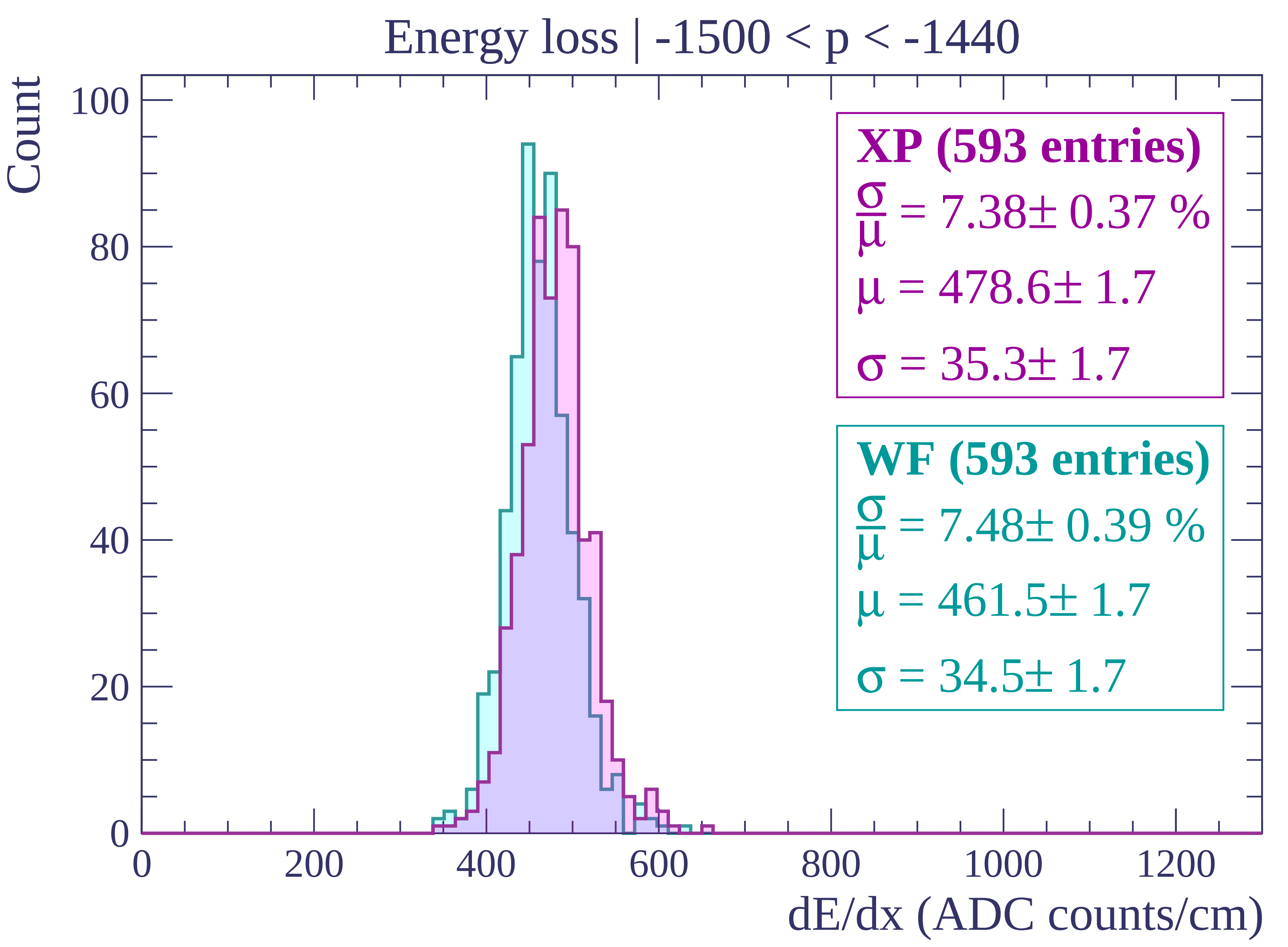
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (623 entries)
σ/μ = 7.99± 0.41 %
μ = 482.5± 1.9
σ = 38.5± 1.8
WF (623 entries)
σ/μ = 7.90± 0.35 %
μ = 465.9± 1.7
σ = 36.8± 1.5

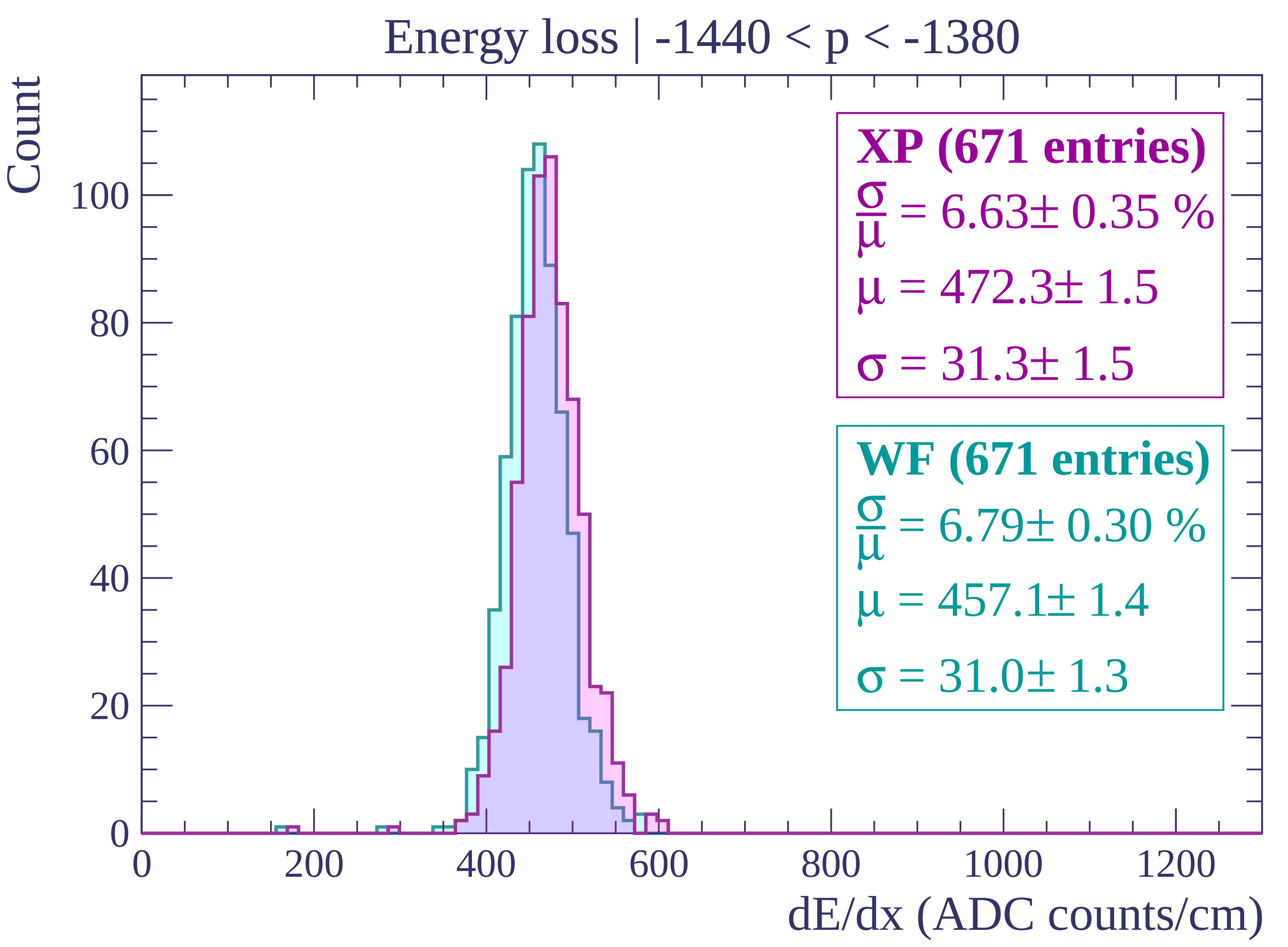


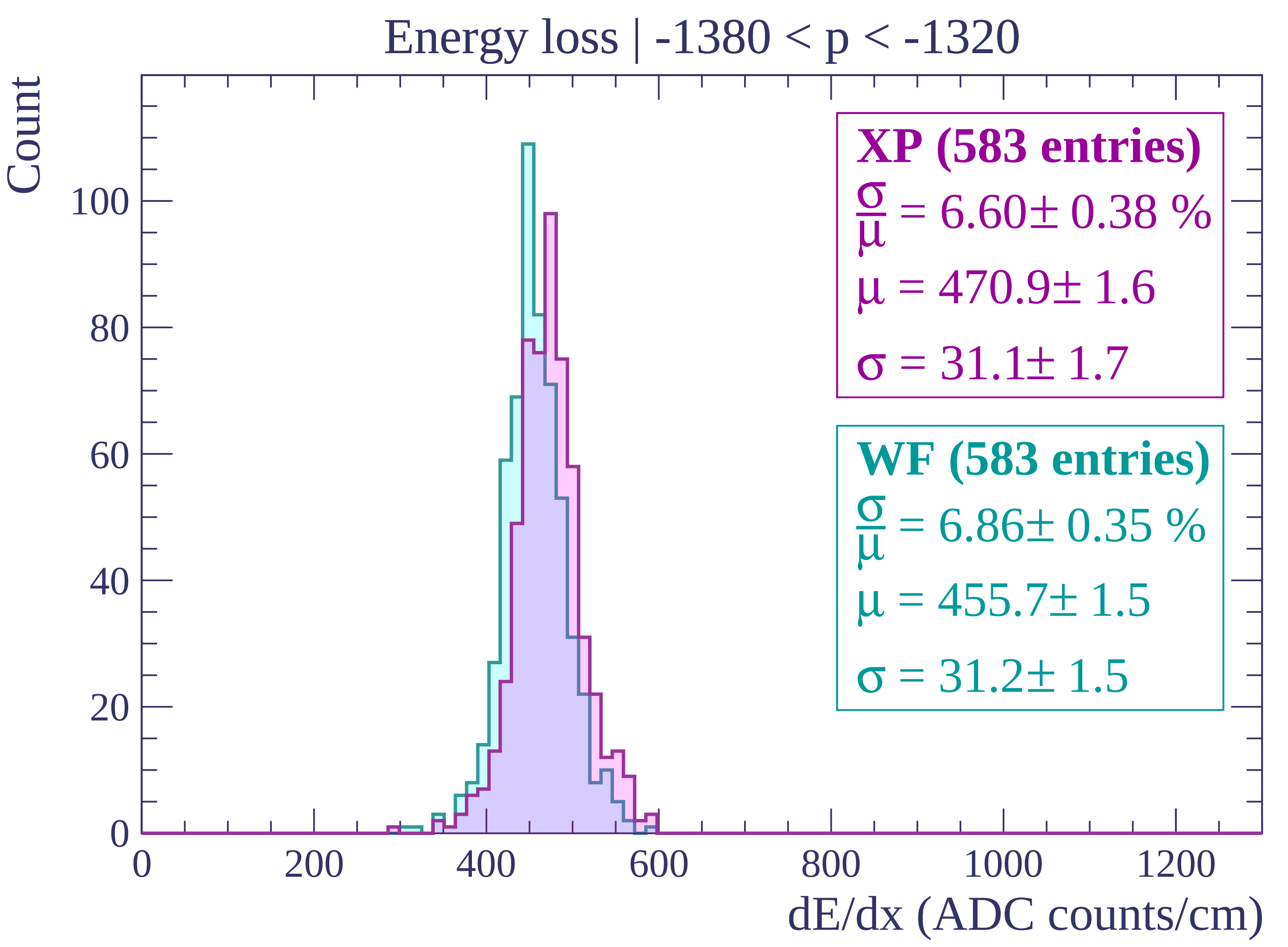
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (594 entries)
σ/μ = 7.96± 0.38 %
μ = 484.5± 1.8
σ = 38.6± 1.7
WF (594 entries)
σ/μ = 7.66± 0.34 %
μ = 467.3± 1.6
σ = 35.8± 1.5

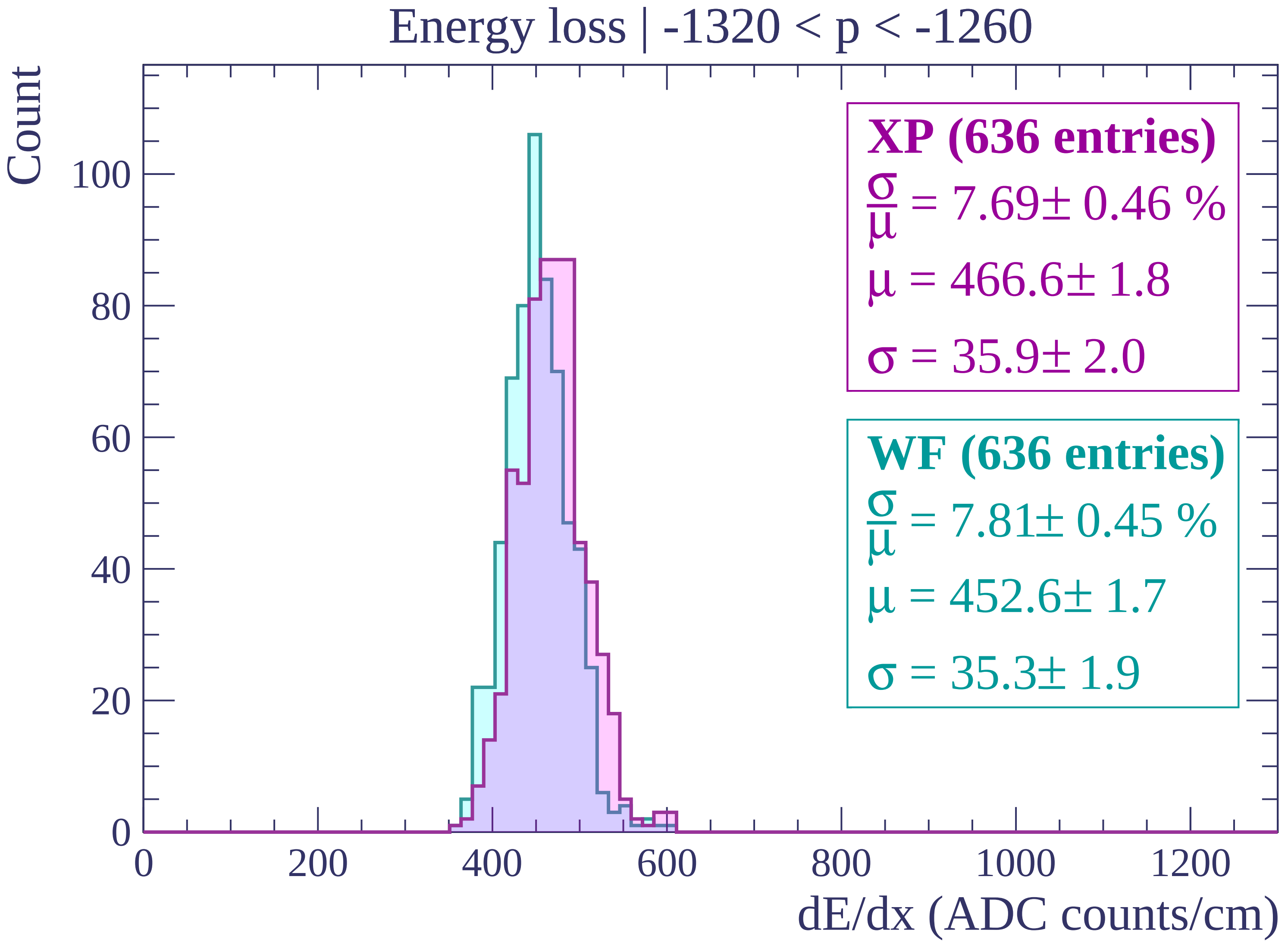


Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (638 entries)
σ/μ = 8.33± 0.41 %
μ = 479.8± 1.9
σ = 40.0± 1.8
WF (638 entries)
σ/μ = 8.12± 0.34 %
μ = 463.3± 1.7
σ = 37.6± 1.4

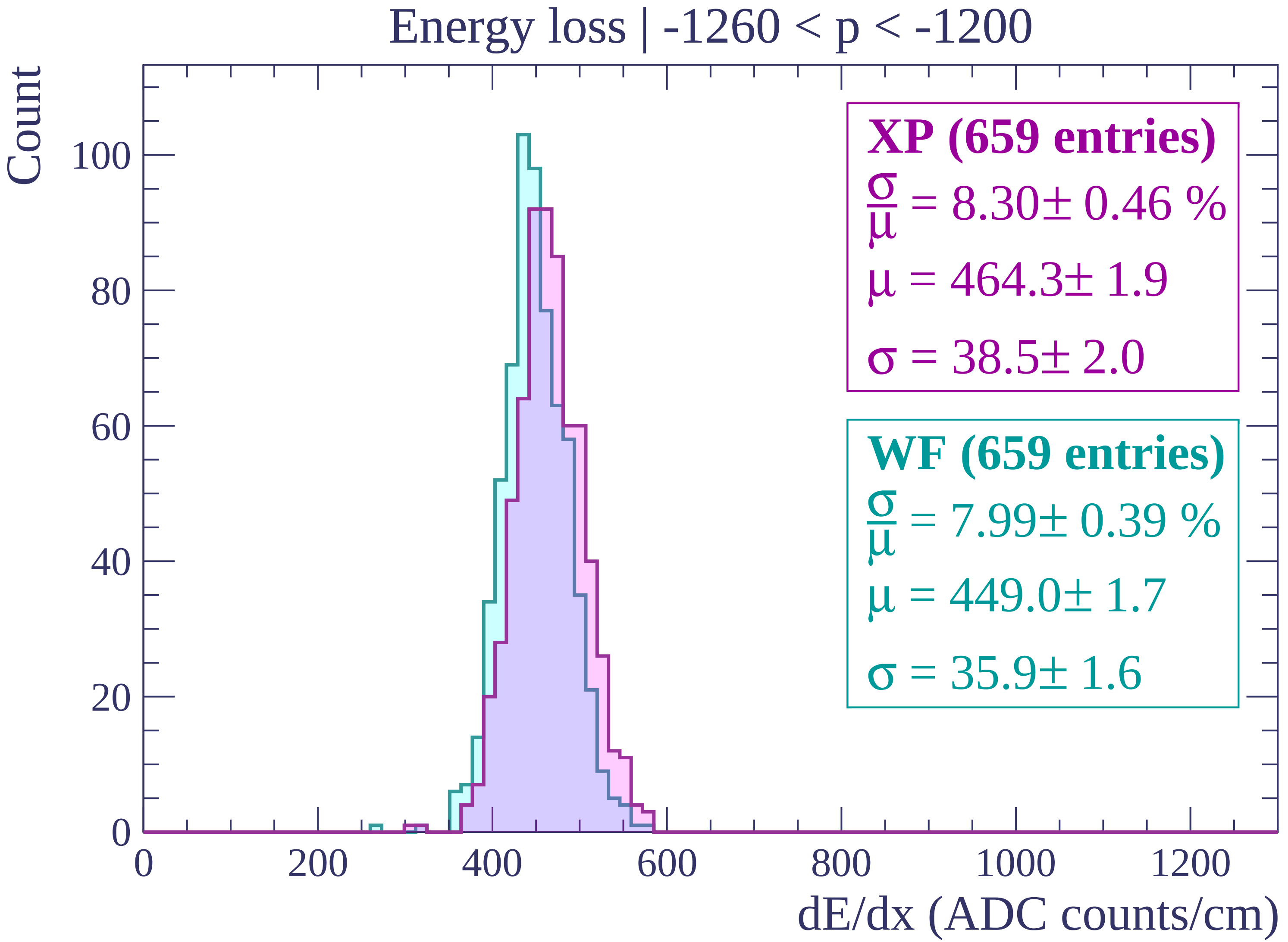

Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (593 entries)
$\frac{\sigma}{\mu}$ = 7.38± 0.37 %
μ = 478.6± 1.7
σ = 35.3± 1.7
WF (593 entries)
$\frac{\sigma}{\mu}$ = 7.48± 0.39 %
μ = 461.5± 1.7
σ = 34.5± 1.7


Energy loss | -1440 < p < -1380
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (671 entries)
σ/μ = 6.63± 0.35 %
μ = 472.3± 1.5
σ = 31.3± 1.5
WF (671 entries)
σ/μ = 6.79± 0.30 %
μ = 457.1± 1.4
σ = 31.0± 1.3

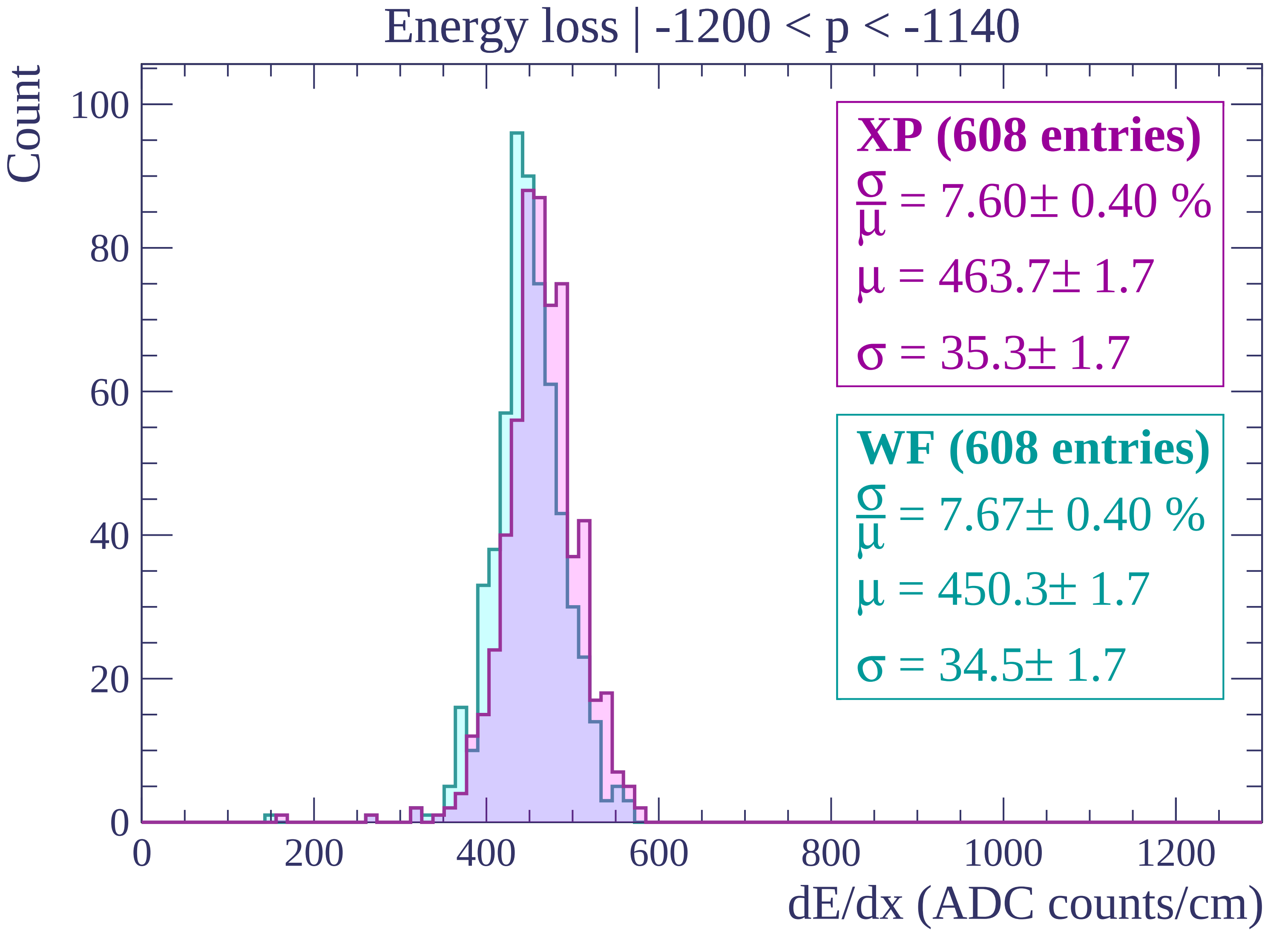

Energy loss | -1380 < p < -1320
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (583 entries)
σ/μ = 6.60± 0.38 %
μ = 470.9± 1.6
σ = 31.1± 1.7
WF (583 entries)
σ/μ = 6.86± 0.35 %
μ = 455.7± 1.5
σ = 31.2± 1.5

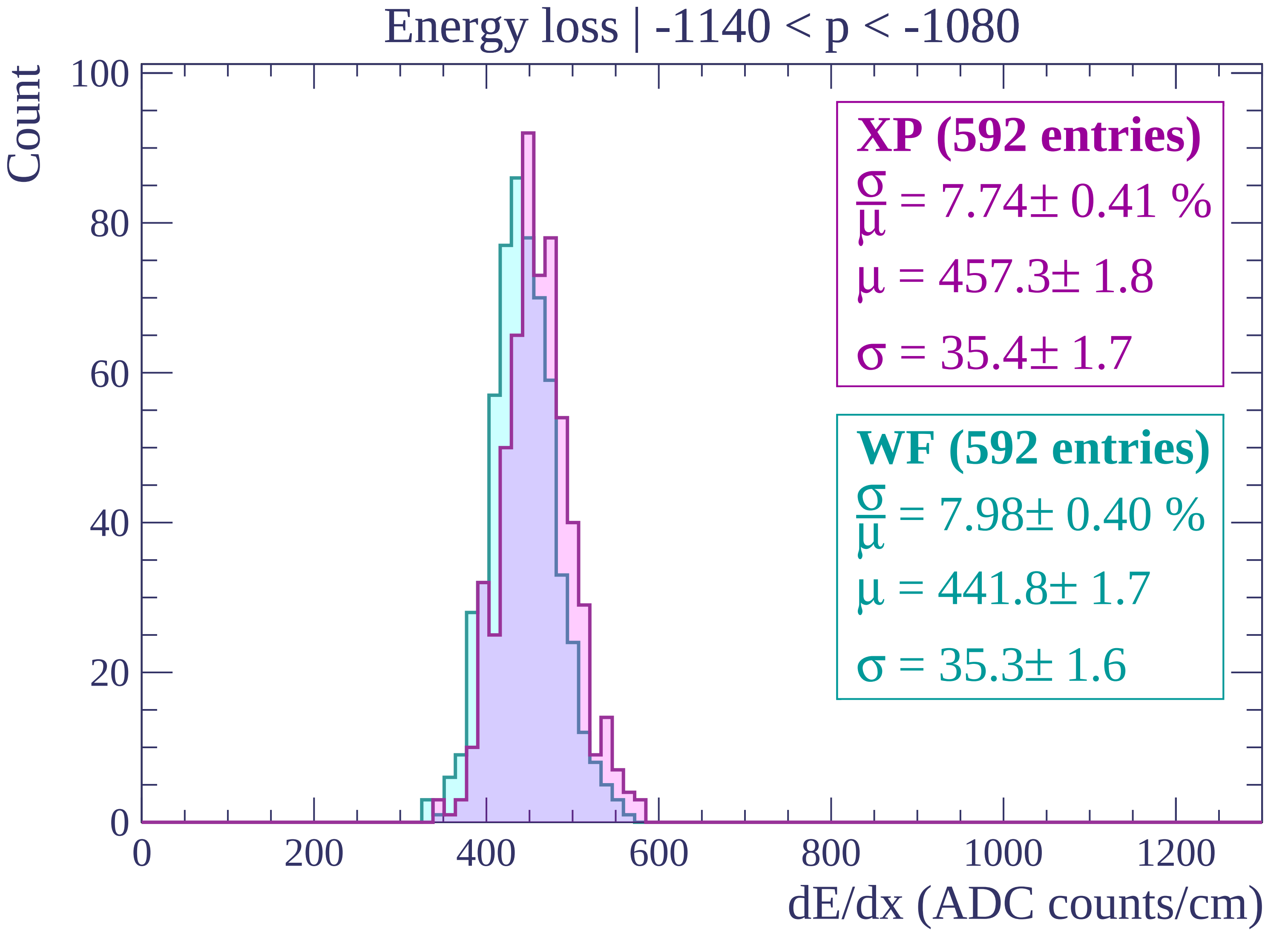


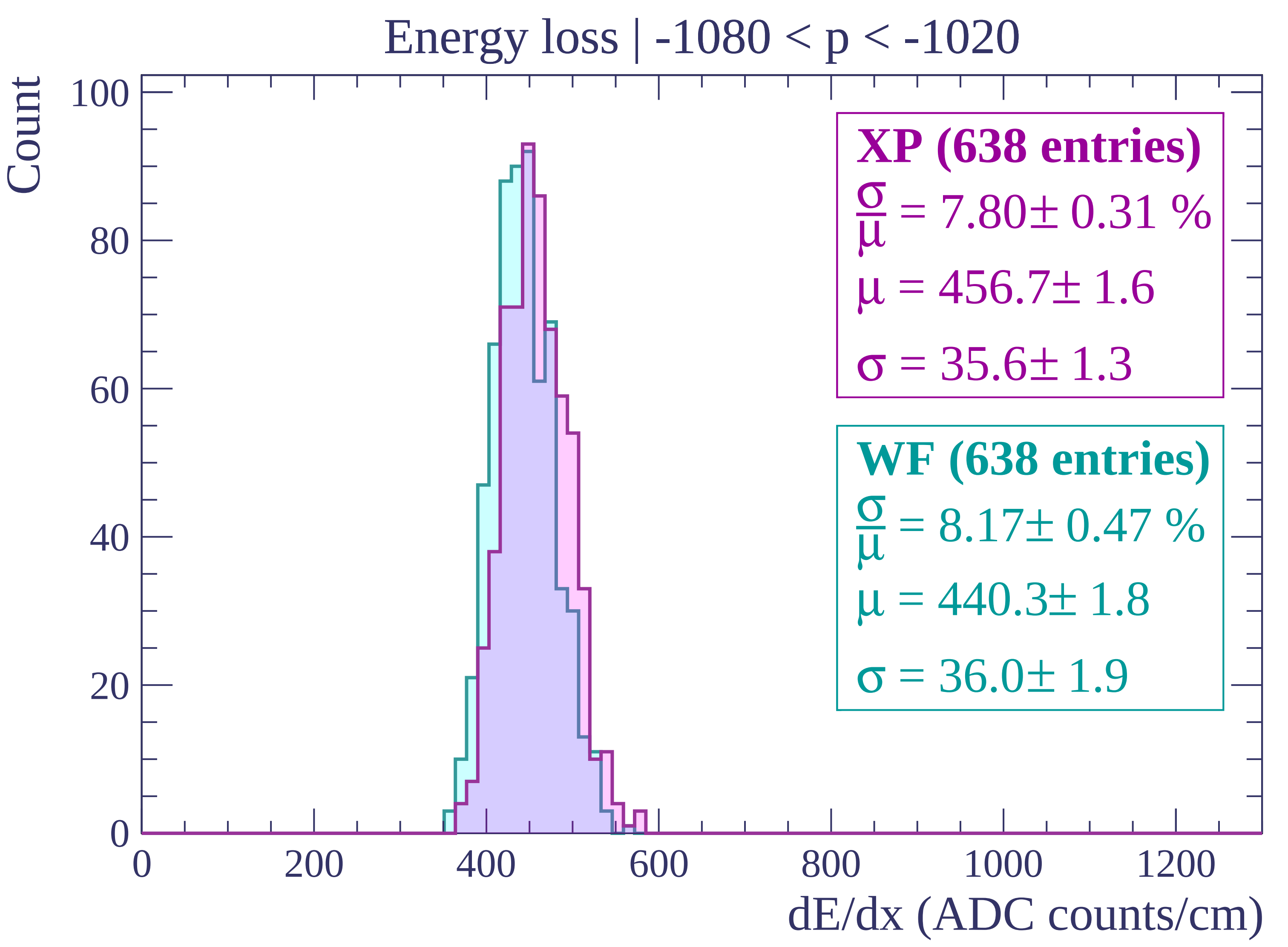
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (636 entries)
σ/μ = 7.69± 0.46 %
μ = 466.6± 1.8
σ = 35.9± 2.0
WF (636 entries)
σ/μ = 7.81± 0.45 %
μ = 452.6± 1.7
σ = 35.3± 1.9

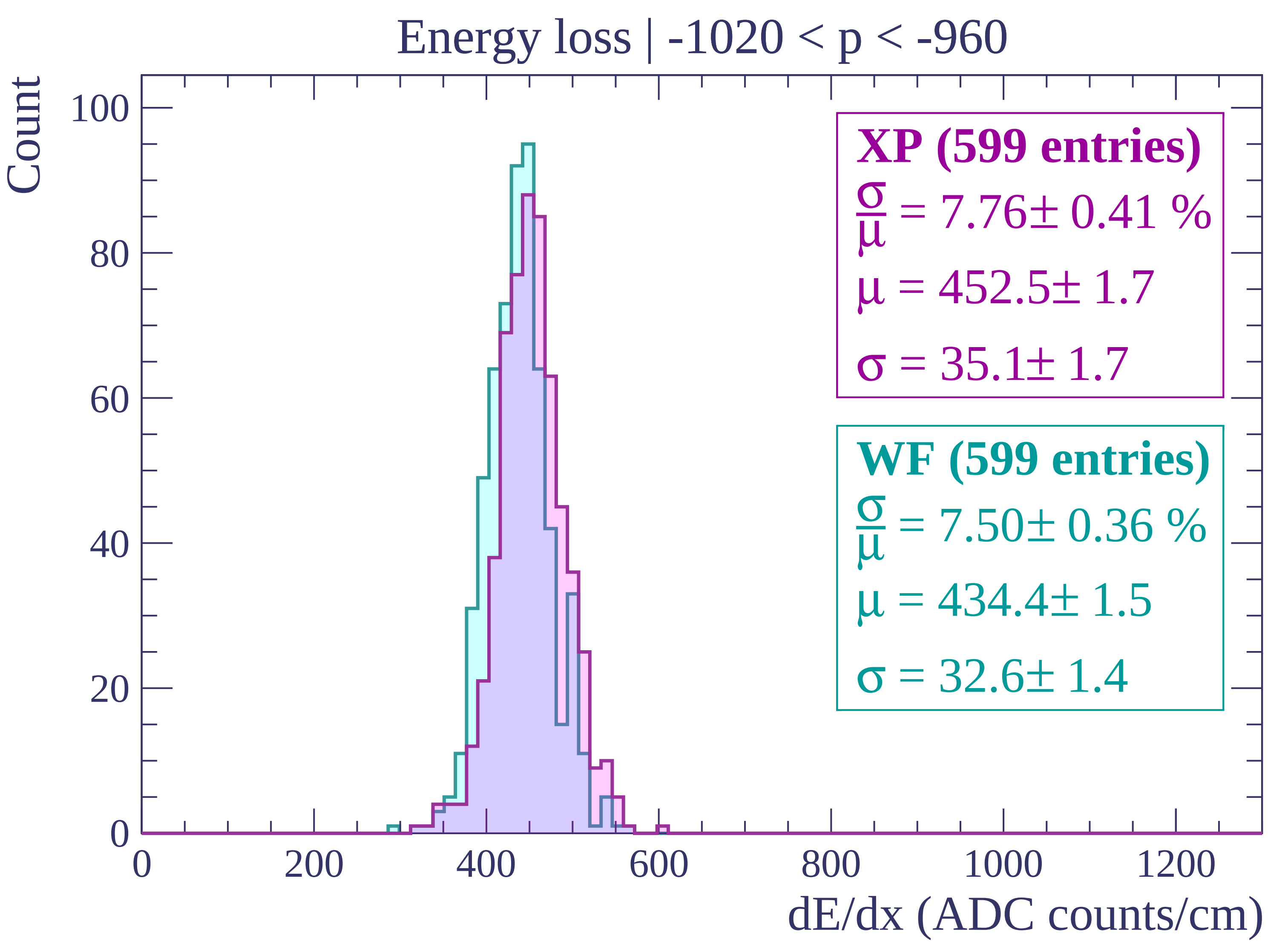


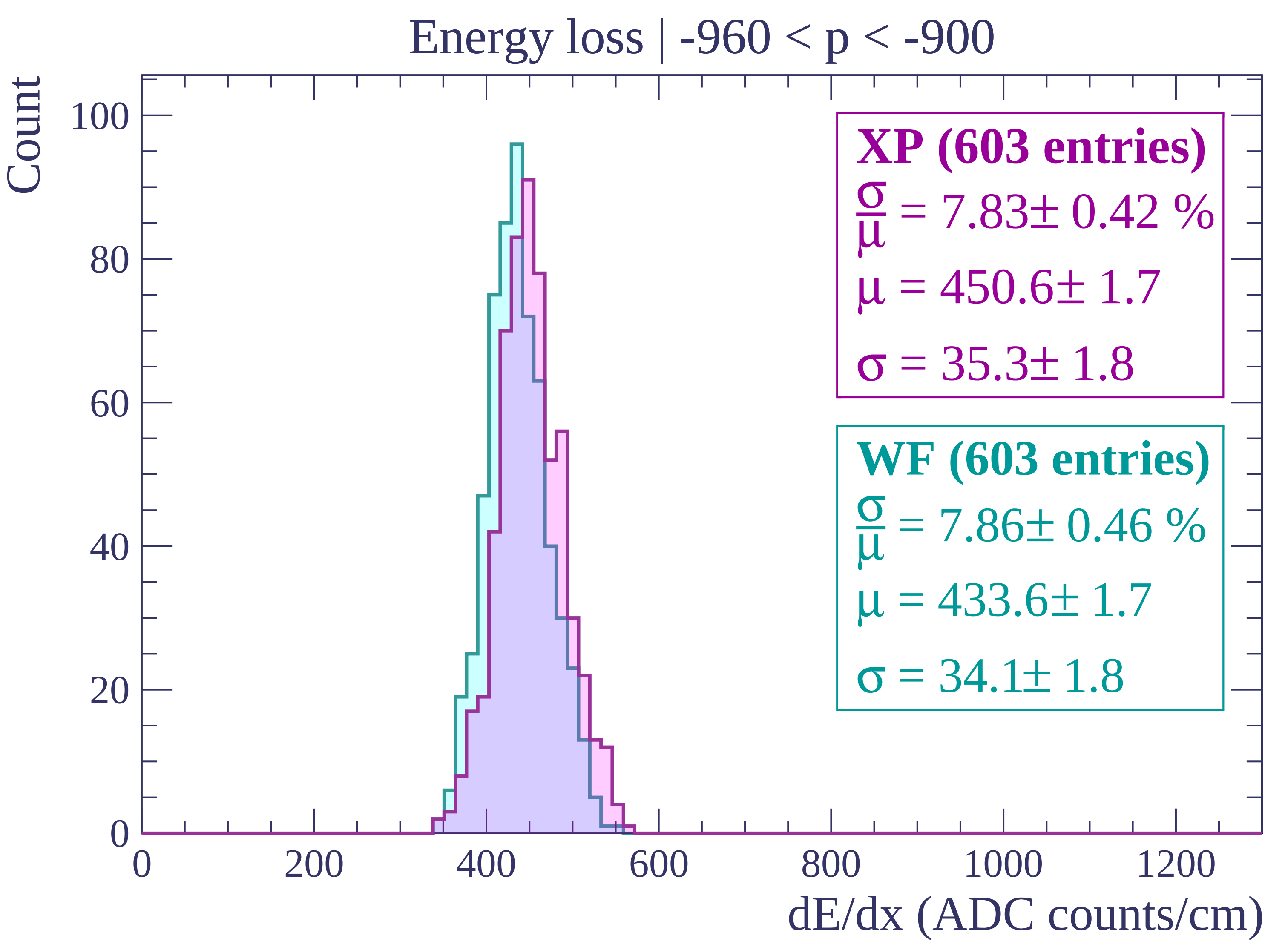
Energy loss | -1260 < p < -1200
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (659 entries)
σ/μ = 8.30± 0.46 %
μ = 464.3± 1.9
σ = 38.5± 2.0
WF (659 entries)
σ/μ = 7.99± 0.39 %
μ = 449.0± 1.7
σ = 35.9± 1.6

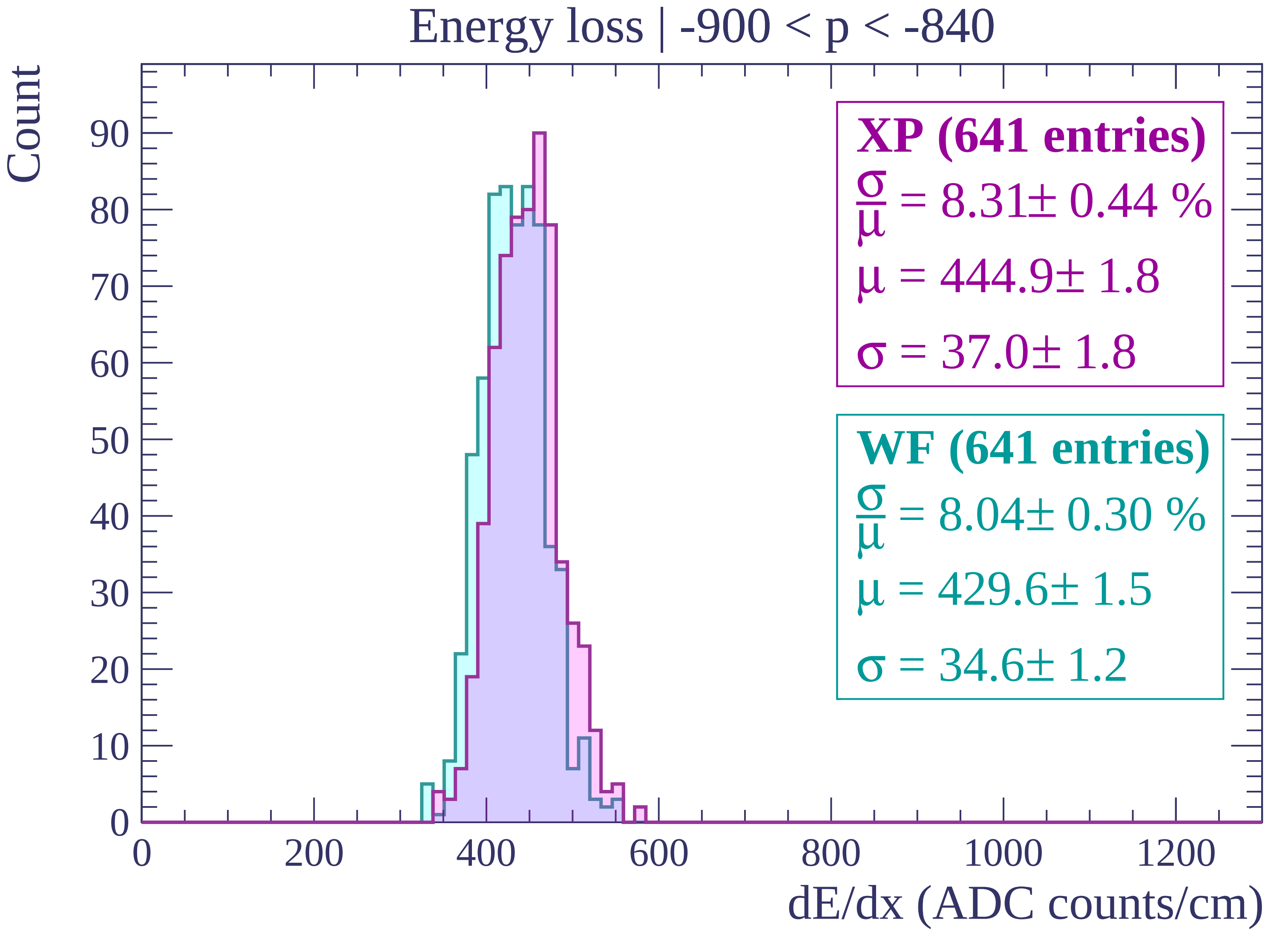

Energy loss | -1200 < p < -1140
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (608 entries)
σ/μ = 7.60± 0.40 %
μ = 463.7± 1.7
σ = 35.3± 1.7
WF (608 entries)
σ/μ = 7.67± 0.40 %
μ = 450.3± 1.7
σ = 34.5± 1.7


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (592 entries)
$\frac{\sigma}{\mu}$ = 7.74± 0.41 %
μ = 457.3± 1.8
σ = 35.4± 1.7
WF (592 entries)
$\frac{\sigma}{\mu}$ = 7.98± 0.40 %
μ = 441.8± 1.7
σ = 35.3± 1.6

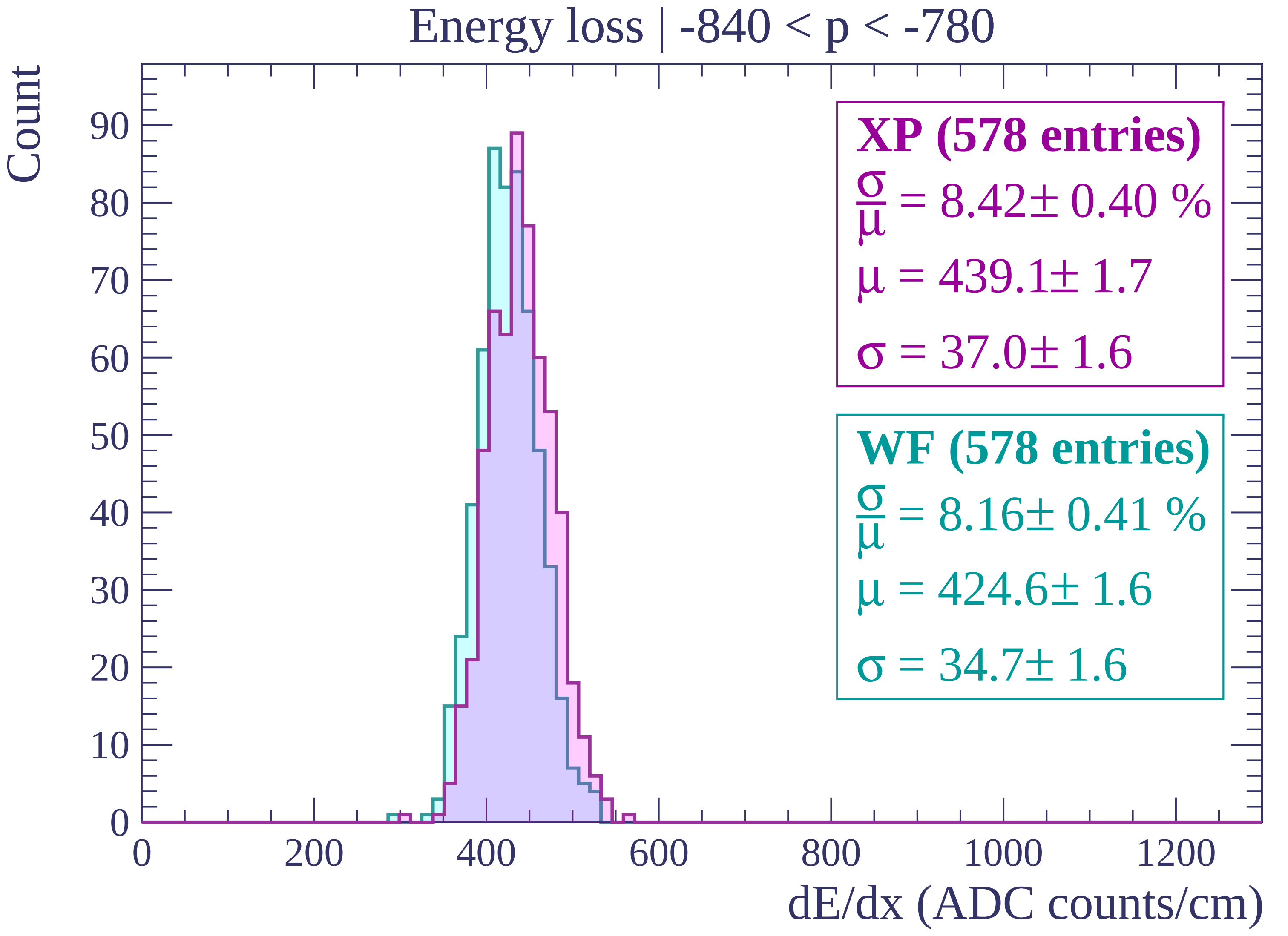

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (638 entries)
σ/μ = 7.80± 0.31 %
μ = 456.7± 1.6
σ = 35.6± 1.3
WF (638 entries)
σ/μ = 8.17± 0.47 %
μ = 440.3± 1.8
σ = 36.0± 1.9


Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (599 entries)
σ/μ = 7.76± 0.41 %
μ = 452.5± 1.7
σ = 35.1± 1.7
WF (599 entries)
σ/μ = 7.50± 0.36 %
μ = 434.4± 1.5
σ = 32.6± 1.4

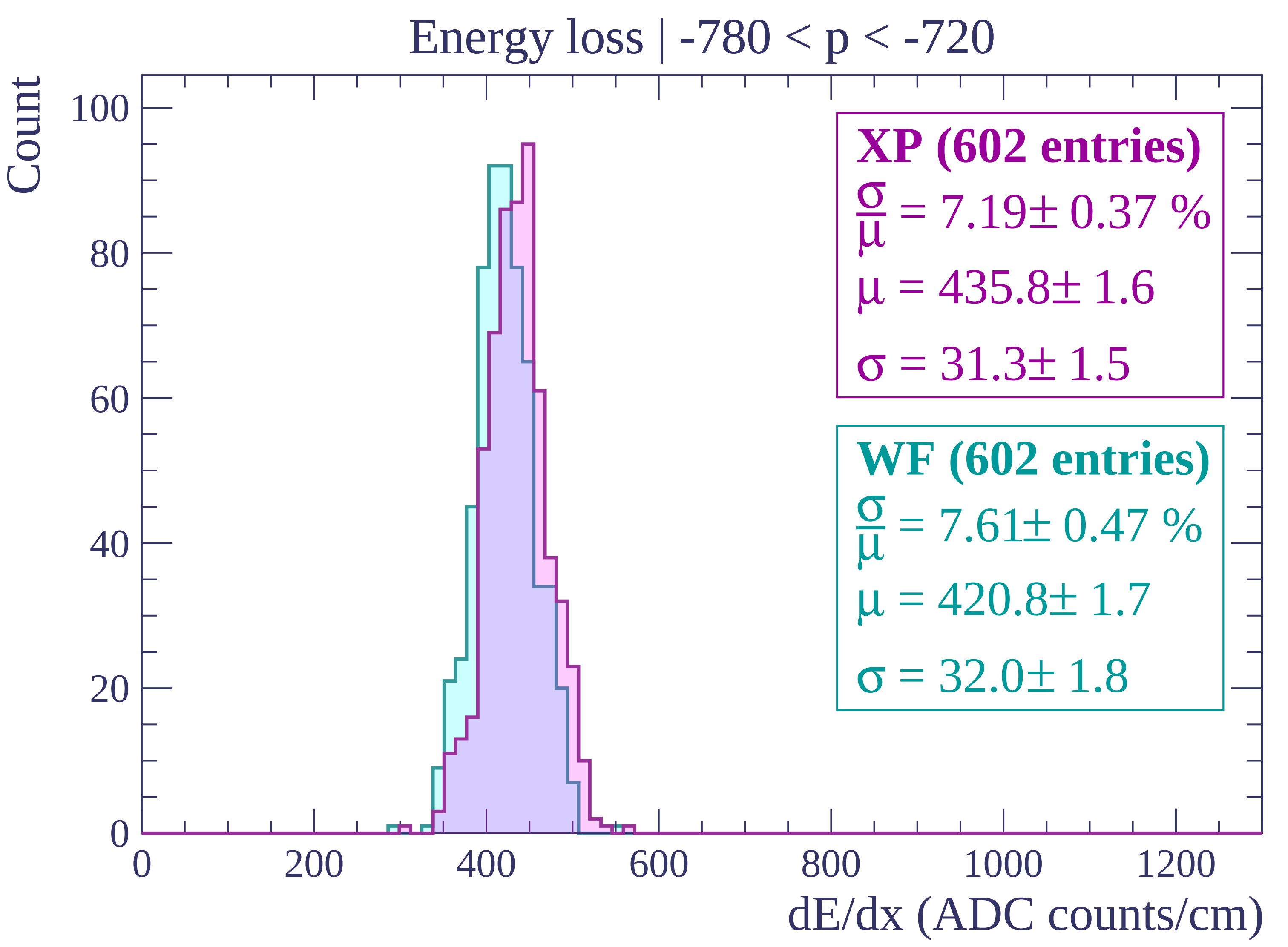

Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (603 entries)
$\frac{\sigma}{\mu}$ = 7.83± 0.42 %
μ = 450.6± 1.7
σ = 35.3± 1.8
WF (603 entries)
$\frac{\sigma}{\mu}$ = 7.86± 0.46 %
μ = 433.6± 1.7
σ = 34.1± 1.8



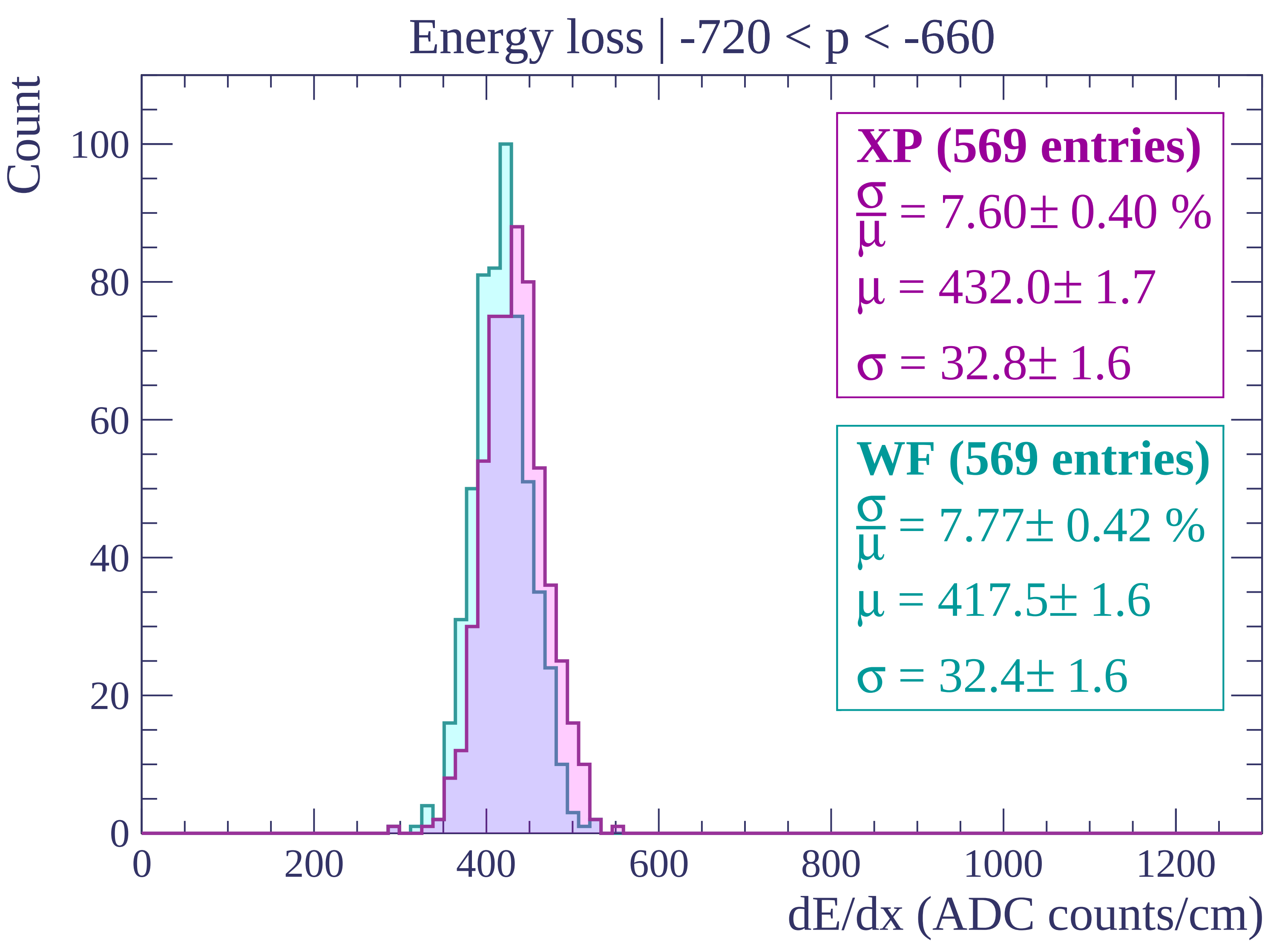
Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (641 entries)
σ/μ = 8.31± 0.44 %
μ = 444.9± 1.8
σ = 37.0± 1.8
WF (641 entries)
σ/μ = 8.04± 0.30 %
μ = 429.6± 1.5
σ = 34.6± 1.2

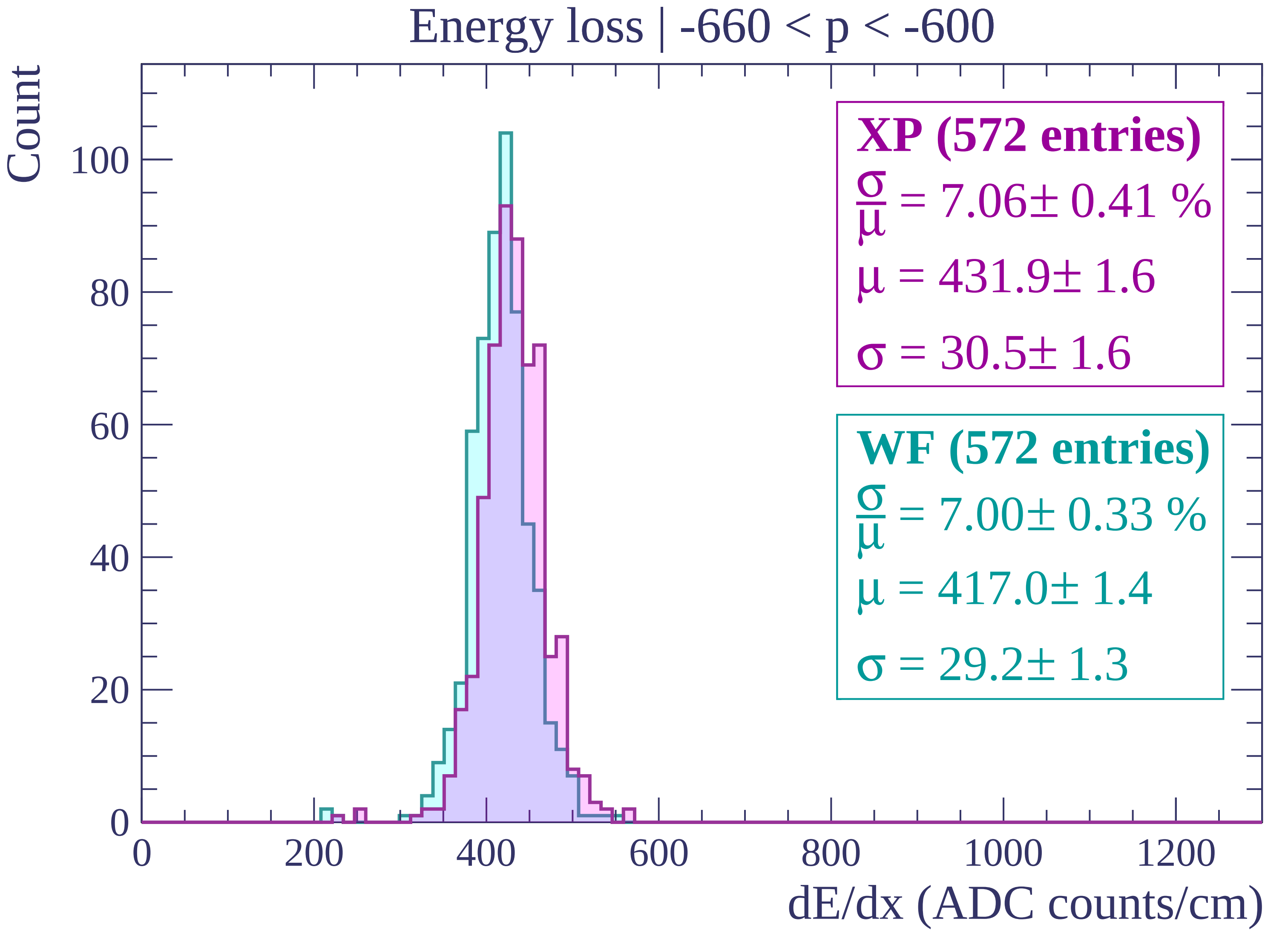


Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (578 entries)
σ/μ = 8.42± 0.40 %
μ = 439.1± 1.7
σ = 37.0± 1.6
WF (578 entries)
σ/μ = 8.16± 0.41 %
μ = 424.6± 1.6
σ = 34.7± 1.6

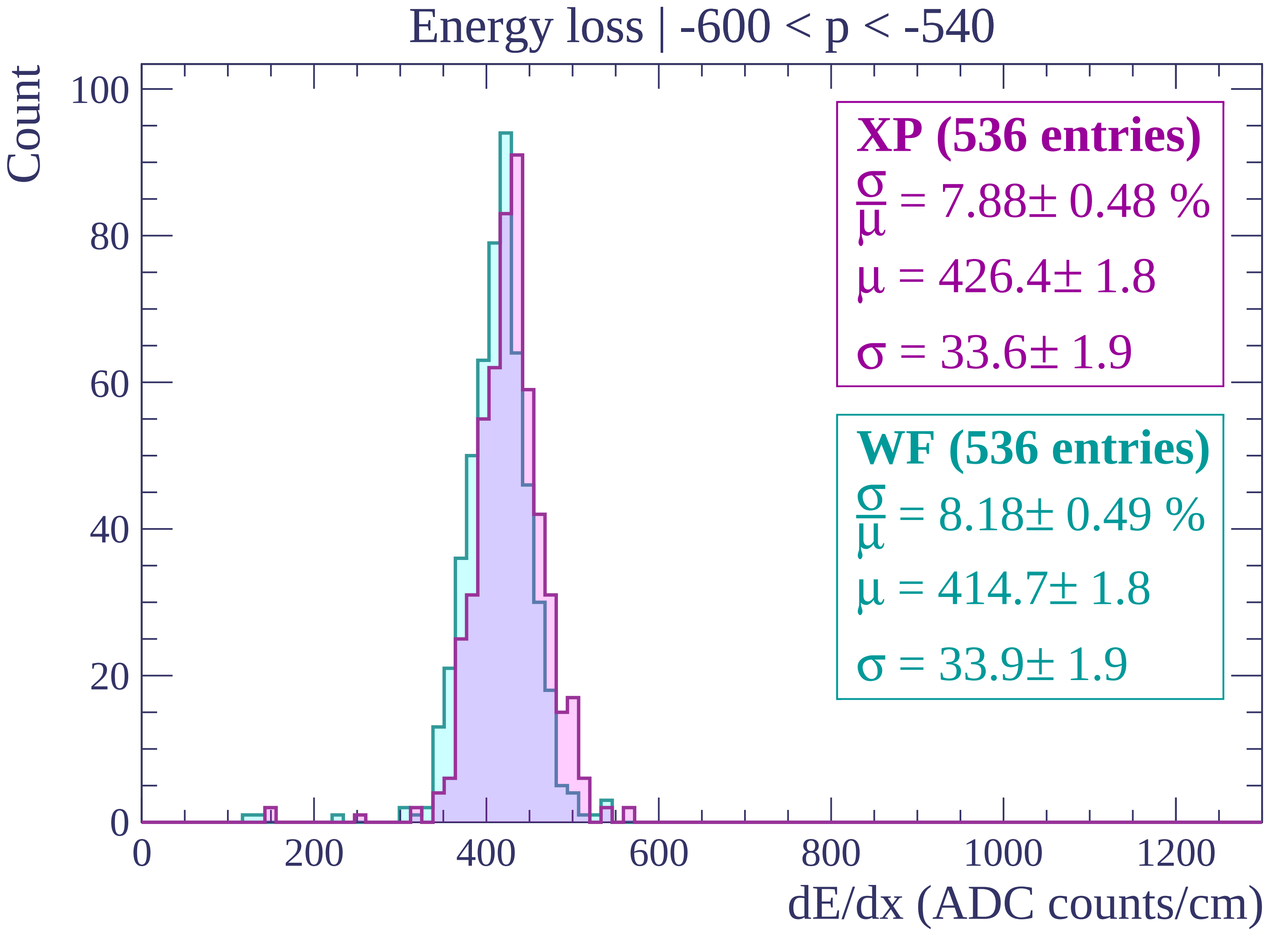


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (602 entries)
$\frac{\sigma}{\mu}$ = 7.19± 0.37 %
μ = 435.8± 1.6
σ = 31.3± 1.5
WF (602 entries)
$\frac{\sigma}{\mu}$ = 7.61± 0.47 %
μ = 420.8± 1.7
σ = 32.0± 1.8

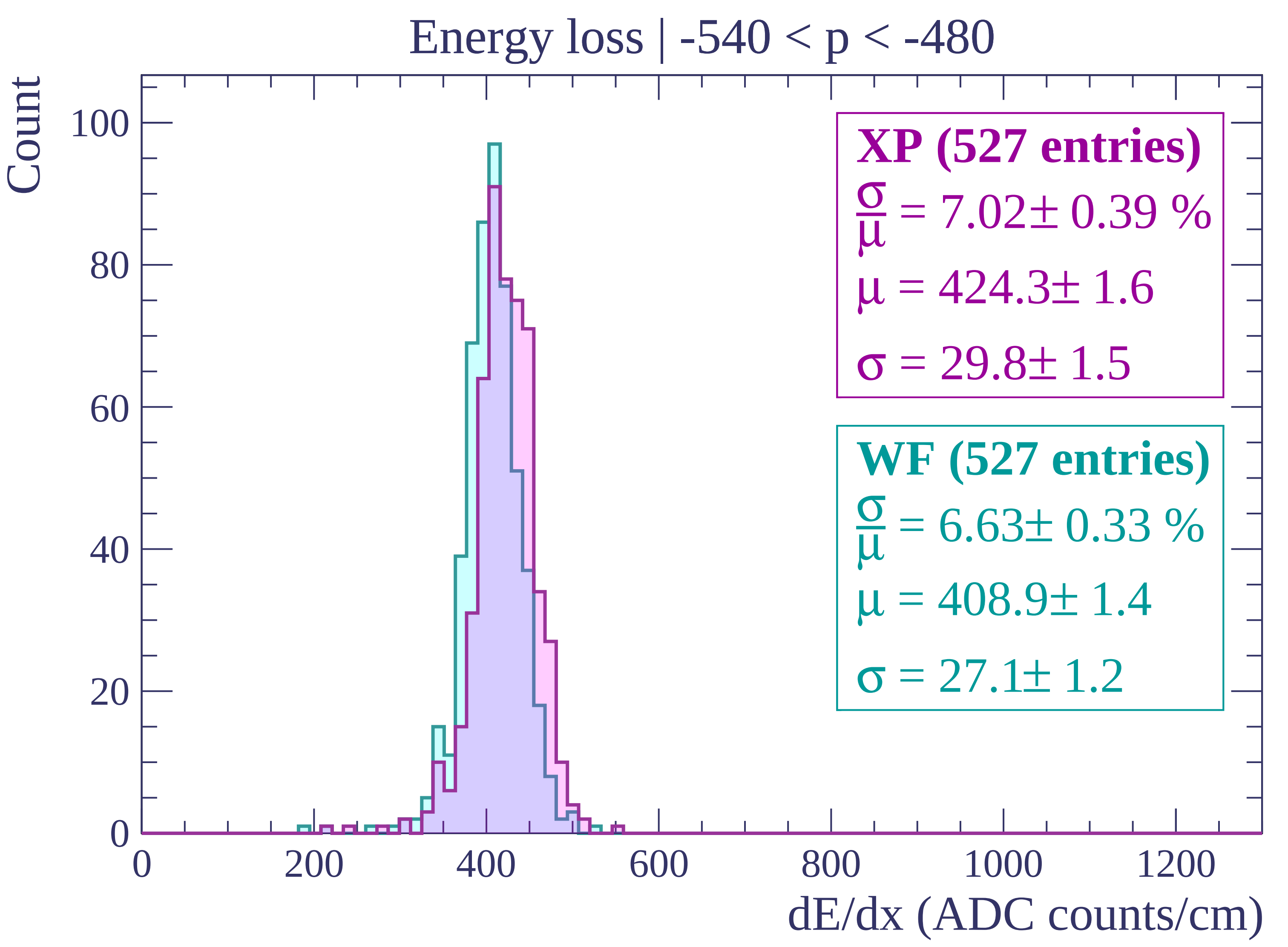

Energy loss | -720 < p < -660
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (569 entries)
σ/μ = 7.60± 0.40 %
μ = 432.0± 1.7
σ = 32.8± 1.6
WF (569 entries)
σ/μ = 7.77± 0.42 %
μ = 417.5± 1.6
σ = 32.4± 1.6

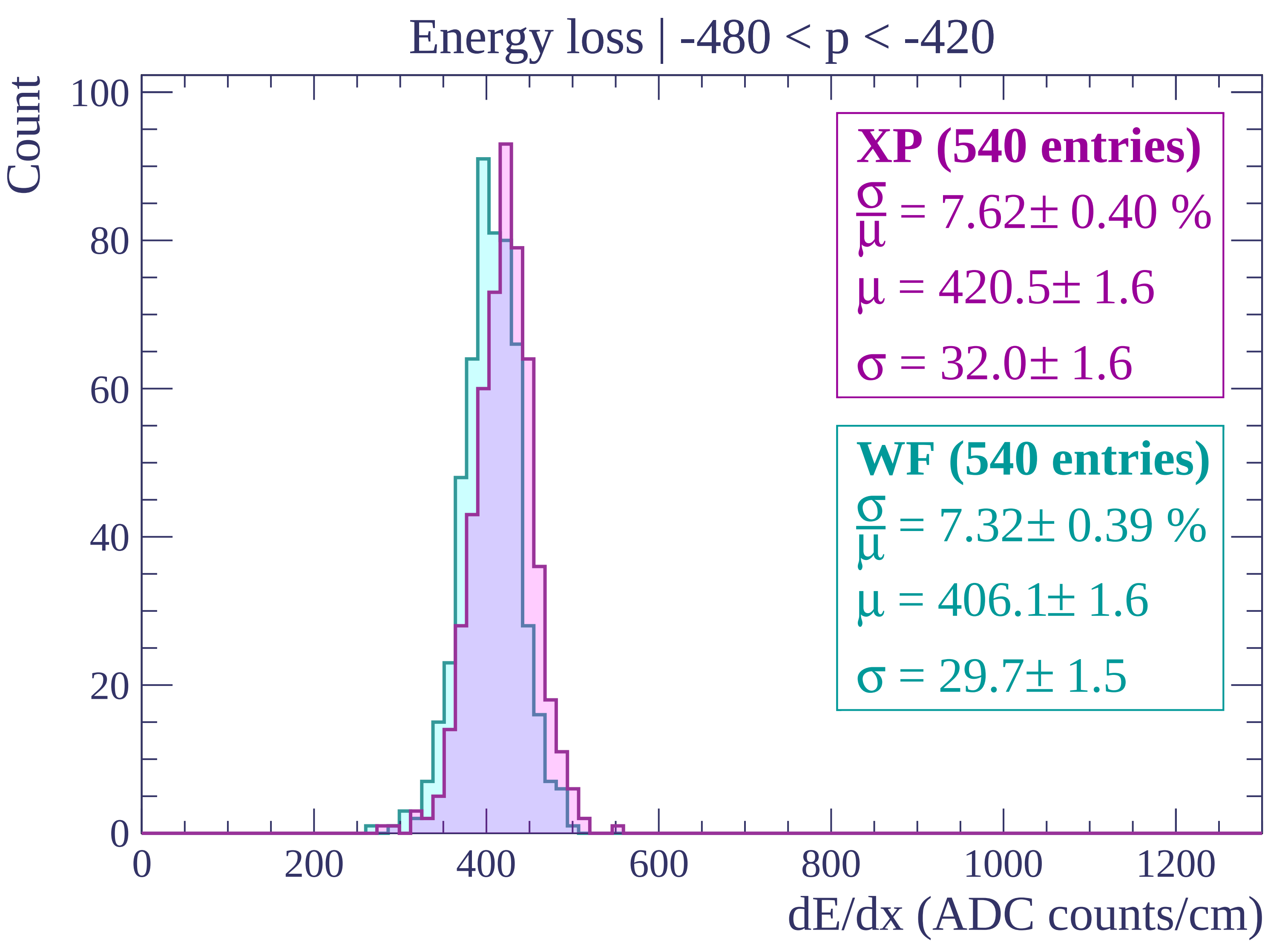


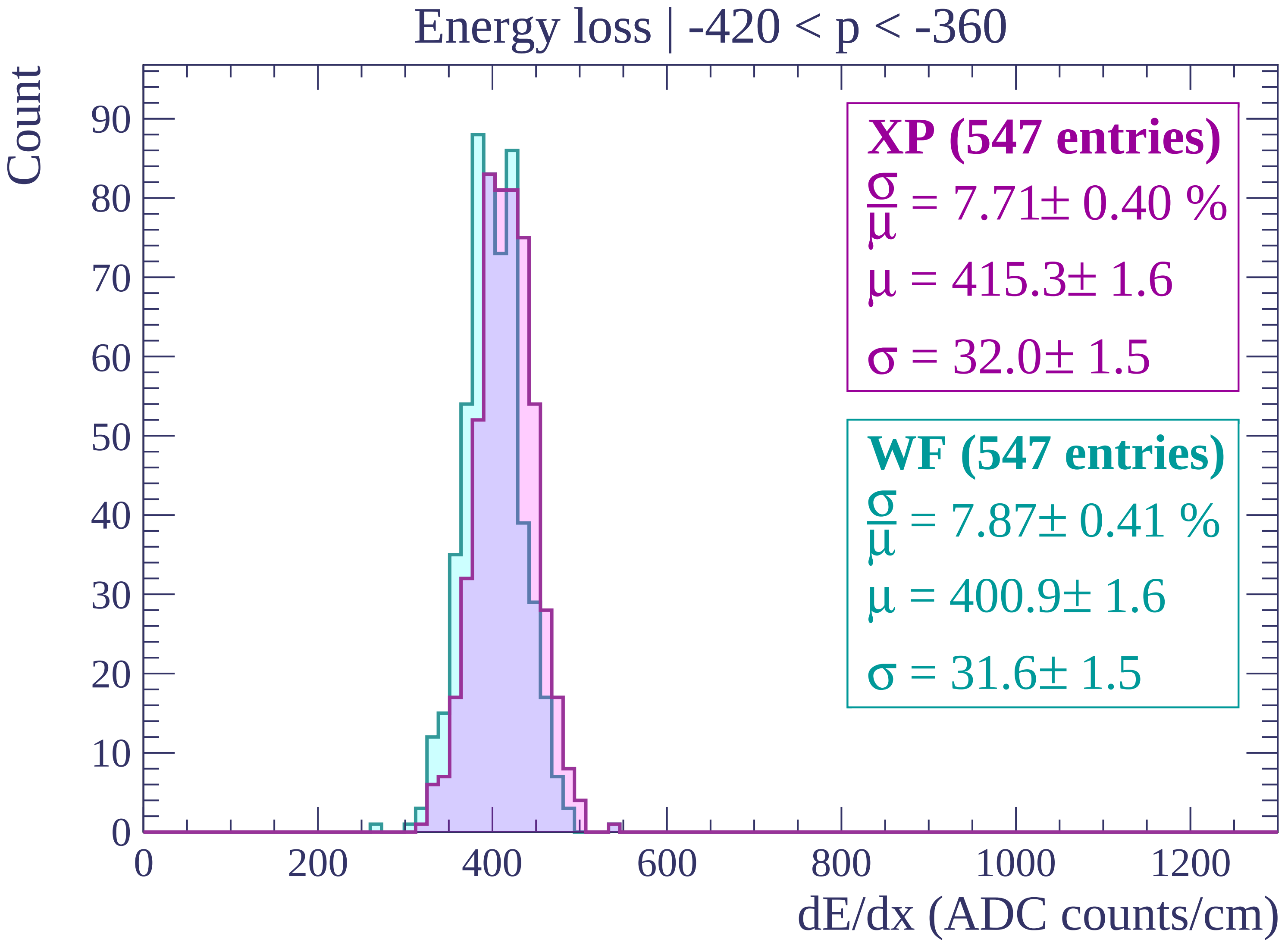
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
XP (572 entries)
σ/μ = 7.06± 0.41 %
μ = 431.9± 1.6
σ = 30.5± 1.6
WF (572 entries)
σ/μ = 7.00± 0.33 %
μ = 417.0± 1.4
σ = 29.2± 1.3
0
20
40
60
80
100
0
200
400
600
800
1000
1200



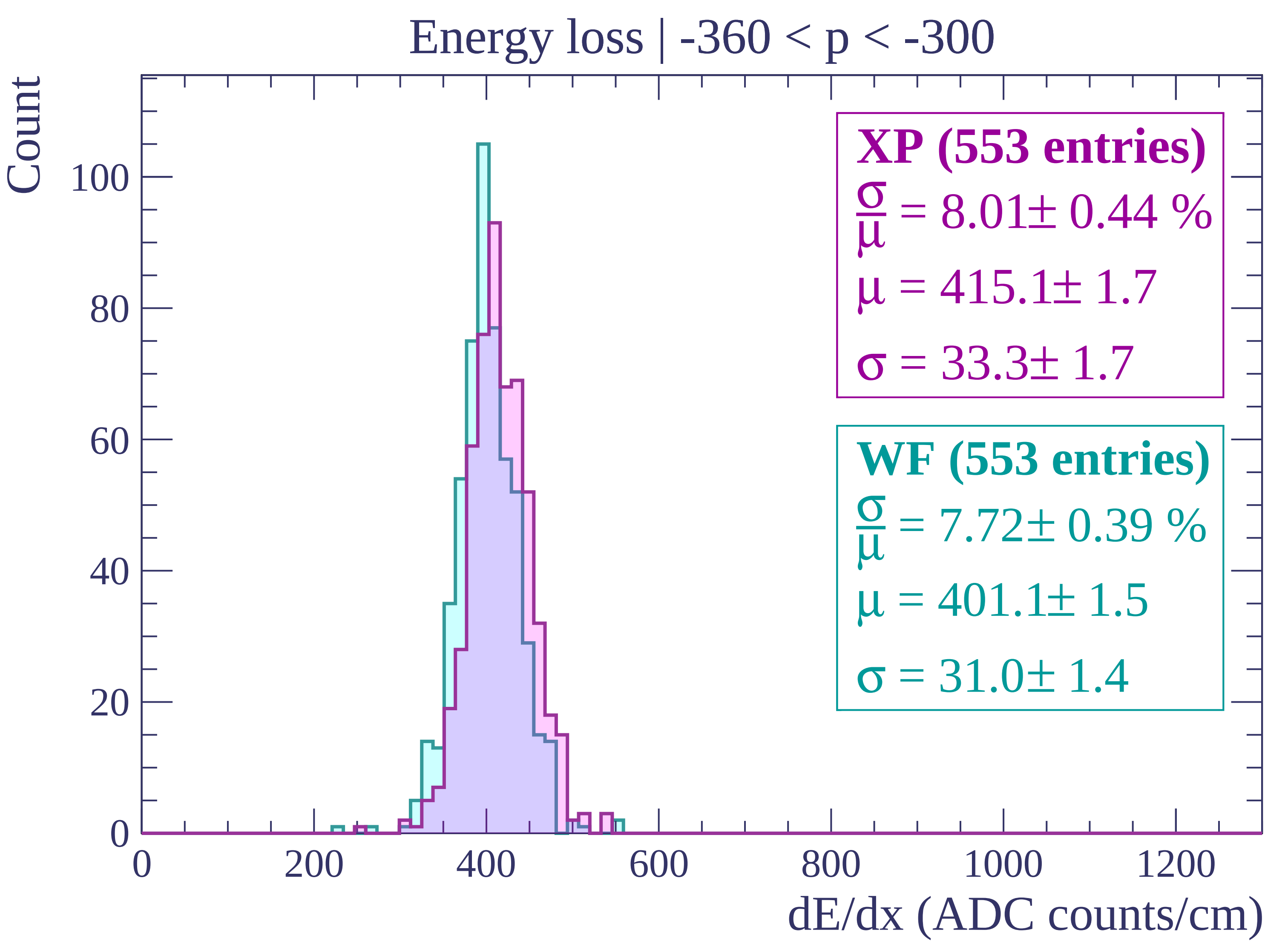
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (536 entries)
σ/μ = 7.88± 0.48 %
μ = 426.4± 1.8
σ = 33.6± 1.9
WF (536 entries)
σ/μ = 8.18± 0.49 %
μ = 414.7± 1.8
σ = 33.9± 1.9


Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (527 entries)
σ/μ = 7.02± 0.39 %
μ = 424.3± 1.6
σ = 29.8± 1.5
WF (527 entries)
σ/μ = 6.63± 0.33 %
μ = 408.9± 1.4
σ = 27.1± 1.2

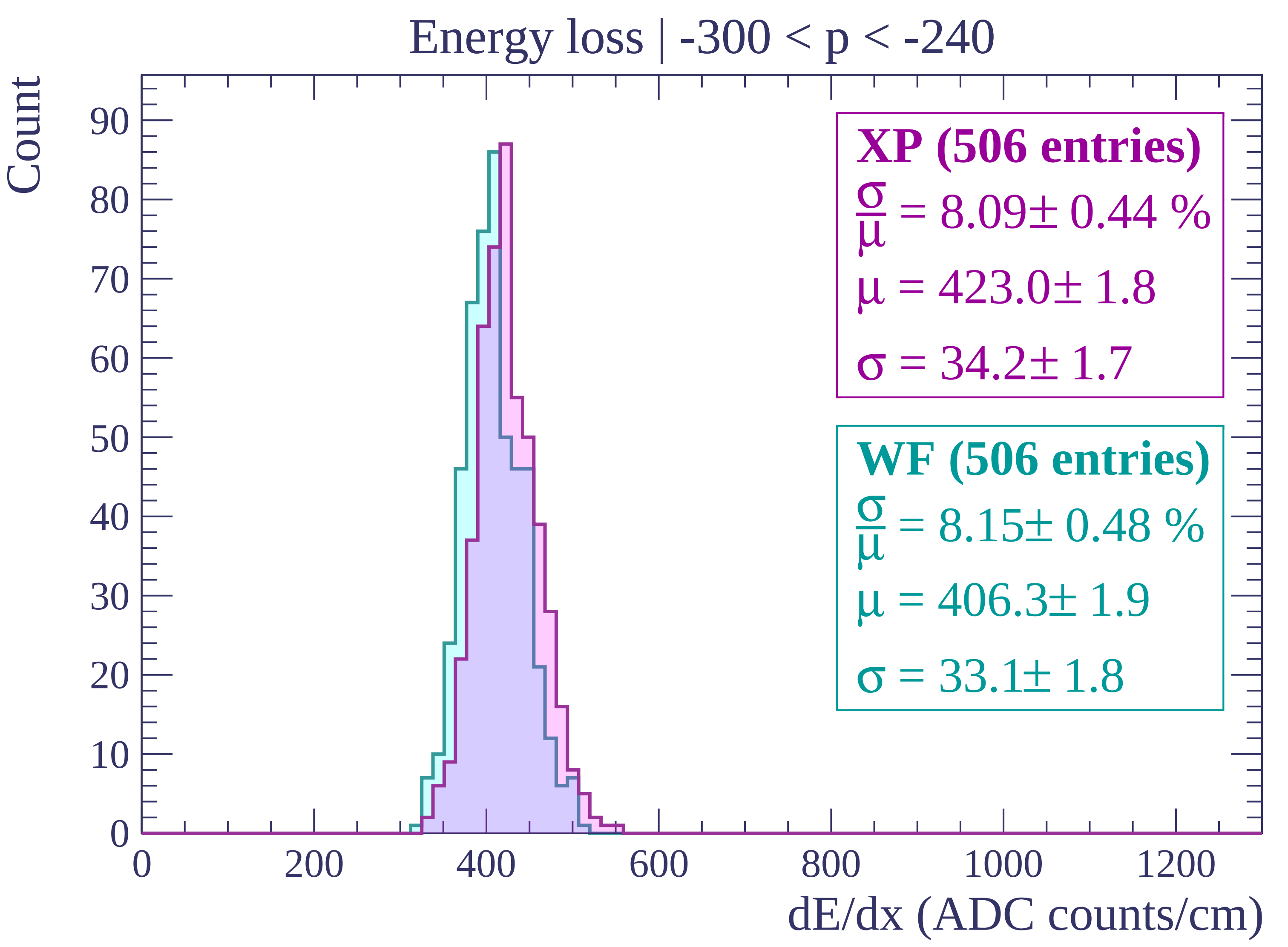

Energy loss | -480 < p < -420
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (540 entries)
σ/μ = 7.62± 0.40 %
μ = 420.5± 1.6
σ = 32.0± 1.6
WF (540 entries)
σ/μ = 7.32± 0.39 %
μ = 406.1± 1.6
σ = 29.7± 1.5

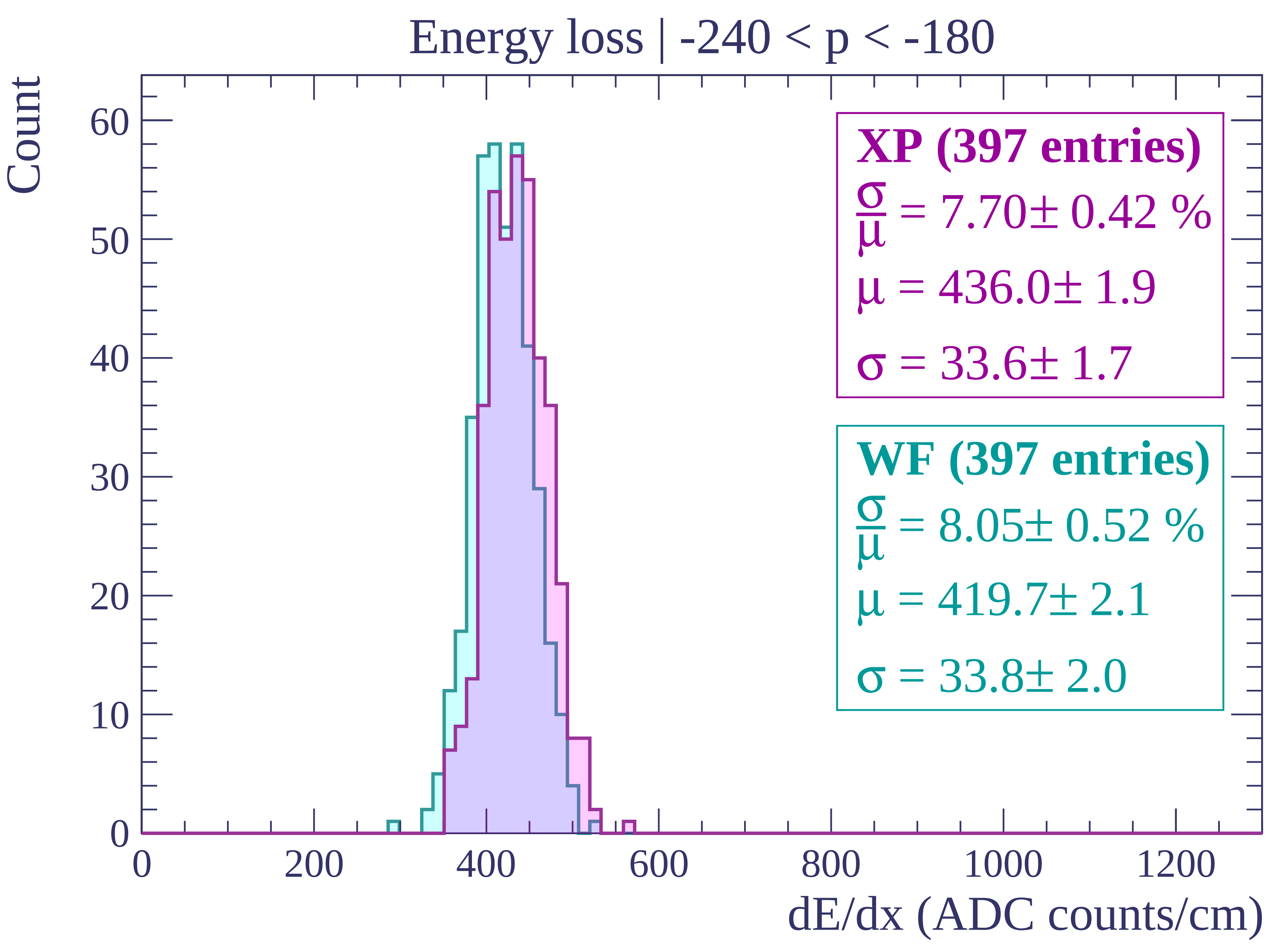


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (547 entries)
σ/μ = 7.71± 0.40 %
μ = 415.3± 1.6
σ = 32.0± 1.5
WF (547 entries)
σ/μ = 7.87± 0.41 %
μ = 400.9± 1.6
σ = 31.6± 1.5

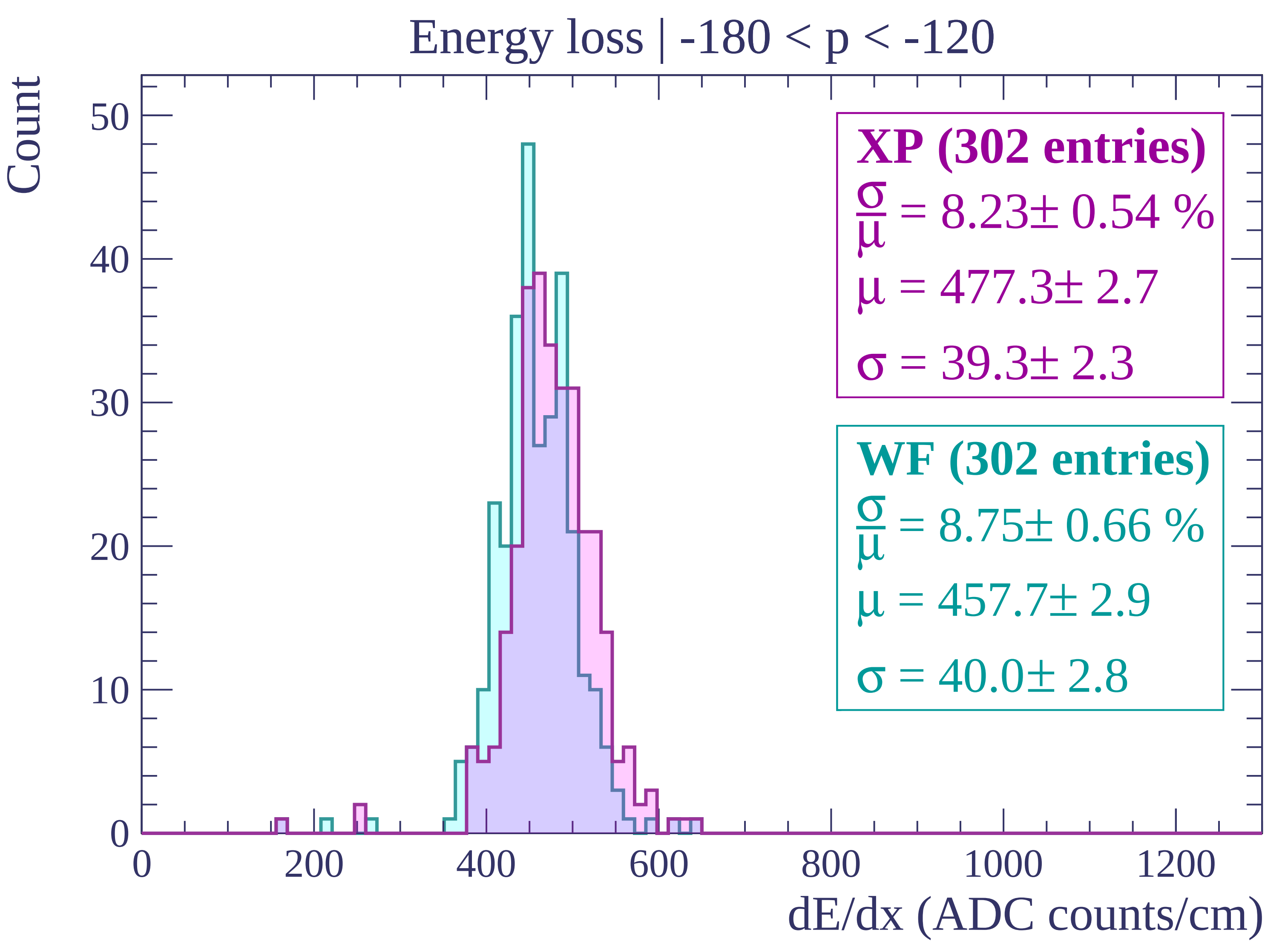


Energy loss | -360 < p < -300
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (553 entries)
$\frac{\sigma}{\mu}$ = 8.01± 0.44 %
μ = 415.1± 1.7
σ = 33.3± 1.7
WF (553 entries)
$\frac{\sigma}{\mu}$ = 7.72± 0.39 %
μ = 401.1± 1.5
σ = 31.0± 1.4

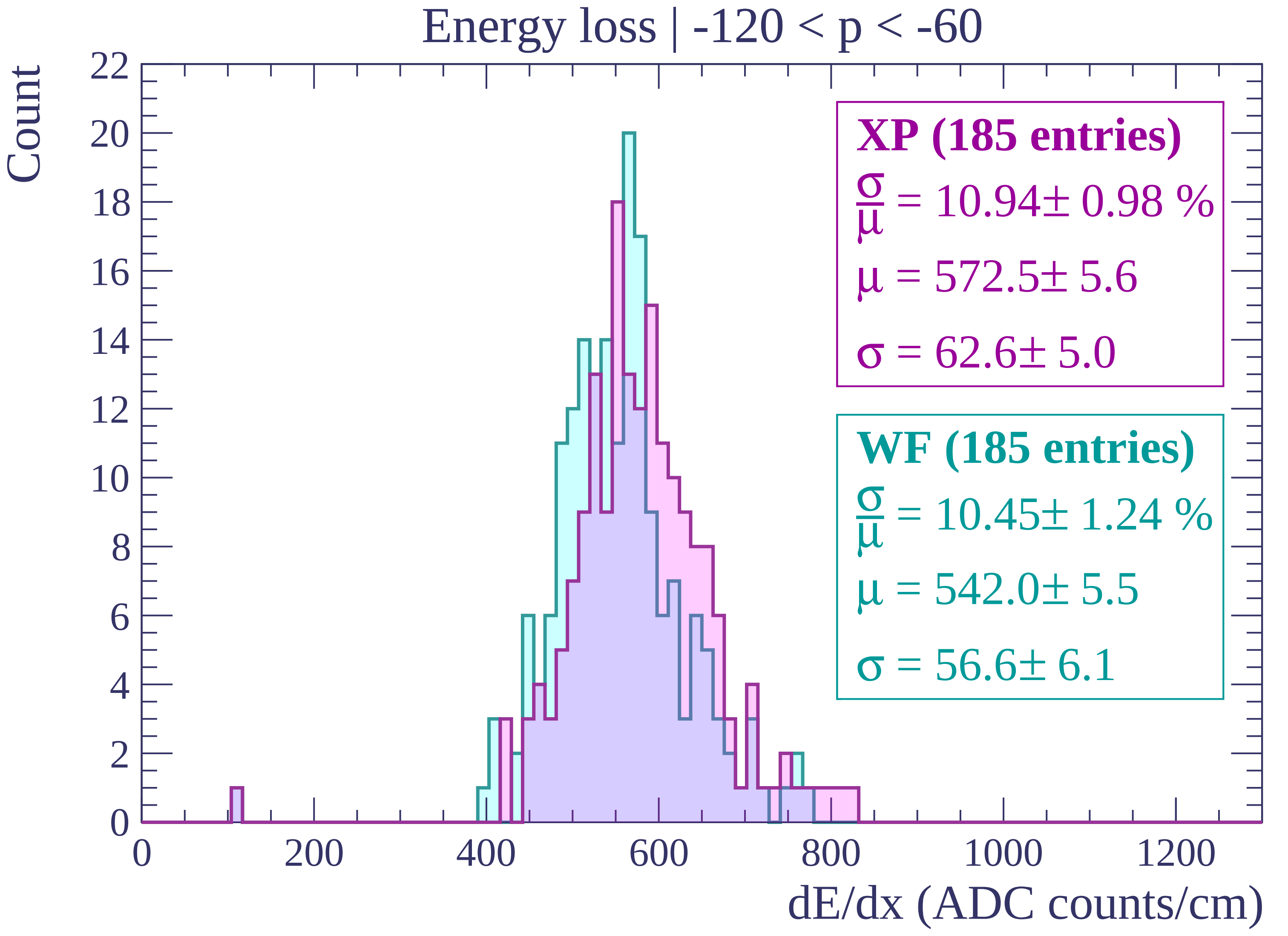

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (506 entries)
σ/μ = 8.09± 0.44 %
μ = 423.0± 1.8
σ = 34.2± 1.7
WF (506 entries)
σ/μ = 8.15± 0.48 %
μ = 406.3± 1.9
σ = 33.1± 1.8



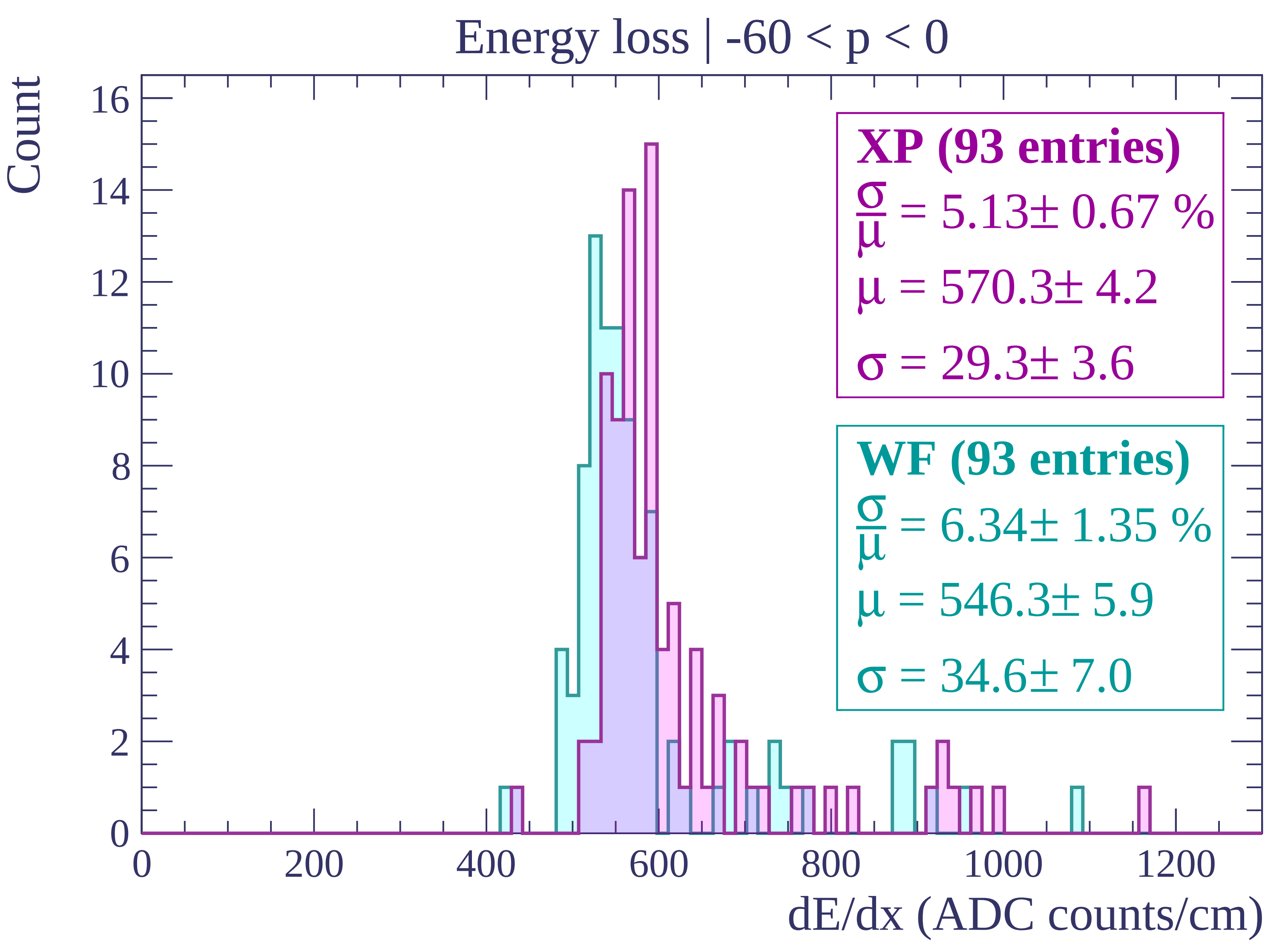
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (397 entries)
σ/μ = 7.70± 0.42 %
μ = 436.0± 1.9
σ = 33.6± 1.7
WF (397 entries)
σ/μ = 8.05± 0.52 %
μ = 419.7± 2.1
σ = 33.8± 2.0



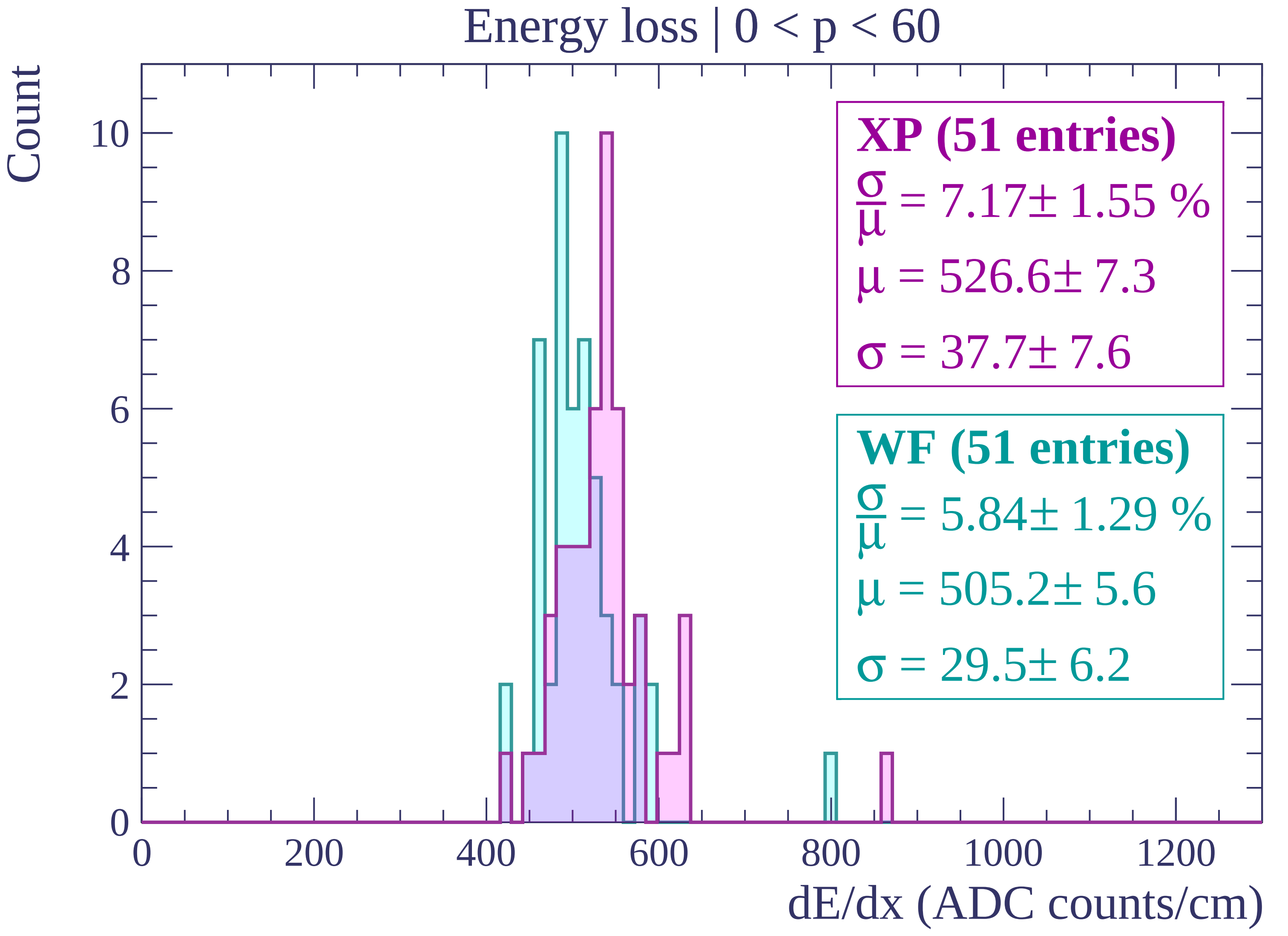
Energy loss | -180 < p < -120
Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (302 entries)
σ/μ = 8.23± 0.54 %
μ = 477.3± 2.7
σ = 39.3± 2.3
WF (302 entries)
σ/μ = 8.75± 0.66 %
μ = 457.7± 2.9
σ = 40.0± 2.8



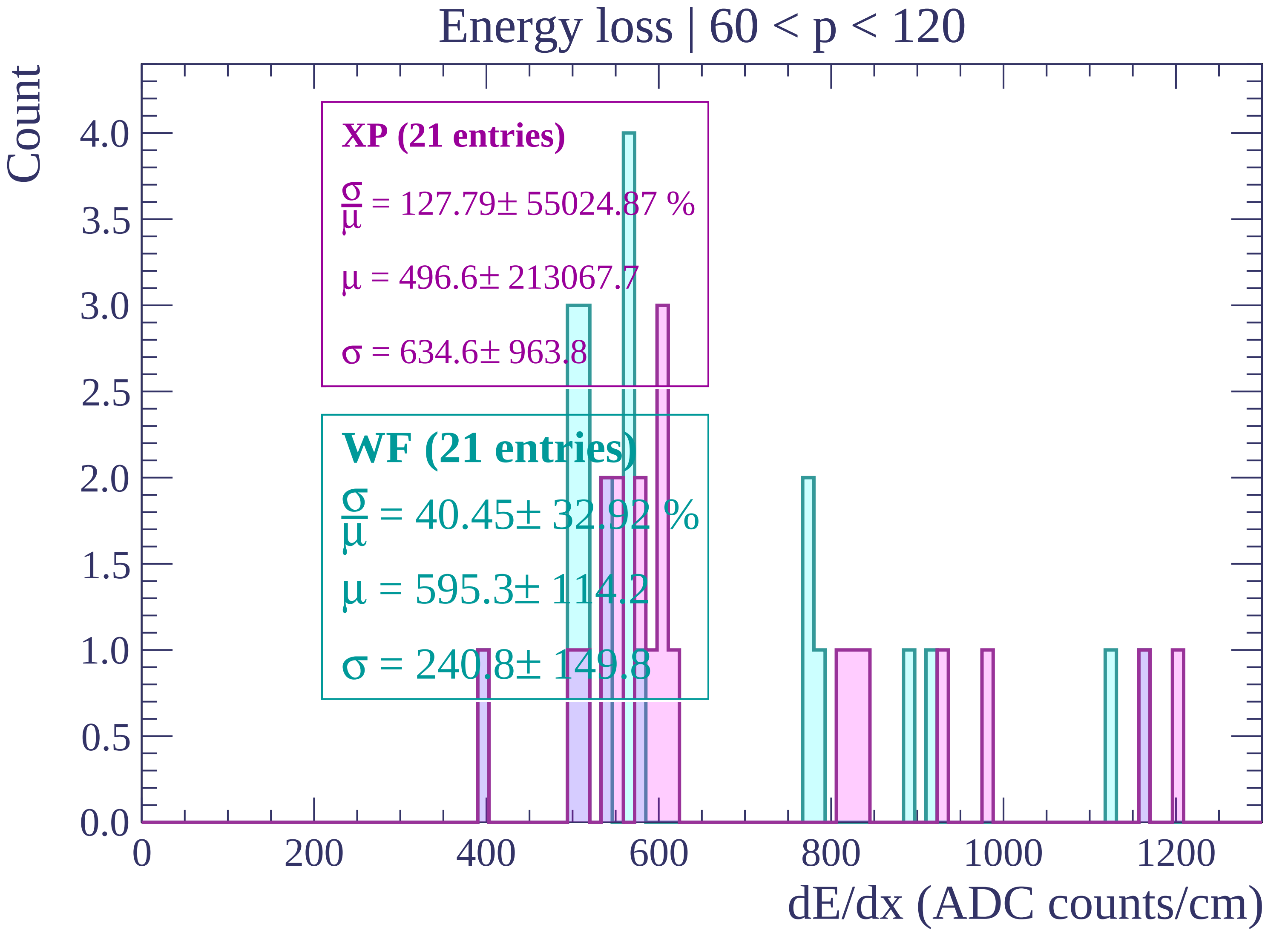
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
XP (185 entries)
σ/μ = 10.94± 0.98 %
μ = 572.5± 5.6
σ = 62.6± 5.0
WF (185 entries)
σ/μ = 10.45± 1.24 %
μ = 542.0± 5.5
σ = 56.6± 6.1
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200

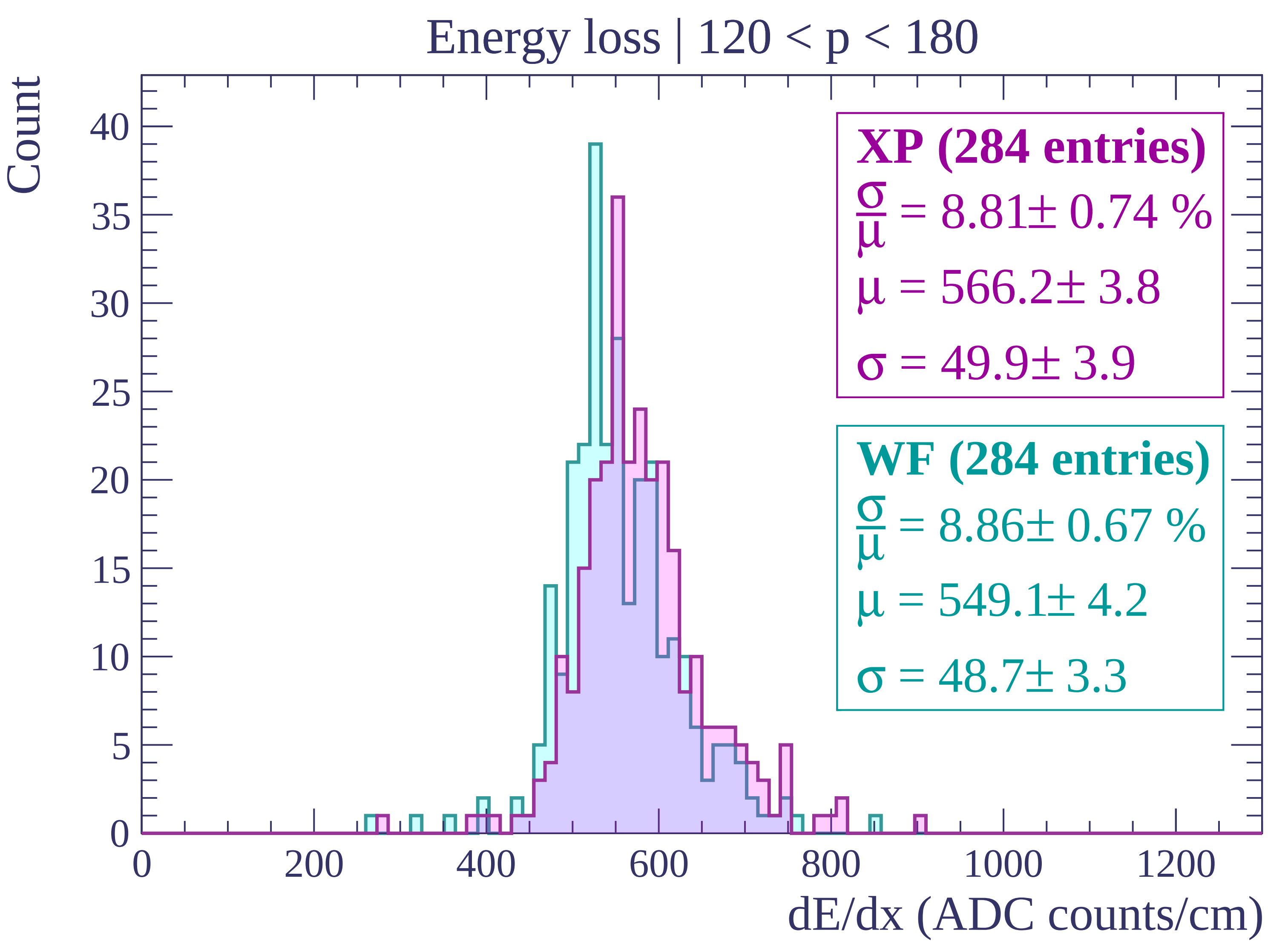


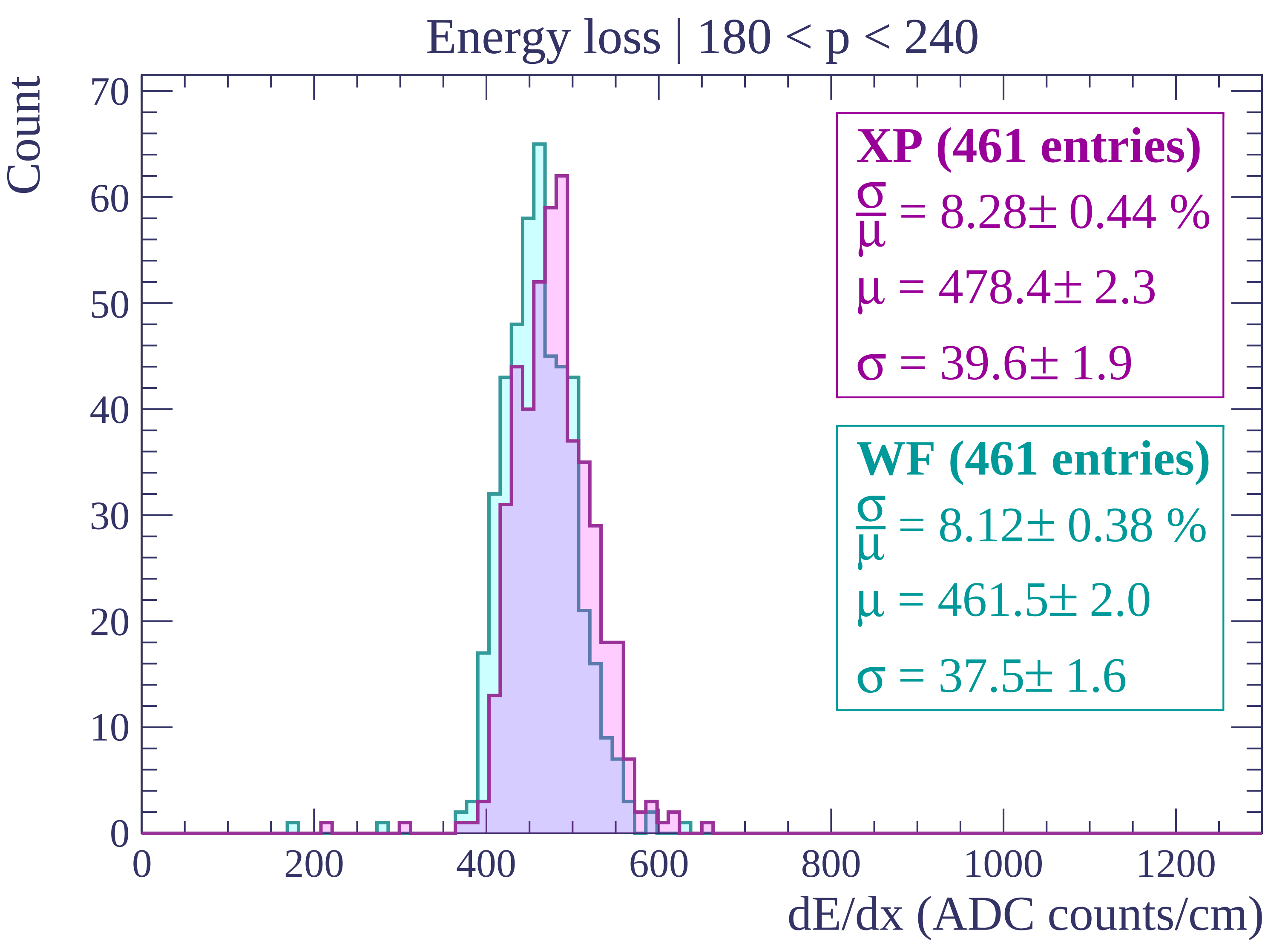
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
XP (93 entries)
$\frac{\sigma}{\mu}$ = 5.13± 0.67 %
μ = 570.3± 4.2
σ = 29.3± 3.6
WF (93 entries)
$\frac{\sigma}{\mu}$ = 6.34± 1.35 %
μ = 546.3± 5.9
σ = 34.6± 7.0
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16



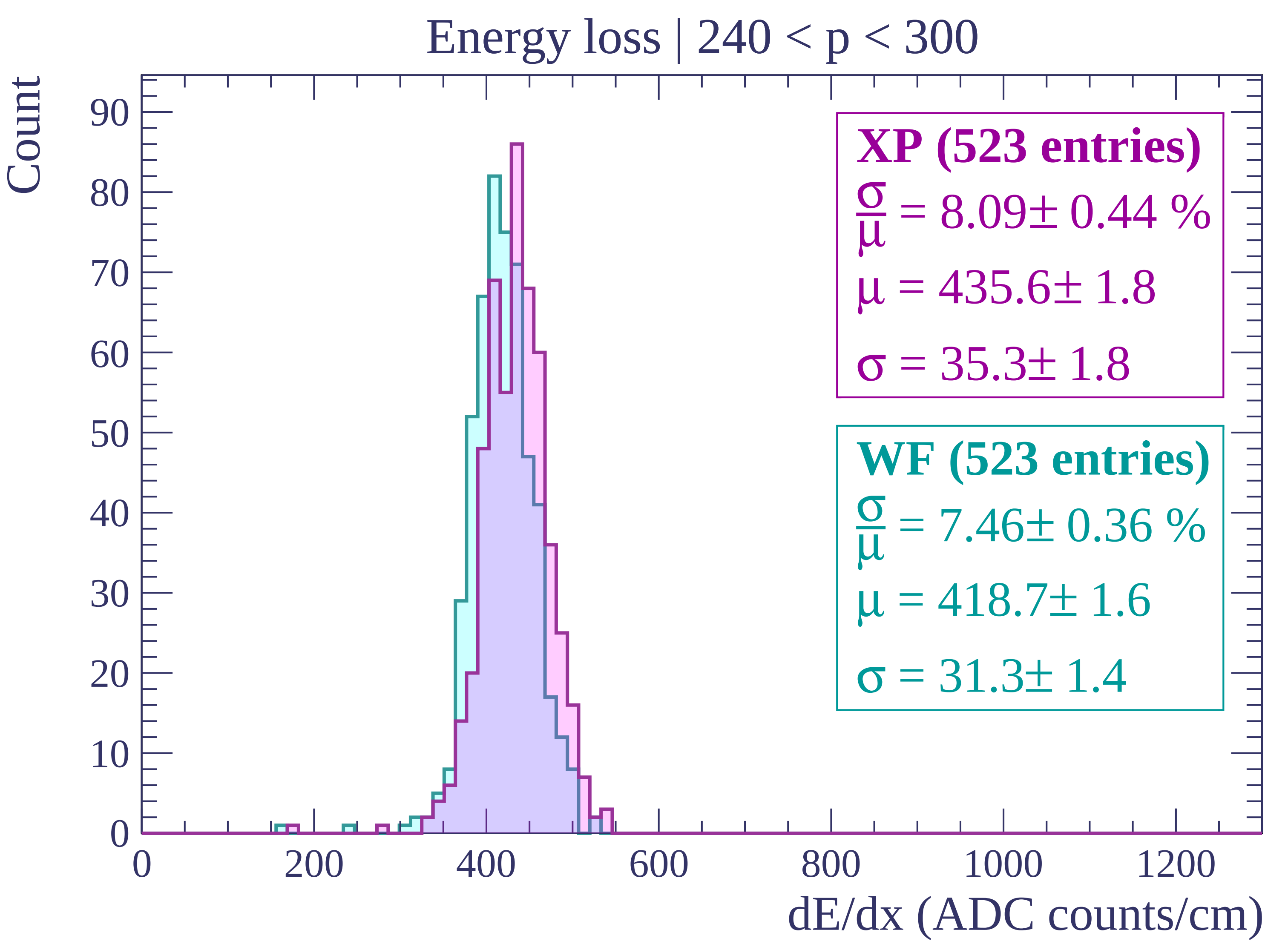
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
XP (51 entries)
σ/μ = 7.17± 1.55 %
μ = 526.6± 7.3
σ = 37.7± 7.6
WF (51 entries)
σ/μ = 5.84± 1.29 %
μ = 505.2± 5.6
σ = 29.5± 6.2

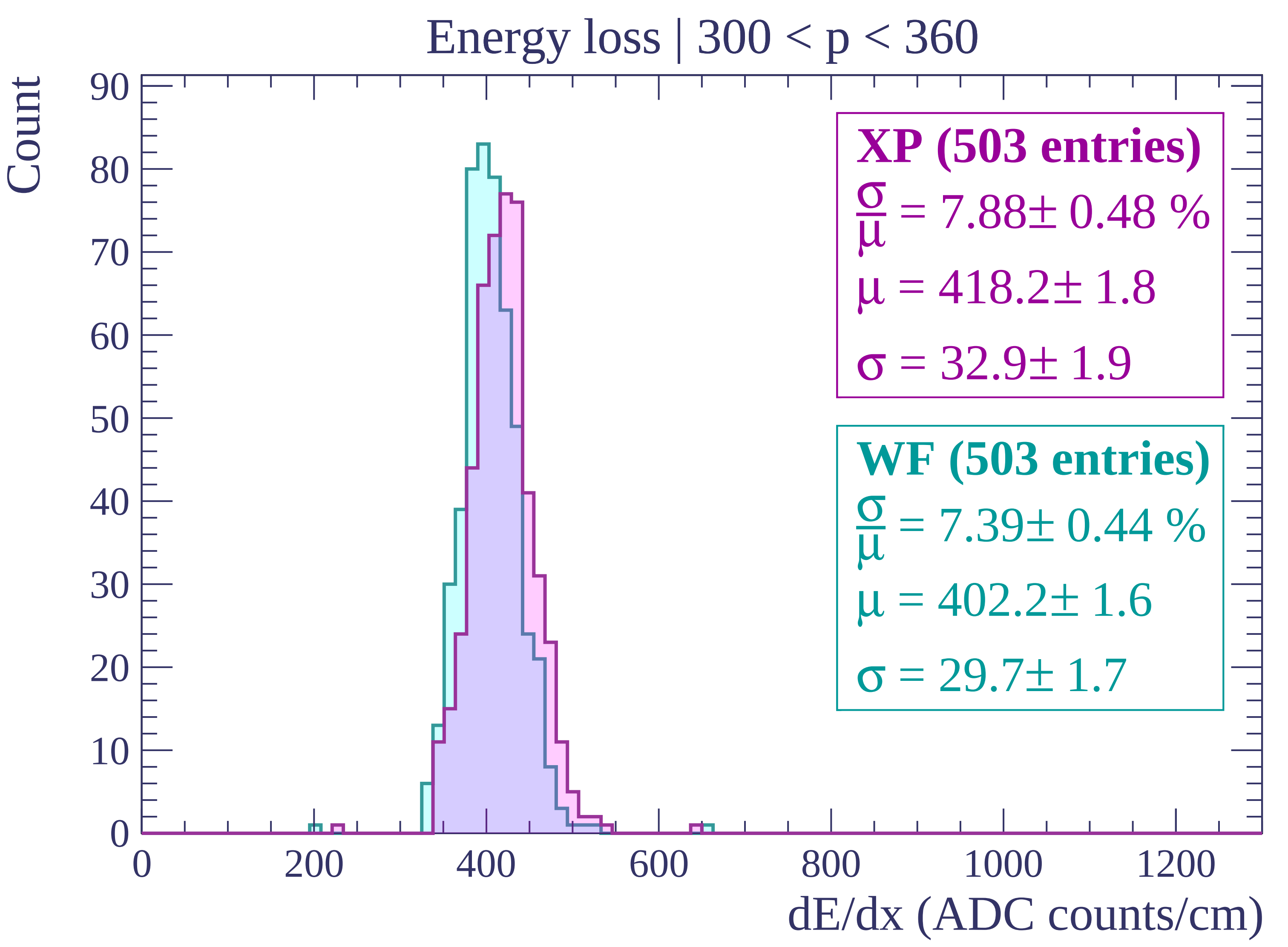

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP (21 entries)
σ/μ = 127.79± 55024.87 %
μ = 496.6± 213067.7
σ = 634.6± 963.8
WF (21 entries)
σ/μ = 40.45± 32.92 %
μ = 595.3± 114.2
σ = 240.8± 149.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200

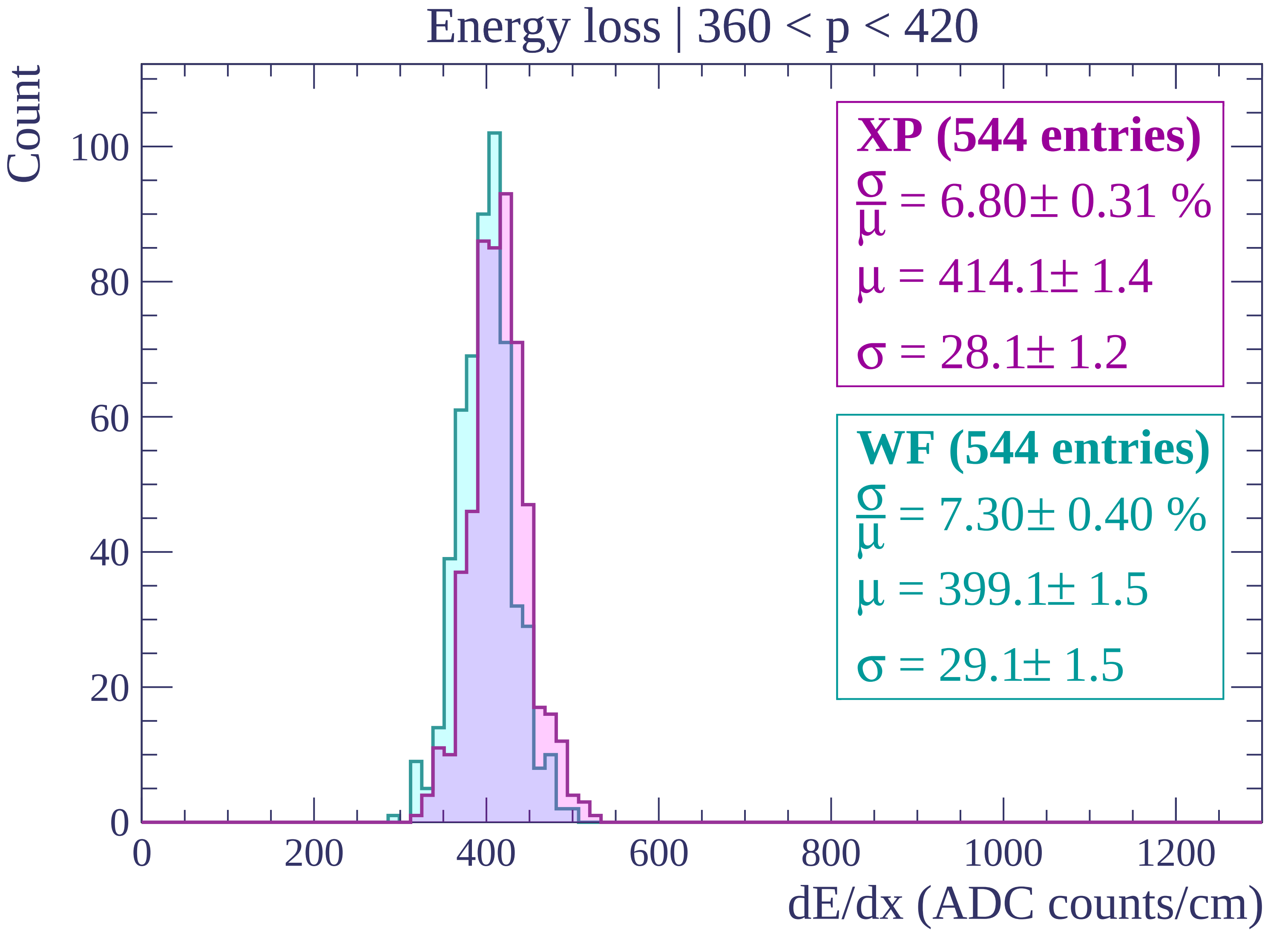

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (284 entries)
$\frac{\sigma}{\mu}$ = 8.81± 0.74 %
μ = 566.2± 3.8
σ = 49.9± 3.9
WF (284 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.67 %
μ = 549.1± 4.2
σ = 48.7± 3.3
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200



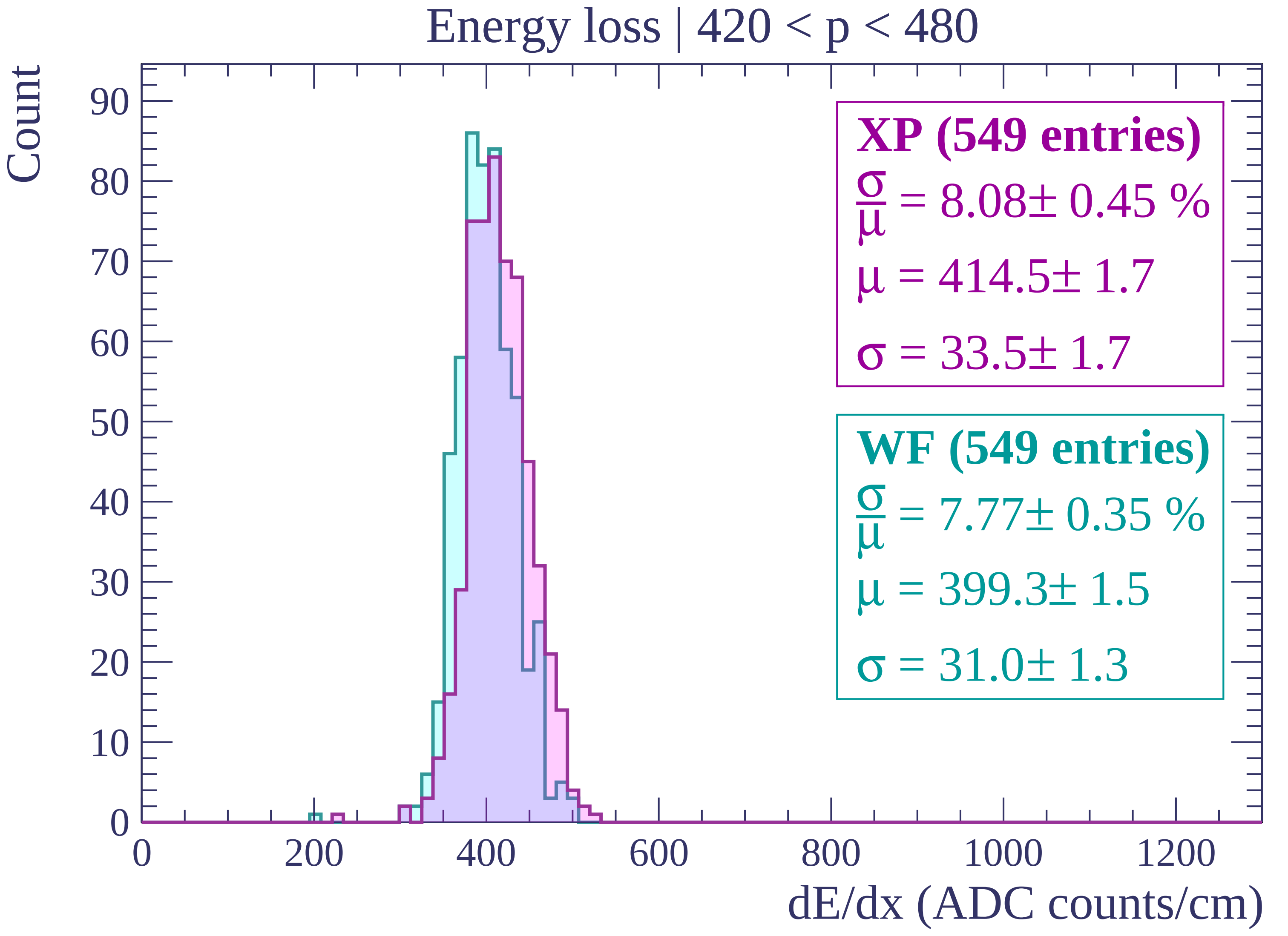
Energy loss | 180 < p < 240
Count
XP (461 entries)
σ/μ = 8.28± 0.44 %
μ = 478.4± 2.3
σ = 39.6± 1.9
WF (461 entries)
σ/μ = 8.12± 0.38 %
μ = 461.5± 2.0
σ = 37.5± 1.6
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70


Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
XP (523 entries)
σ/μ = 8.09± 0.44 %
μ = 435.6± 1.8
σ = 35.3± 1.8
WF (523 entries)
σ/μ = 7.46± 0.36 %
μ = 418.7± 1.6
σ = 31.3± 1.4

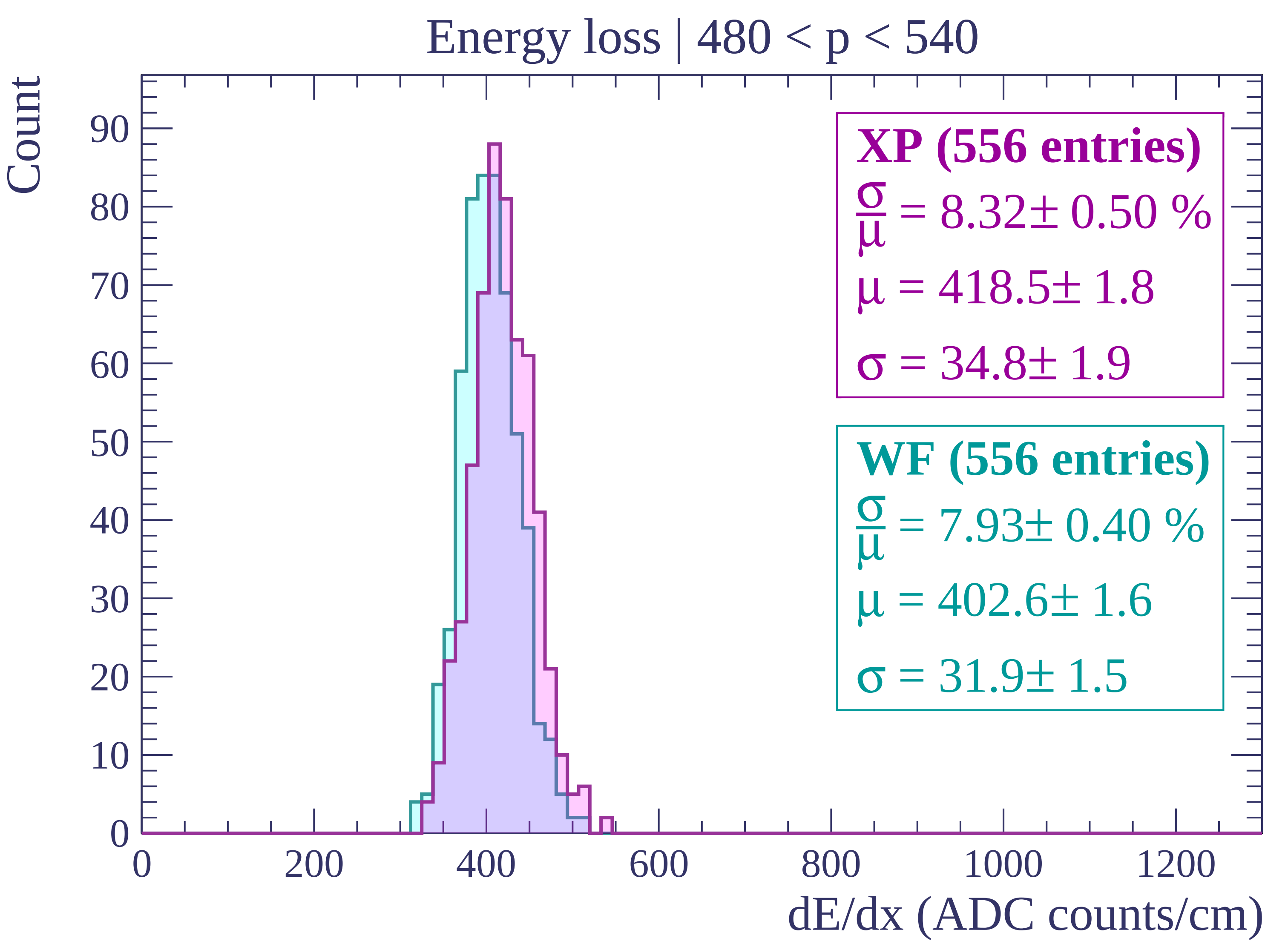

Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (503 entries)
σ/μ = 7.88± 0.48 %
μ = 418.2± 1.8
σ = 32.9± 1.9
WF (503 entries)
σ/μ = 7.39± 0.44 %
μ = 402.2± 1.6
σ = 29.7± 1.7



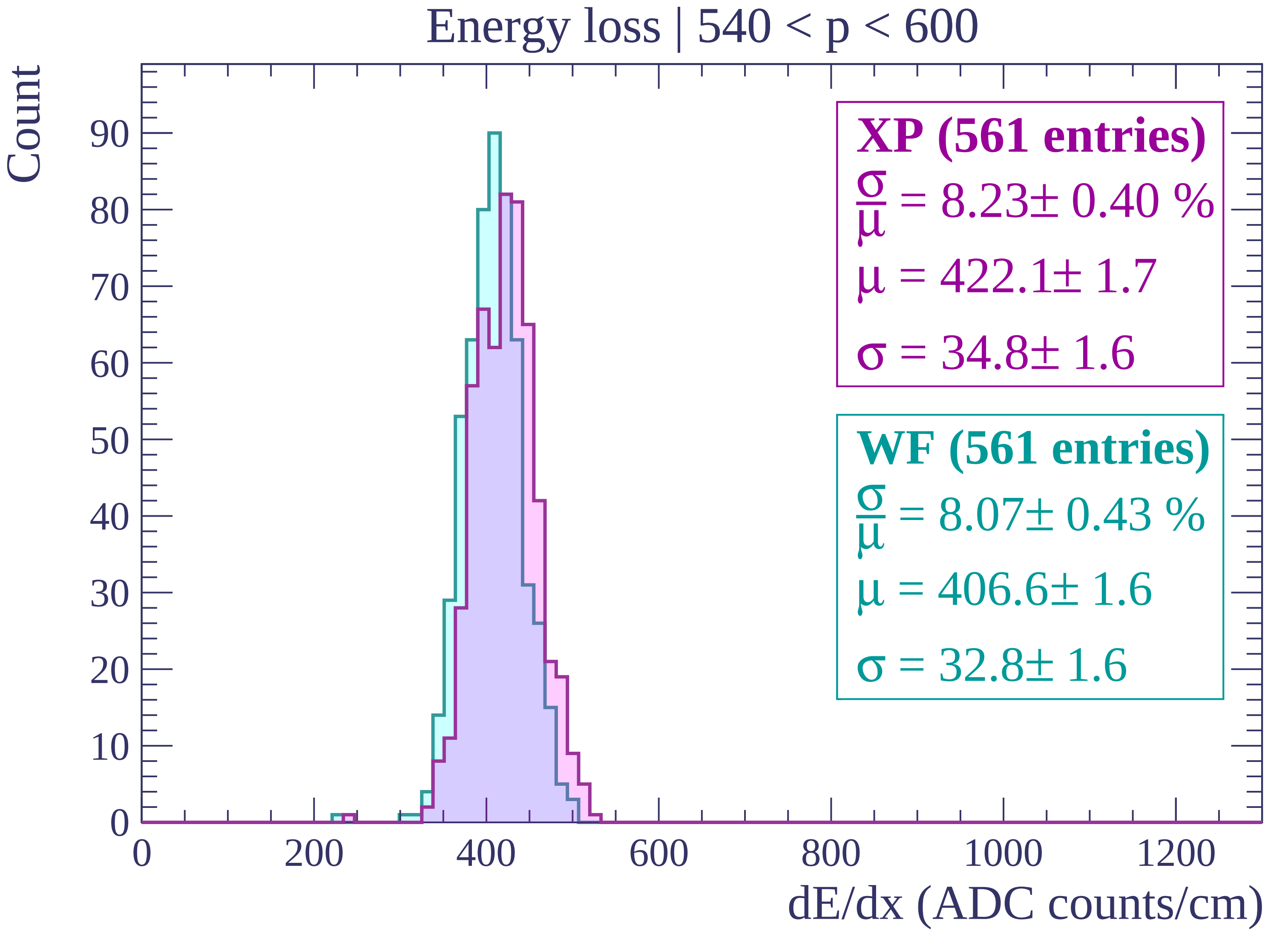
Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (544 entries)
$\frac{\sigma}{\mu}$ = 6.80± 0.31 %
μ = 414.1± 1.4
σ = 28.1± 1.2
WF (544 entries)
$\frac{\sigma}{\mu}$ = 7.30± 0.40 %
μ = 399.1± 1.5
σ = 29.1± 1.5



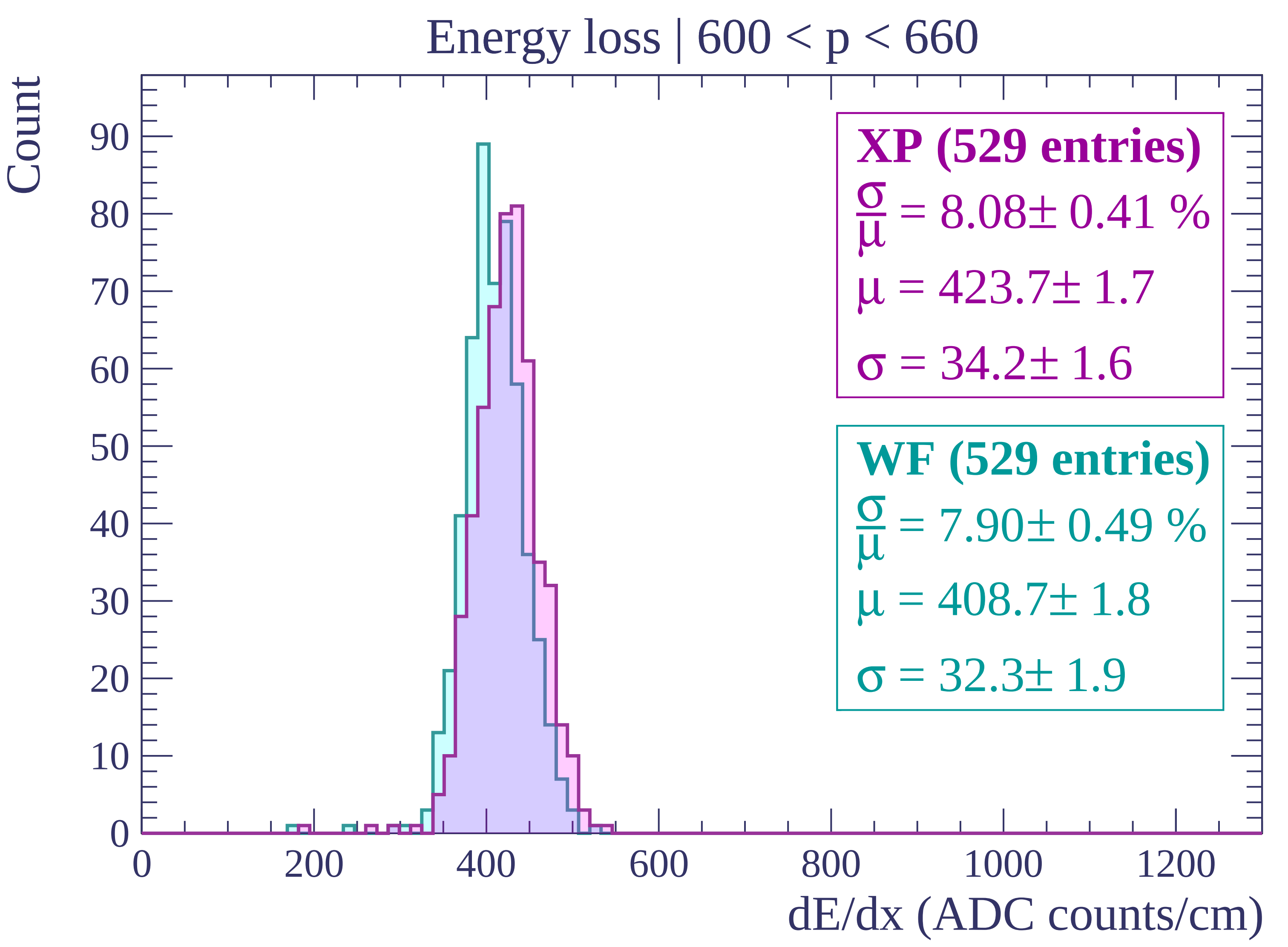
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (549 entries)
σ/μ = 8.08± 0.45 %
μ = 414.5± 1.7
σ = 33.5± 1.7
WF (549 entries)
σ/μ = 7.77± 0.35 %
μ = 399.3± 1.5
σ = 31.0± 1.3



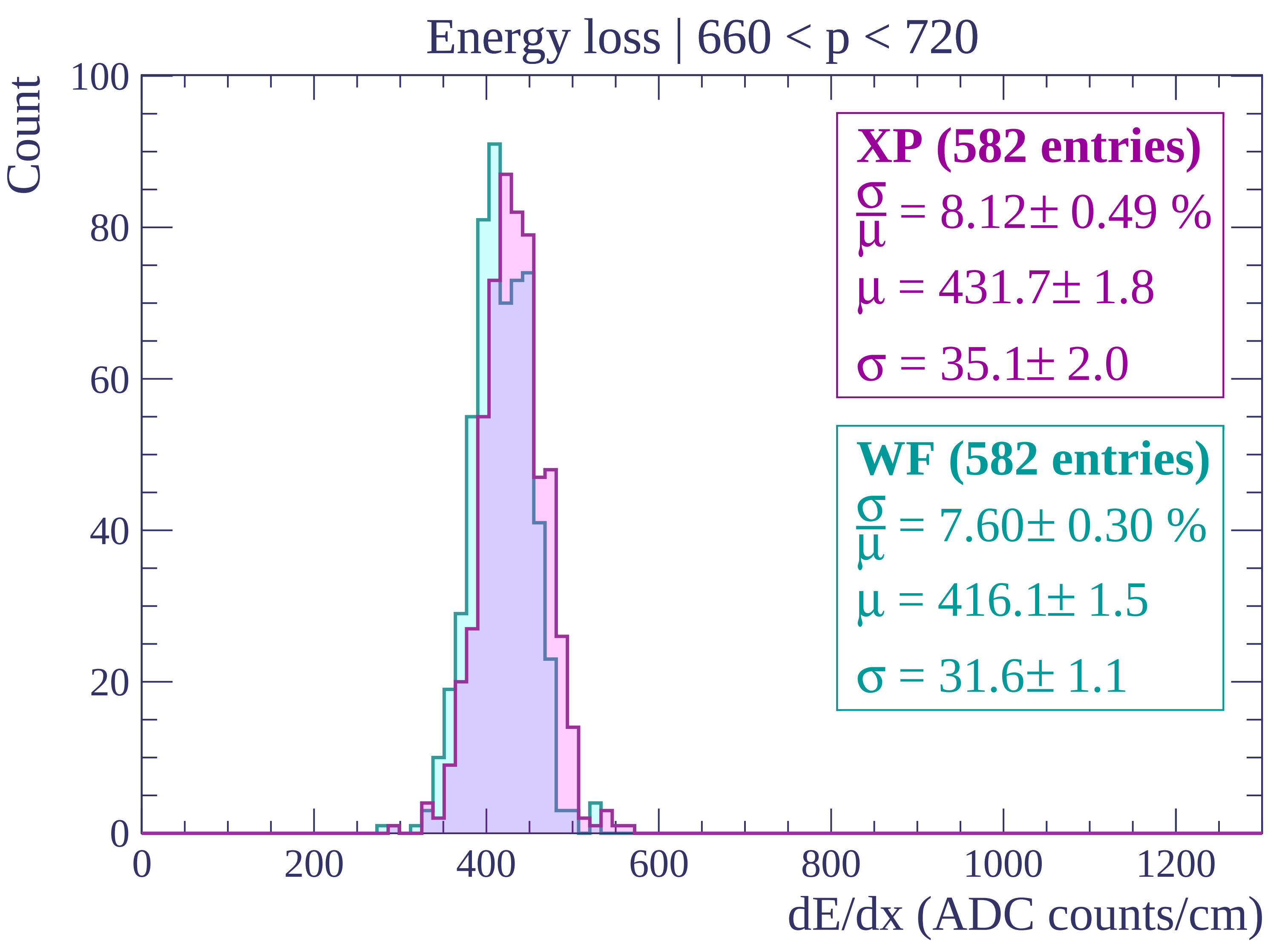
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (556 entries)
σ/μ = 8.32± 0.50 %
μ = 418.5± 1.8
σ = 34.8± 1.9
WF (556 entries)
σ/μ = 7.93± 0.40 %
μ = 402.6± 1.6
σ = 31.9± 1.5
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



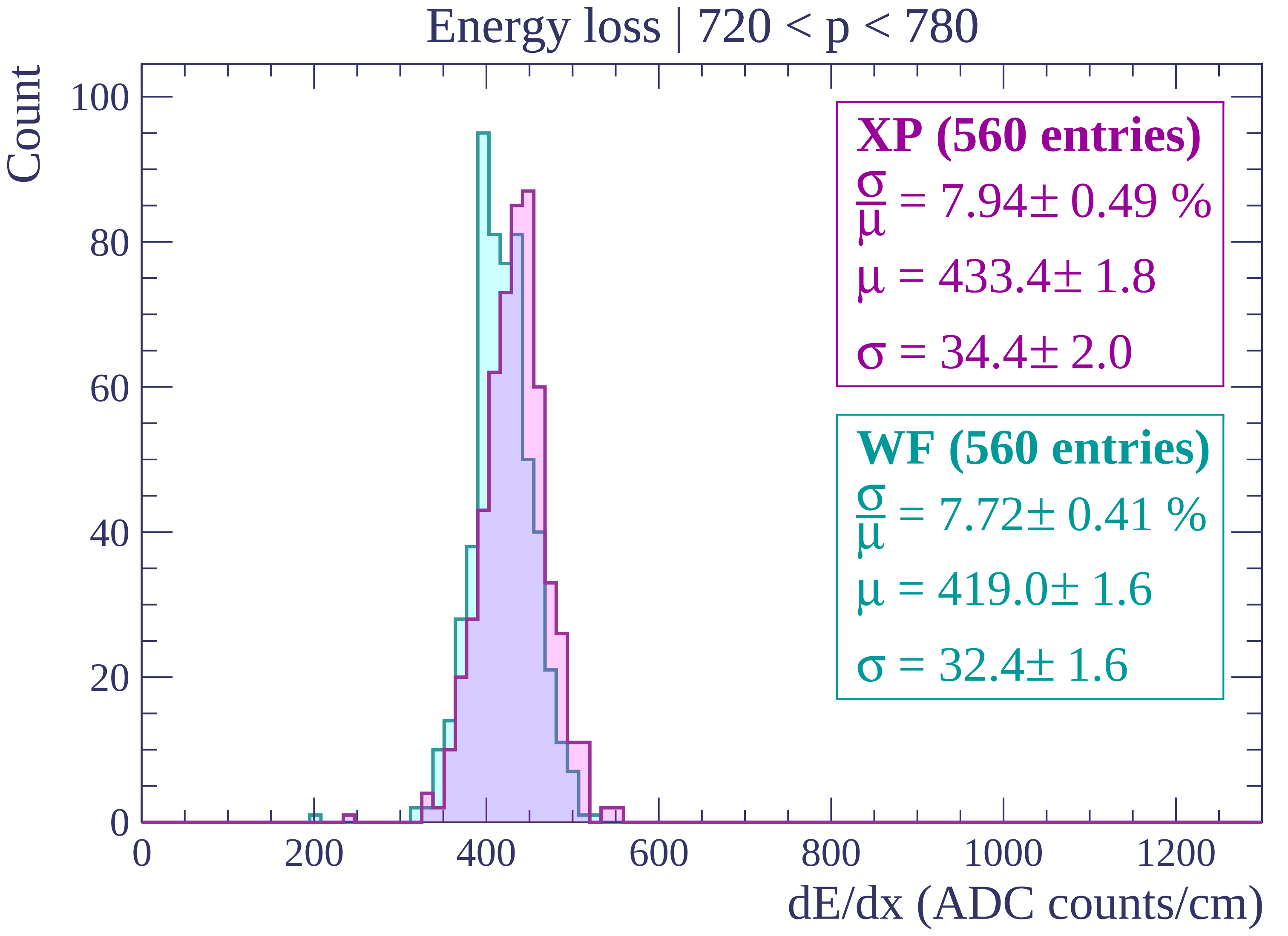
Energy loss | 540 < p < 600
Count
XP (561 entries)
σ/μ = 8.23± 0.40 %
μ = 422.1± 1.7
σ = 34.8± 1.6
WF (561 entries)
σ/μ = 8.07± 0.43 %
μ = 406.6± 1.6
σ = 32.8± 1.6
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

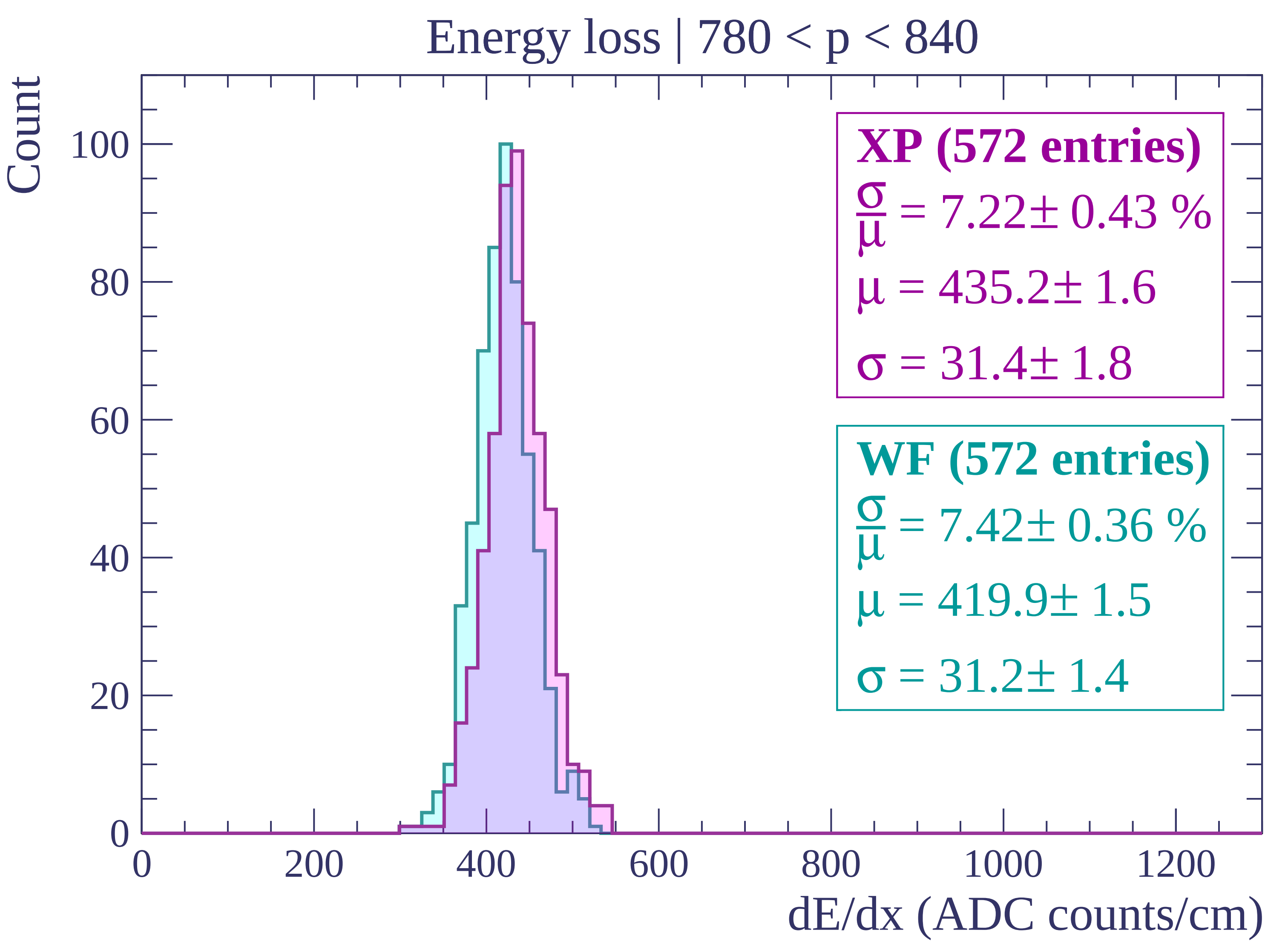


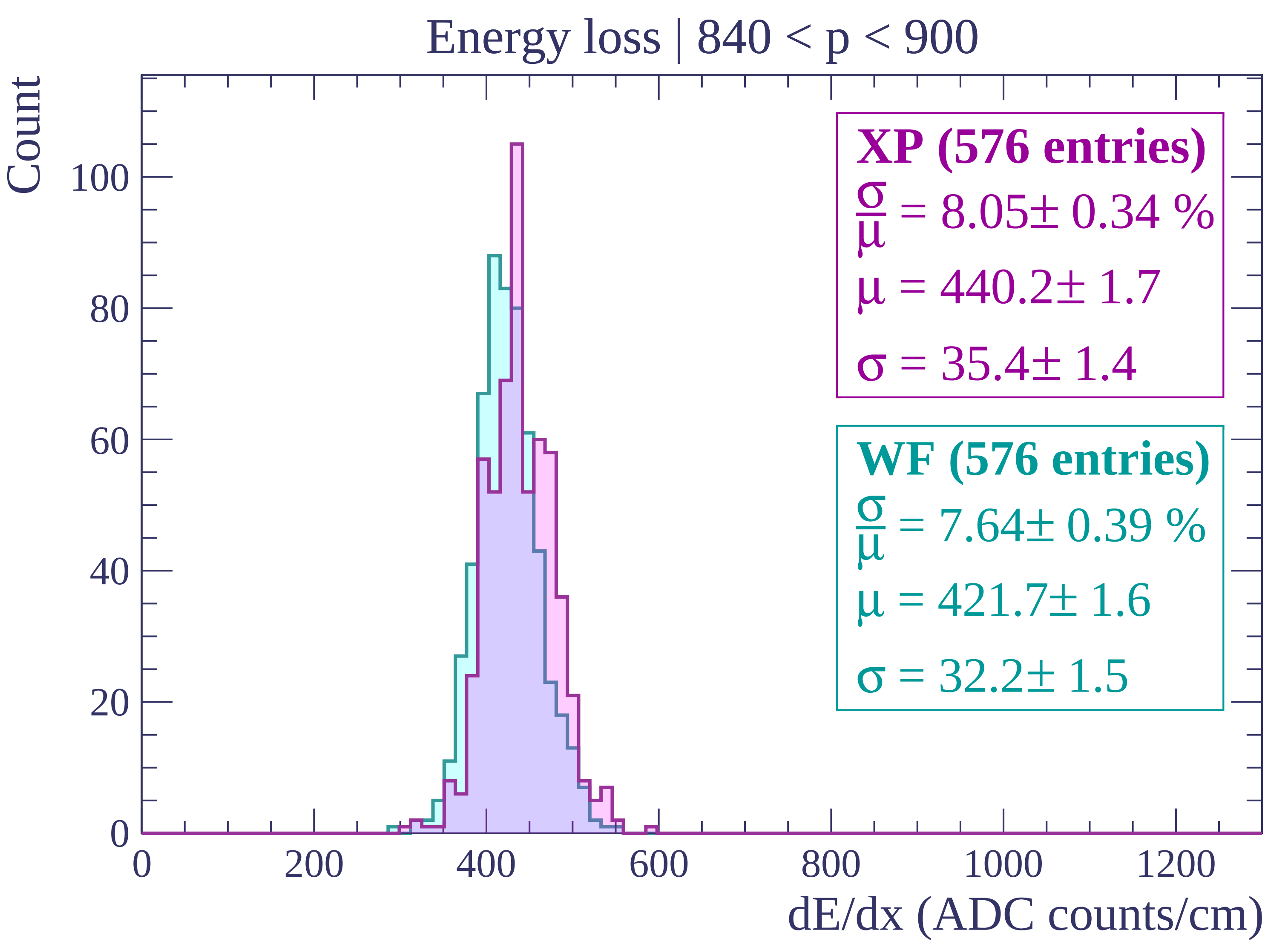
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (529 entries)
σ/μ = 8.08± 0.41 %
μ = 423.7± 1.7
σ = 34.2± 1.6
WF (529 entries)
σ/μ = 7.90± 0.49 %
μ = 408.7± 1.8
σ = 32.3± 1.9

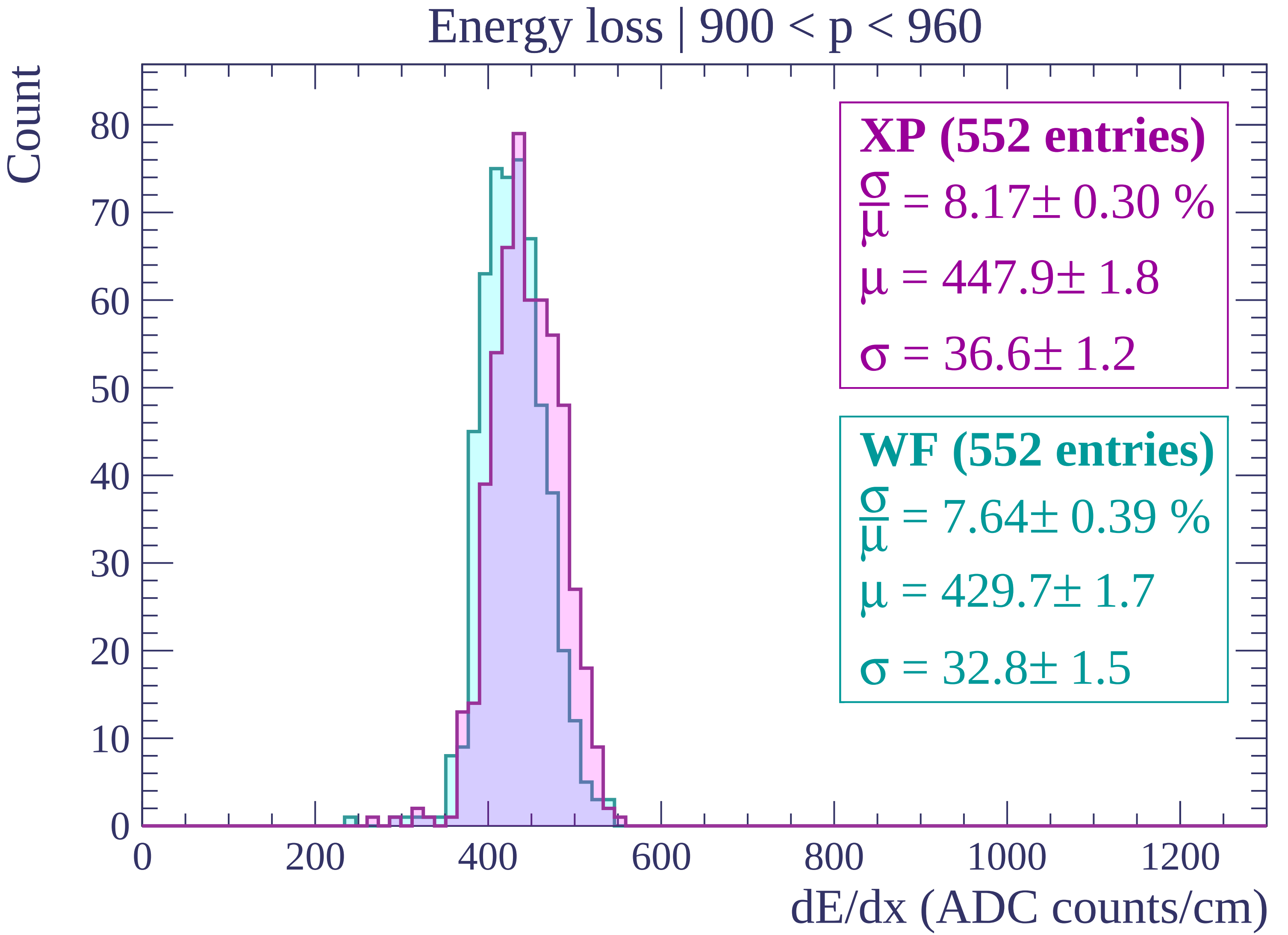


Energy loss | 660 < p < 720
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (582 entries)
σ/μ = 8.12± 0.49 %
μ = 431.7± 1.8
σ = 35.1± 2.0
WF (582 entries)
σ/μ = 7.60± 0.30 %
μ = 416.1± 1.5
σ = 31.6± 1.1

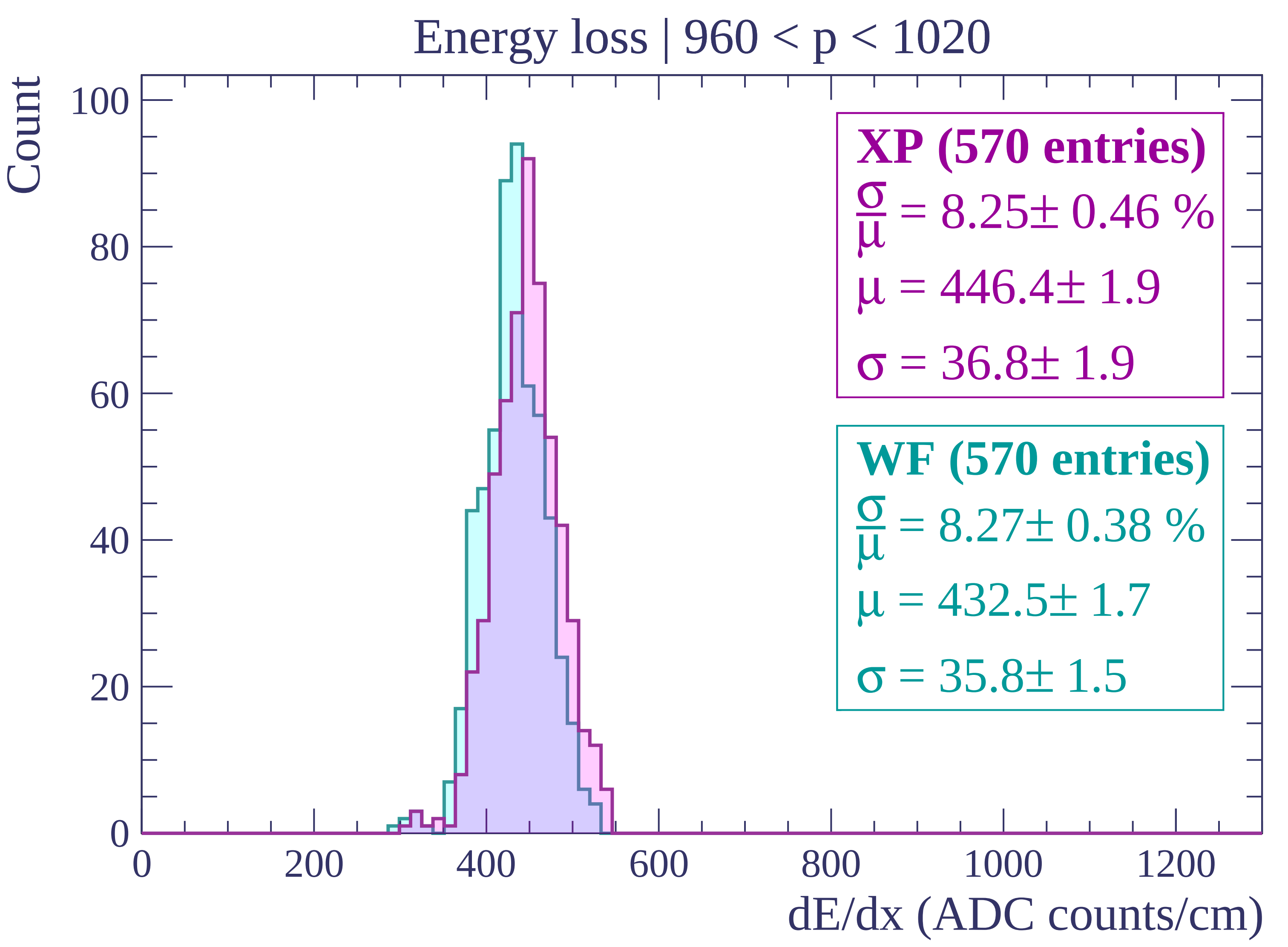

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (560 entries)
σ/μ = 7.94± 0.49 %
μ = 433.4± 1.8
σ = 34.4± 2.0
WF (560 entries)
σ/μ = 7.72± 0.41 %
μ = 419.0± 1.6
σ = 32.4± 1.6


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (572 entries)
σ/μ = 7.22± 0.43 %
μ = 435.2± 1.6
σ = 31.4± 1.8
WF (572 entries)
σ/μ = 7.42± 0.36 %
μ = 419.9± 1.5
σ = 31.2± 1.4

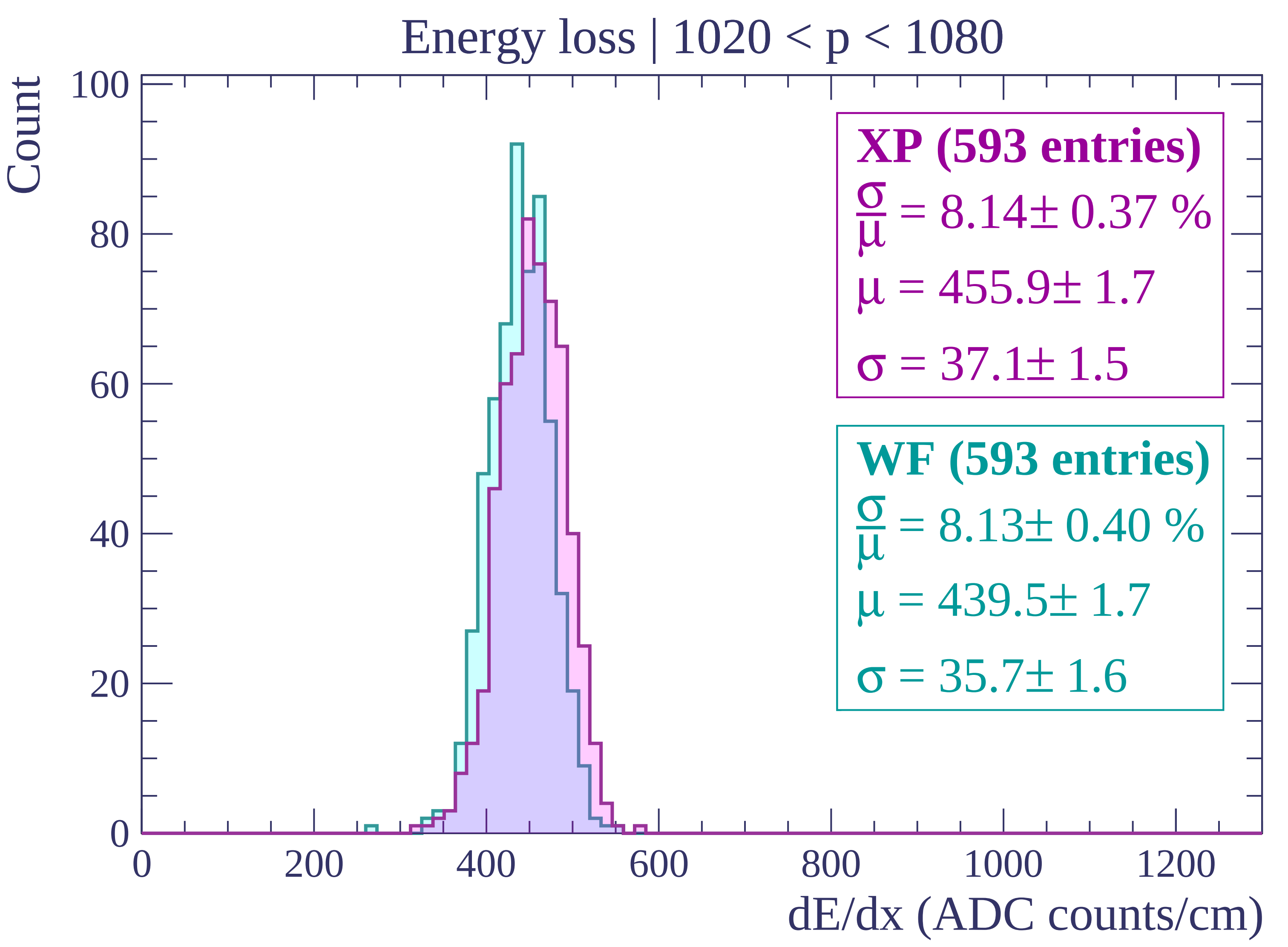

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (576 entries)
σ/μ = 8.05± 0.34 %
μ = 440.2± 1.7
σ = 35.4± 1.4
WF (576 entries)
σ/μ = 7.64± 0.39 %
μ = 421.7± 1.6
σ = 32.2± 1.5

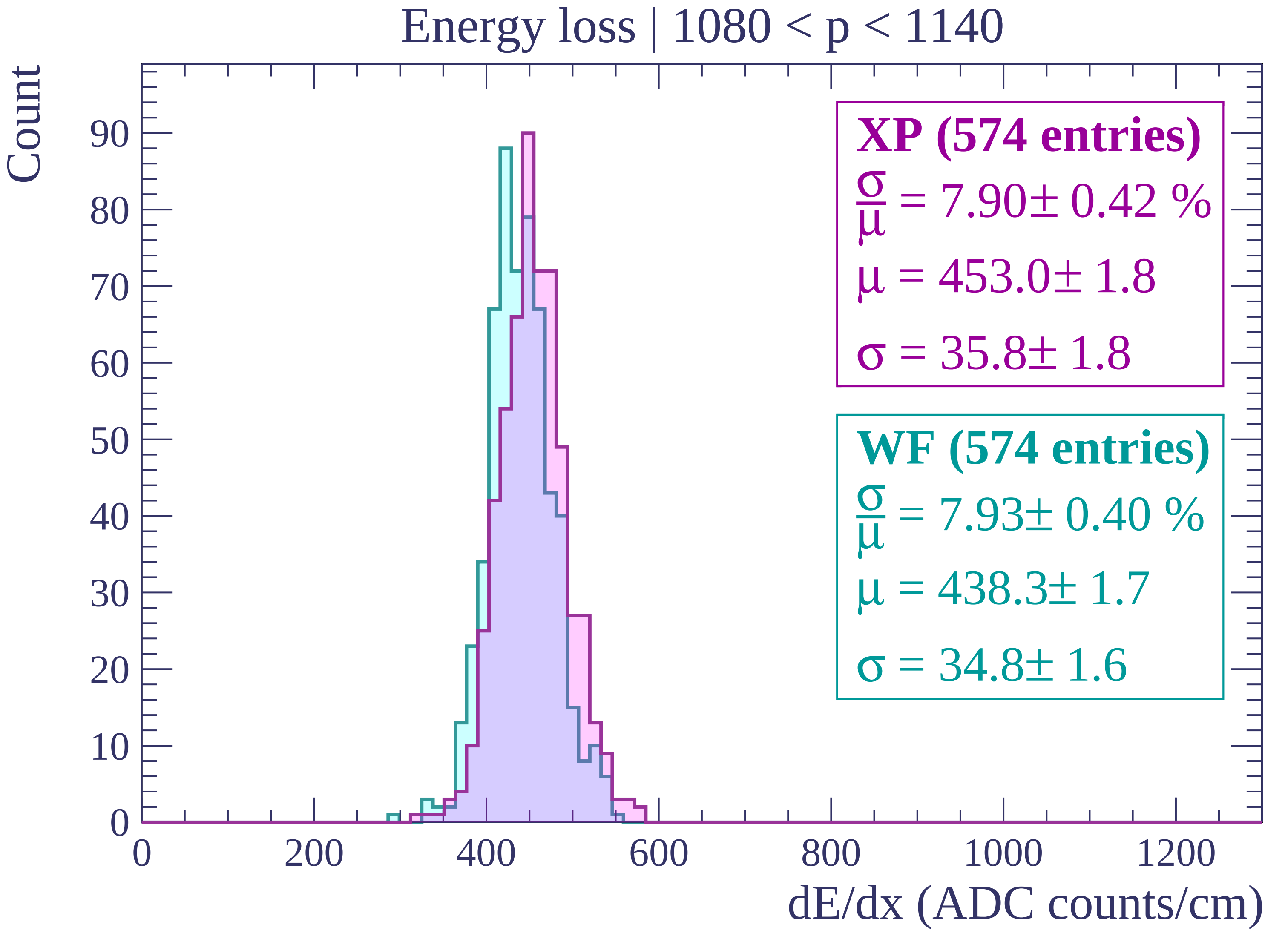


Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (552 entries)
σ/μ = 8.17± 0.30 %
μ = 447.9± 1.8
σ = 36.6± 1.2
WF (552 entries)
σ/μ = 7.64± 0.39 %
μ = 429.7± 1.7
σ = 32.8± 1.5

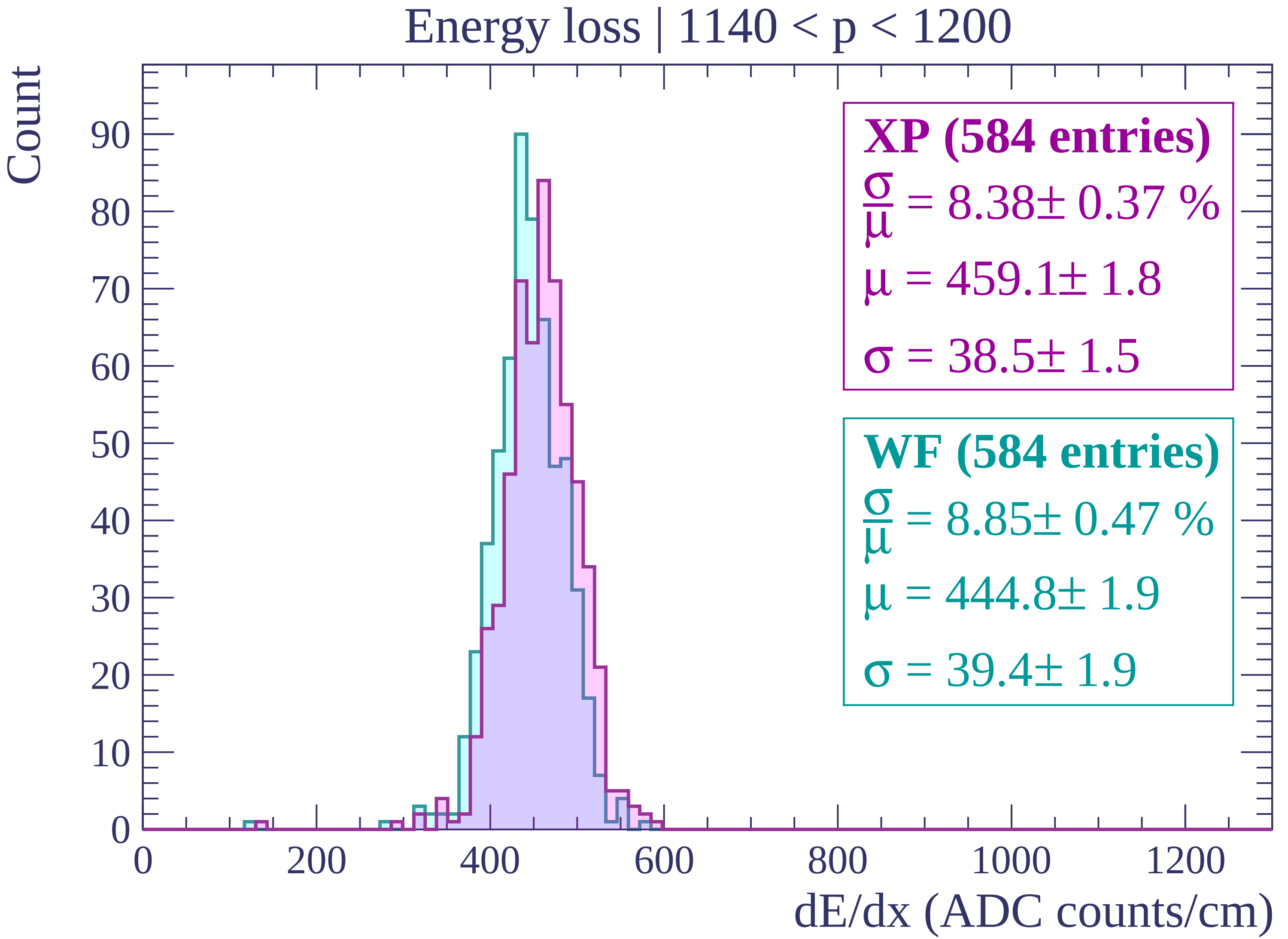


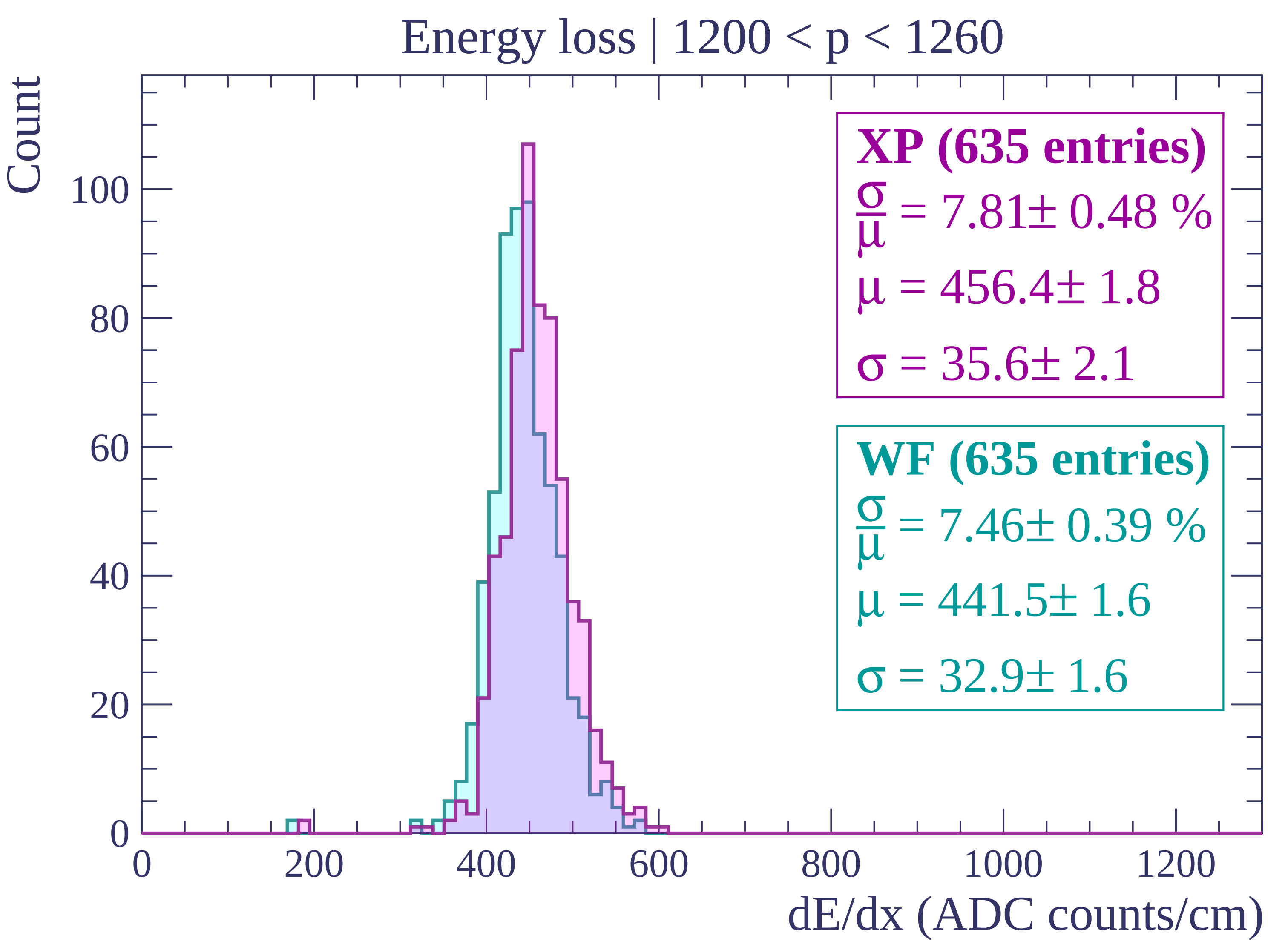
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (570 entries)
σ/μ = 8.25± 0.46 %
μ = 446.4± 1.9
σ = 36.8± 1.9
WF (570 entries)
σ/μ = 8.27± 0.38 %
μ = 432.5± 1.7
σ = 35.8± 1.5


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (593 entries)
σ/μ = 8.14± 0.37 %
μ = 455.9± 1.7
σ = 37.1± 1.5
WF (593 entries)
σ/μ = 8.13± 0.40 %
μ = 439.5± 1.7
σ = 35.7± 1.6

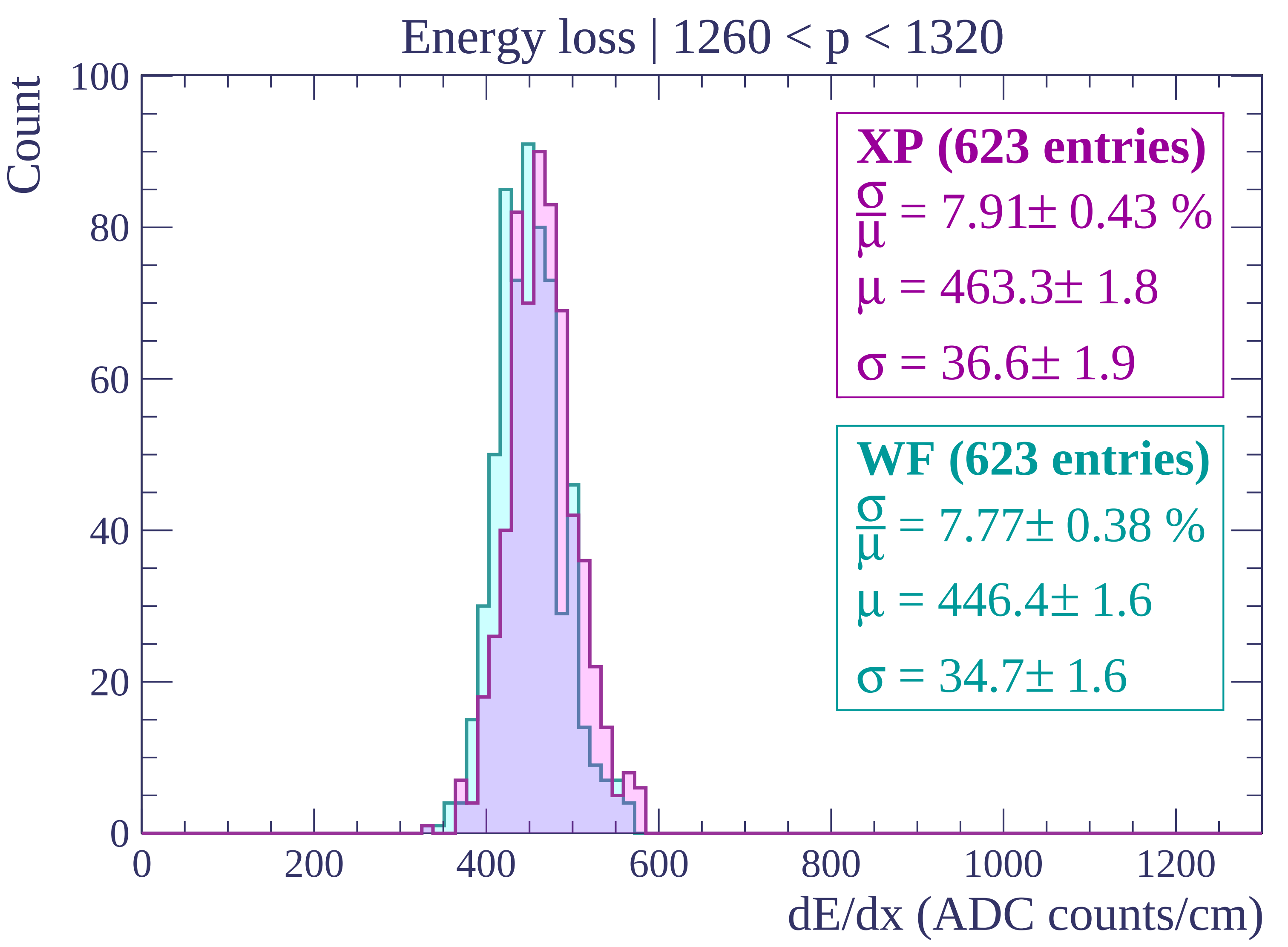


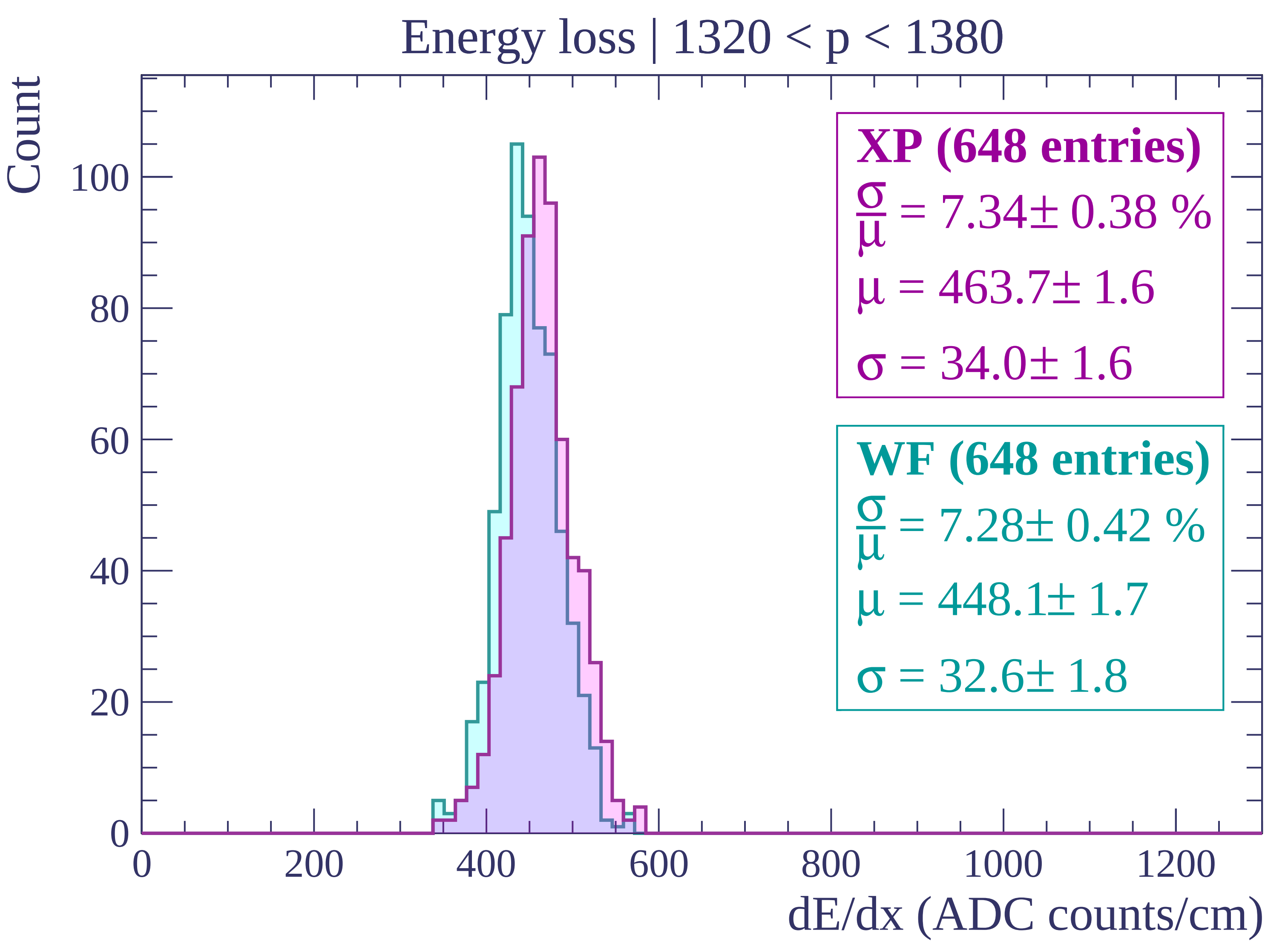
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
XP (574 entries)
$\frac{\sigma}{\mu}$ = 7.90± 0.42 %
μ = 453.0± 1.8
σ = 35.8± 1.8
WF (574 entries)
$\frac{\sigma}{\mu}$ = 7.93± 0.40 %
μ = 438.3± 1.7
σ = 34.8± 1.6
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (584 entries)
σ/μ = 8.38± 0.37 %
μ = 459.1± 1.8
σ = 38.5± 1.5
WF (584 entries)
σ/μ = 8.85± 0.47 %
μ = 444.8± 1.9
σ = 39.4± 1.9
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200

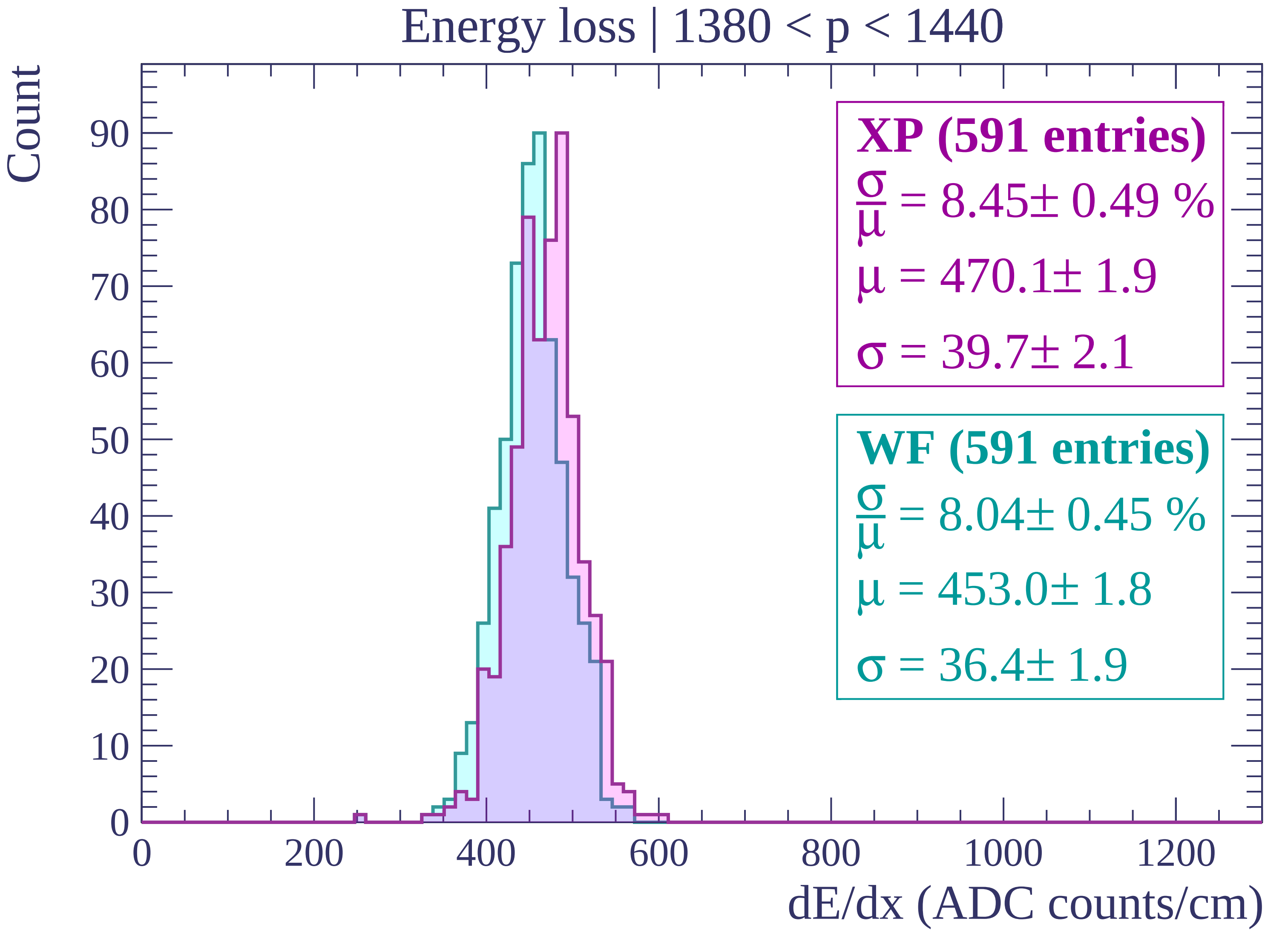


Energy loss | 1200 < p < 1260
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (635 entries)
$\frac{\sigma}{\mu}$ = 7.81± 0.48 %
μ = 456.4± 1.8
σ = 35.6± 2.1
WF (635 entries)
$\frac{\sigma}{\mu}$ = 7.46± 0.39 %
μ = 441.5± 1.6
σ = 32.9± 1.6

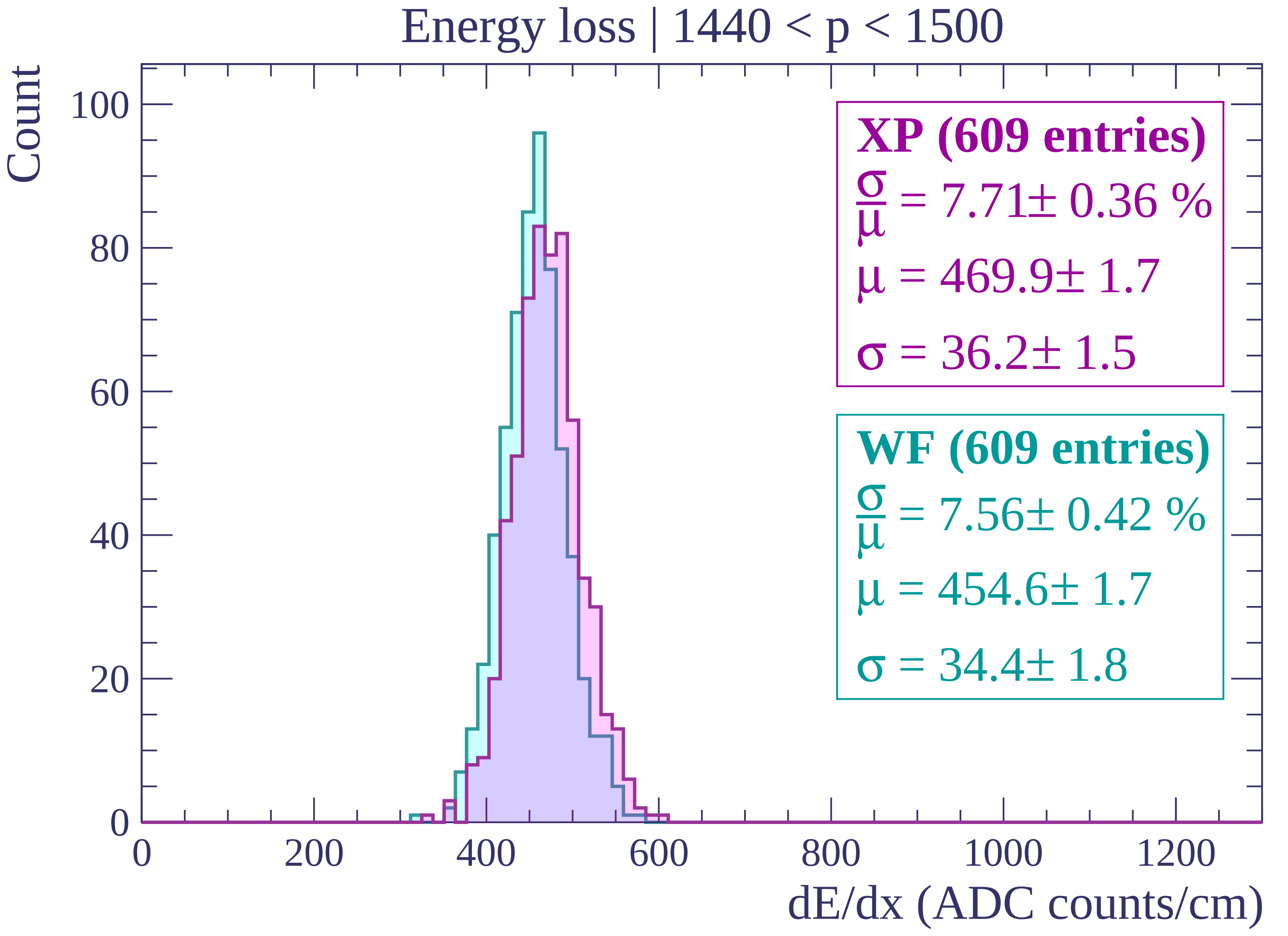

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
XP (623 entries)
σ/μ = 7.91± 0.43 %
μ = 463.3± 1.8
σ = 36.6± 1.9
WF (623 entries)
σ/μ = 7.77± 0.38 %
μ = 446.4± 1.6
σ = 34.7± 1.6
0
20
40
60
80
100
0
200
400
600
800
1000
1200



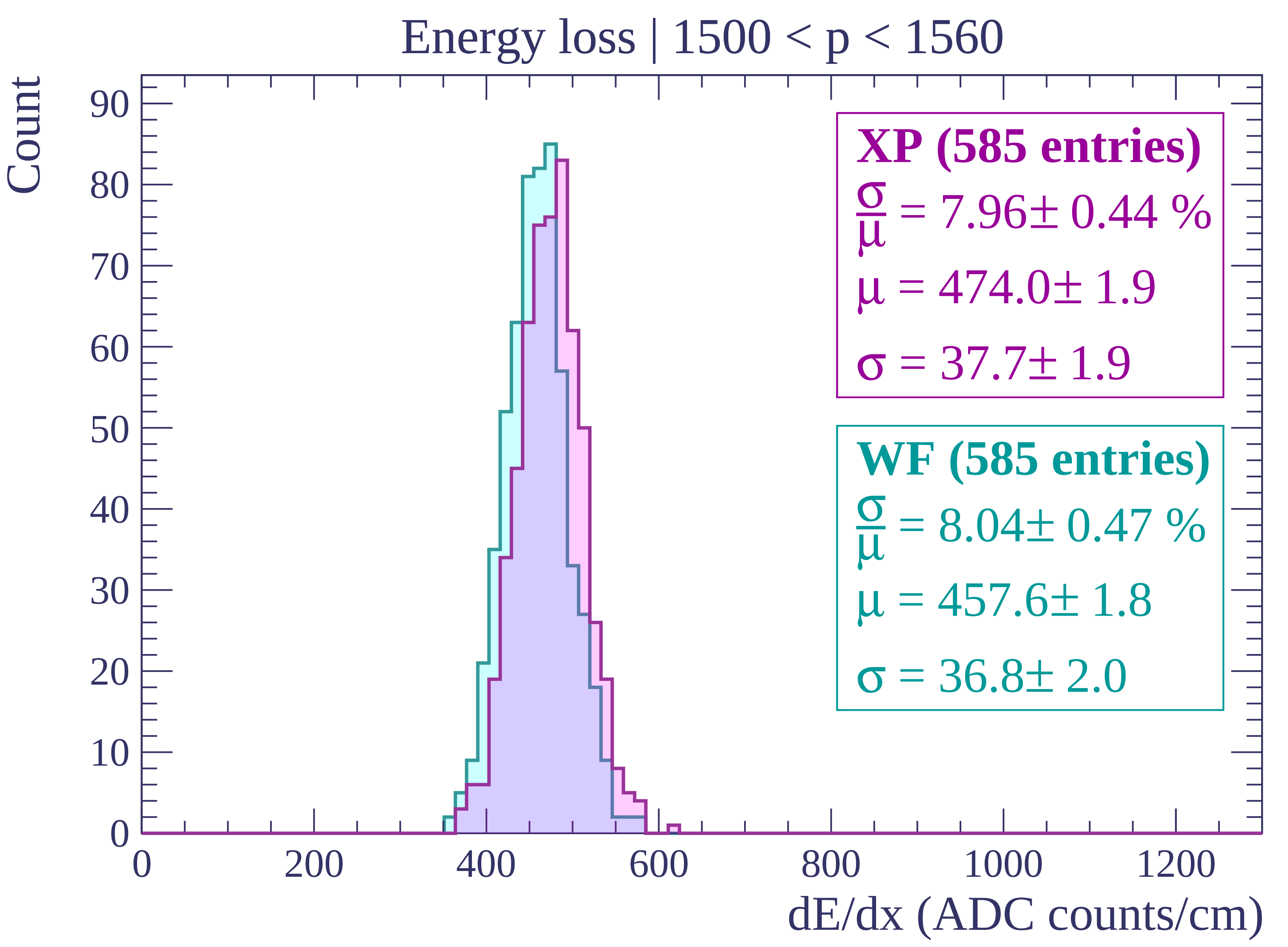
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (648 entries)
$\frac{\sigma}{\mu}$ = 7.34± 0.38 %
μ = 463.7± 1.6
σ = 34.0± 1.6
WF (648 entries)
$\frac{\sigma}{\mu}$ = 7.28± 0.42 %
μ = 448.1± 1.7
σ = 32.6± 1.8



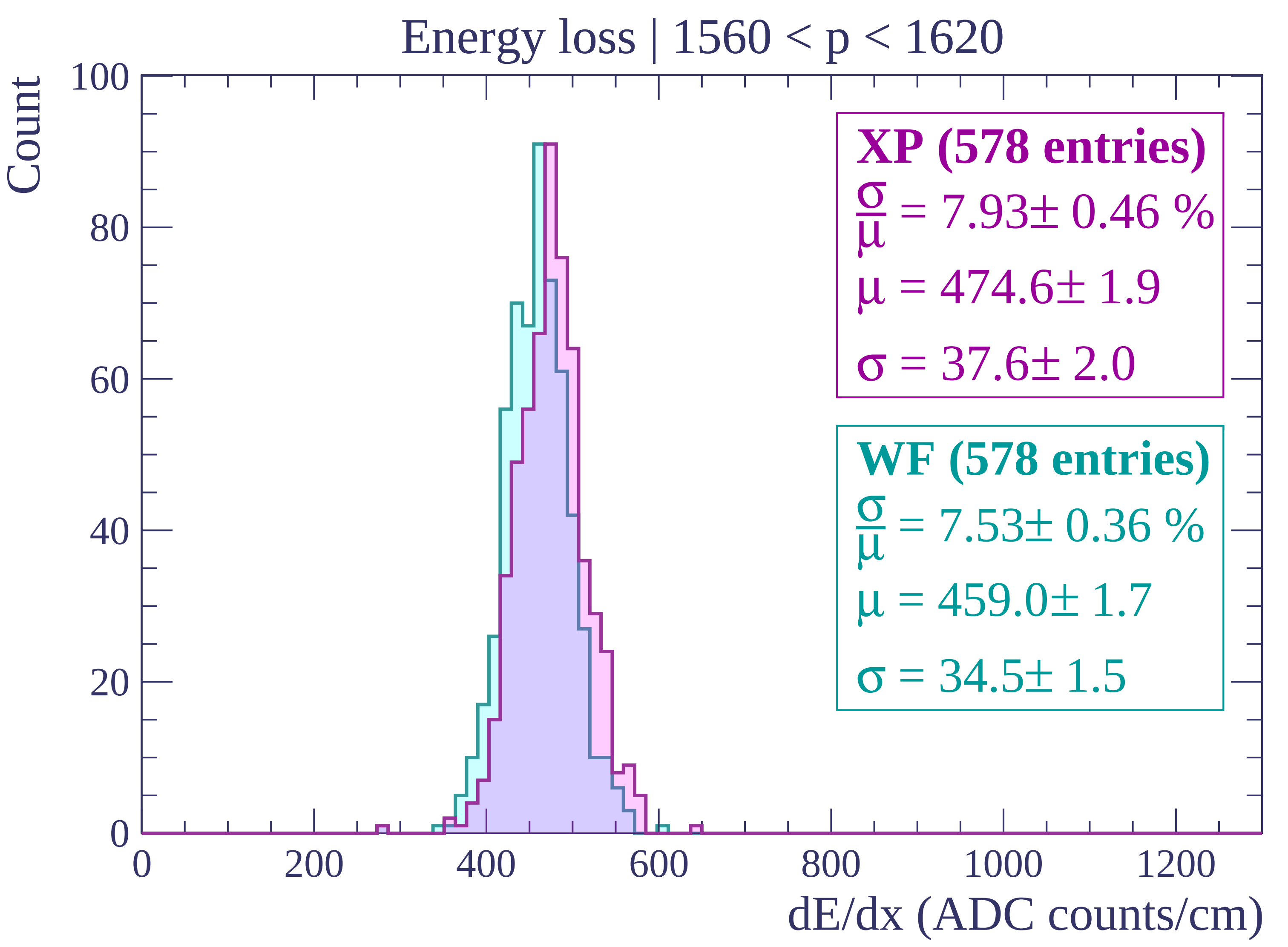
Energy loss | 1380 < p < 1440
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (591 entries)
$\frac{\sigma}{\mu}$ = 8.45± 0.49 %
μ = 470.1± 1.9
σ = 39.7± 2.1
WF (591 entries)
$\frac{\sigma}{\mu}$ = 8.04± 0.45 %
μ = 453.0± 1.8
σ = 36.4± 1.9

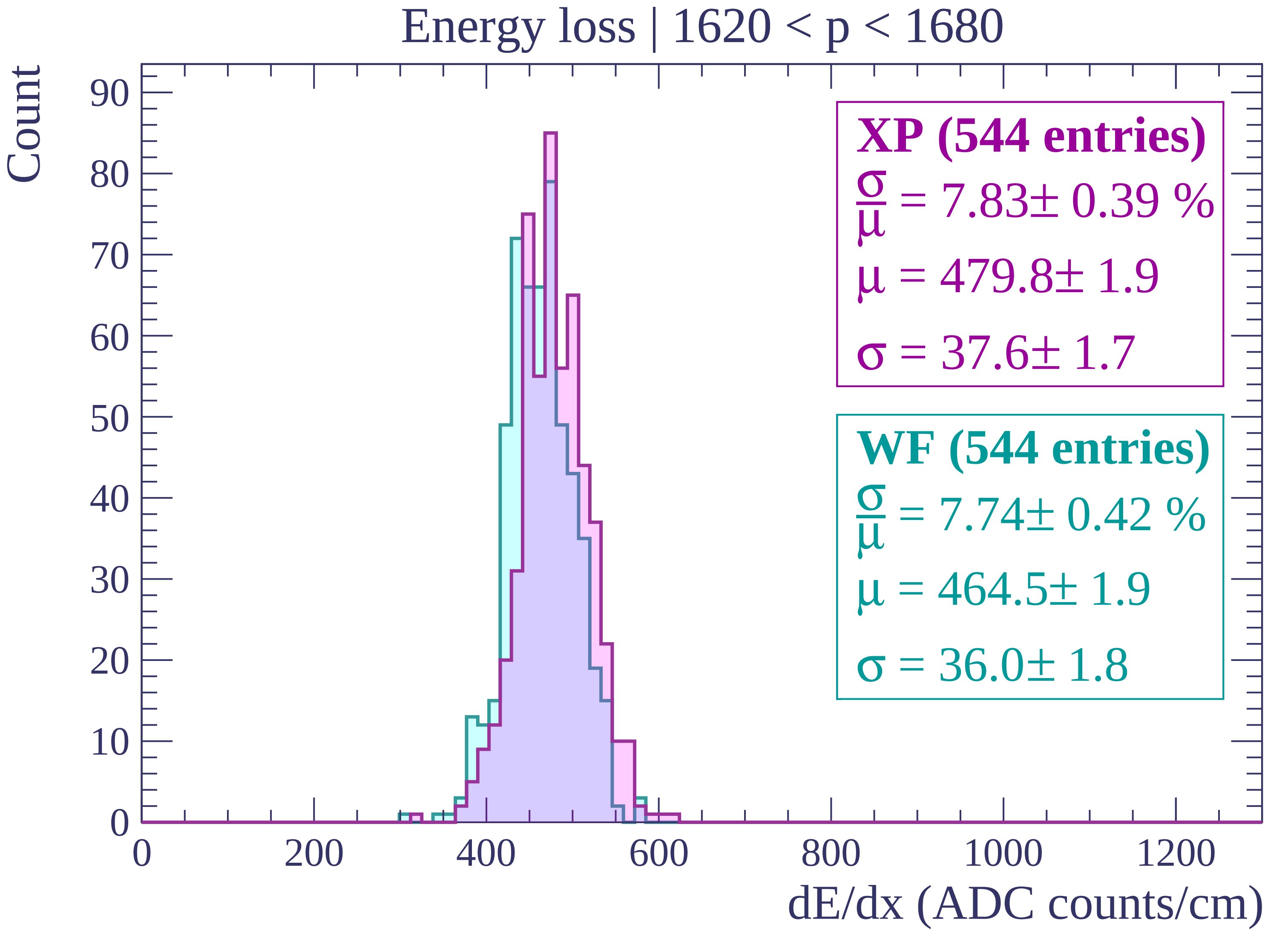


Energy loss | 1440 < p < 1500
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (609 entries)
σ/μ = 7.71± 0.36 %
μ = 469.9± 1.7
σ = 36.2± 1.5
WF (609 entries)
σ/μ = 7.56± 0.42 %
μ = 454.6± 1.7
σ = 34.4± 1.8

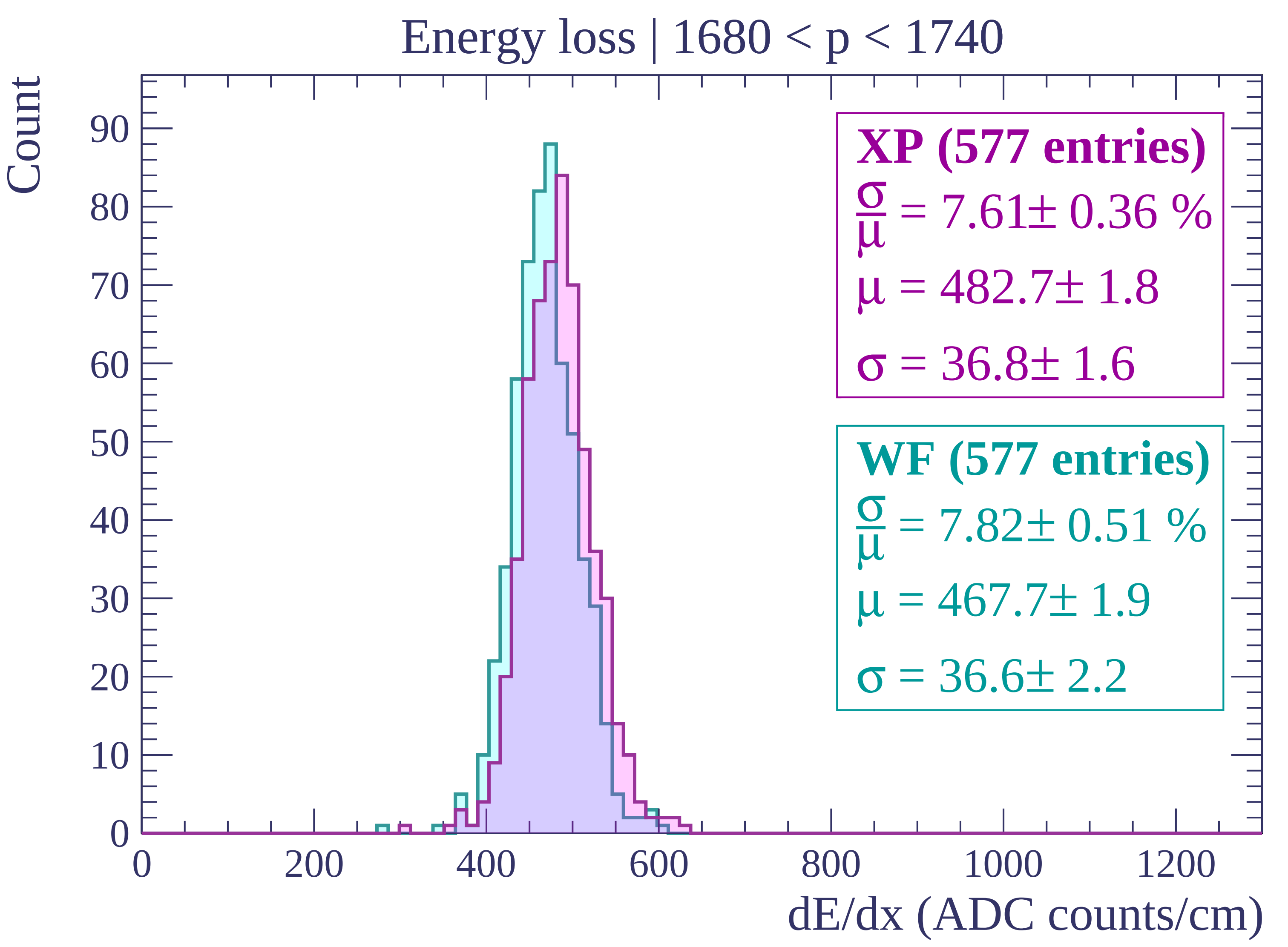


Energy loss | 1500 < p < 1560
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (585 entries)
$\frac{\sigma}{\mu}$ = 7.96± 0.44 %
μ = 474.0± 1.9
σ = 37.7± 1.9
WF (585 entries)
$\frac{\sigma}{\mu}$ = 8.04± 0.47 %
μ = 457.6± 1.8
σ = 36.8± 2.0

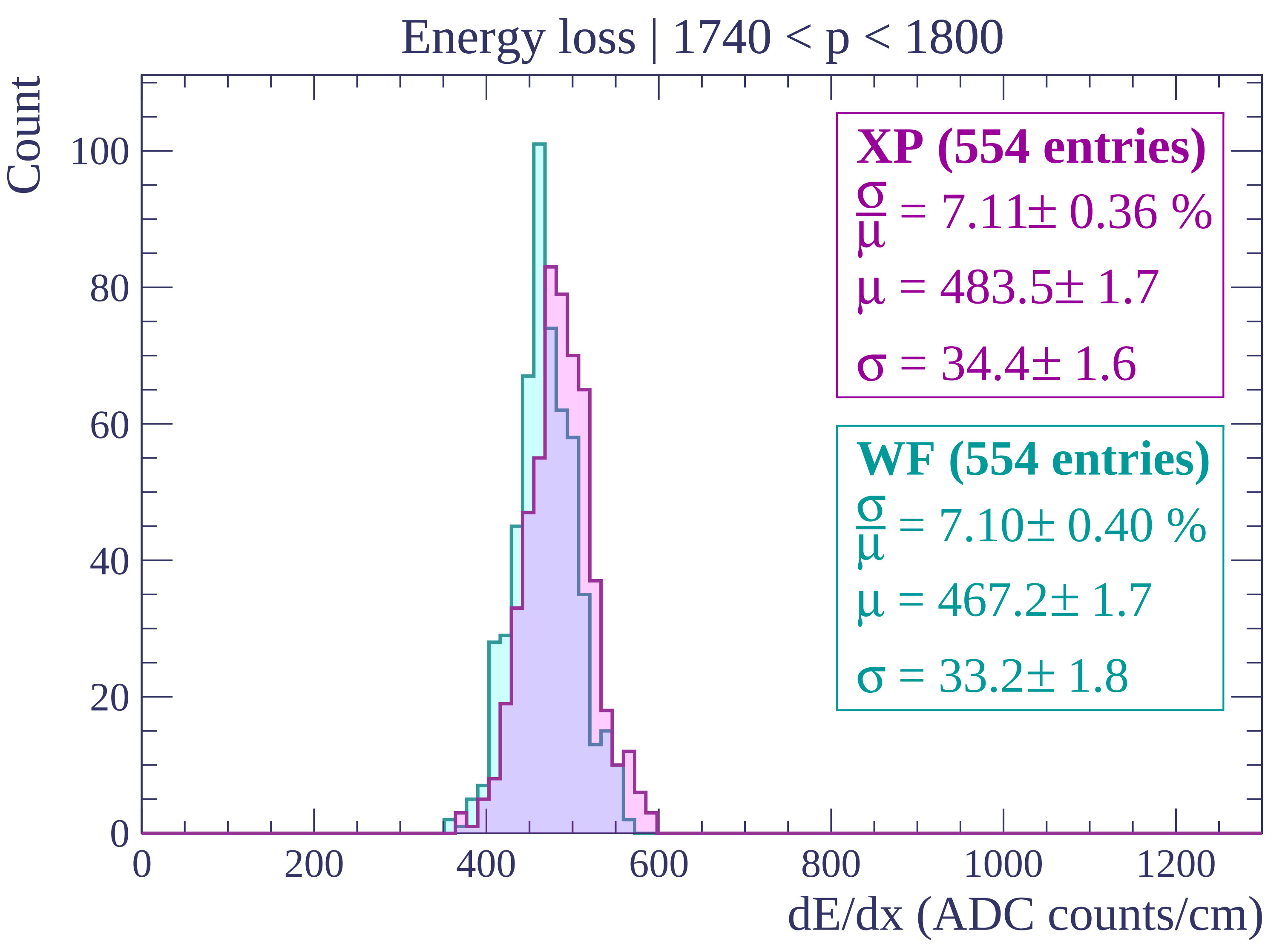

Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (578 entries)
$\frac{\sigma}{\mu}$ = 7.93± 0.46 %
μ = 474.6± 1.9
σ = 37.6± 2.0
WF (578 entries)
$\frac{\sigma}{\mu}$ = 7.53± 0.36 %
μ = 459.0± 1.7
σ = 34.5± 1.5

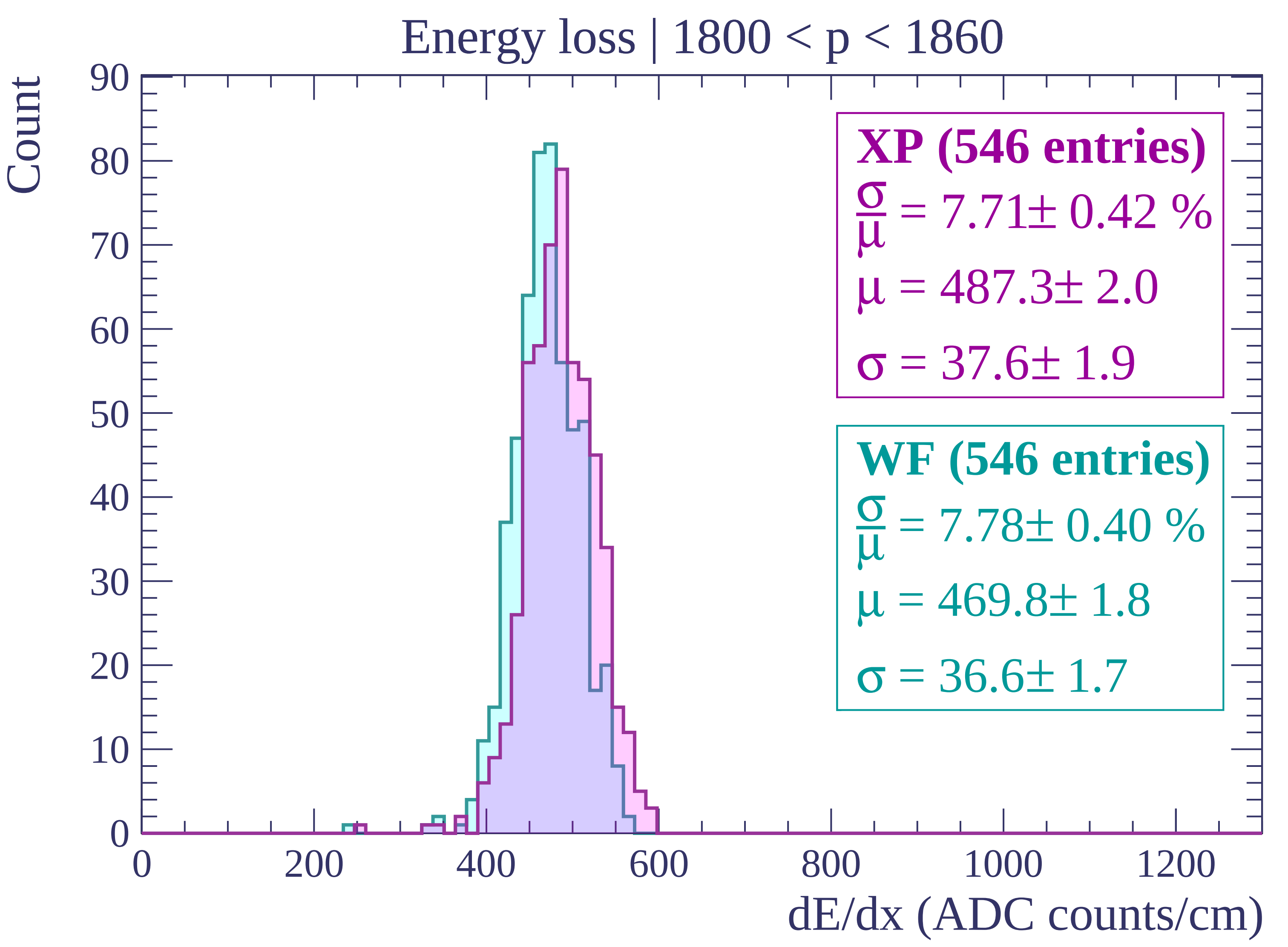


Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (544 entries)
$\frac{\sigma}{\mu}$ = 7.83± 0.39 %
μ = 479.8± 1.9
σ = 37.6± 1.7
WF (544 entries)
$\frac{\sigma}{\mu}$ = 7.74± 0.42 %
μ = 464.5± 1.9
σ = 36.0± 1.8

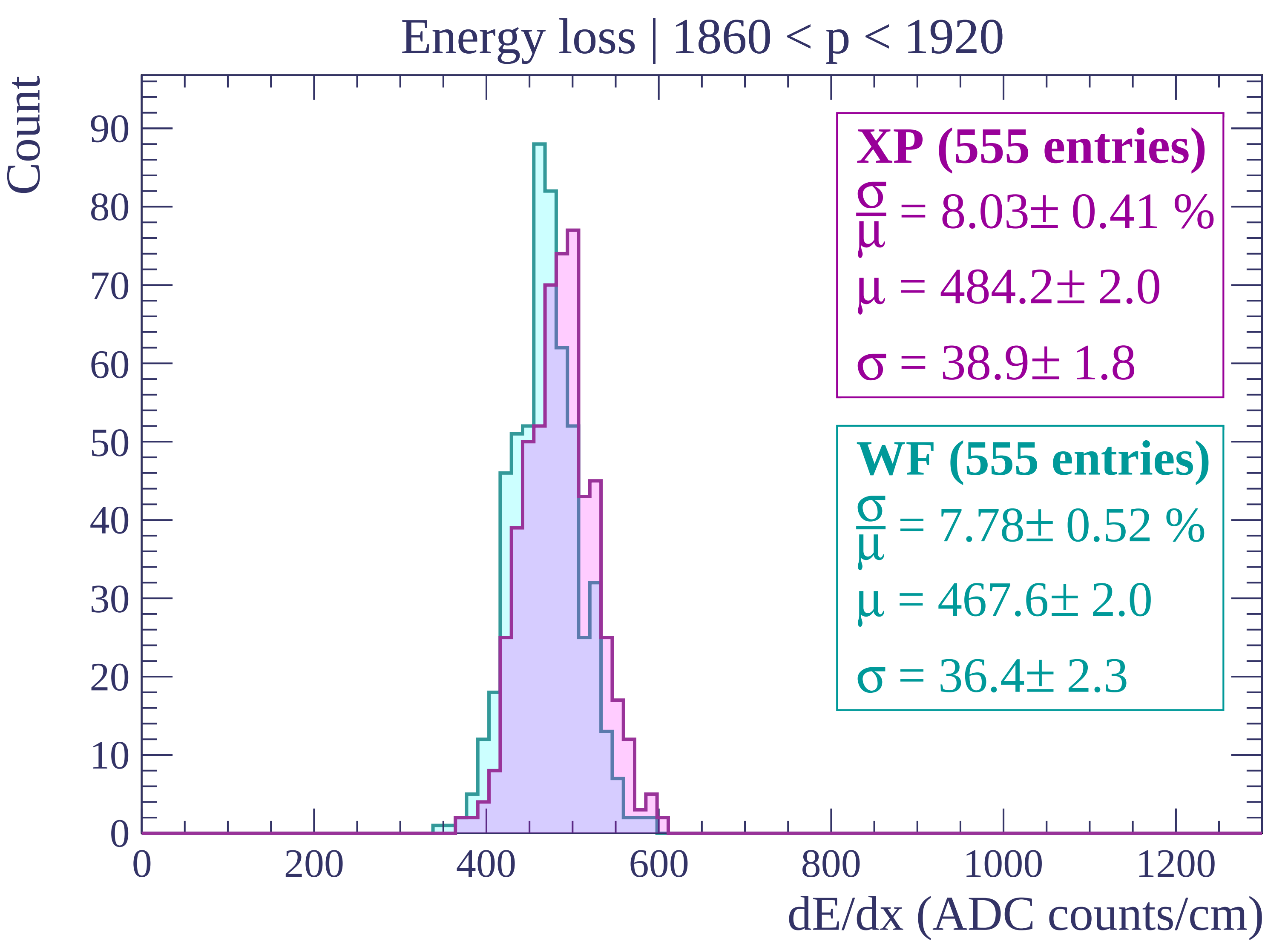


Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (577 entries)
$\frac{\sigma}{\mu}$ = 7.61± 0.36 %
μ = 482.7± 1.8
σ = 36.8± 1.6
WF (577 entries)
$\frac{\sigma}{\mu}$ = 7.82± 0.51 %
μ = 467.7± 1.9
σ = 36.6± 2.2

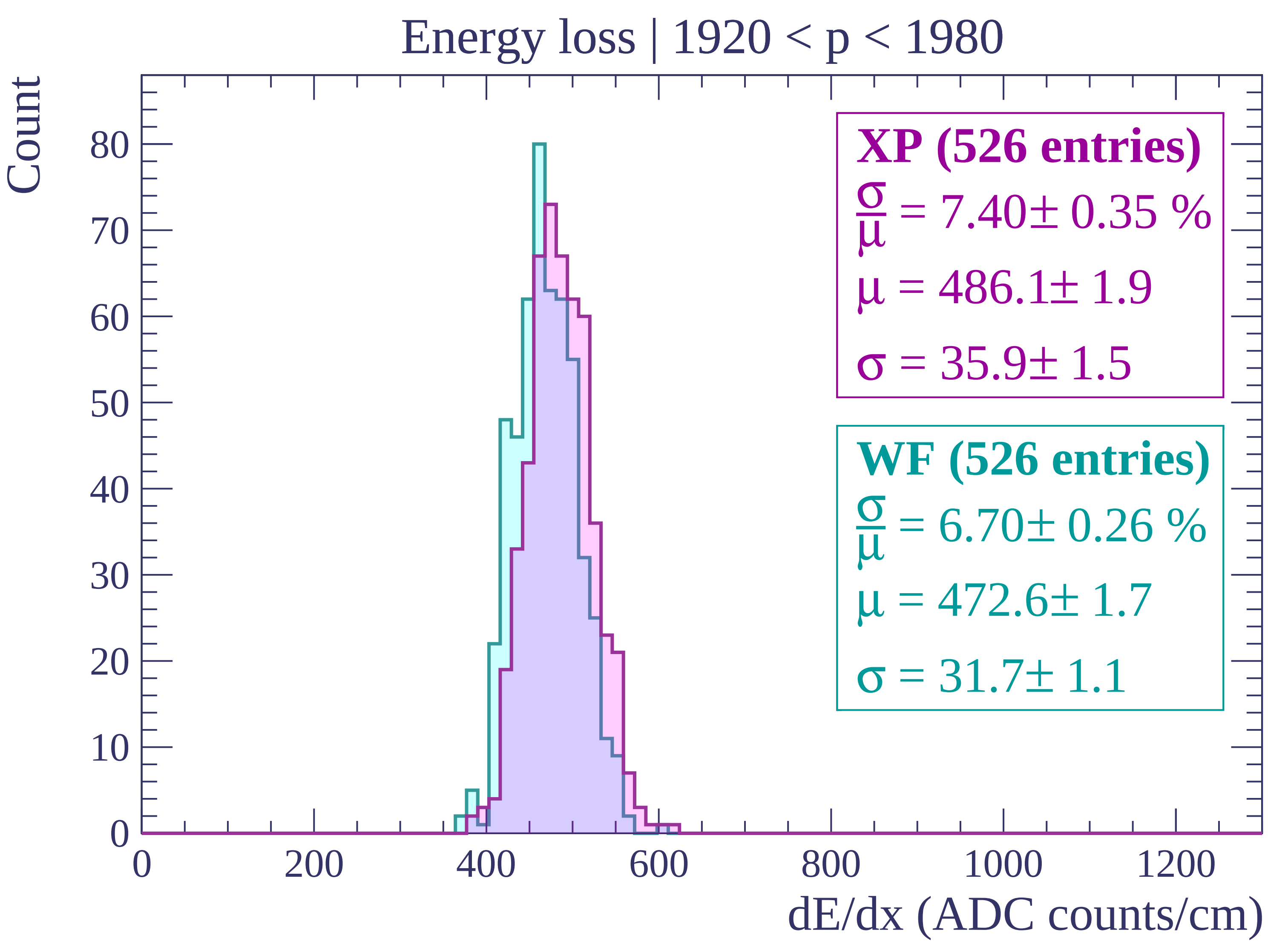

Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (554 entries)
$\frac{\sigma}{\mu}$ = 7.11± 0.36 %
μ = 483.5± 1.7
σ = 34.4± 1.6
WF (554 entries)
$\frac{\sigma}{\mu}$ = 7.10± 0.40 %
μ = 467.2± 1.7
σ = 33.2± 1.8

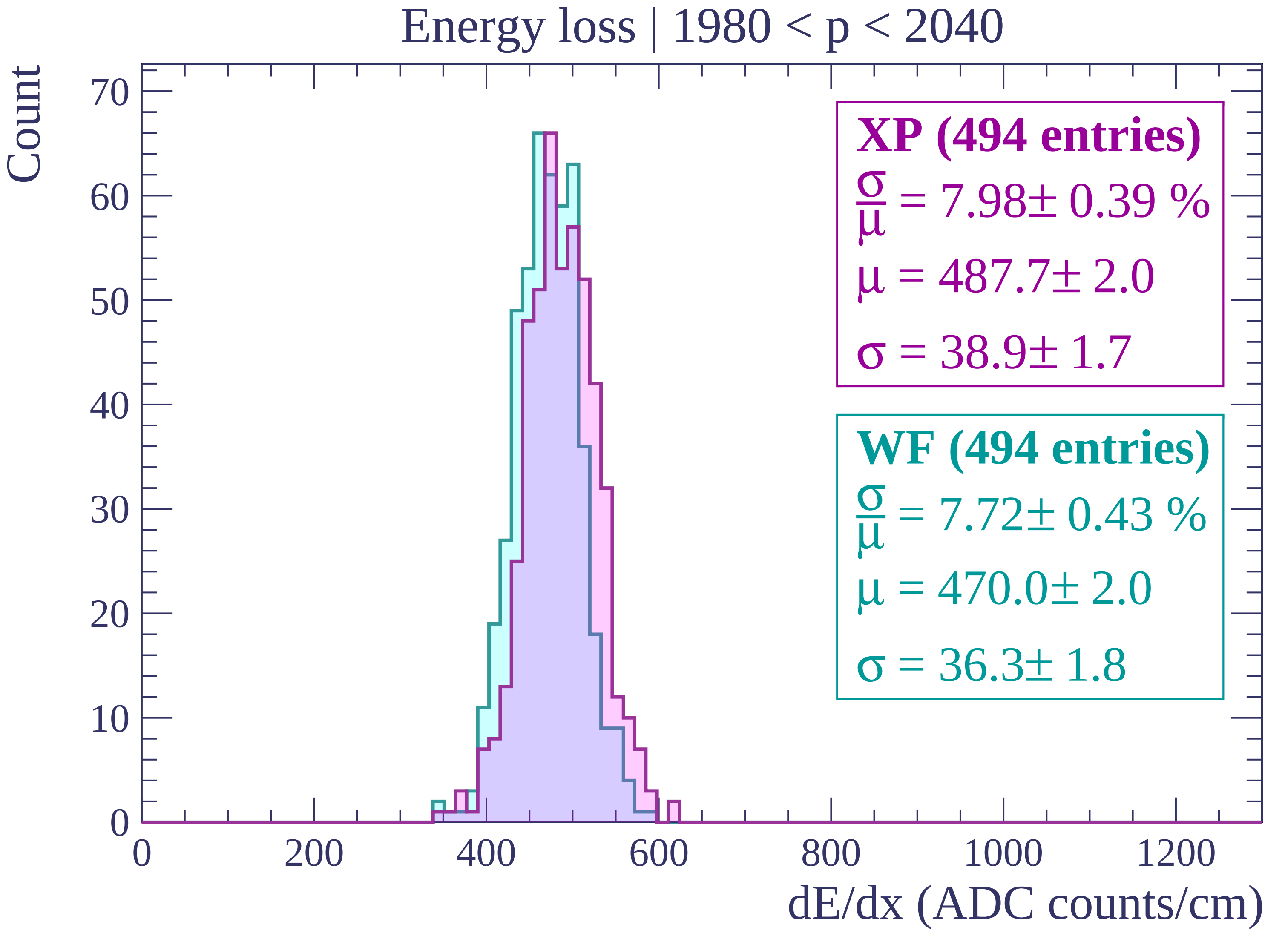


Energy loss | 1800 < p < 1860
Count
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (546 entries)
σ/μ = 7.71± 0.42 %
μ = 487.3± 2.0
σ = 37.6± 1.9
WF (546 entries)
σ/μ = 7.78± 0.40 %
μ = 469.8± 1.8
σ = 36.6± 1.7

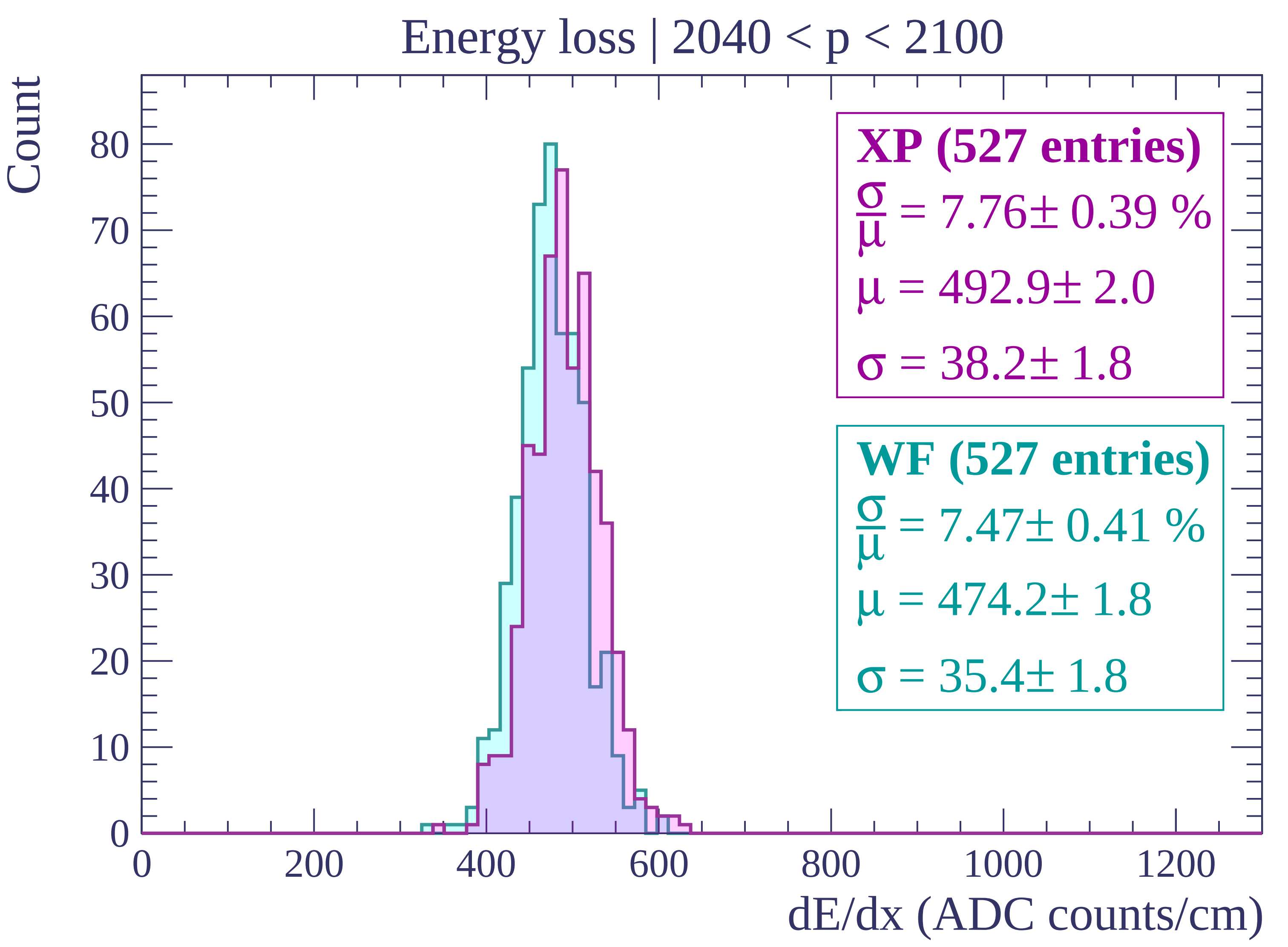


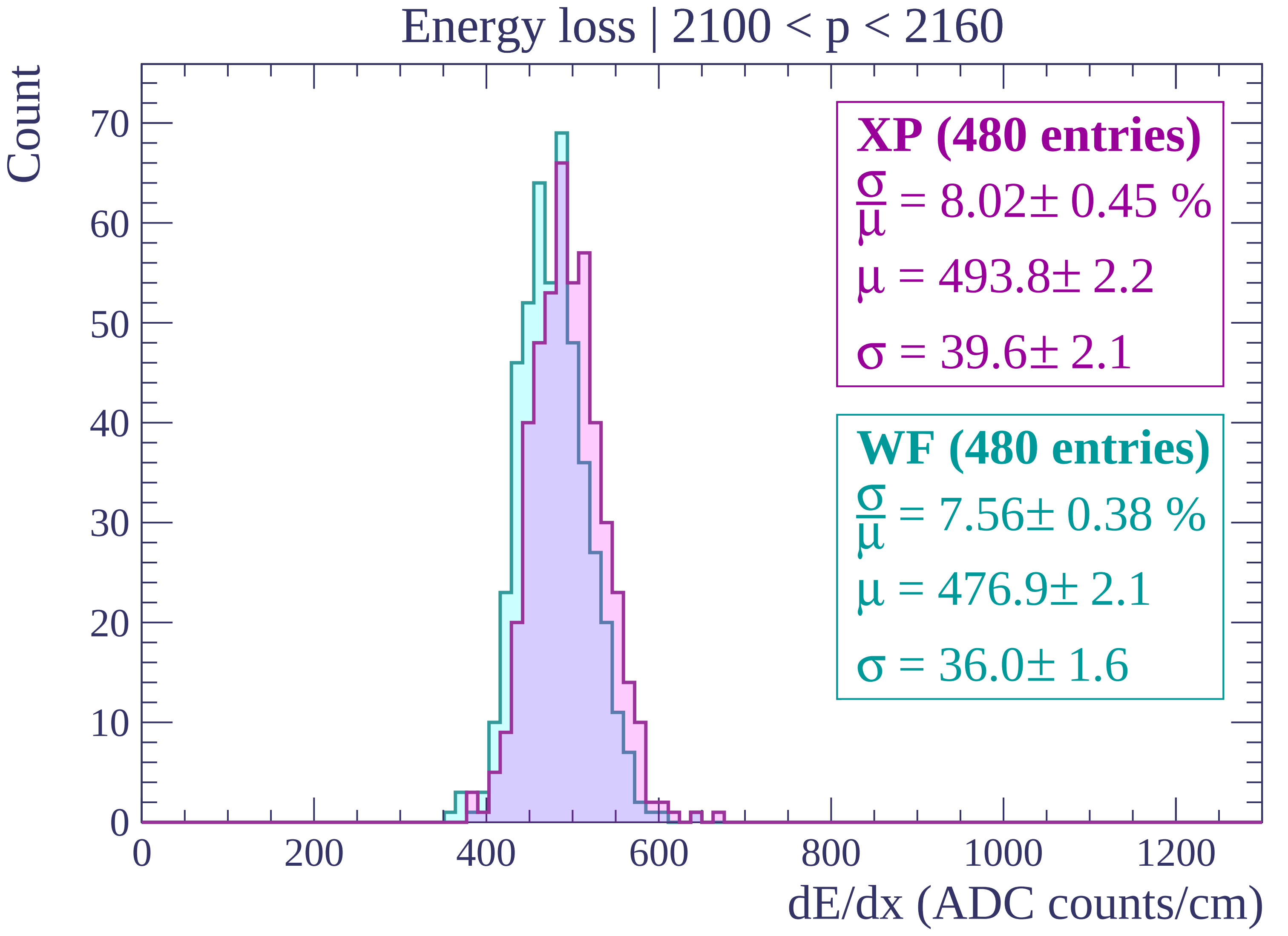
Energy loss | 1860 < p < 1920
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (555 entries)
$\frac{\sigma}{\mu}$ = 8.03± 0.41 %
μ = 484.2± 2.0
σ = 38.9± 1.8
WF (555 entries)
$\frac{\sigma}{\mu}$ = 7.78± 0.52 %
μ = 467.6± 2.0
σ = 36.4± 2.3

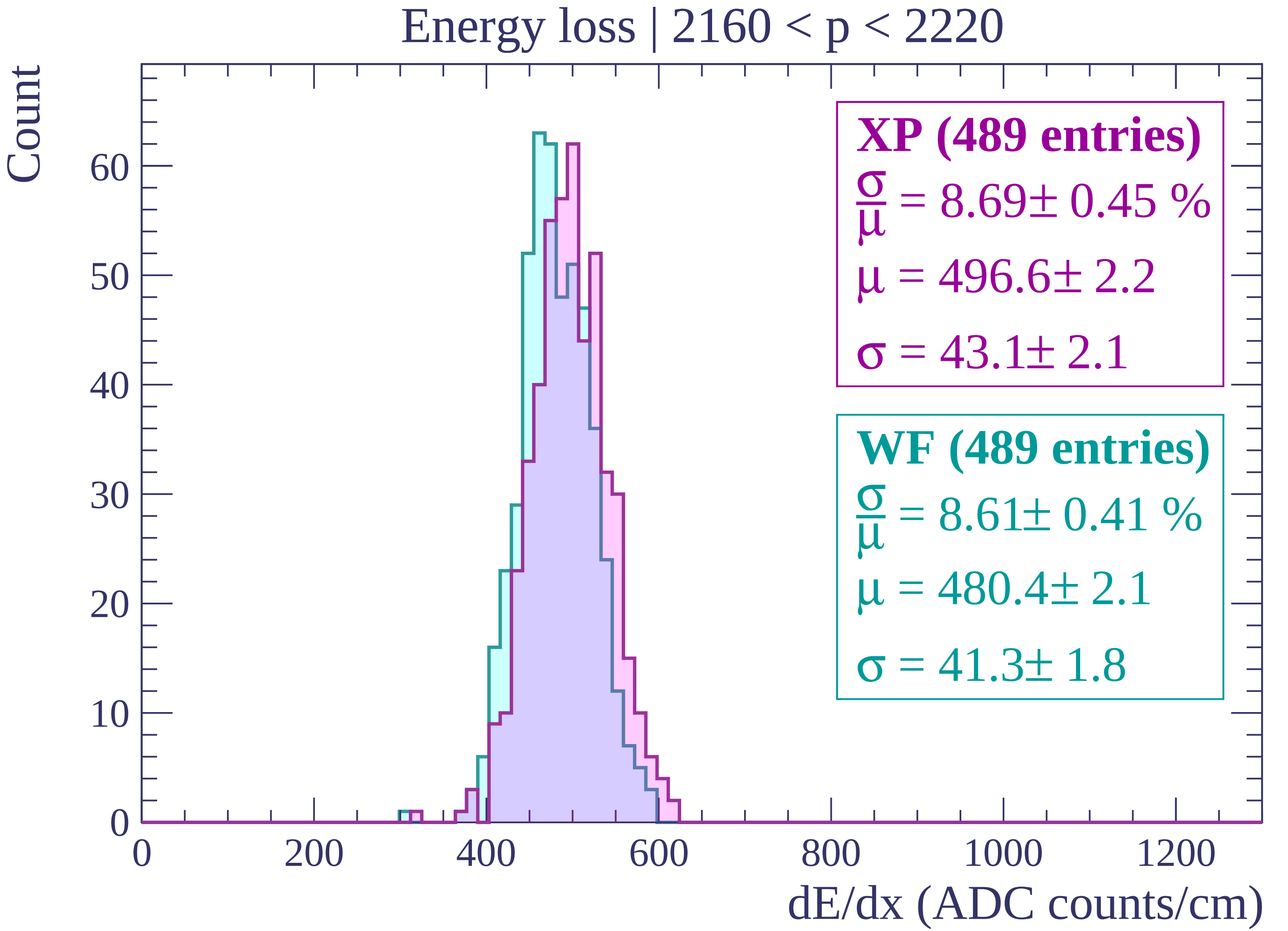

Energy loss | 1920 < p < 1980
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (526 entries)
σ/μ = 7.40± 0.35 %
μ = 486.1± 1.9
σ = 35.9± 1.5
WF (526 entries)
σ/μ = 6.70± 0.26 %
μ = 472.6± 1.7
σ = 31.7± 1.1

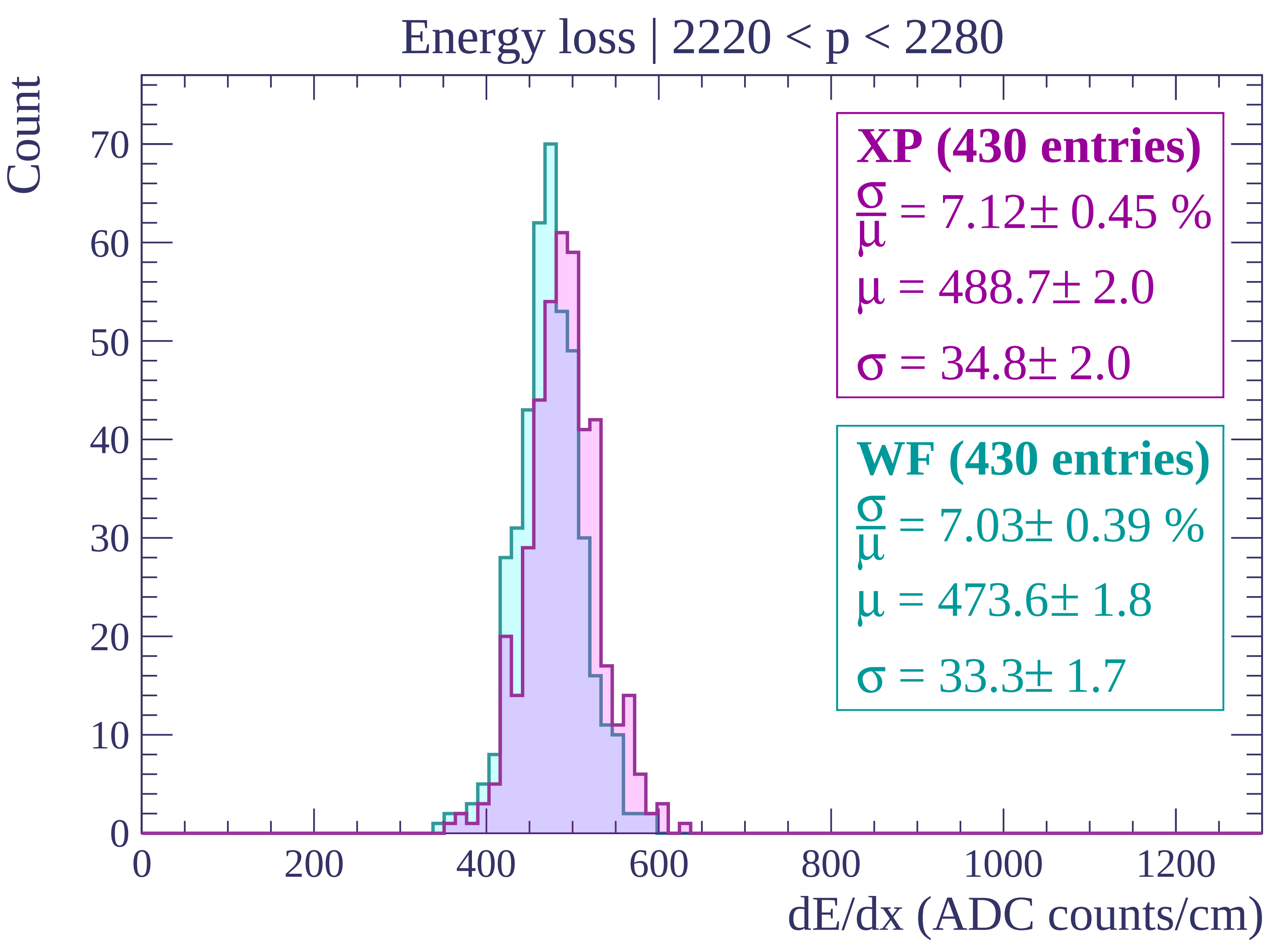


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (494 entries)
σ/μ = 7.98± 0.39 %
μ = 487.7± 2.0
σ = 38.9± 1.7
WF (494 entries)
σ/μ = 7.72± 0.43 %
μ = 470.0± 2.0
σ = 36.3± 1.8

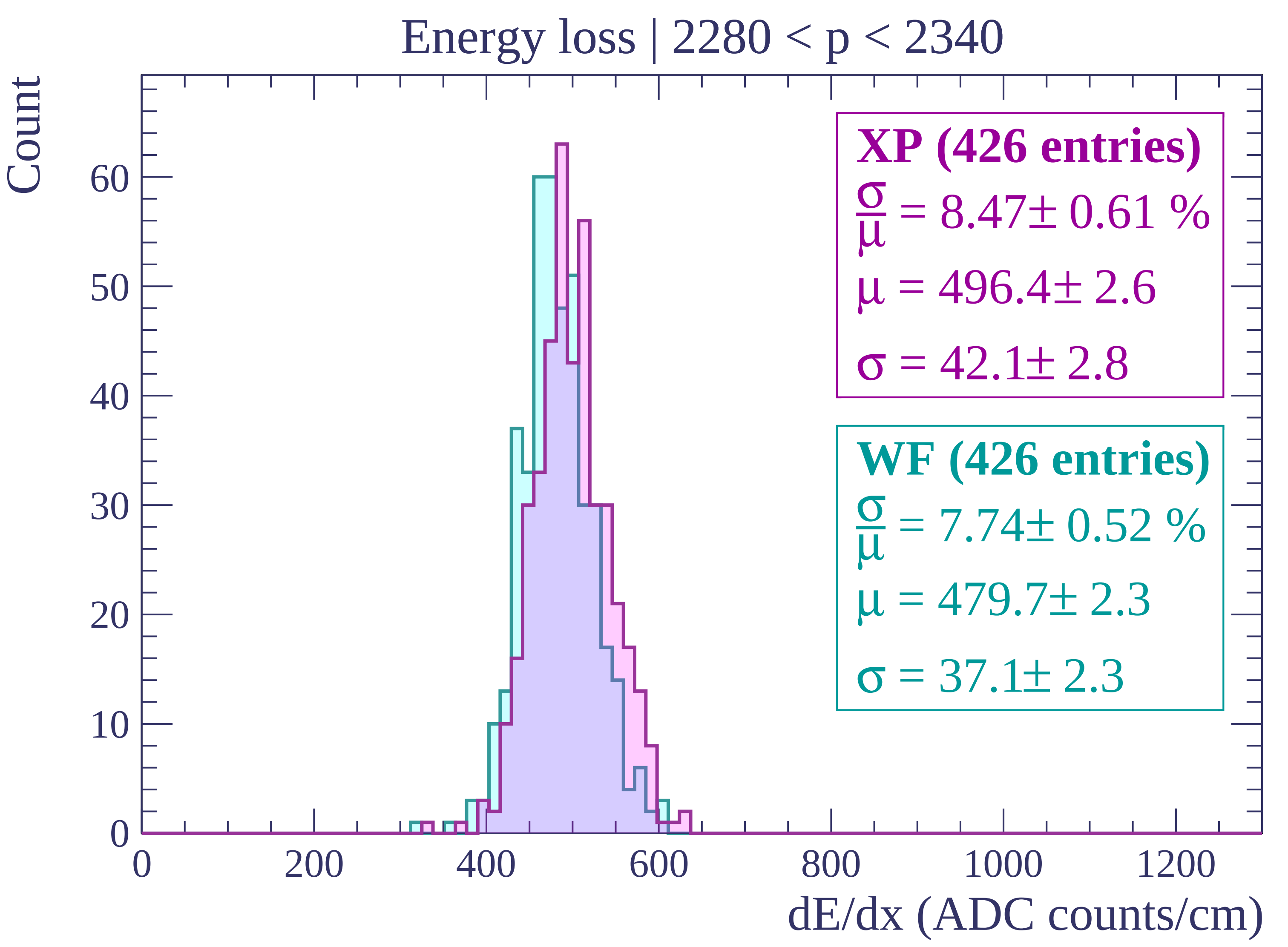

Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (527 entries)
$\frac{\sigma}{\mu}$ = 7.76± 0.39 %
μ = 492.9± 2.0
σ = 38.2± 1.8
WF (527 entries)
$\frac{\sigma}{\mu}$ = 7.47± 0.41 %
μ = 474.2± 1.8
σ = 35.4± 1.8
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80


Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
XP (480 entries)
$\frac{\sigma}{\mu}$ = 8.02± 0.45 %
μ = 493.8± 2.2
σ = 39.6± 2.1
WF (480 entries)
$\frac{\sigma}{\mu}$ = 7.56± 0.38 %
μ = 476.9± 2.1
σ = 36.0± 1.6
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70

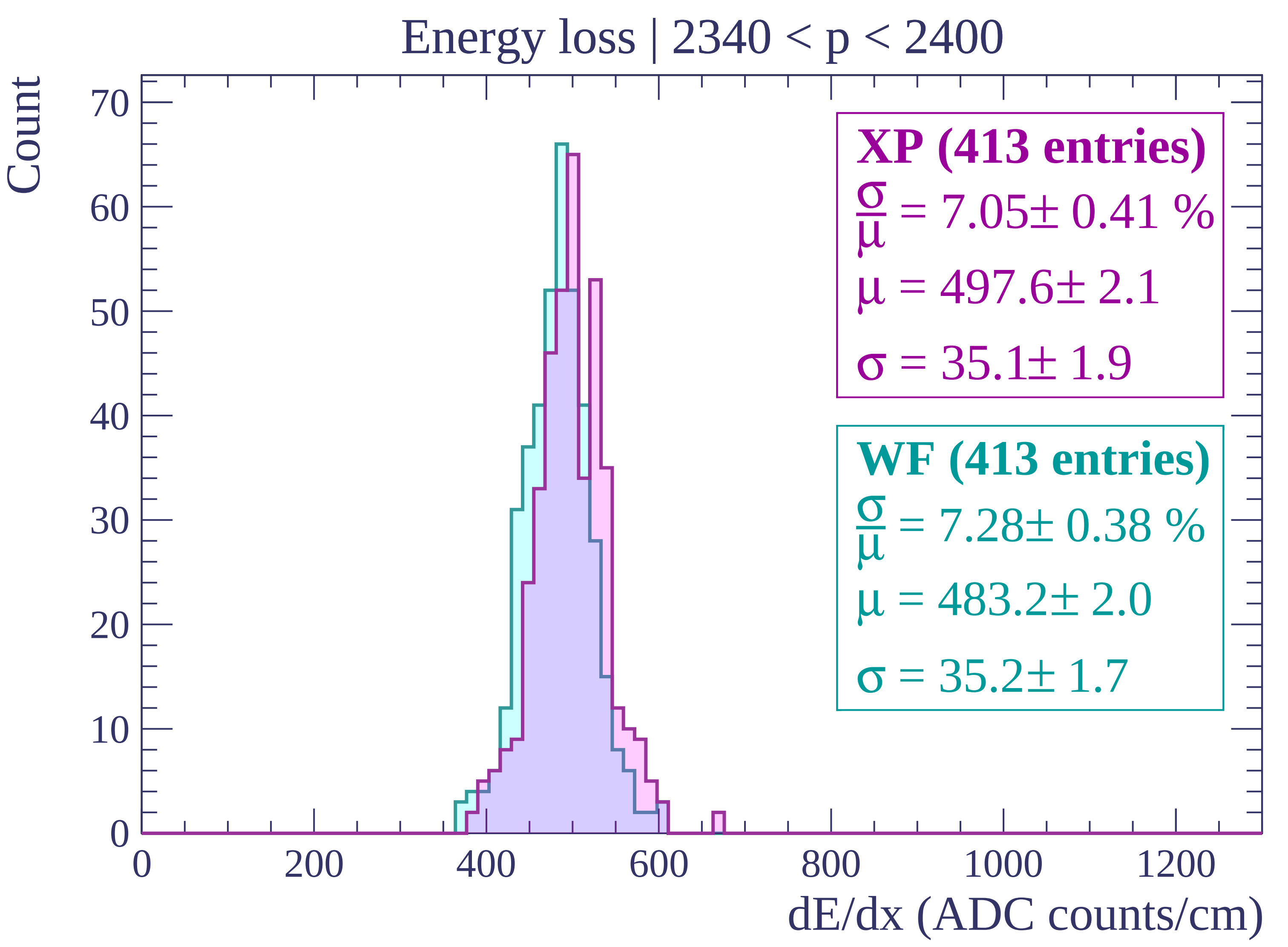

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (489 entries)
σ/μ = 8.69± 0.45 %
μ = 496.6± 2.2
σ = 43.1± 2.1
WF (489 entries)
σ/μ = 8.61± 0.41 %
μ = 480.4± 2.1
σ = 41.3± 1.8

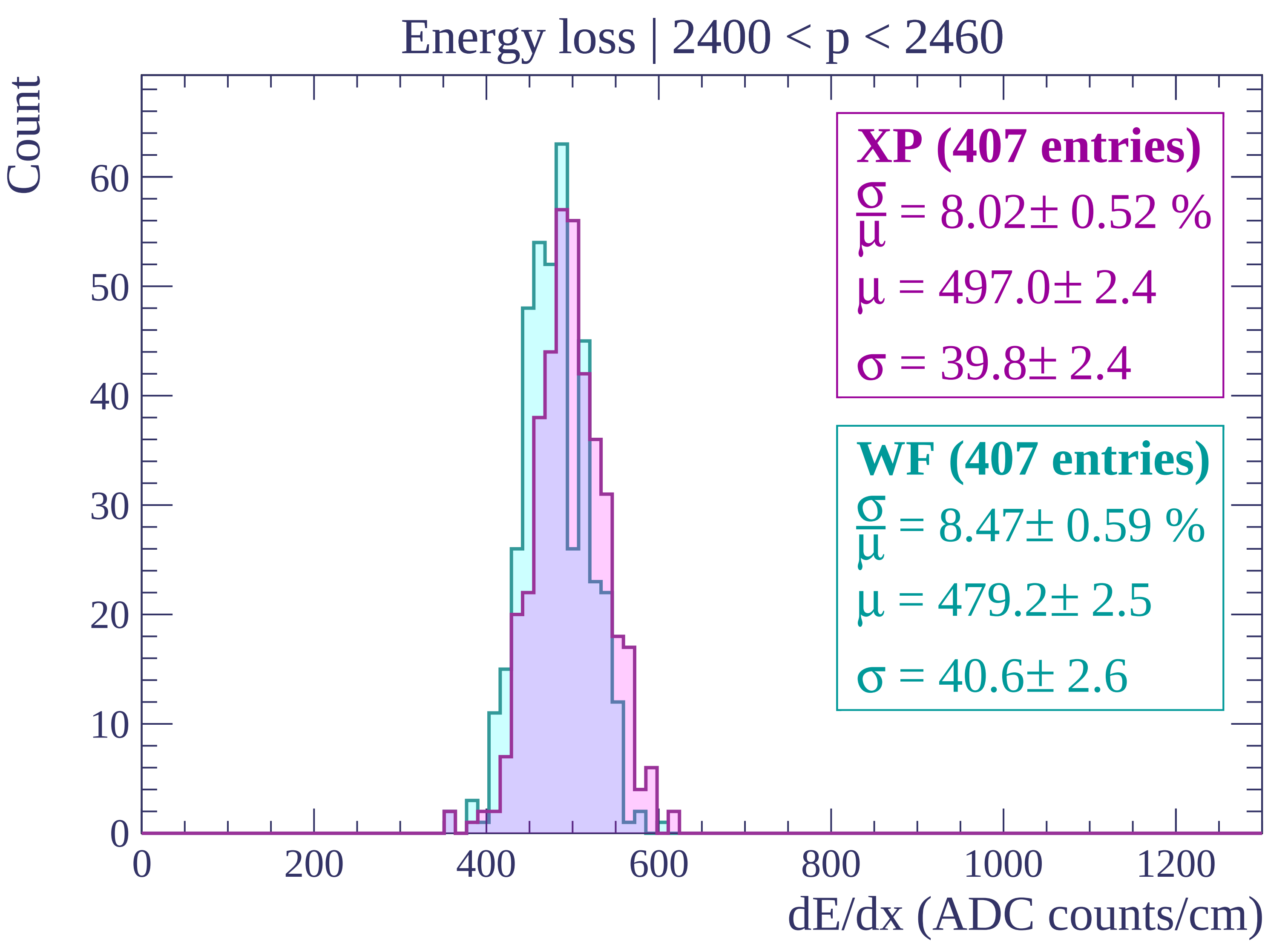


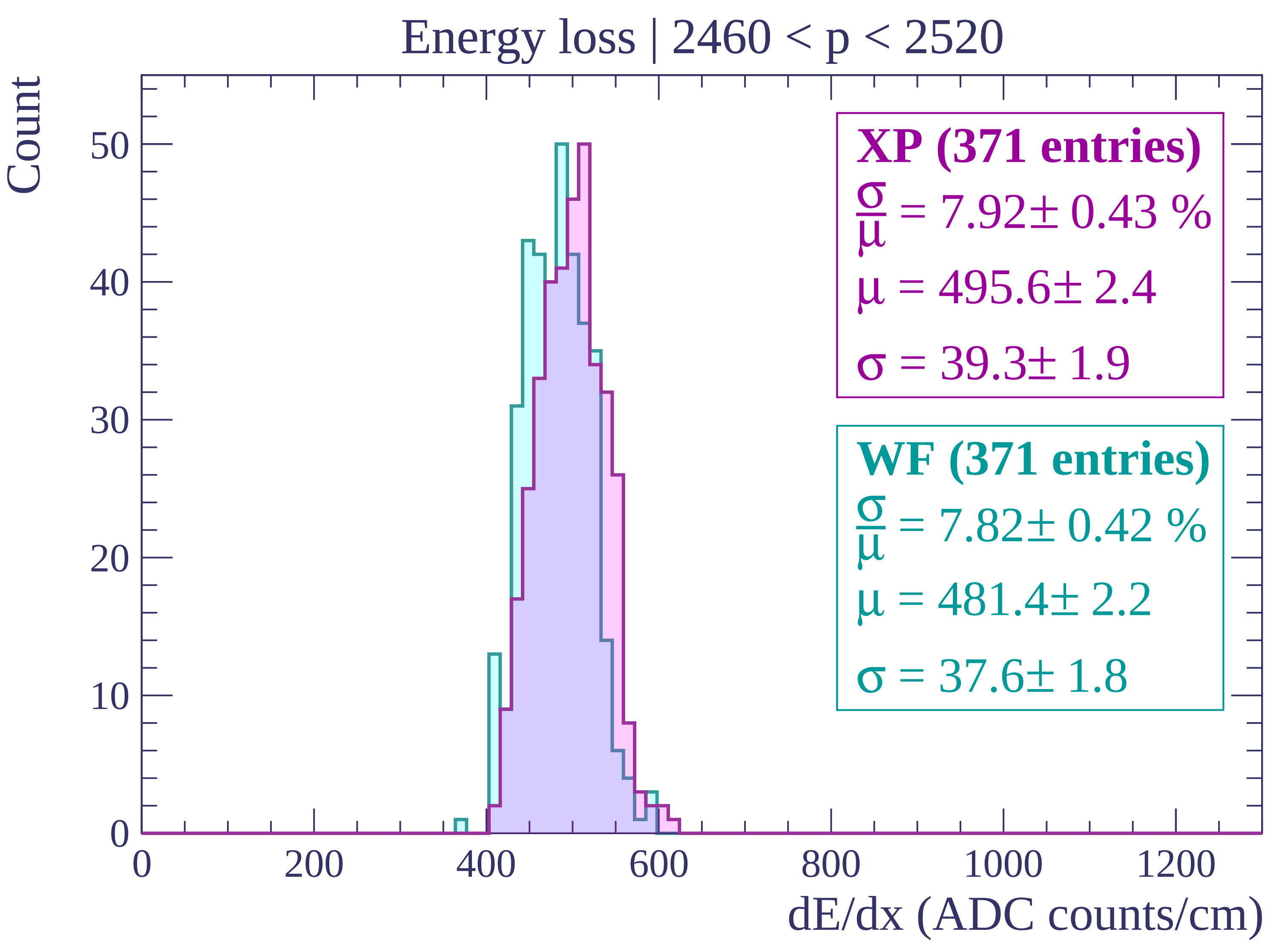
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (430 entries)
$\frac{\sigma}{\mu}$ = 7.12± 0.45 %
μ = 488.7± 2.0
σ = 34.8± 2.0
WF (430 entries)
$\frac{\sigma}{\mu}$ = 7.03± 0.39 %
μ = 473.6± 1.8
σ = 33.3± 1.7



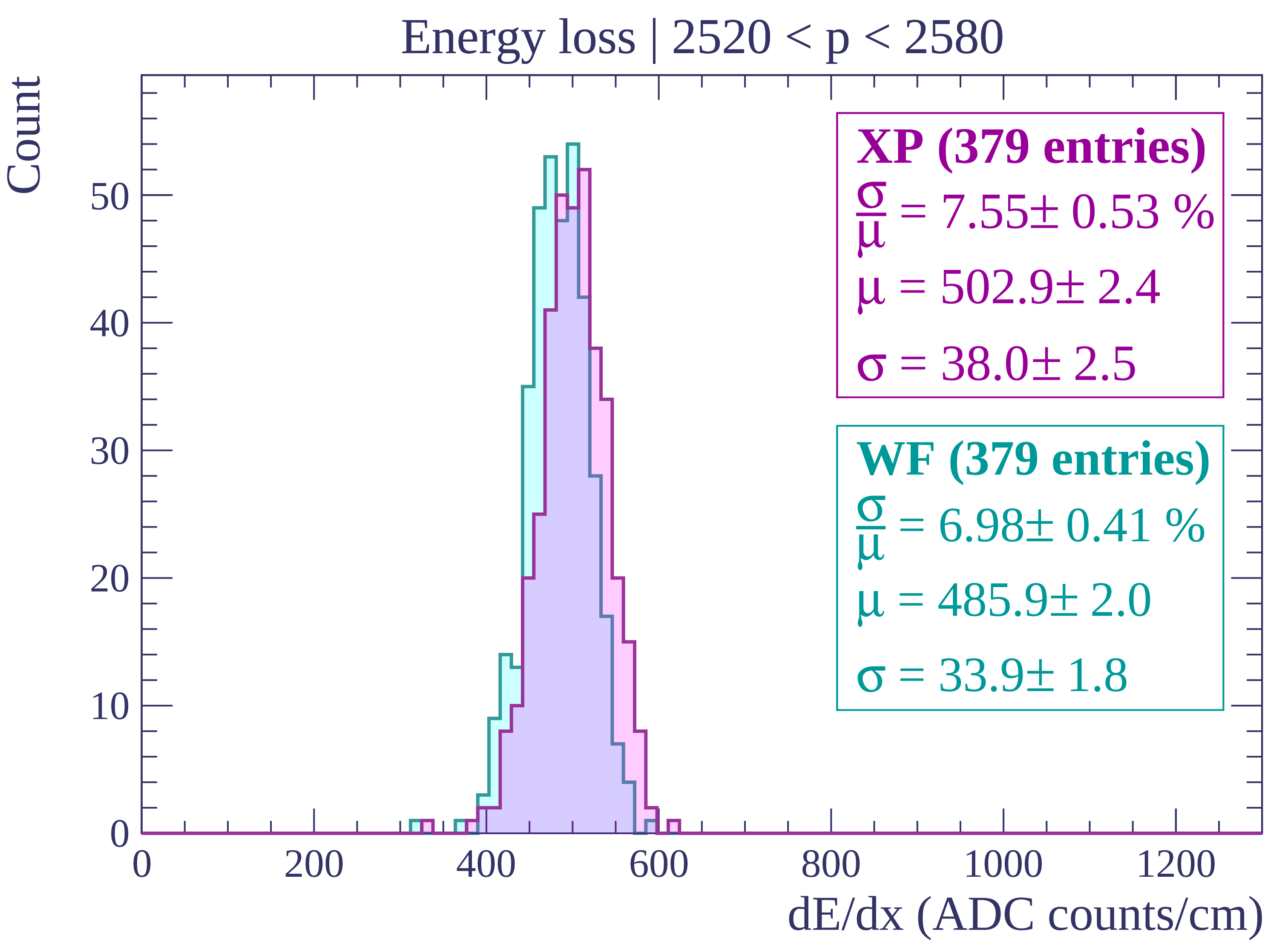
Energy loss | 2280 < p < 2340
Count
XP (426 entries)
σ/μ = 8.47± 0.61 %
μ = 496.4± 2.6
σ = 42.1± 2.8
WF (426 entries)
σ/μ = 7.74± 0.52 %
μ = 479.7± 2.3
σ = 37.1± 2.3
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

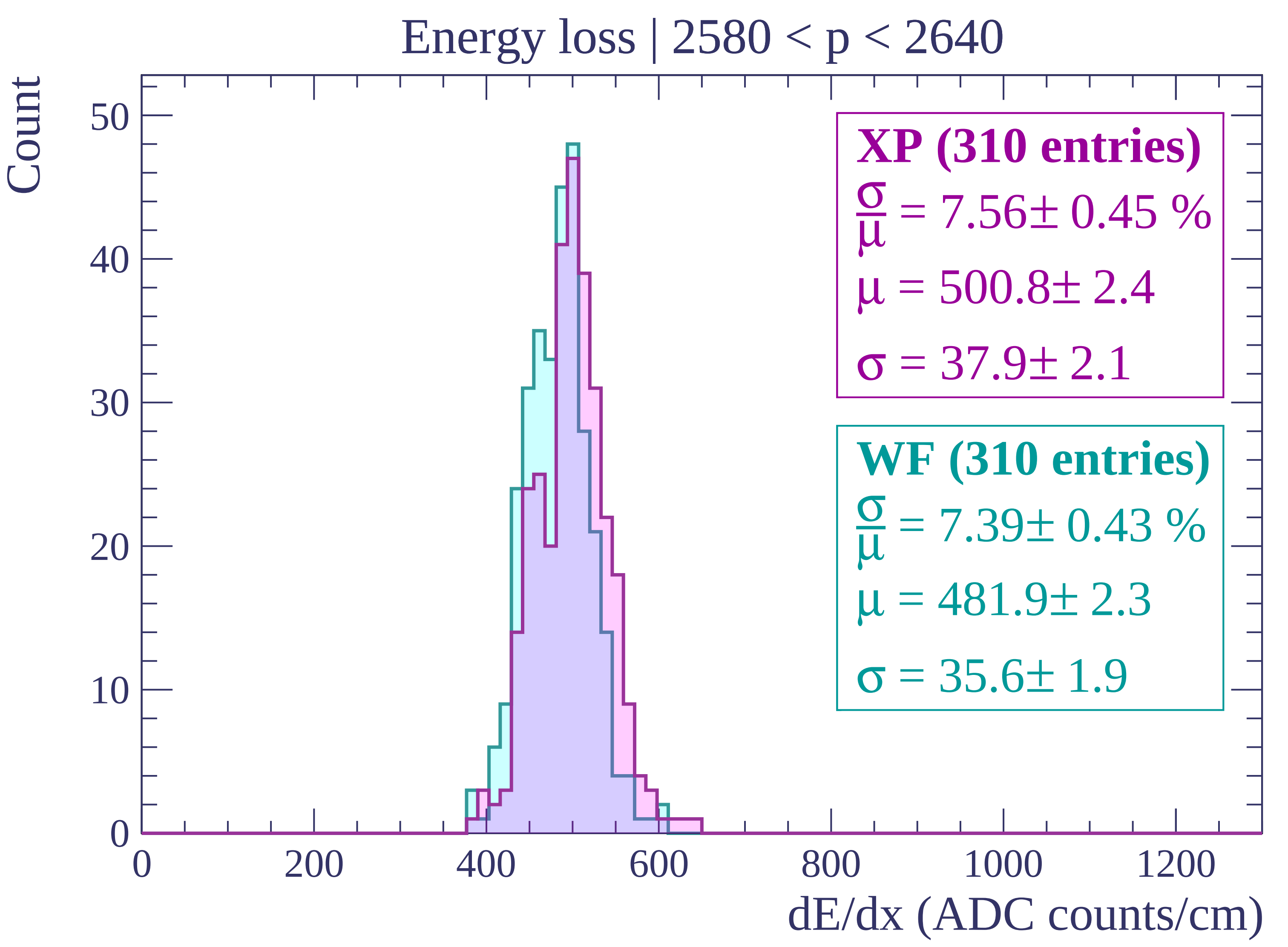

Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (413 entries)
σ/μ = 7.05± 0.41 %
μ = 497.6± 2.1
σ = 35.1± 1.9
WF (413 entries)
σ/μ = 7.28± 0.38 %
μ = 483.2± 2.0
σ = 35.2± 1.7


Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (407 entries)
σ/μ = 8.02± 0.52 %
μ = 497.0± 2.4
σ = 39.8± 2.4
WF (407 entries)
σ/μ = 8.47± 0.59 %
μ = 479.2± 2.5
σ = 40.6± 2.6

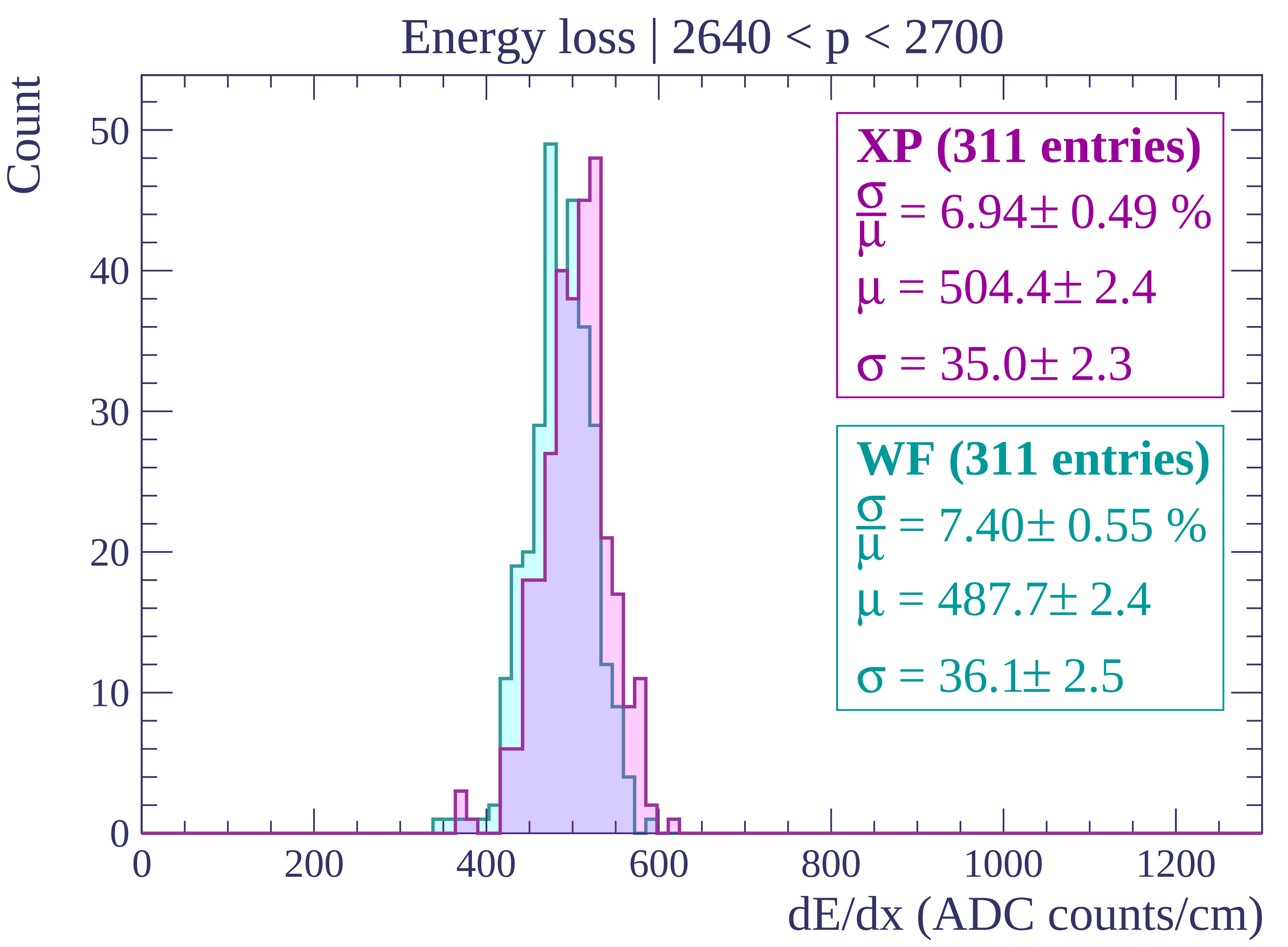


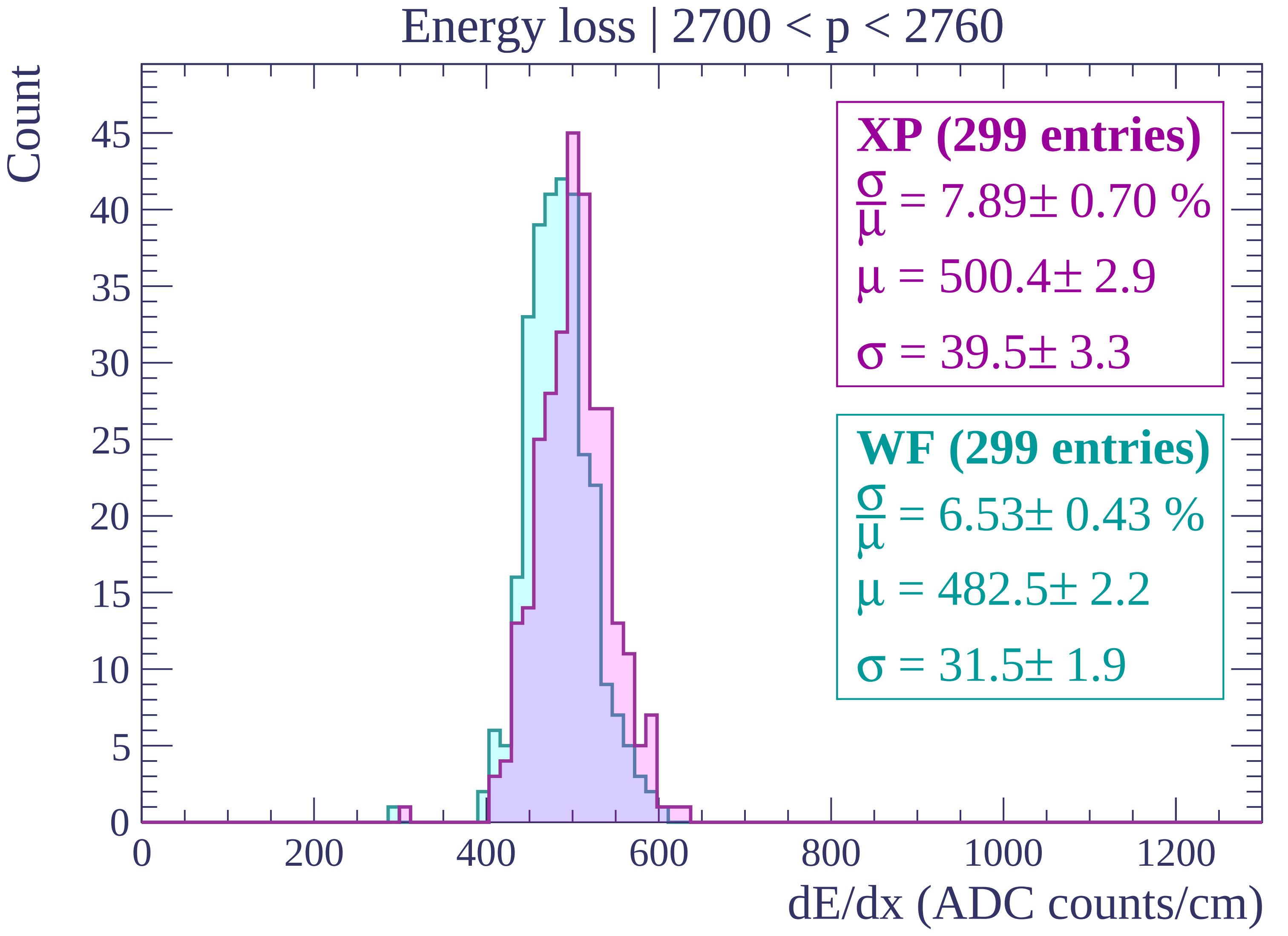
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (371 entries)
σ/μ = 7.92± 0.43 %
μ = 495.6± 2.4
σ = 39.3± 1.9
WF (371 entries)
σ/μ = 7.82± 0.42 %
μ = 481.4± 2.2
σ = 37.6± 1.8

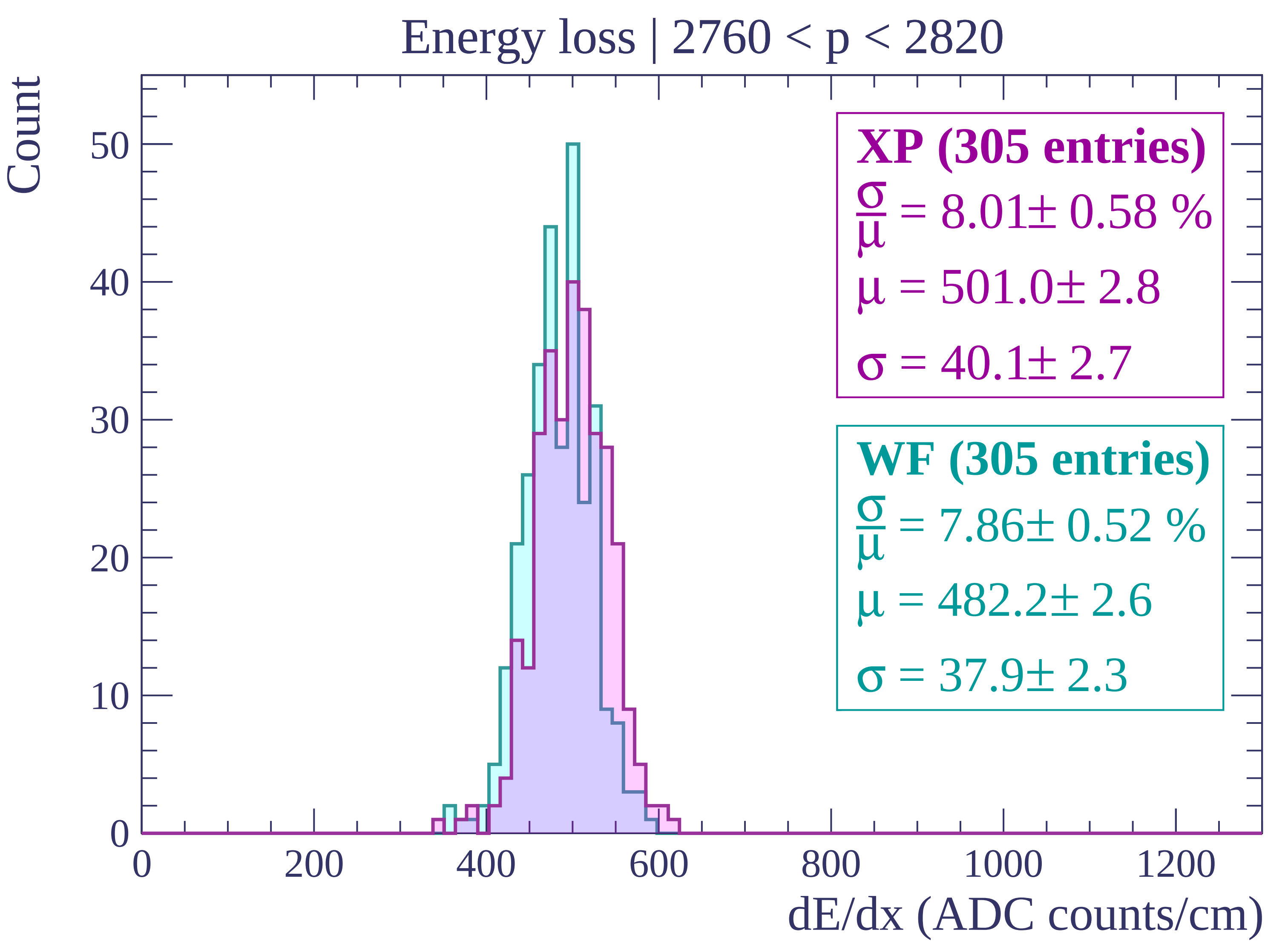

Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP (379 entries)
$\frac{\sigma}{\mu}$ = 7.55± 0.53 %
μ = 502.9± 2.4
σ = 38.0± 2.5
WF (379 entries)
$\frac{\sigma}{\mu}$ = 6.98± 0.41 %
μ = 485.9± 2.0
σ = 33.9± 1.8



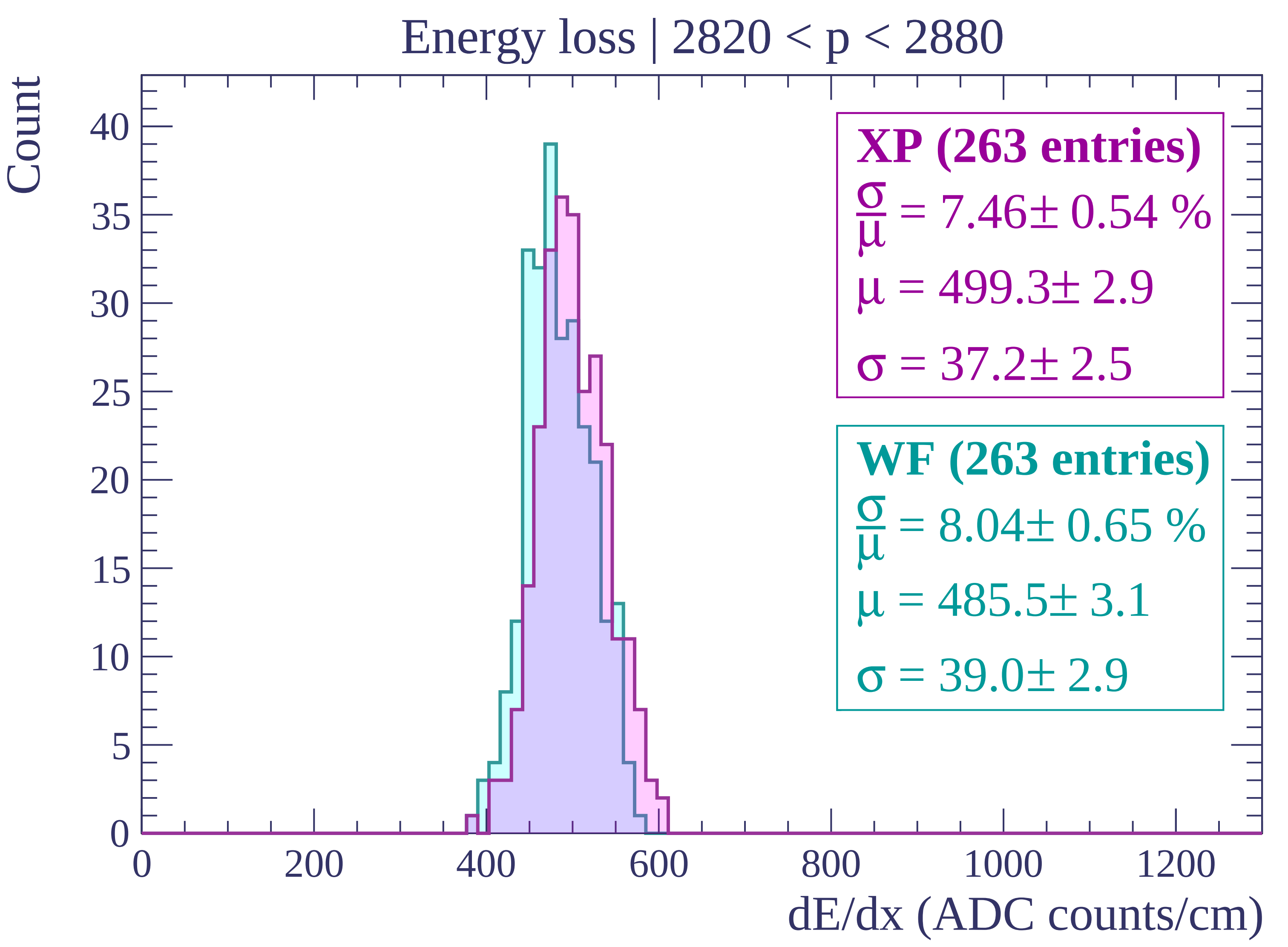
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (310 entries)
$\frac{\sigma}{\mu}$ = 7.56± 0.45 %
μ = 500.8± 2.4
σ = 37.9± 2.1
WF (310 entries)
$\frac{\sigma}{\mu}$ = 7.39± 0.43 %
μ = 481.9± 2.3
σ = 35.6± 1.9


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
XP (311 entries)
σ/μ = 6.94± 0.49 %
μ = 504.4± 2.4
σ = 35.0± 2.3
WF (311 entries)
σ/μ = 7.40± 0.55 %
μ = 487.7± 2.4
σ = 36.1± 2.5
0
10
20
30
40
50
0
200
400
600
800
1000
1200

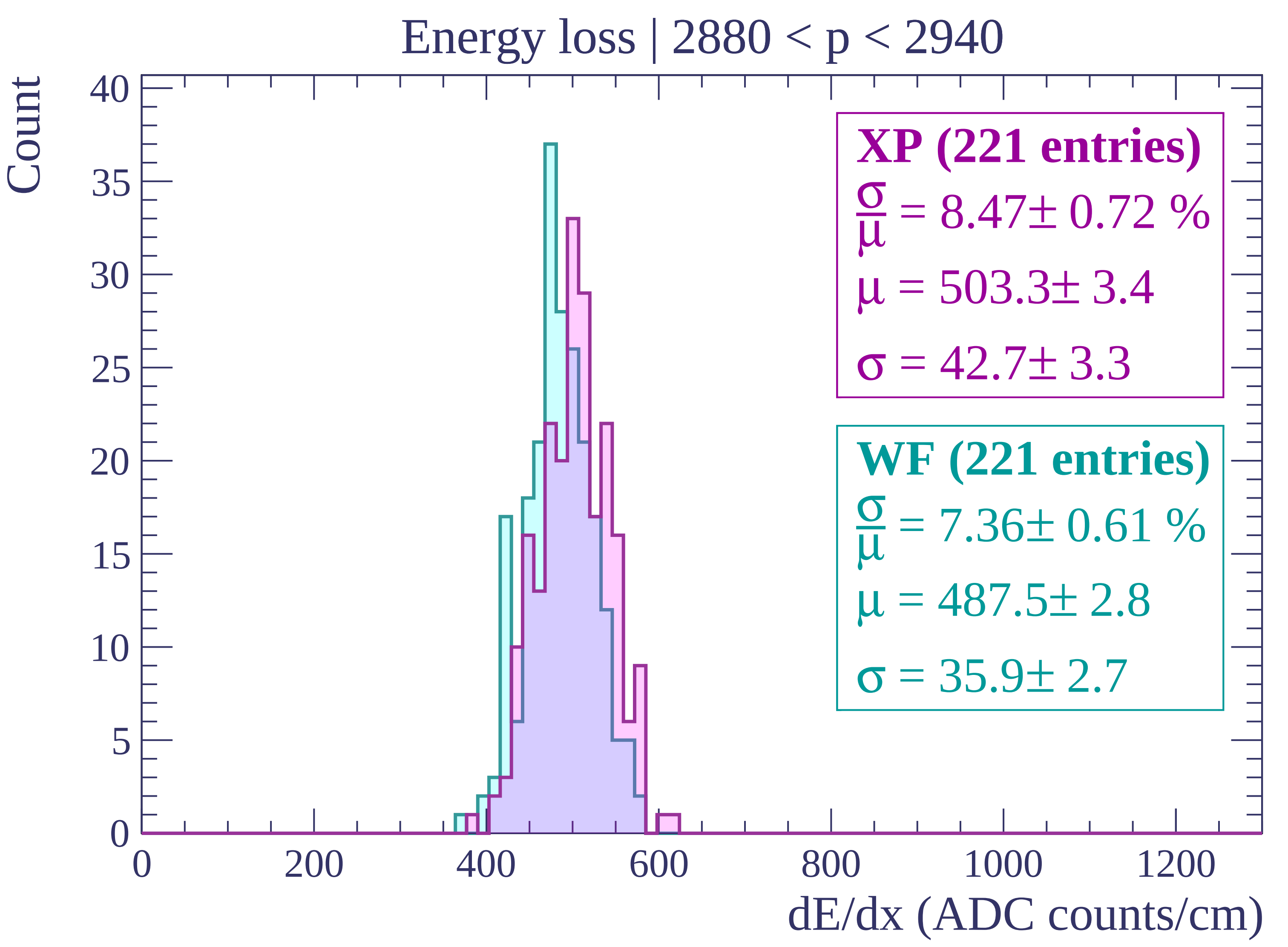

Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (299 entries)
σ/μ = 7.89± 0.70 %
μ = 500.4± 2.9
σ = 39.5± 3.3
WF (299 entries)
σ/μ = 6.53± 0.43 %
μ = 482.5± 2.2
σ = 31.5± 1.9

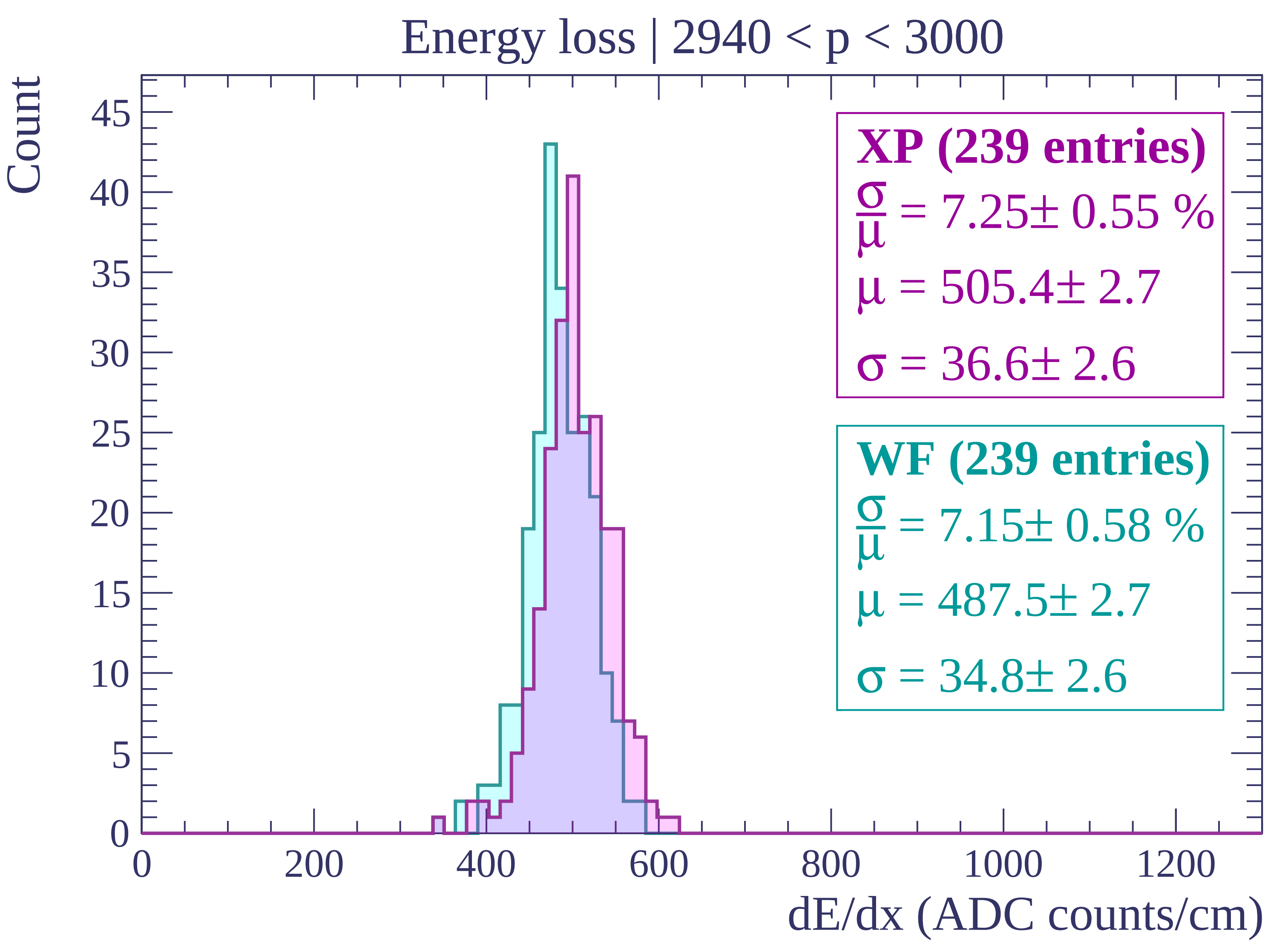


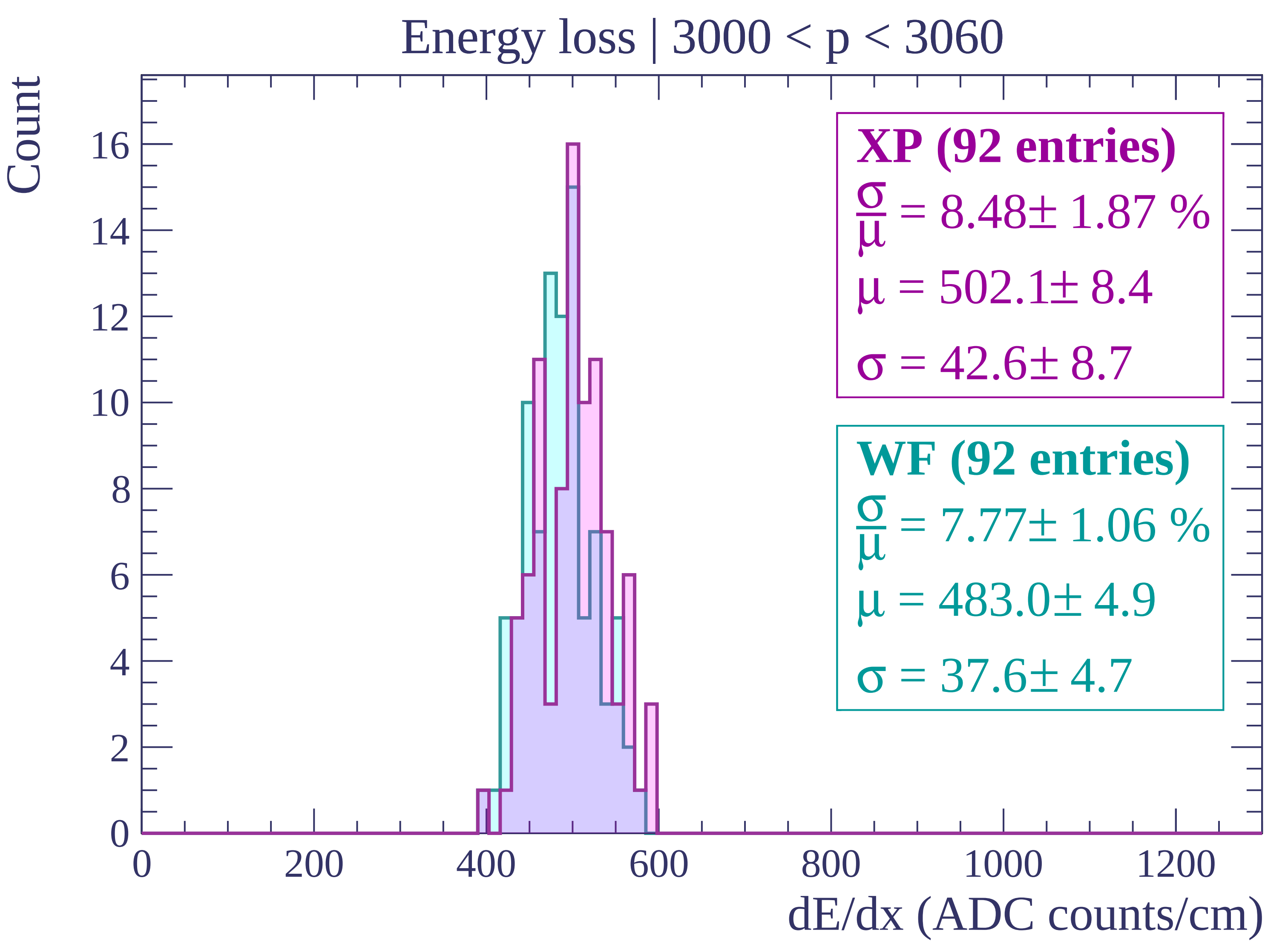
Energy loss | 2760 < p < 2820
Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (305 entries)
σ/μ = 8.01± 0.58 %
μ = 501.0± 2.8
σ = 40.1± 2.7
WF (305 entries)
σ/μ = 7.86± 0.52 %
μ = 482.2± 2.6
σ = 37.9± 2.3

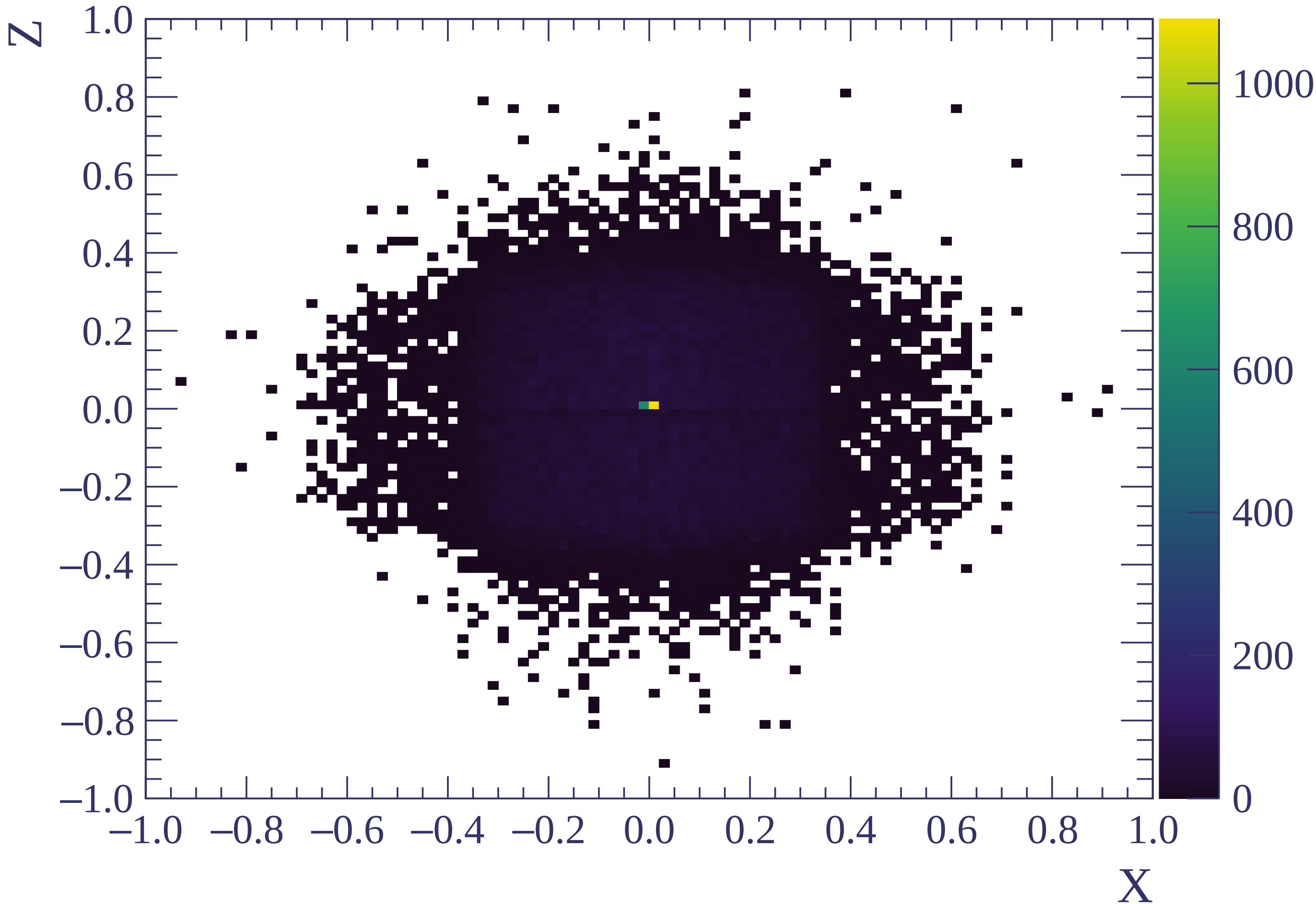


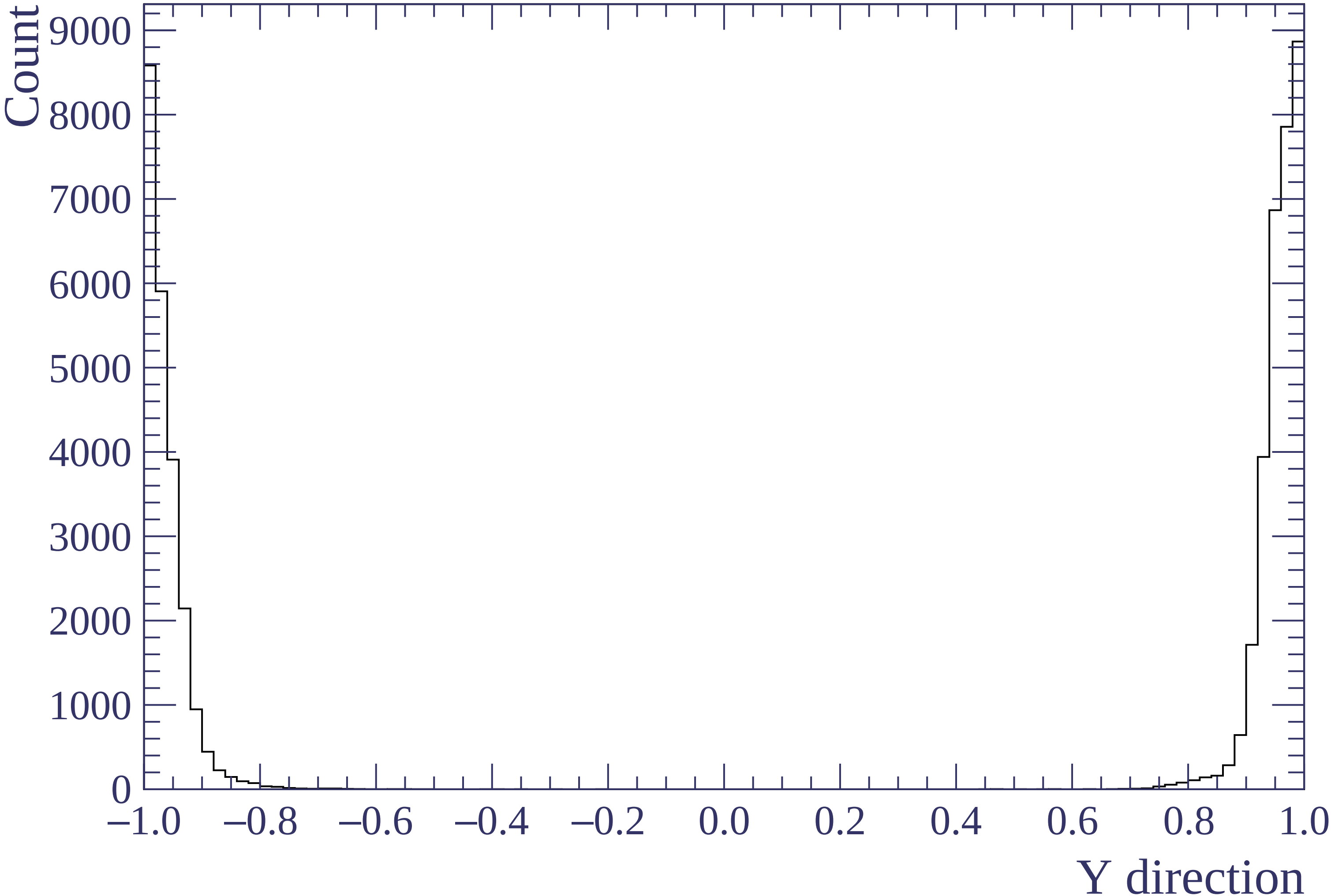
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (263 entries)
σ/μ = 7.46± 0.54 %
μ = 499.3± 2.9
σ = 37.2± 2.5
WF (263 entries)
σ/μ = 8.04± 0.65 %
μ = 485.5± 3.1
σ = 39.0± 2.9


Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (221 entries)
$\frac{\sigma}{\mu}$ = 8.47± 0.72 %
μ = 503.3± 3.4
σ = 42.7± 3.3
WF (221 entries)
$\frac{\sigma}{\mu}$ = 7.36± 0.61 %
μ = 487.5± 2.8
σ = 35.9± 2.7

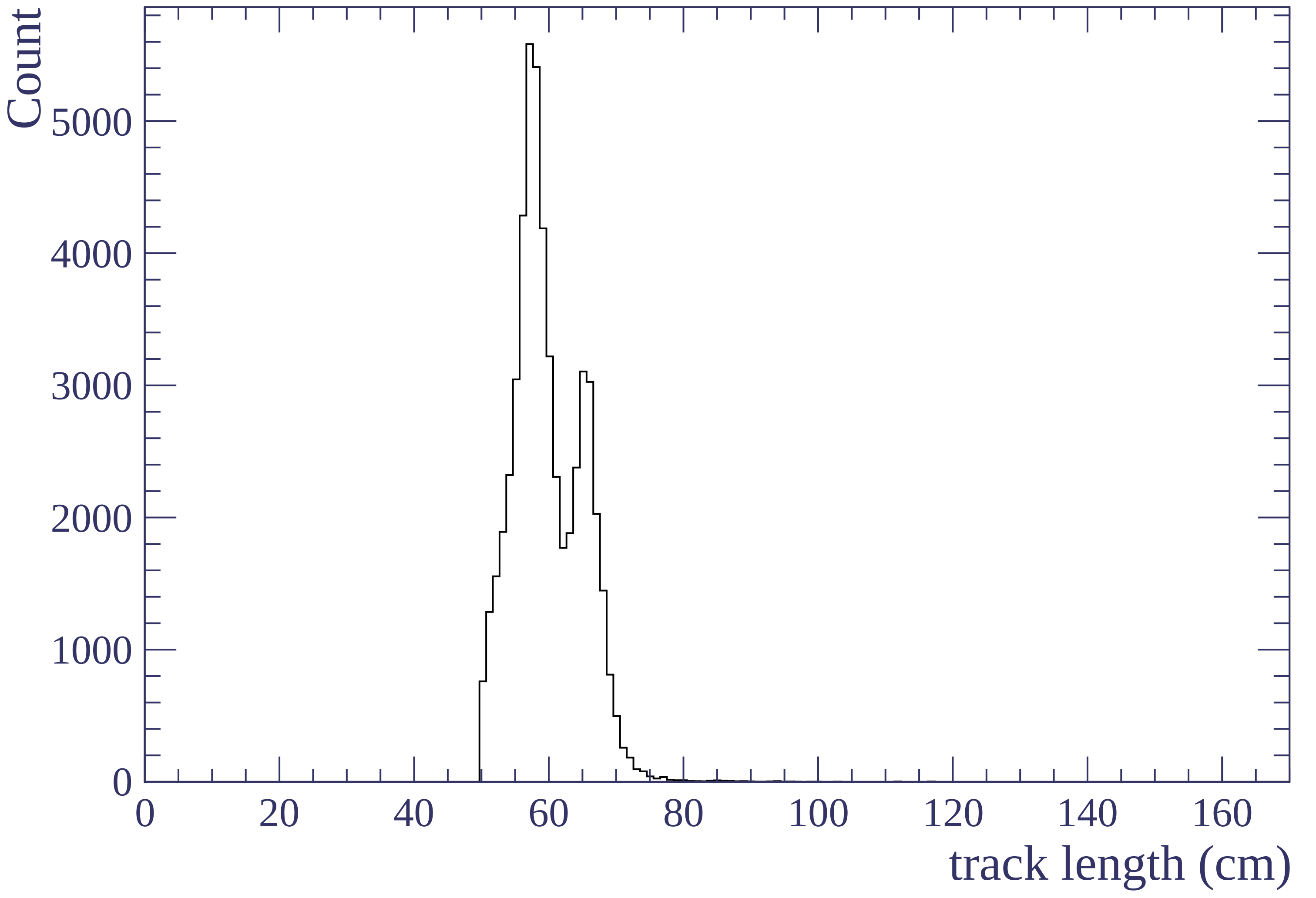

Energy loss | 2940 < p < 3000
Count
45
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (239 entries)
σ/μ = 7.25± 0.55 %
μ = 505.4± 2.7
σ = 36.6± 2.6
WF (239 entries)
σ/μ = 7.15± 0.58 %
μ = 487.5± 2.7
σ = 34.8± 2.6


Energy loss | 3000 < p < 3060
Count
XP (92 entries)
σ/μ = 8.48± 1.87 %
μ = 502.1± 8.4
σ = 42.6± 8.7
WF (92 entries)
σ/μ = 7.77± 1.06 %
μ = 483.0± 4.9
σ = 37.6± 4.7
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

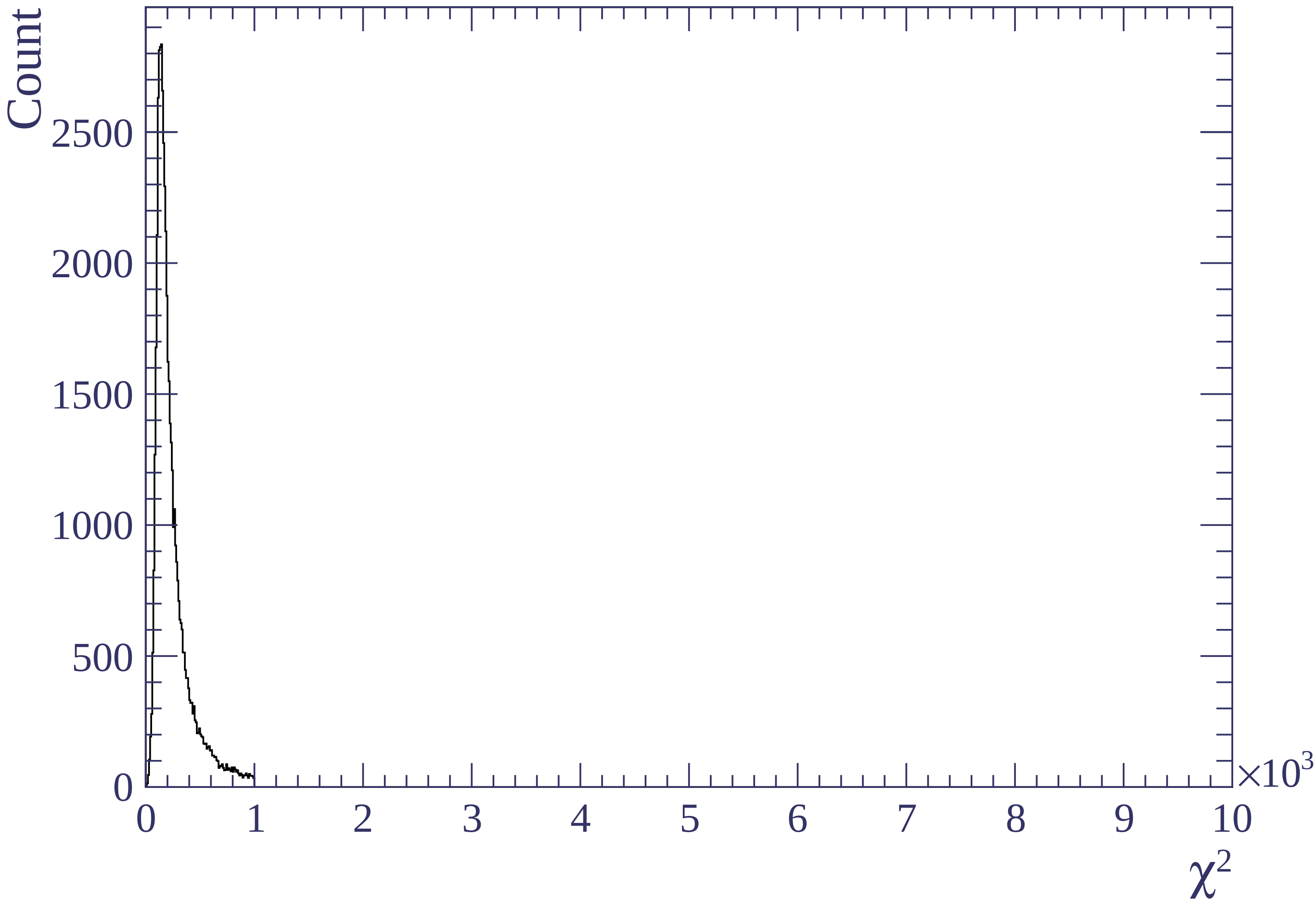

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
1000
800
600
400
200
0

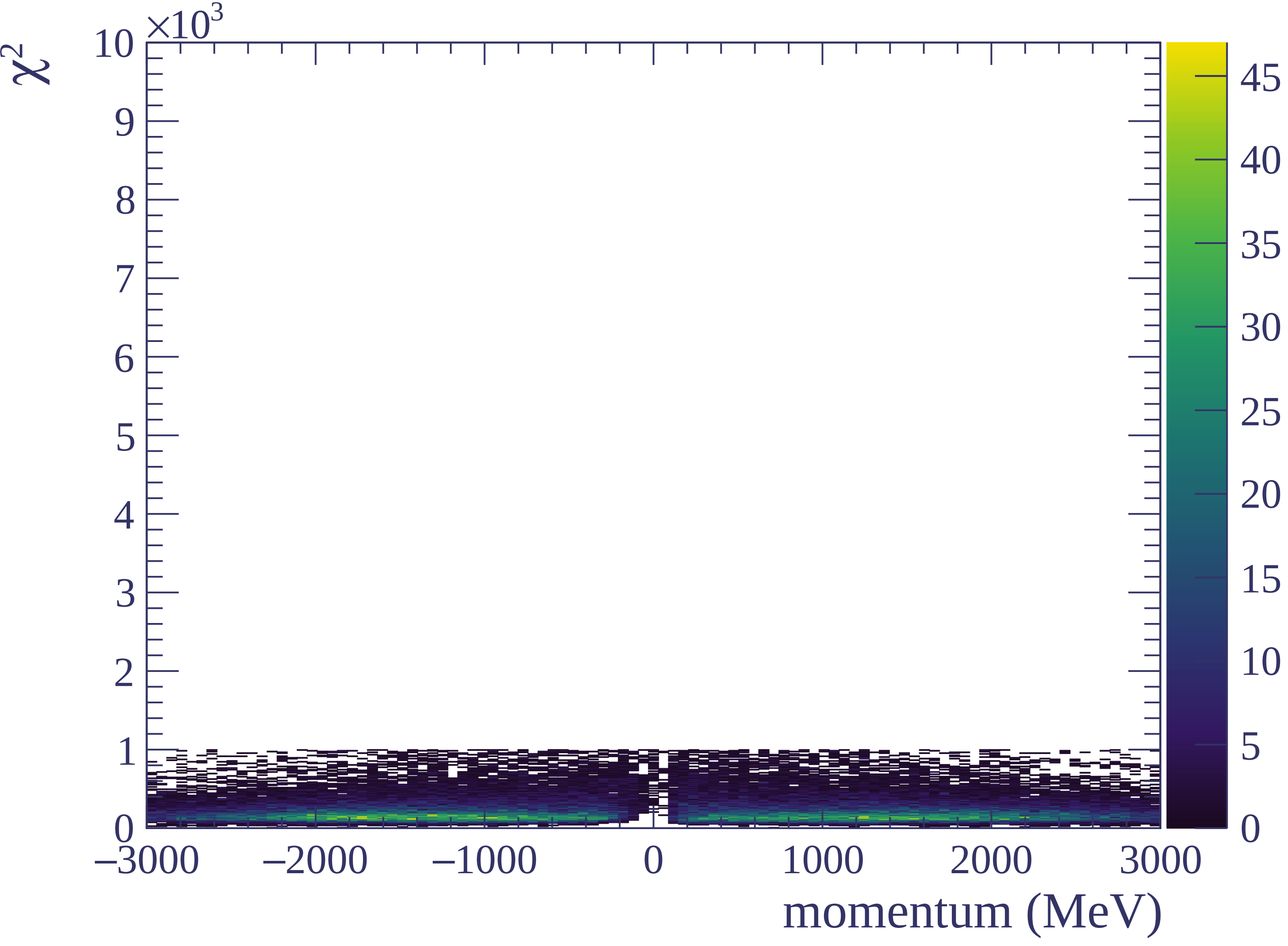


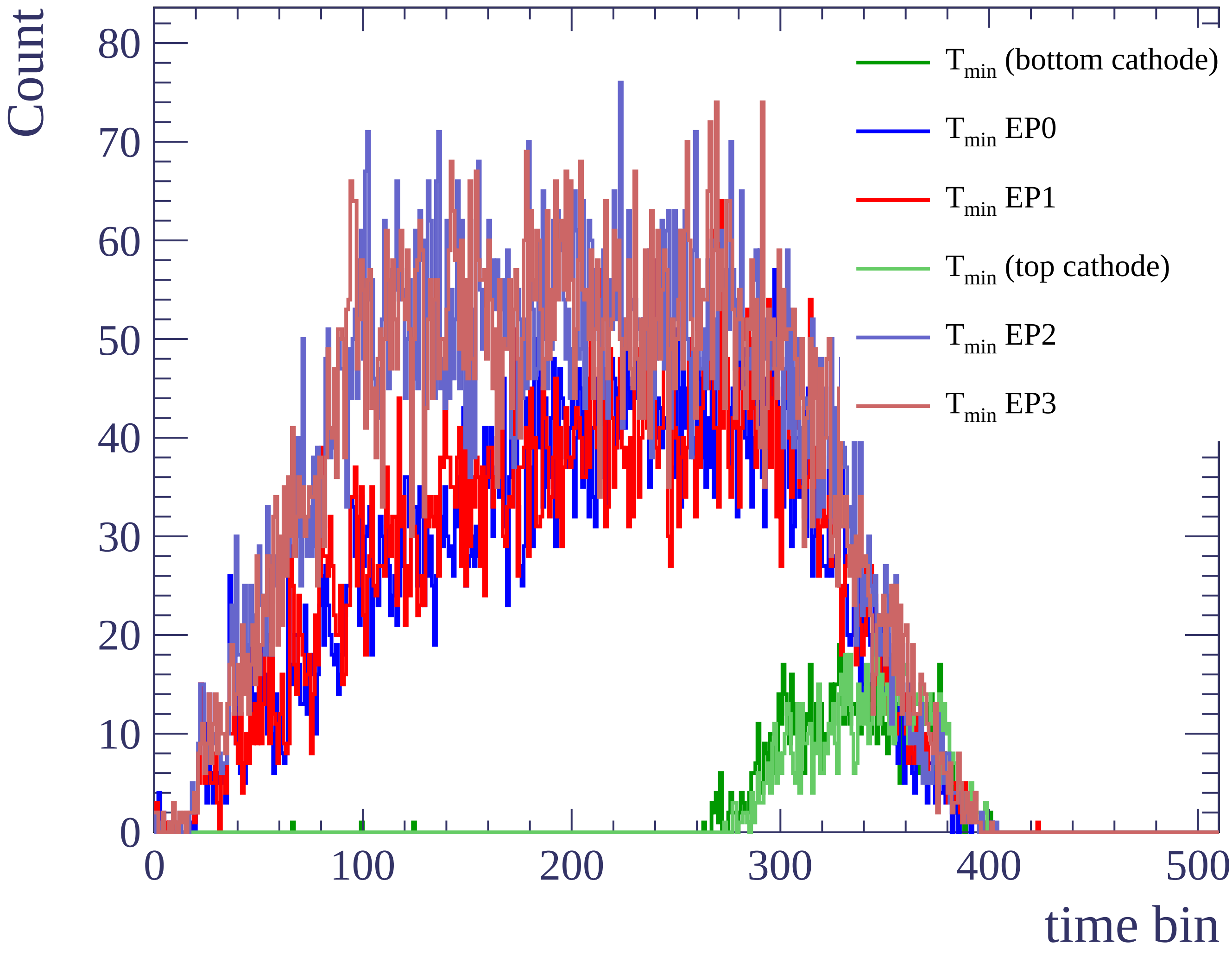
Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

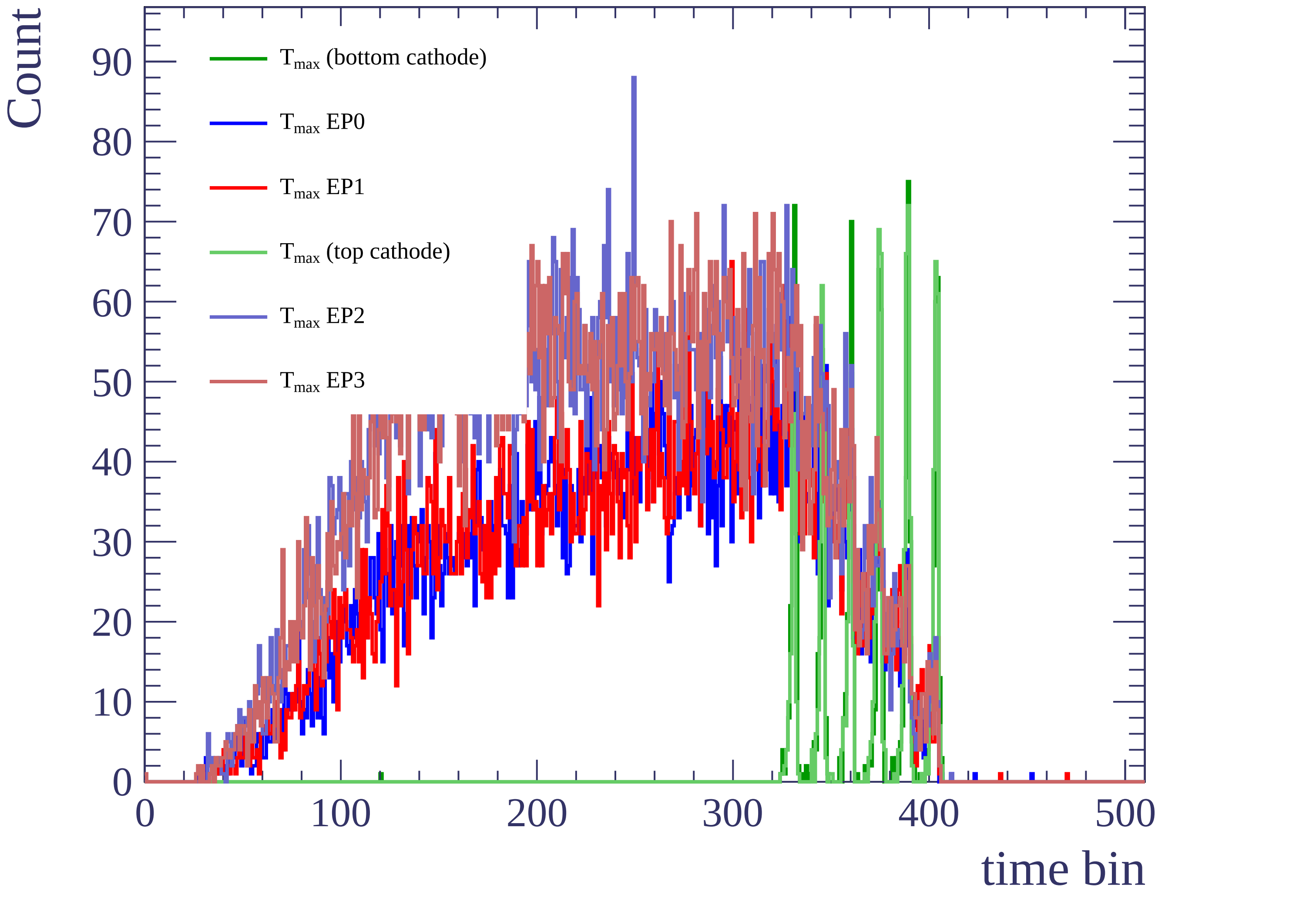


Count
0
1000
2000
3000
4000
5000
0
20
40
60
80
100
120
140
160
track length (cm)


Count
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
10
$\times 10^3$
χ^2


χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
45
40
35
30
25
20
15
10
5
0


Count
time bin
0
10
20
30
40
50
60
70
80
0
100
200
300
400
500
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80
90